Symbolic gate atlas

Exact identities. Explicit phase policies. Native entangler counts.

327 targets × 32 basis panels

Each target has one page with 32 basis panels in alphabetical order. Blue labels denote literal matrix equality. Brown labels denote a projective circuit with its exact scalar factor. Both appear when the projective construction is strictly cheaper. Physical GP operations are forbidden.
E counts native two-qubit primitives, L counts one-qubit operations. Scores are displayed rounded; the archive stores exact expressions. Only an explicit matching certificate establishes a minimum entangler count. A verified upper bound does not establish minimum local cost. Proof conditions, source hashes and attempt histories remain in the archive.
Open cases are shown as open. Proved obstructions preserve the supplied basis and may be conditional for parameterized families. Register constructors use the recorded integer-width or matrix fixtures. These pages do not claim a solution for every register size.
Circuits are drawn by Qiskit's LaTeX drawer. Long circuits continue on successive rows in time order. An explicit $[G]^{\times n}$ box denotes n consecutive identical primitive operations; their full physical count remains in E.
Long exact expressions use numbered η_j labels defined in the expression glossary. No parameter is rounded or dropped.
The separate Clifford volume lists every exact phase in each distinct one-qubit alphabet, with 24 circuits per page. Identical alphabets shared by multiple presets are printed once.

Archive SHA256: 98f19ce6d2d1b4af05dccd65fdda73f0f49f159d08acec0476eb1d0fe6669033

Alphabetical target index

Links lead to the 32-basis page for each target.

ACX 4
B 5
Barenco 6
C3SX 7
C3X 8
C4X 9
Canonical 10
CCP 11
CCX 12
CCXPow 13
CCY 14
CCZ 15
CCZPow 16
CCZS 17
CH 18
CNOT 19
Controlled 20
CP 21
CR 22
CROT 23
CRX 24
CRY 25
CRZ 26
CS 27
CSdg 28
CSWAP 29
CSX 30
CSXdg 31
CT 32
CTdg 33
CU 34
CU1 35
CU3 36
CX 37
CXPow 38
CY 39
CZ 40
CZPow 41
CZPowGate 42
CZS 43
DB 44
DCX 45
Deutsch 46
Diagonal 47
ECR 48
fFredkin 49
fSim 50
FSWAP 51
FSWAPPow 52
Givens 53
GP 54
GPi 55
GPi2 56
H 57
H_q1 58
H_q2 59
H_q3 60
H_q4 61
H_q5 62
H_q6 63
H_q7 64
HPow 65
HS 66
HS_q1 67
HS_q2 68
HS_q3 69
HS_q4 70
HS_q5 71
HS_q6 72
HS_q7 73
HSdg 74
HSdg_q1 75
HSdg_q2 76
HSdg_q3 77
HSdg_q4 78
HSdg_q5 79
HSdg_q6 80
HSdg_q7 81
HSX 82
HSX_q1 83
HSX_q2 84
HSX_q3 85
HSX_q4 86
HSX_q5 87
HSX_q6 88
HSX_q7 89
HSY 90
HSY_q1 91
HSY_q2 92
HSY_q3 93
HSY_q4 94
HSY_q5 95
HSY_q6 96
HSY_q7 97
HX 98
HX_q1 99
HX_q2 100
HX_q3 101
HX_q4 102
HX_q5 103
HX_q6 104
HX_q7 105
HY 106
HY_q1 107
HY_q2 108
HY_q3 109
HY_q4 110
HY_q5 111
HY_q6 112
HY_q7 113
HZ 114
HZ_q1 115
HZ_q2 116
HZ_q3 117
HZ_q4 118
HZ_q5 119
HZ_q6 120
HZ_q7 121
I 122
I_q1 123
I_q2 124
I_q3 125
I_q4 126
I_q5 127
I_q6 128
I_q7 129
Identity 130
IonQMS 131
IonQMSMax 132
IonQZZ 133
iSWAP 134
iSWAP_pow 135
ISWAPPow 136
LinearFunction 137
Magic 138
Margolus 139
MCMT 140
MCPhase 141
MCX 142
MS 143
P 144
Pauli 145
PauliRotation 146
Peres 147
Permutation 148
PhasedFSim 149
PhasedISWAPPow 150
PhasedX 151
PhasedXPow 152
PhasedXZ 153
QFT 154
QFTdg 155
R 156
RC3X 157
RCCX 158
RV 159
Rx 160
Rxmy 161
Rxx 162
Rxy 163
Rxz 164
Ry 165
Ryy 166
Rz 167
Rzx 168
Rzz 169
S 170
S_q1 171
S_q2 172
S_q3 173
S_q4 174
S_q5 175
S_q6 176
S_q7 177
Sdg 178
Sdg_q1 179
Sdg_q2 180
Sdg_q3 181
Sdg_q4 182
Sdg_q5 183
Sdg_q6 184
Sdg_q7 185
SH 186
SH_q1 187
SH_q2 188
SH_q3 189
SH_q4 190
SH_q5 191
SH_q6 192
SH_q7 193
SHX 194
SHX_q1 195
SHX_q2 196
SHX_q3 197
SHX_q4 198
SHX_q5 199
SHX_q6 200
SHX_q7 201
SHY 202
SHY_q1 203
SHY_q2 204
SHY_q3 205
SHY_q4 206
SHY_q5 207
SHY_q6 208
SHY_q7 209
SHZ 210
SHZ_q1 211
SHZ_q2 212
SHZ_q3 213
SHZ_q4 214
SHZ_q5 215
SHZ_q6 216
SHZ_q7 217
SiSWAP 218
SiSWAPdg 219
SqrtH 220
SqrtHdg 221
SSWAP 222
SSWAPdg 223
SWAP 224
SWAPPow 225
SX 226
Sx 227
SX_q1 228
Sx_q1 229
SX_q2 230
Sx_q2 231
SX_q3 232
Sx_q3 233
SX_q4 234
Sx_q4 235
SX_q5 236
Sx_q5 237
SX_q6 238
Sx_q6 239
SX_q7 240
Sx_q7 241
Sxdg 242
Sxdg_q1 243
Sxdg_q2 244
Sxdg_q3 245
Sxdg_q4 246
Sxdg_q5 247
Sxdg_q6 248
Sxdg_q7 249
SxY 250
SxY_q1 251
SxY_q2 252
SxY_q3 253
SxY_q4 254
SxY_q5 255
SxY_q6 256
SxY_q7 257
SxZ 258
SxZ_q1 259
SxZ_q2 260
SxZ_q3 261
SxZ_q4 262
SxZ_q5 263
SxZ_q6 264
SxZ_q7 265
SY 266
Sy 267
SY_q1 268
SY_q2 269
SY_q3 270
SY_q4 271
SY_q5 272
SY_q6 273
SY_q7 274
SYC 275
Sydg 276
T 277
Tdg 278
TK2 279
Toffoli 280
U 281
U1 282
U2 283
U3 284
UC 285
UCRX 286
UCRY 287
UCRZ 288
Unitary 289
WalshHadamard 290
X 291
X_q1 292
X_q2 293
X_q3 294
X_q4 295
X_q5 296
X_q6 297
X_q7 298
XPow 299
XX 300
XX90 301
xx_minus_yy 302
xx_plus_yy 303
XXPow 304
Y 305
Y_q1 306
Y_q2 307
Y_q3 308
Y_q4 309
Y_q5 310
Y_q6 311
Y_q7 312
YPow 313
YY 314
YY90 315
YYPow 316
Z 317
Z_q1 318
Z_q2 319
Z_q3 320
Z_q4 321
Z_q5 322
Z_q6 323
Z_q7 324
ZPow 325
ZZ 326
ZZ90 327
ZZMax 328
ZZPhase 329
ZZPow 330

Native basis key

Every panel preserves its supplied alphabet. Native units and macro counts matter.

Preset	Entangler	Local alphabet	Scope / units
aqt_rxx	Rxx	R, Rz	exact continuous
berkeley_b	B	U3	exact continuous
clifford_only	CX	H, S	Clifford analysis/control; not universal.
clifford_t_cx	CX	H, S, T, Tdg	Dense universal; exact only for representable targets, not arbitrary symbolic angles.
clifford_t_cz	CZ	H, S, T, Tdg	approximate discrete
cx_h_phase	CX	H, P	exact continuous
cx_rz_ry	CX	Rz, Ry	exact continuous
cx_u	CX	U3	exact continuous
cz_rz_rx	CZ	Rz, Rx	exact continuous
cz_u	CZ	U3	exact continuous
google_cz	CZ	PhasedXZ	exact continuous
google_fsim	fSim	PhasedXZ	exact continuous
google_sqrt_iswap	SiSWAP	PhasedXZ	exact continuous
google_syc	SYC	PhasedXZ	exact continuous
ibm_cx	CX	Rz, Sx, X	exact continuous
ibm_cz	CZ	Rz, Sx, X	exact continuous
ibm_ecr	ECR	Rz, Sx, X	exact continuous
ionq_aria	IonQMS	GPi, GPi2	Native turns; interaction theta restricted to [0,1/4] on hardware.
ionq_aria_full	IonQMSMax	GPi, GPi2	exact continuous
ionq_forte	IonQZZ	GPi, GPi2	exact continuous
iqm_cz	CZ	R	R(theta,phi) represents PRX in radians; IQM API units require conversion.
iswap_u	iSWAP	U3	exact continuous
quantinuum_zzmax	ZZMax	PhasedX, Rz	exact continuous
quantinuum_zzphase	ZZPhase	PhasedX, Rz	exact continuous
rigetti_cz	CZ	Rz, Sx, X	Sx differs from native Rx(pi/2) by a tracked global phase.
sqrt_iswap_u	SiSWAP	U3	exact continuous
sqrt_swap_u	SSWAP	U3	exact continuous
tket_tk2	TK2	PhasedX, Rz	Compiler macro count; expand TK2 before comparing physical pulse counts.
xx90_u	XX90	U3	exact continuous
xx_rz_ry	Rxx	Rz, Ry	exact continuous
zx_u	Rzx	U3	exact continuous
zz_u	Rzz	U3	exact continuous

ACX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

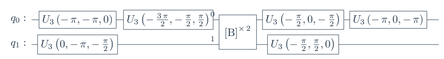
PROJECTIVE E=1 / L=7 / score ≈ 19.226 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

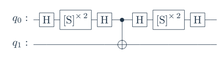
berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — upper bound



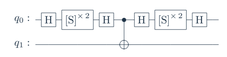
clifford_only

EXACT PHASE E=1 / L=8 / score ≈ 20.000 — proved minimum



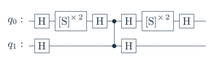
clifford_t_cx

EXACT PHASE E=1 / L=8 / score ≈ 20.000 — proved minimum



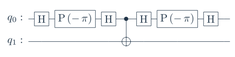
clifford_t_cz

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



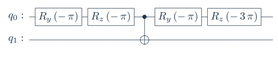
cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.004 — proved minimum



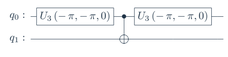
cx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.008 — proved minimum



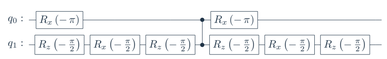
cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.012 — proved minimum



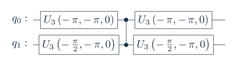
cz_rz_rx

EXACT PHASE E=1 / L=8 / score ≈ 21.216 — proved minimum



cz_u

EXACT PHASE E=1 / L=4 / score ≈ 12.424 — proved minimum



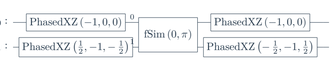
google_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.828 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 21.836 — upper bound



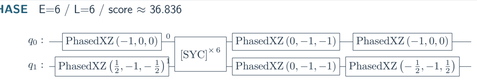
PROJECTIVE E=2 / L=6 / score ≈ 21.236 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

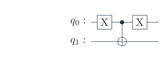
google_syc

EXACT PHASE E=6 / L=6 / score ≈ 36.836 — upper bound



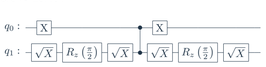
ibm_cx

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



ibm_cz

EXACT PHASE E=1 / L=8 / score ≈ 20.404 — proved minimum



ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.202 — proved minimum

ionq_aria

PROJECTIVE E=1 / L=7 / score ≈ 19.420 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=7 / score ≈ 19.018 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

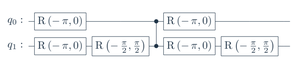
PROJECTIVE E=1 / L=8 / score ≈ 21.818 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.824 — proved minimum



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved minimum



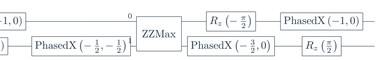
PROJECTIVE E=2 / L=3 / score ≈ 15.018 — proved minimum



$U = \eta_1 C$. Scalar metadata; no physical GP.

quantinuum_zzmax

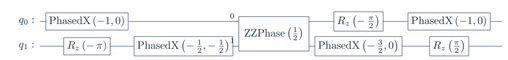
PROJECTIVE E=1 / L=7 / score ≈ 19.022 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

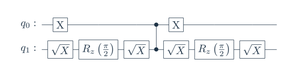
PROJECTIVE E=1 / L=7 / score ≈ 19.224 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=8 / score ≈ 20.404 — proved minimum



sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 21.236 — upper bound



$U = e^{-i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=7 / score ≈ 19.230 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

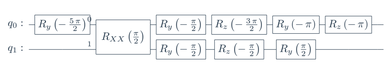
xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.230 — proved minimum



xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 21.418 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

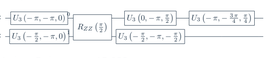
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.026 — upper bound

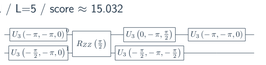


zz_u

EXACT PHASE E=1 / L=5 / score ≈ 15.832 — upper bound



PROJECTIVE E=1 / L=5 / score ≈ 15.032 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.

B

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.904 — proved minimum



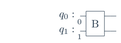
PROJECTIVE E=2 / L=16 / score ≈ 43.858 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum

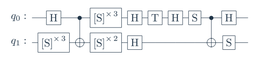


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

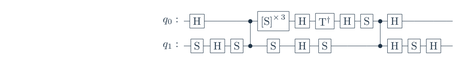
PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved minimum



$U = e^{-\frac{i\pi}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

clifford_t_cz

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved minimum



$U = e^{-\frac{i\pi}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

cx_h_phase

EXACT PHASE E=2 / L=19 / score ≈ 47.220 — proved minimum



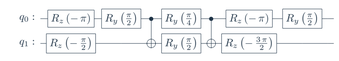
PROJECTIVE E=2 / L=11 / score ≈ 31.212 — proved minimum



$U = e^{-\frac{i\pi}{8}}C$. Scalar metadata; no physical GP.

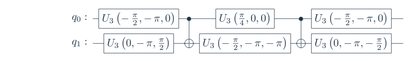
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.416 — proved minimum



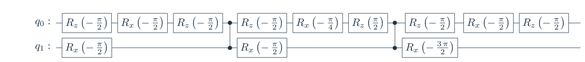
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.436 — proved minimum



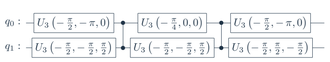
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.624 — proved minimum



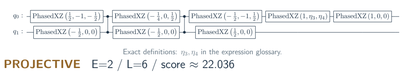
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved minimum



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.709 — proved minimum



Exact definitions: η_0, η_1 in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved minimum



$U = e^{\frac{3i\pi}{8}}C$. Scalar metadata; no physical GP.

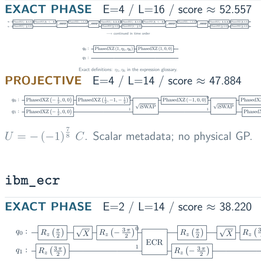
google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 13.220 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 52.557 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 47.884 — upper bound



$U = -(-1)^{\frac{7}{8}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.757 — upper bound



Exact definitions: η_3, η_4 in the expression glossary.

PROJECTIVE E=12 / L=14 / score ≈ 79.084 — upper bound



$U = (-1)^{\frac{3}{8}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.008 — proved minimum



PROJECTIVE E=2 / L=9 / score ≈ 26.402 — proved minimum

$U = \sqrt[4]{-1}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved minimum



PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved minimum



$U = (-1)^{\frac{3}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.220 — proved minimum



PROJECTIVE E=2 / L=14 / score ≈ 37.816 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.646 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 46.246 — upper bound



$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.842 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 45.442 — upper bound

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.442 — upper bound

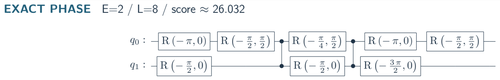


PROJECTIVE E=2 / L=19 / score ≈ 51.042 — upper bound

$U = iC$. Scalar metadata; no physical GP.

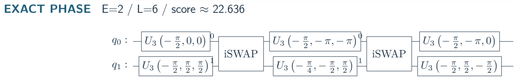
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.032 — proved minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.286 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.028 — proved minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.656 — proved minimum

PROJECTIVE E=2 / L=15 / score ≈ 41.852 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved minimum



PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved minimum



$U = (-1)^{\frac{3}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — upper bound



Exact definitions: η_{13}, η_2 in the expression glossary

PROJECTIVE E=4 / L=14 / score ≈ 49.884 — upper bound



$U = -\sqrt[8]{-1}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — upper bound



Exact definitions: η_{13}, η_2 in the expression glossary

PROJECTIVE E=4 / L=14 / score ≈ 49.884 — upper bound



$U = -(-1)^{\frac{5}{8}}C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.606 — proved minimum



xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — proved minimum



Exact definitions: η_{11}, η_{12} in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved minimum



$U = \eta_{13}C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.672 — proved minimum



PROJECTIVE E=2 / L=30 / score ≈ 74.264 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

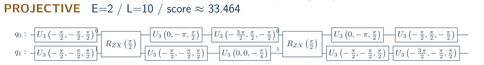
zx_u

EXACT PHASE E=2 / L=12 / score ≈ 37.476 — upper bound



Exact definitions: η_{14}, η_8 in the expression glossary.

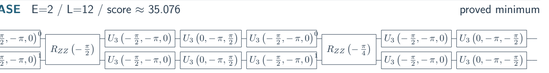
PROJECTIVE E=2 / L=10 / score ≈ 33.464 — upper bound



$U = \sqrt[8]{-1}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.076 — proved minimum



Barenco (α, ϕ, θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=15 / score ≈ 41.250 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=13 / score ≈ 45.678 upper bound



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

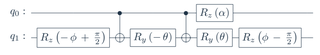
cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 32.816 upper bound



cx_rz_ry

PROJECTIVE E=2 / L=5 / score ≈ 18.410 upper bound



$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.430 upper bound



cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 37.628 upper bound



$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.254 upper bound



google_cz

EXACT PHASE E=2 / L=9 / score ≈ 30.105 upper bound



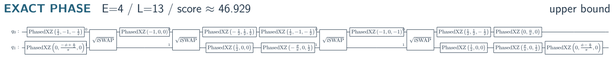
google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 30.113 upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=13 / score ≈ 46.929 upper bound

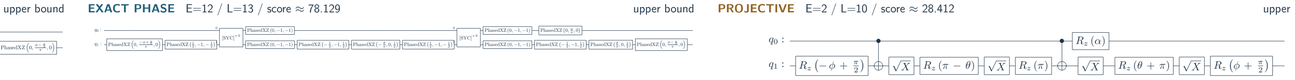


google_syc

EXACT PHASE E=12 / L=13 / score ≈ 78.129 upper bound

ibm_cx

PROJECTIVE E=2 / L=10 / score ≈ 28.412 upper bound



$U = -e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

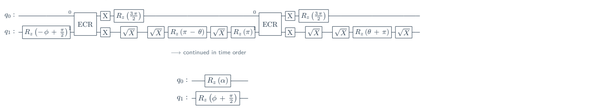
PROJECTIVE E=2 / L=20 / score ≈ 49.220 upper bound



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

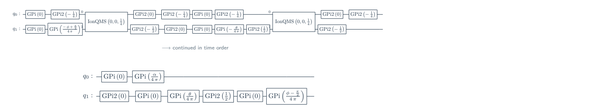
PROJECTIVE E=2 / L=18 / score ≈ 44.816 upper bound



$U = -e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

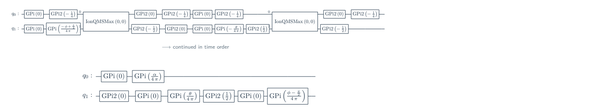
PROJECTIVE E=2 / L=24 / score ≈ 63.311 upper bound



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

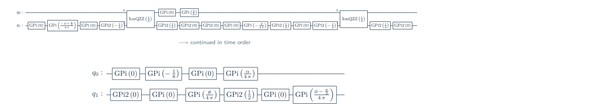
PROJECTIVE E=2 / L=24 / score ≈ 62.507 upper bound



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

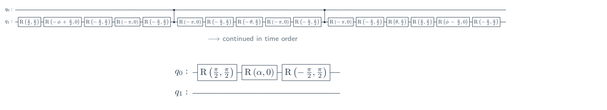
PROJECTIVE E=2 / L=26 / score ≈ 68.107 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

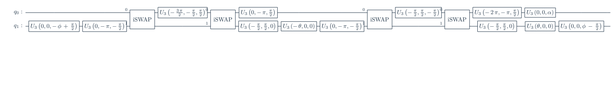
PROJECTIVE E=2 / L=19 / score ≈ 50.876 upper bound



$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 upper bound



quantinuum_zzmax

PROJECTIVE E=2 / L=14 / score ≈ 39.501 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

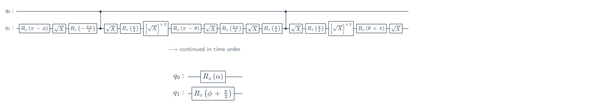
PROJECTIVE E=2 / L=14 / score ≈ 39.905 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

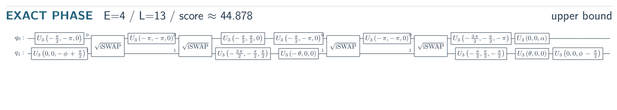
PROJECTIVE E=2 / L=20 / score ≈ 49.220 upper bound



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

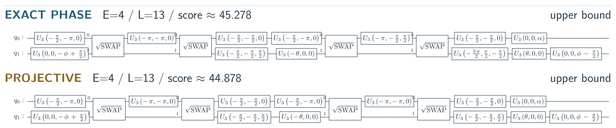
sqrt_iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 44.878 upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 upper bound



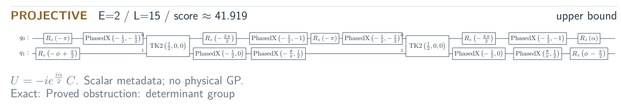
PROJECTIVE E=4 / L=13 / score ≈ 44.878 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

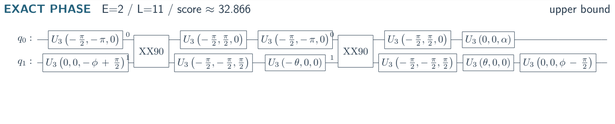
PROJECTIVE E=2 / L=15 / score ≈ 41.919 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

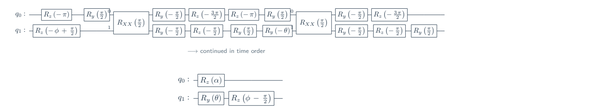
xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 32.866 upper bound



xx_rz_ry

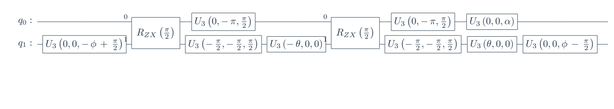
PROJECTIVE E=2 / L=19 / score ≈ 49.242 upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

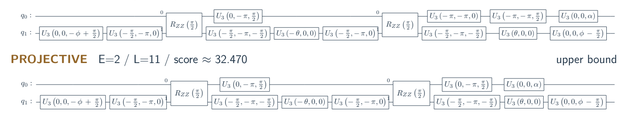
zx_u

EXACT PHASE E=2 / L=9 / score ≈ 28.458 upper bound



zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.476 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 32.470 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

C3SX

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

berkeley_b

Exact: Open: time limit reached
Projective: Open: time limit reached

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz
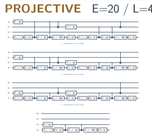
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

cx_h_phase

Exact: Open: time limit reached
Projective: Open: time limit reached

cx_rz_ry

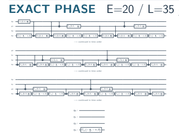
PROJECTIVE E=20 / L=42 / score ≈ 183.684 upper bound



$U = -\sqrt[32]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=20 / L=35 / score ≈ 169.810 upper bound

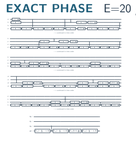


cz_rz_rx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

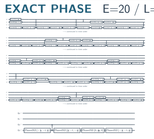
cz_u

EXACT PHASE E=20 / L=75 / score ≈ 258.050 upper bound



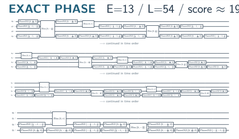
google_cz

EXACT PHASE E=20 / L=75 / score ≈ 268.850 upper bound



google_fsim

EXACT PHASE E=13 / L=54 / score ≈ 197.376 upper bound



google_sqrt_iswap

Exact: Open: time limit reached
Projective: Open: time limit reached

google_syc

Exact: Open: time limit reached
Projective: Open: time limit reached

ibm_cx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria_full

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_forte
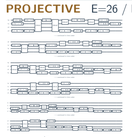
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iqm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iswap_u

PROJECTIVE E=26 / L=80 / score ≈ 295.280 upper bound



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

quantinuum_zzmax

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

quantinuum_zzphase
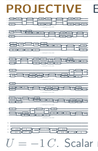
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

sqrt_iswap_u

PROJECTIVE E=40 / L=115 / score ≈ 434.290 upper bound



$U = -1\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

sqrt_swap_u
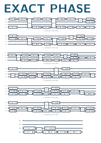
Exact: Open: time limit reached
Projective: Open: time limit reached

tket_tk2

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

xx90_u

EXACT PHASE E=20 / L=95 / score ≈ 314.170 upper bound

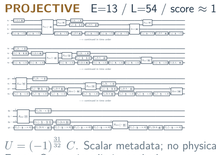


xx_rz_ry

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

zx_u

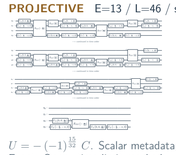
PROJECTIVE E=13 / L=54 / score ≈ 188.750 upper bound



$U = (-1)^{\frac{31}{32}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

zz_u

PROJECTIVE E=13 / L=46 / score ≈ 169.902 upper bound



$U = -(-1)^{\frac{15}{32}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

C3X

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

berkeley_b

Exact: Open: time limit reached
Projective: Open: time limit reached

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

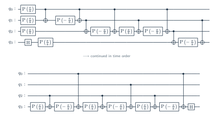
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

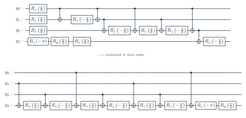
cx_h_phase

EXACT PHASE E=14 / L=17 / score ≈ 99.030 upper bound



cx_rz_ry

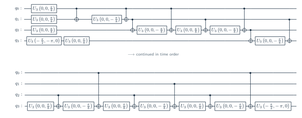
PROJECTIVE E=14 / L=19 / score ≈ 103.438 upper bound



$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=14 / L=17 / score ≈ 99.502 upper bound

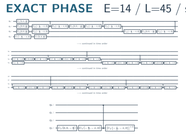


cz_rz_rx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

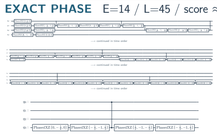
cz_u

EXACT PHASE E=14 / L=45 / score ≈ 161.270 upper bound



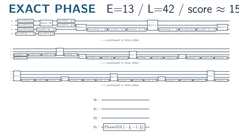
google_cz

EXACT PHASE E=14 / L=45 / score ≈ 167.270 upper bound



google_fsim

EXACT PHASE E=13 / L=42 / score ≈ 156.704 upper bound



google_sqrt_iswap

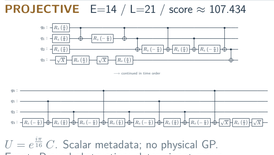
Exact: Open: time limit reached
Projective: Open: time limit reached

google_syc

Exact: Open: time limit reached
Projective: Open: time limit reached

ibm_cx

PROJECTIVE E=14 / L=21 / score ≈ 107.434 upper bound



$U = e^{\frac{15}{16}i}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

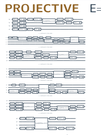
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria_full

PROJECTIVE E=14 / L=104 / score ≈ 294.064 upper bound
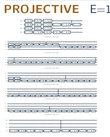


$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iqm_cz

PROJECTIVE E=14 / L=105 / score ≈ 299.420 upper bound
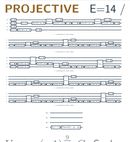


$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

Exact: Open: time limit reached
Projective: Open: time limit reached

quantinuum_zzmax

PROJECTIVE E=14 / L=78 / score ≈ 236.016 upper bound



$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

sqrt_iswap_u

Exact: Open: time limit reached
Projective: Open: time limit reached

sqrt_swap_u

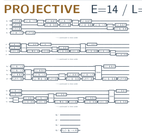
Exact: Open: time limit reached
Projective: Open: time limit reached

tket_tk2

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

xx90_u

PROJECTIVE E=14 / L=59 / score ≈ 200.554 upper bound



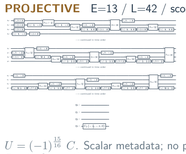
$U = -1\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

xx_rz_ry

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

zx_u

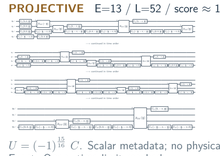
PROJECTIVE E=13 / L=42 / score ≈ 158.078 upper bound



$U = (-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

zz_u

PROJECTIVE E=13 / L=52 / score ≈ 177.738 upper bound



$U = (-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

C4X

5 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

berkeley_b

Exact: Open: time limit reached
Projective: Open: time limit reached

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

cx_h_phase
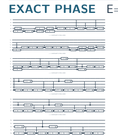
Exact: Open: time limit reached
Projective: Open: time limit reached

cx_rz_ry

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

cx_u

EXACT PHASE E=36 / L=69 / score ≈ 313.214 upper bound



cz_rz_rx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

cz_u

Exact: Open: time limit reached
Projective: Open: time limit reached

google_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

google_fsim

Exact: Open: time limit reached
Projective: Open: time limit reached

google_sqrt_iswap

Exact: Open: time limit reached
Projective: Open: time limit reached

google_syc

Exact: Open: time limit reached
Projective: Open: time limit reached

ibm_cx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria_full

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_forte

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iqm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iswap_u

Exact: Open: time limit reached
Projective: Open: time limit reached

quantinuum_zzmax

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

quantinuum_zzphase

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

sqrt_iswap_u

Exact: Open: time limit reached
Projective: Open: time limit reached

sqrt_swap_u

Exact: Open: time limit reached
Projective: Open: time limit reached

tket_tk2

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

xx90_u

Exact: Open: time limit reached
Projective: Open: time limit reached

xx_rz_ry

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

zx_u

Exact: Open: time limit reached
Projective: Open: time limit reached

zz_u

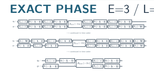
Exact: Open: time limit reached
Projective: Open: time limit reached

Canonical (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — upper bound



PROJECTIVE E=3 / L=20 / score ≈ 56.872 — upper bound



$U = \eta_{15}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 65.102 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

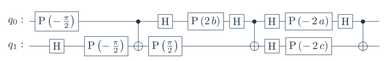
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

PROJECTIVE E=3 / L=12 / score ≈ 36.612 — upper bound



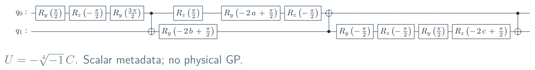
$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — upper bound



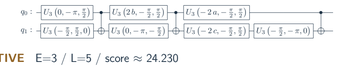
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — upper bound



PROJECTIVE E=3 / L=5 / score ≈ 24.230 — upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — upper bound



PROJECTIVE E=3 / L=11 / score ≈ 35.822 — upper bound



$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 34.660 — upper bound



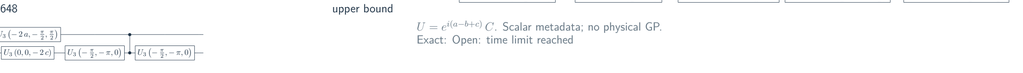
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_cz

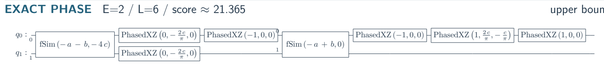
PROJECTIVE E=3 / L=8 / score ≈ 30.439 — upper bound



$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 21.365 — upper bound



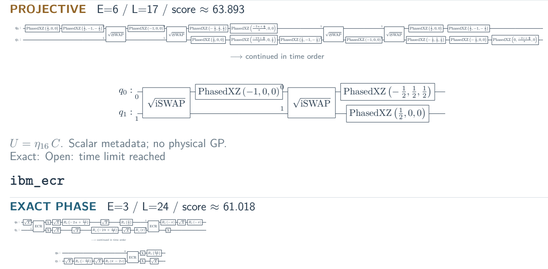
PROJECTIVE E=2 / L=4 / score ≈ 16.693 — upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

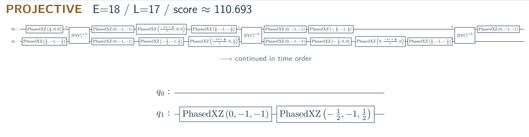
PROJECTIVE E=6 / L=17 / score ≈ 63.893 — upper bound



$U = \eta_{16}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_syc

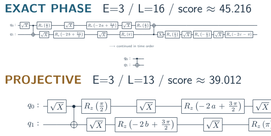
PROJECTIVE E=18 / L=17 / score ≈ 110.693 — upper bound



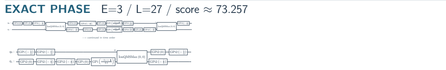
$U = \eta_{17}\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — upper bound



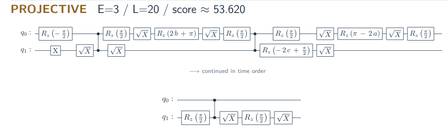
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

PROJECTIVE E=3 / L=20 / score ≈ 53.620 — upper bound



$U = -1\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — upper bound

PROJECTIVE E=3 / L=22 / score ≈ 57.214 — upper bound



$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 74.463 — upper bound



PROJECTIVE E=3 / L=27 / score ≈ 74.263 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 73.257 — upper bound

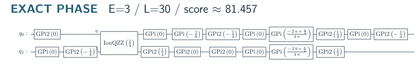
PROJECTIVE E=3 / L=27 / score ≈ 73.057 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=3 / L=30 / score ≈ 81.457 — upper bound



iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — upper bound



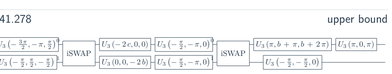
PROJECTIVE E=3 / L=13 / score ≈ 40.452 — upper bound



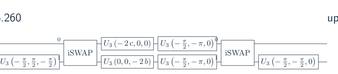
$U = -\left(-1\right)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.278 — upper bound



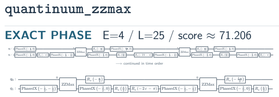
PROJECTIVE E=3 / L=10 / score ≈ 35.260 — upper bound



$U = \frac{\sqrt{2}(1+i)e^{ib}}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 71.206 — upper bound



PROJECTIVE E=3 / L=10 / score ≈ 34.293 — upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 53.049 — upper bound



rigetti_cz

PROJECTIVE E=3 / L=20 / score ≈ 53.620 — upper bound



$U = -1\, C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 67.914 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — upper bound



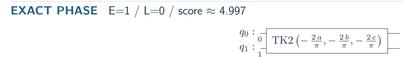
PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = -\sqrt[4]{-1}e^{ic}\, C$. Scalar metadata; no physical GP.

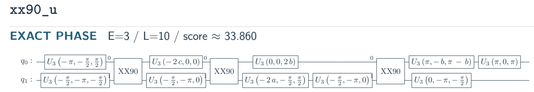
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.997 — conditional minimum



xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 33.860 — upper bound



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — upper bound



$U = ie^{-ib}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=41 / score ≈ 98.888 — upper bound



PROJECTIVE E=3 / L=31 / score ≈ 80.068 — upper bound



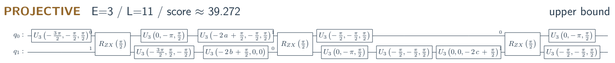
$U = \eta_{15}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — upper bound



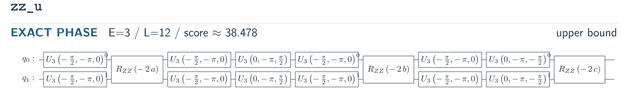
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — upper bound

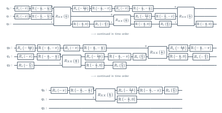


CCP (θ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

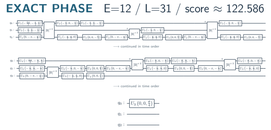
PROJECTIVE E=6 / L=37 / score ≈ 108.122 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

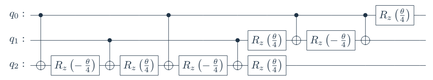
cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — upper bound



cx_rz_ry

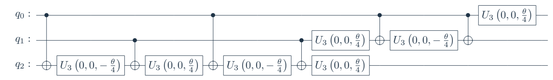
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

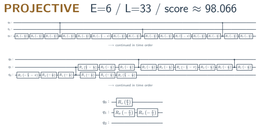
cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — upper bound



cz_rz_rx

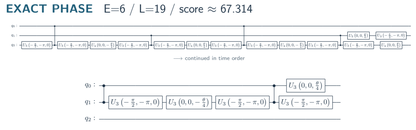
PROJECTIVE E=6 / L=33 / score ≈ 98.066 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

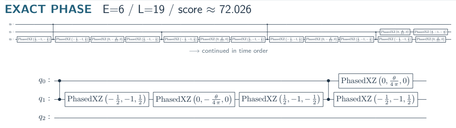
cz_u

EXACT PHASE E=6 / L=19 / score ≈ 67.314 — upper bound



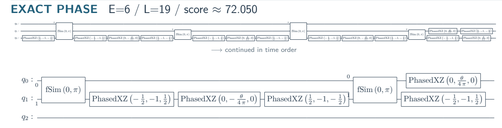
google_cz

EXACT PHASE E=6 / L=19 / score ≈ 72.026 — upper bound



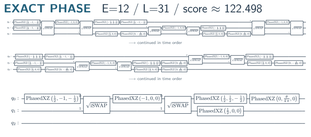
google_fsim

EXACT PHASE E=6 / L=19 / score ≈ 72.050 — upper bound



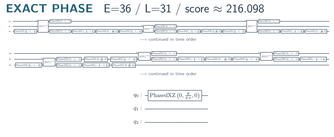
google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 122.498 — upper bound



google_syc

EXACT PHASE E=36 / L=31 / score ≈ 216.098 — upper bound



ibm_cx

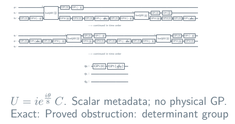
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

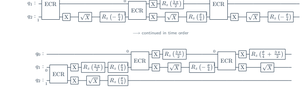
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=30 / score ≈ 87.824 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=44 / score ≈ 128.436 — upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=44 / score ≈ 126.024 — upper bound

$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 142.824 — upper bound

$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

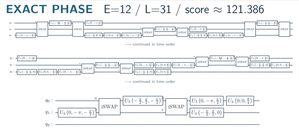
PROJECTIVE E=6 / L=45 / score ≈ 127.380 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 121.386 — upper bound



quantinuum_zzmax

PROJECTIVE E=6 / L=31 / score ≈ 94.686 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=31 / score ≈ 95.898 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

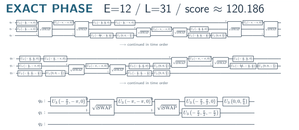
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

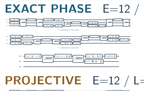
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — upper bound



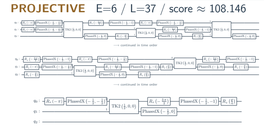
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

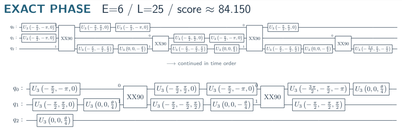
PROJECTIVE E=6 / L=37 / score ≈ 108.146 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

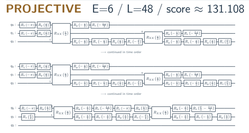
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — upper bound



xx_rz_ry

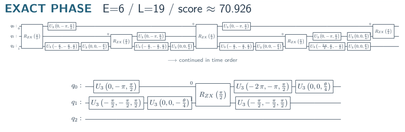
PROJECTIVE E=6 / L=48 / score ≈ 131.108 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

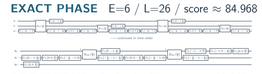
zx_u

EXACT PHASE E=6 / L=19 / score ≈ 70.926 — upper bound



zz_u

EXACT PHASE E=6 / L=26 / score ≈ 84.968 — upper bound



PROJECTIVE E=6 / L=25 / score ≈ 82.962 — upper bound



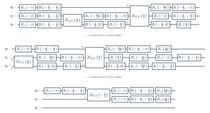
$U = -iC$. Scalar metadata; no physical GP.

CCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

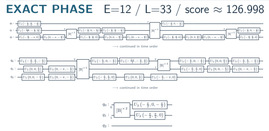
PROJECTIVE E=5 / L=40 / score ≈ 112.926 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — upper bound

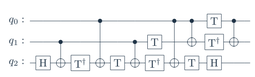


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

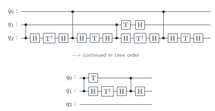
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — proved minimum



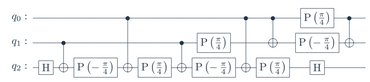
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved minimum



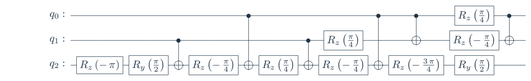
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — proved minimum



cx_rz_ry

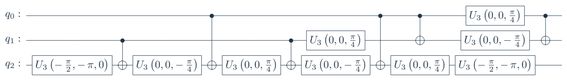
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group
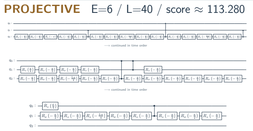
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 45.254 — proved minimum



cz_rz_rx

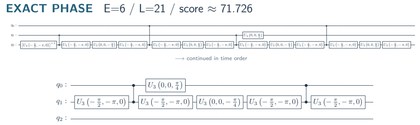
PROJECTIVE E=6 / L=40 / score ≈ 113.280 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

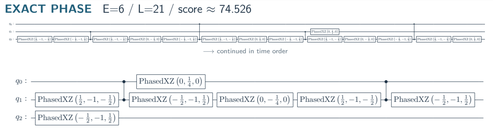
cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.726 — proved minimum



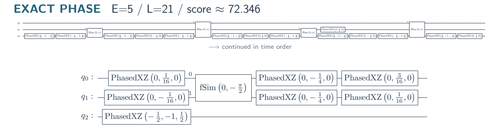
google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 — proved minimum



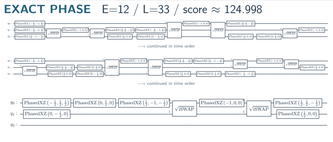
google_fsim

EXACT PHASE E=5 / L=21 / score ≈ 72.346 — upper bound



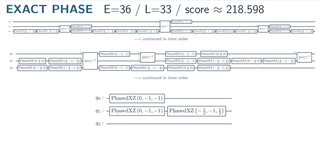
google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound



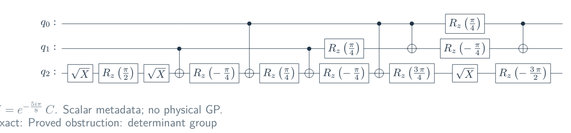
google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — upper bound



ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 — proved minimum



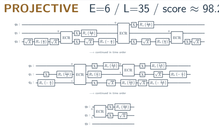
$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

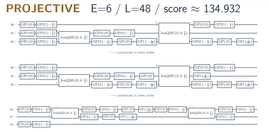
PROJECTIVE E=6 / L=35 / score ≈ 98.228 — proved minimum



$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

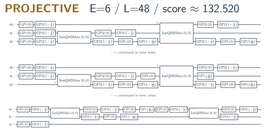
PROJECTIVE E=6 / L=48 / score ≈ 134.932 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

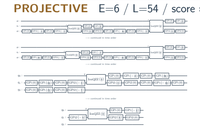
PROJECTIVE E=6 / L=48 / score ≈ 132.520 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

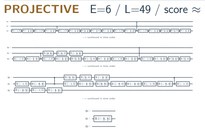
PROJECTIVE E=6 / L=54 / score ≈ 149.320 — upper bound



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=33 / score ≈ 118.198 — upper bound



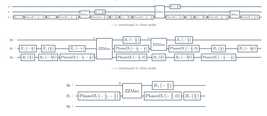
PROJECTIVE E=10 / L=31 / score ≈ 114.186 — upper bound



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

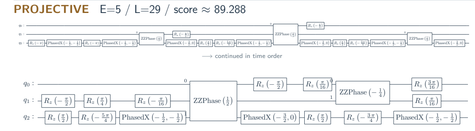
PROJECTIVE E=6 / L=35 / score ≈ 103.498 — proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

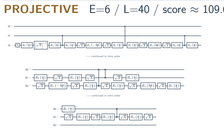
PROJECTIVE E=5 / L=29 / score ≈ 89.288 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

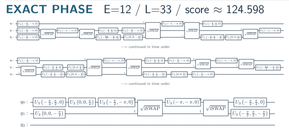
PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved minimum



$U = e^{\frac{1\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

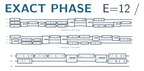
sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound



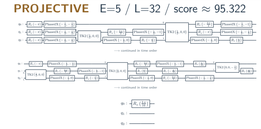
PROJECTIVE E=12 / L=33 / score ≈ 124.598 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

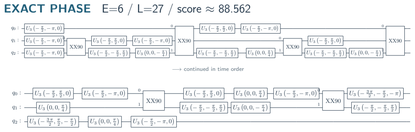
PROJECTIVE E=5 / L=32 / score ≈ 95.322 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

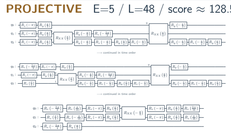
xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — proved minimum



xx_rz_ry

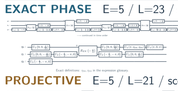
PROJECTIVE E=5 / L=48 / score ≈ 128.506 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=23 / score ≈ 76.548 — upper bound



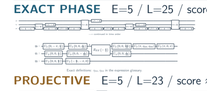
PROJECTIVE E=5 / L=21 / score ≈ 72.536 — upper bound



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=25 / score ≈ 80.160 — upper bound



PROJECTIVE E=5 / L=23 / score ≈ 76.148 — upper bound



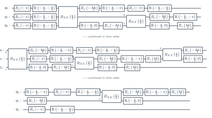
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CCXPow $(t, global_{shift})$

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

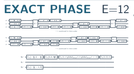
PROJECTIVE E=6 / L=41 / score ≈ 116.934 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=35 / score ≈ 131.010 — upper bound



PROJECTIVE E=12 / L=33 / score ≈ 126.998 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

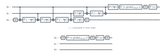
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=17 / score ≈ 60.822 — upper bound



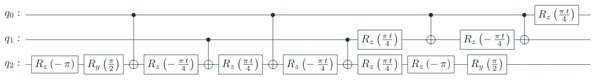
PROJECTIVE E=6 / L=9 / score ≈ 44.814 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=6 / L=11 / score ≈ 49.222 — upper bound



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=11 / score ≈ 49.266 — upper bound



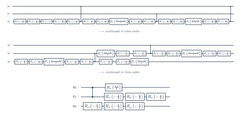
PROJECTIVE E=6 / L=9 / score ≈ 45.254 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

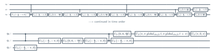
PROJECTIVE E=6 / L=38 / score ≈ 108.876 — upper bound



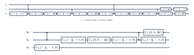
$U = -e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=23 / score ≈ 75.738 — upper bound



PROJECTIVE E=6 / L=21 / score ≈ 71.726 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=6 / L=23 / score ≈ 78.538 — upper bound



PROJECTIVE E=6 / L=21 / score ≈ 74.526 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=6 / L=23 / score ≈ 78.562 — upper bound



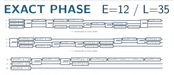
PROJECTIVE E=6 / L=21 / score ≈ 74.550 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — upper bound



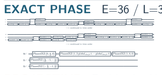
PROJECTIVE E=12 / L=33 / score ≈ 124.998 — upper bound



$U = ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=36 / L=35 / score ≈ 222.610 — upper bound



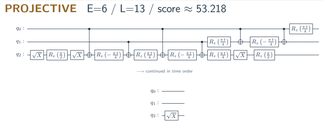
PROJECTIVE E=36 / L=33 / score ≈ 218.598 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

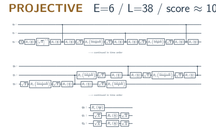
PROJECTIVE E=6 / L=13 / score ≈ 53.218 — upper bound



$U = e^{\frac{i\pi t\left(8 g l o b a l_{shift}+1\right)}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

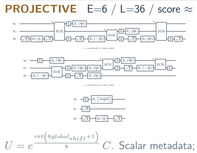
PROJECTIVE E=6 / L=38 / score ≈ 105.642 — upper bound



$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

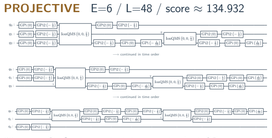
PROJECTIVE E=6 / L=36 / score ≈ 100.228 — upper bound



$U = e^{\frac{i\pi t\left(8 g l o b a l_{shift}+1\right)}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

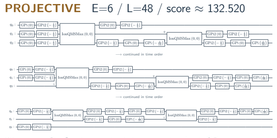
PROJECTIVE E=6 / L=48 / score ≈ 134.932 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

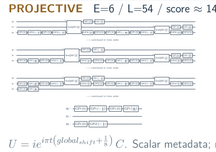
PROJECTIVE E=6 / L=48 / score ≈ 132.520 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

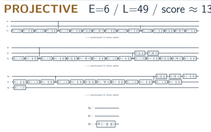
PROJECTIVE E=6 / L=54 / score ≈ 149.320 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

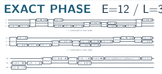
PROJECTIVE E=6 / L=49 / score ≈ 136.196 — upper bound



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.810 — upper bound



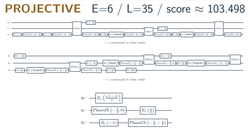
PROJECTIVE E=12 / L=33 / score ≈ 125.798 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

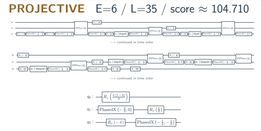
PROJECTIVE E=6 / L=35 / score ≈ 103.498 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

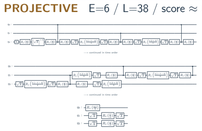
PROJECTIVE E=6 / L=35 / score ≈ 104.710 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

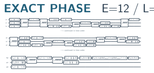
PROJECTIVE E=6 / L=38 / score ≈ 105.642 — upper bound



$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 128.610 — upper bound



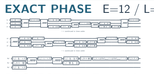
PROJECTIVE E=12 / L=33 / score ≈ 124.598 — upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — upper bound



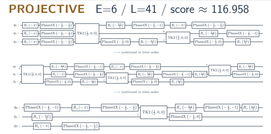
PROJECTIVE E=12 / L=33 / score ≈ 124.598 — upper bound



$U = -ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=41 / score ≈ 116.958 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=29 / score ≈ 92.574 — upper bound



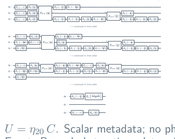
PROJECTIVE E=6 / L=27 / score ≈ 88.562 — upper bound



$U = -e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=6 / L=52 / score ≈ 139.516 — upper bound



$U = \eta_{20}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=23 / score ≈ 79.350 — upper bound



PROJECTIVE E=6 / L=21 / score ≈ 75.338 — upper bound



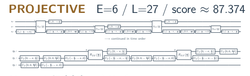
$U = -e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=30 / score ≈ 93.392 — upper bound



PROJECTIVE E=6 / L=27 / score ≈ 87.374 — upper bound



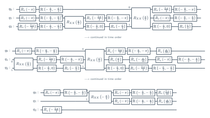
$U = -ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

CCY

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

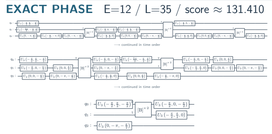
PROJECTIVE E=5 / L=41 / score ≈ 115.328 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=35 / score ≈ 131.410 — upper bound

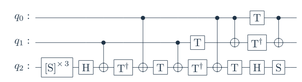


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

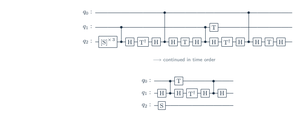
clifford_t_cx

EXACT PHASE E=6 / L=13 / score ≈ 50.000 — proved minimum



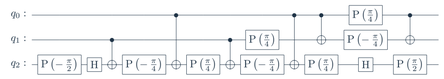
clifford_t_cz

EXACT PHASE E=6 / L=23 / score ≈ 70.000 — proved minimum



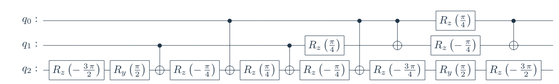
cx_h_phase

EXACT PHASE E=6 / L=11 / score ≈ 49.218 — proved minimum



cx_rz_ry

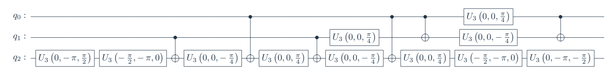
PROJECTIVE E=6 / L=11 / score ≈ 49.622 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

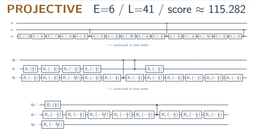
cx_u

EXACT PHASE E=6 / L=11 / score ≈ 49.666 — proved minimum



cz_rz_rx

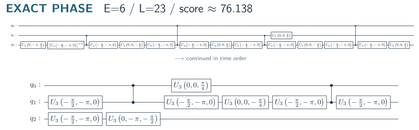
PROJECTIVE E=6 / L=41 / score ≈ 115.282 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

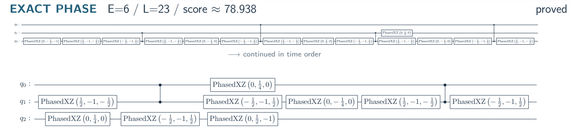
cz_u

EXACT PHASE E=6 / L=23 / score ≈ 76.138 — proved minimum



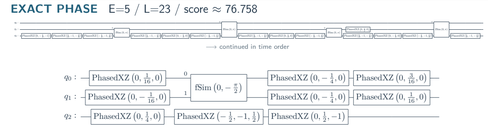
google_cz

EXACT PHASE E=6 / L=23 / score ≈ 78.938 — proved minimum



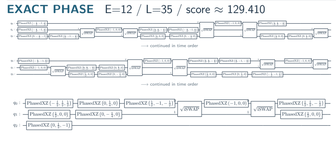
google_fsim

EXACT PHASE E=5 / L=23 / score ≈ 76.758 — upper bound



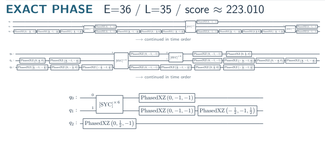
google_sqrt_iswap

EXACT PHASE E=12 / L=35 / score ≈ 129.410 — upper bound



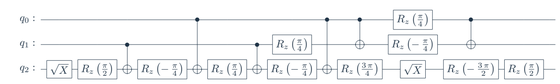
google_syc

EXACT PHASE E=36 / L=35 / score ≈ 223.010 — upper bound



ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.420 — proved minimum



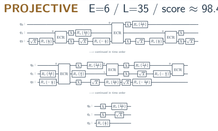
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

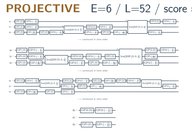
PROJECTIVE E=6 / L=35 / score ≈ 98.430 — proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

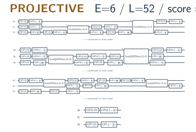
PROJECTIVE E=6 / L=52 / score ≈ 144.140 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

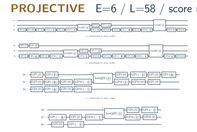
PROJECTIVE E=6 / L=52 / score ≈ 141.728 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

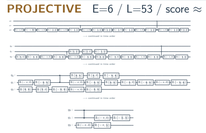
PROJECTIVE E=6 / L=58 / score ≈ 158.528 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

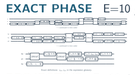
PROJECTIVE E=6 / L=53 / score ≈ 145.012 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=35 / score ≈ 122.610 — upper bound



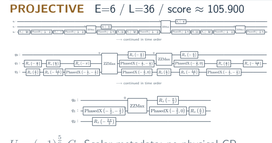
PROJECTIVE E=10 / L=33 / score ≈ 118.598 — upper bound



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

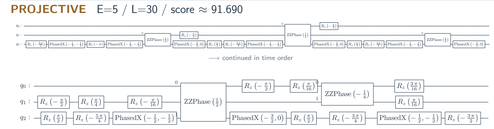
PROJECTIVE E=6 / L=36 / score ≈ 105.900 — proved minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=5 / L=30 / score ≈ 91.690 — upper bound



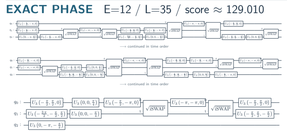
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

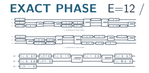
sqrt_iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — upper bound

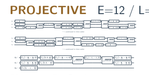


sqrt_swap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.410 — upper bound



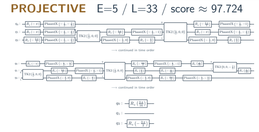
PROJECTIVE E=12 / L=35 / score ≈ 129.010 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

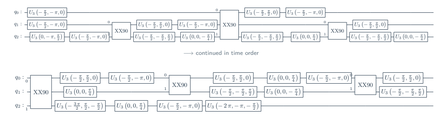
PROJECTIVE E=5 / L=33 / score ≈ 97.724 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

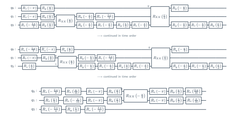
xx90_u

EXACT PHASE E=6 / L=29 / score ≈ 92.974 — proved minimum



xx_rz_ry

PROJECTIVE E=5 / L=49 / score ≈ 130.908 — upper bound



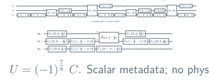
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=25 / score ≈ 80.960 — upper bound



PROJECTIVE E=5 / L=23 / score ≈ 76.948 — upper bound



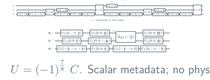
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=27 / score ≈ 84.572 — upper bound



PROJECTIVE E=5 / L=25 / score ≈ 80.560 — upper bound



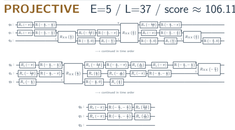
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CCZ

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

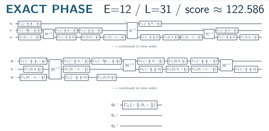
PROJECTIVE E=5 / L=37 / score ≈ 106.116 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — upper bound



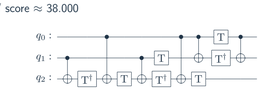
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

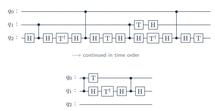
clifford_t_cx

EXACT PHASE E=6 / L=7 / score ≈ 38.000 — proved minimum



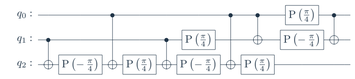
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved minimum



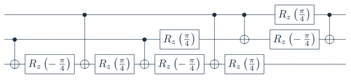
cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — proved minimum



cx_rz_ry

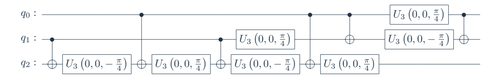
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — proved minimum



cz_rz_rx

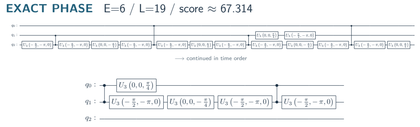
PROJECTIVE E=6 / L=36 / score ≈ 104.672 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

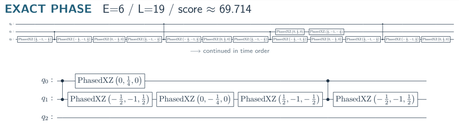
cz_u

EXACT PHASE E=6 / L=19 / score ≈ 67.314 — proved minimum



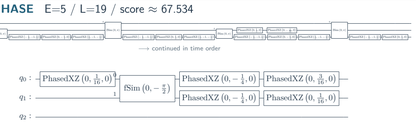
google_cz

EXACT PHASE E=6 / L=19 / score ≈ 69.714 — proved minimum



google_fsim

EXACT PHASE E=5 / L=19 / score ≈ 67.534 — upper bound

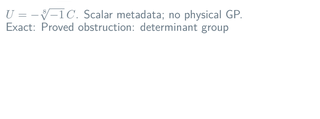


google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — upper bound

google_syc

EXACT PHASE E=36 / L=31 / score ≈ 213.786 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cx

PROJECTIVE E=6 / L=7 / score ≈ 40.814 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

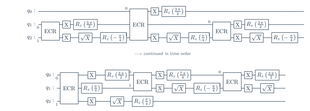
PROJECTIVE E=6 / L=36 / score ≈ 101.440 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

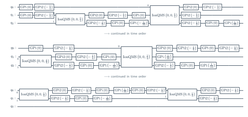
PROJECTIVE E=6 / L=30 / score ≈ 87.824 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

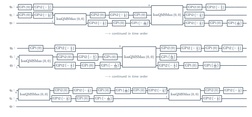
PROJECTIVE E=6 / L=44 / score ≈ 126.124 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

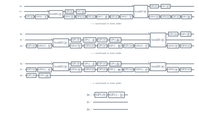
PROJECTIVE E=6 / L=44 / score ≈ 123.712 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 140.512 — upper bound



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

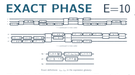
PROJECTIVE E=6 / L=45 / score ≈ 127.380 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=31 / score ≈ 113.786 — upper bound



PROJECTIVE E=10 / L=29 / score ≈ 109.774 — upper bound



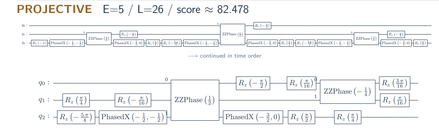
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

Exact: Proved obstruction: determinant group
Projective: Open: construction not found

quantinuum_zzphase

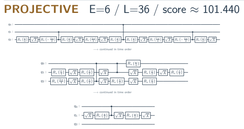
PROJECTIVE E=5 / L=26 / score ≈ 82.478 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

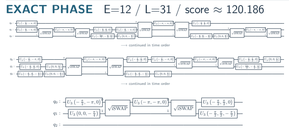
PROJECTIVE E=6 / L=36 / score ≈ 101.440 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

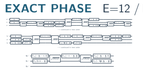
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — upper bound



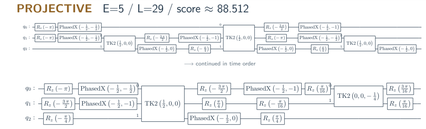
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

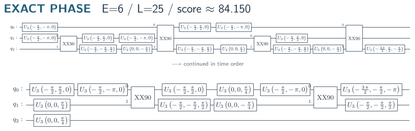
PROJECTIVE E=5 / L=29 / score ≈ 88.512 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

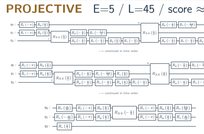
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — proved minimum



xx_rz_ry

PROJECTIVE E=5 / L=45 / score ≈ 122.100 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=21 / score ≈ 72.136 — upper bound



PROJECTIVE E=5 / L=19 / score ≈ 68.124 — upper bound



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=23 / score ≈ 75.748 — upper bound



PROJECTIVE E=5 / L=21 / score ≈ 71.736 — upper bound



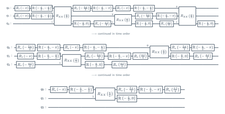
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CCZPow $(t, global_{shift})$

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

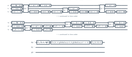
PROJECTIVE E=6 / L=37 / score ≈ 108.122 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.598 — upper bound



PROJECTIVE E=12 / L=31 / score ≈ 122.586 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

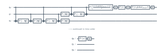
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

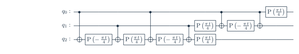
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=14 / score ≈ 54.820 — upper bound



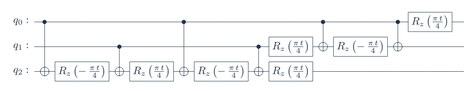
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

cx_rz_ry

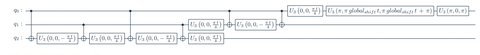
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



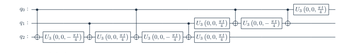
$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=9 / score ≈ 44.854 — upper bound



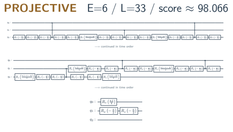
PROJECTIVE E=6 / L=7 / score ≈ 40.842 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

cz_rz_rx

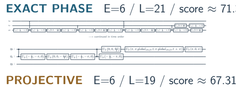
PROJECTIVE E=6 / L=33 / score ≈ 98.066 — upper bound



$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.326 — upper bound



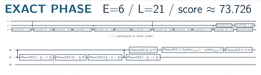
PROJECTIVE E=6 / L=19 / score ≈ 67.314 — upper bound



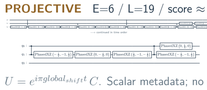
$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=6 / L=21 / score ≈ 73.726 — upper bound



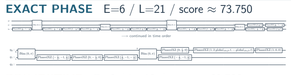
PROJECTIVE E=6 / L=19 / score ≈ 69.714 — upper bound



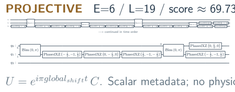
$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 73.750 — upper bound



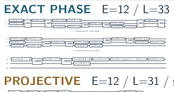
PROJECTIVE E=6 / L=19 / score ≈ 69.738 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.198 — upper bound



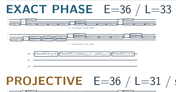
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



$U = ie^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 217.798 — upper bound



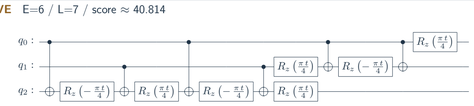
PROJECTIVE E=36 / L=31 / score ≈ 213.786 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

ibm_cx

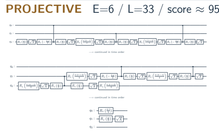
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

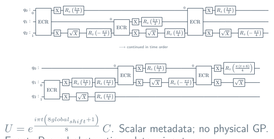
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = \eta_{21}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_eagle_cz

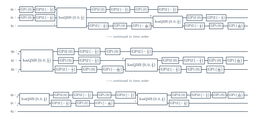
PROJECTIVE E=6 / L=30 / score ≈ 87.824 — upper bound



$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

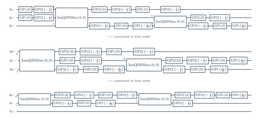
PROJECTIVE E=6 / L=44 / score ≈ 126.124 — upper bound



$U = \eta_{20}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

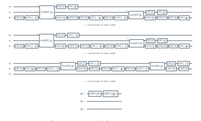
PROJECTIVE E=6 / L=44 / score ≈ 123.712 — upper bound



$U = \eta_{20}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

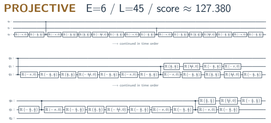
PROJECTIVE E=6 / L=50 / score ≈ 140.512 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

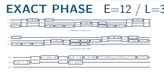
PROJECTIVE E=6 / L=45 / score ≈ 127.380 — upper bound



$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.398 — upper bound



PROJECTIVE E=12 / L=31 / score ≈ 121.386 — upper bound



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

quantinuum_zzmax

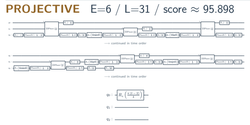
PROJECTIVE E=6 / L=31 / score ≈ 94.686 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

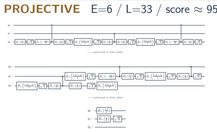
PROJECTIVE E=6 / L=31 / score ≈ 95.898 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

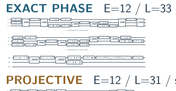
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = \eta_{21}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.198 — upper bound



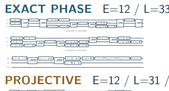
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



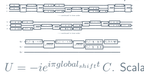
$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — upper bound



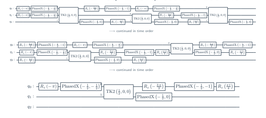
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



$U = -ie^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=37 / score ≈ 108.146 — upper bound



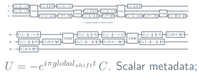
$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.162 — upper bound



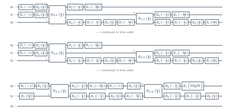
PROJECTIVE E=6 / L=25 / score ≈ 84.150 — upper bound



$U = -e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=6 / L=48 / score ≈ 131.108 — upper bound



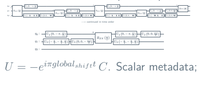
$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 74.938 — upper bound



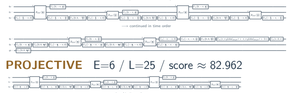
PROJECTIVE E=6 / L=19 / score ≈ 70.926 — upper bound



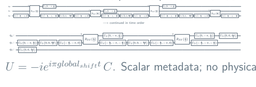
$U = -e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 88.980 — upper bound



PROJECTIVE E=6 / L=25 / score ≈ 82.962 — upper bound



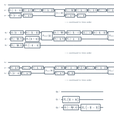
$U = -ie^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

CCZS (θ, ϕ, γ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

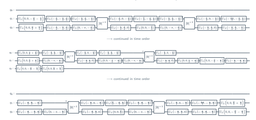
PROJECTIVE E=6 / L=51 / score ≈ 139.370 — upper bound



$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=47 / score ≈ 162.082 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

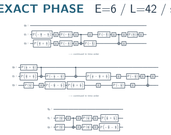
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

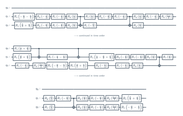
cx_h_phase

EXACT PHASE E=6 / L=42 / score ≈ 114.044 — upper bound



cx_rz_ry

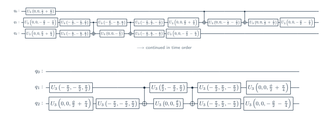
PROJECTIVE E=6 / L=42 / score ≈ 118.084 — upper bound



$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

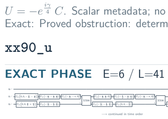
cx_u

EXACT PHASE E=6 / L=23 / score ≈ 80.338 — upper bound



cz_rz_rx

PROJECTIVE E=6 / L=53 / score ≈ 142.306 — upper bound



$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

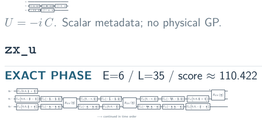
cz_u

EXACT PHASE E=6 / L=35 / score ≈ 106.810 — upper bound



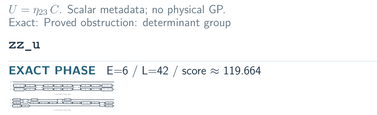
google_cz

EXACT PHASE E=6 / L=35 / score ≈ 110.164 — upper bound



google_fsim

EXACT PHASE E=4 / L=15 / score ≈ 55.539 — upper bound

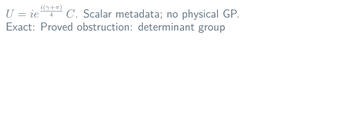


google_sqrt_iswap

EXACT PHASE E=12 / L=47 / score ≈ 160.636 — upper bound

google_syc

EXACT PHASE E=36 / L=47 / score ≈ 254.236 — upper bound



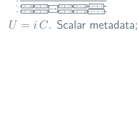
ibm_cx

PROJECTIVE E=6 / L=34 / score ≈ 97.236 — upper bound

$U = -e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=67 / score ≈ 165.660 — upper bound



$U = -e^{\frac{5i}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=54 / score ≈ 137.640 — upper bound

$U = -e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=72 / score ≈ 194.534 — upper bound

$U = \eta_{22} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=72 / score ≈ 192.122 — upper bound

$U = \eta_{22} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=78 / score ≈ 208.922 — upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=73 / score ≈ 191.692 — upper bound

$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=8 / L=31 / score ≈ 105.586 — upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=49 / score ≈ 134.803 — upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=49 / score ≈ 136.015 — upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=67 / score ≈ 165.660 — upper bound

$U = -e^{\frac{5i}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=47 / score ≈ 159.682 — upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=47 / score ≈ 160.082 — upper bound

PROJECTIVE E=12 / L=47 / score ≈ 159.682 — upper bound

$U = -i C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=4 / L=20 / score ≈ 64.397 — upper bound

$U = \eta_{23} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=41 / score ≈ 123.646 — upper bound

xx_rz_ry

PROJECTIVE E=6 / L=81 / score ≈ 203.774 — upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=35 / score ≈ 110.422 — upper bound

zz_u

EXACT PHASE E=6 / L=42 / score ≈ 119.664 — upper bound

PROJECTIVE E=6 / L=41 / score ≈ 117.658 — upper bound

$U = i C$. Scalar metadata; no physical GP.

CH

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

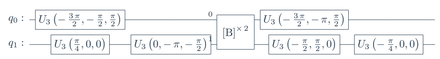
PROJECTIVE E=1 / L=7 / score ≈ 20.426 upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 upper bound

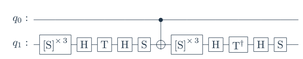


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

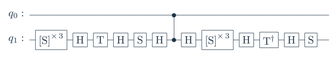
clifford_t_cx

EXACT PHASE E=1 / L=14 / score ≈ 32.000 proved minimum



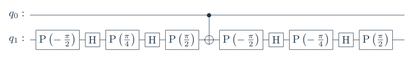
clifford_t_cz

EXACT PHASE E=1 / L=16 / score ≈ 36.000 proved minimum



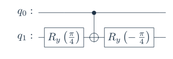
cx_h_phase

EXACT PHASE E=1 / L=10 / score ≈ 25.612 proved minimum



cx_rz_ry

EXACT PHASE E=1 / L=2 / score ≈ 8.804 proved minimum



cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.812 proved minimum



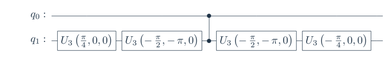
cz_rz_rx

EXACT PHASE E=1 / L=8 / score ≈ 21.816 proved minimum



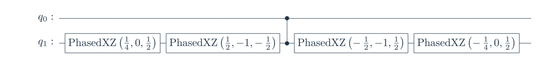
cz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 proved minimum



google_cz

EXACT PHASE E=1 / L=4 / score ≈ 14.024 proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.028 upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 23.036 upper bound



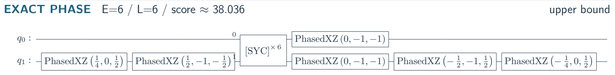
PROJECTIVE E=2 / L=6 / score ≈ 22.436 upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

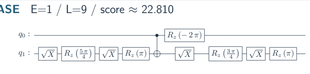
google_syc

EXACT PHASE E=6 / L=6 / score ≈ 38.036 upper bound



ibm_cx

EXACT PHASE E=1 / L=9 / score ≈ 22.810 proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.808 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=1 / L=14 / score ≈ 33.212 proved minimum



PROJECTIVE E=1 / L=13 / score ≈ 31.412 proved minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=12 / score ≈ 29.010 proved minimum

ionq_aria

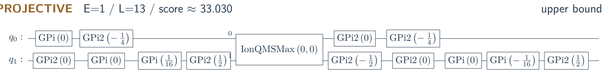
PROJECTIVE E=1 / L=13 / score ≈ 33.432 upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

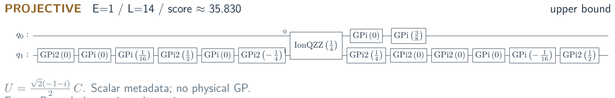
PROJECTIVE E=1 / L=13 / score ≈ 33.030 upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

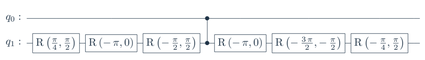
PROJECTIVE E=1 / L=14 / score ≈ 35.830 upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

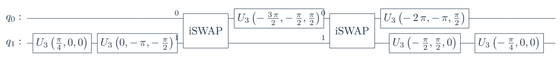
iqm_cz

EXACT PHASE E=1 / L=6 / score ≈ 18.024 proved minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.236 upper bound



quantinuum_zzmax

PROJECTIVE E=1 / L=7 / score ≈ 20.222 proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=7 / score ≈ 20.424 upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=14 / score ≈ 33.212 proved minimum



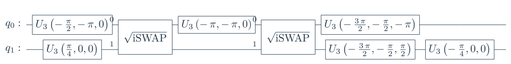
PROJECTIVE E=1 / L=13 / score ≈ 31.412 proved minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 upper bound

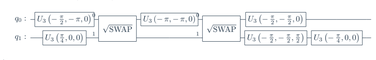


sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.036 upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

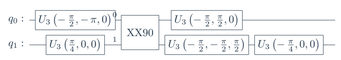
PROJECTIVE E=1 / L=7 / score ≈ 20.430 upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 proved minimum



xx_rz_ry

PROJECTIVE E=1 / L=9 / score ≈ 24.220 upper bound

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.826 upper bound

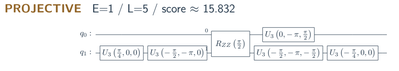


zz_u

EXACT PHASE E=1 / L=6 / score ≈ 18.238 upper bound



PROJECTIVE E=1 / L=5 / score ≈ 15.832 upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

CNOT

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

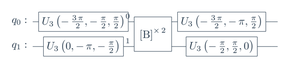
PROJECTIVE E=1 / L=5 / score ≈ 15.218 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — upper bound



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.212 — proved minimum



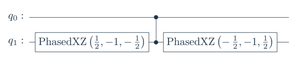
cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved minimum



google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 17.824 — upper bound



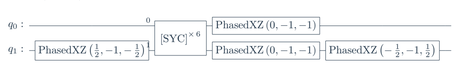
PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — upper bound



ibm_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



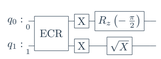
ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



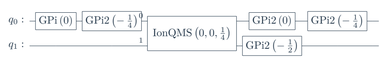
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved minimum



ionq_aria

PROJECTIVE E=1 / L=5 / score ≈ 15.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=6 / score ≈ 17.814 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

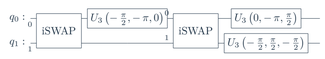
iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved minimum



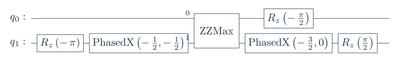
iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved minimum



quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

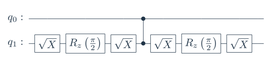
PROJECTIVE E=1 / L=5 / score ≈ 15.216 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

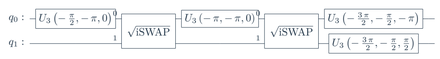
rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



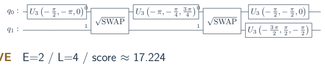
sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — upper bound



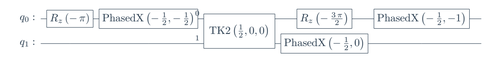
PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

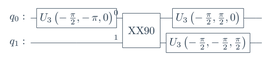
PROJECTIVE E=1 / L=5 / score ≈ 15.222 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved minimum



xx_rz_ry

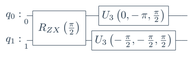
PROJECTIVE E=1 / L=7 / score ≈ 19.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

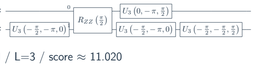
zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.014 — upper bound

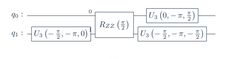


zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — upper bound



PROJECTIVE E=1 / L=3 / score ≈ 11.020 — upper bound



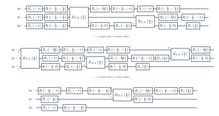
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

Controlled

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=6 / L=41 / score ≈ 118.334 — upper bound



$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 128.398 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

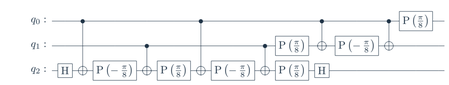
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 46.214 — upper bound



cx_rz_ry

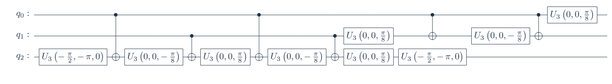
PROJECTIVE E=6 / L=11 / score ≈ 50.622 — upper bound



$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

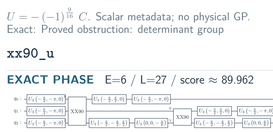
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 46.654 — upper bound



cz_rz_rx

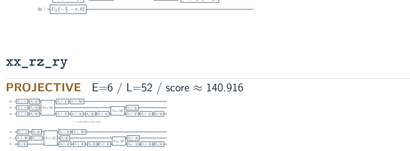
PROJECTIVE E=6 / L=38 / score ≈ 110.276 — upper bound



$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 73.126 — upper bound



google_cz

EXACT PHASE E=6 / L=21 / score ≈ 75.926 — upper bound



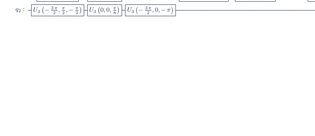
google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 75.950 — upper bound



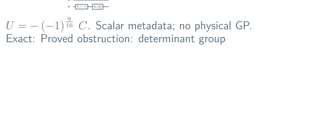
google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 126.398 — upper bound



google_syc

EXACT PHASE E=36 / L=33 / score ≈ 219.998 — upper bound



ibm_cx

PROJECTIVE E=6 / L=13 / score ≈ 54.618 — upper bound

$U = e^{\frac{15}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=38 / score ≈ 107.042 — upper bound



$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=36 / score ≈ 101.628 — upper bound

$U = e^{\frac{17}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 136.332 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 133.920 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 150.720 — upper bound

$U = (-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 137.596 — upper bound

$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 127.198 — upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 104.898 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 106.110 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=38 / score ≈ 107.042 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.998 — upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 126.398 — upper bound

PROJECTIVE E=12 / L=33 / score ≈ 125.998 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=41 / score ≈ 118.358 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 89.962 — upper bound

xx_rz_ry

PROJECTIVE E=6 / L=52 / score ≈ 140.916 — upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 76.738 — upper bound

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 90.780 — upper bound

PROJECTIVE E=6 / L=27 / score ≈ 88.774 — upper bound

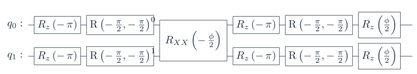
$U = -iC$. Scalar metadata; no physical GP.

CP (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

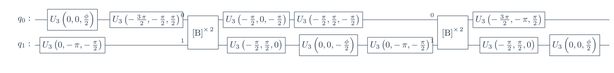
PROJECTIVE E=1 / L=10 / score ≈ 26.230 conditional minimum



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.866 upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

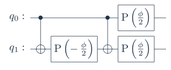
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

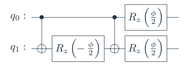
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 upper bound



cx_rz_ry

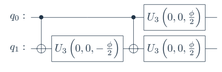
PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 upper bound



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 25.616 upper bound



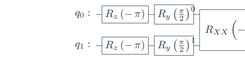
$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 19.430 upper bound



$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.421 upper bound



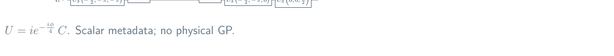
google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 42.057 upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.257 upper bound



ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound



$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 upper bound



$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.635 upper bound



$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

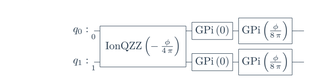
PROJECTIVE E=2 / L=16 / score ≈ 44.831 upper bound



$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

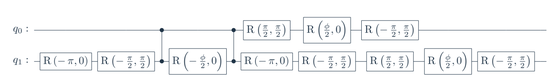
PROJECTIVE E=1 / L=4 / score ≈ 14.601 conditional minimum



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 upper bound



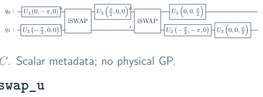
$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 conditional minimum



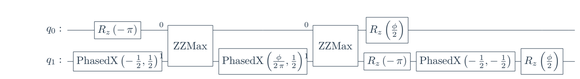
PROJECTIVE E=2 / L=6 / score ≈ 21.036 conditional minimum



$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.950 upper bound



$U = -ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

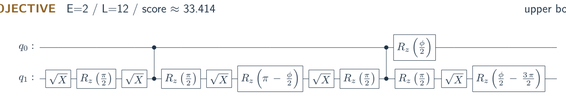
PROJECTIVE E=1 / L=2 / score ≈ 8.936 conditional minimum



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

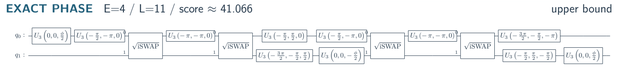
PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound



$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

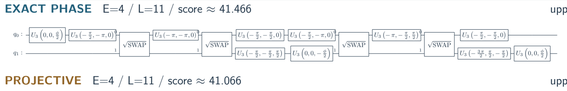
sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.066 upper bound

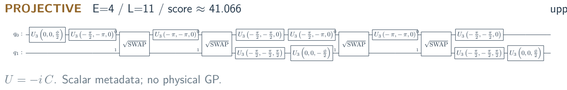


sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 upper bound



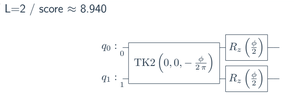
PROJECTIVE E=4 / L=11 / score ≈ 41.066 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

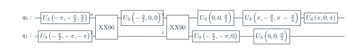
PROJECTIVE E=1 / L=2 / score ≈ 8.940 conditional minimum



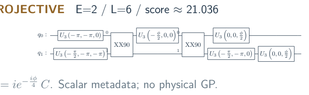
$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 upper bound



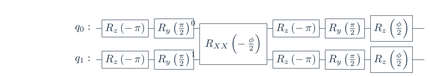
PROJECTIVE E=2 / L=6 / score ≈ 21.036 upper bound



$U = ie^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

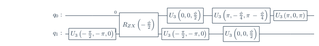
PROJECTIVE E=1 / L=10 / score ≈ 25.422 conditional minimum



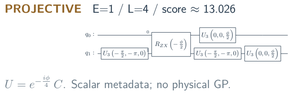
$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 conditional minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.026 conditional minimum



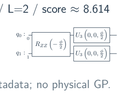
$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 conditional minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.614 conditional minimum



$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

CR (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=16 / score ≈ 47.658 — upper bound



PROJECTIVE E=2 / L=14 / score ≈ 39.648 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=12 / score ≈ 44.072 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

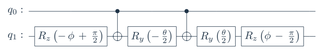
cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound



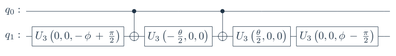
cx_rz_ry

EXACT PHASE E=2 / L=4 / score ≈ 16.808 — upper bound



cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound



cz_rz_rx

EXACT PHASE E=2 / L=13 / score ≈ 35.826 — upper bound



cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — upper bound

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 28.169 — upper bound



google_fsim

EXACT PHASE E=2 / L=8 / score ≈ 28.177 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=12 / score ≈ 44.993 — upper bound

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 76.193 — upper bound



ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — upper bound

PROJECTIVE E=2 / L=9 / score ≈ 26.810 — upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — upper bound



PROJECTIVE E=2 / L=19 / score ≈ 47.618 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=17 / score ≈ 43.214 — upper bound

ionq_aria

EXACT PHASE E=2 / L=22 / score ≈ 59.377 — upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.977 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=22 / score ≈ 58.573 — upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.173 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=24 / score ≈ 64.173 — upper bound

PROJECTIVE E=2 / L=24 / score ≈ 63.773 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=16 / score ≈ 44.464 — upper bound

iswap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — upper bound

quantinuum_zzmax

EXACT PHASE E=5 / L=32 / score ≈ 92.157 — upper bound

PROJECTIVE E=2 / L=13 / score ≈ 37.899 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=15 / score ≈ 46.309 — upper bound

PROJECTIVE E=2 / L=13 / score ≈ 38.303 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — upper bound

PROJECTIVE E=2 / L=19 / score ≈ 47.618 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.272 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — upper bound

PROJECTIVE E=4 / L=12 / score ≈ 43.272 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=34 / score ≈ 97.201 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.317 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — upper bound

xx_rz_ry

EXACT PHASE E=3 / L=22 / score ≈ 59.650 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 47.640 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=8 / score ≈ 26.852 — upper bound

zz_u

EXACT PHASE E=2 / L=11 / score ≈ 32.870 — upper bound

PROJECTIVE E=2 / L=10 / score ≈ 30.864 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

CROT (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=17 / score ≈ 49.260 upper bound



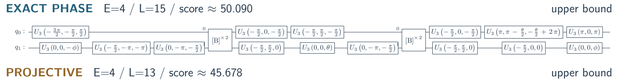
PROJECTIVE E=2 / L=15 / score ≈ 41.250 upper bound



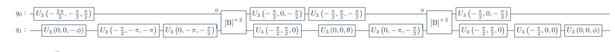
$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 50.090 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.678 upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

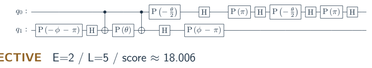
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

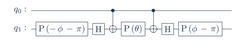
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=13 / score ≈ 34.414 upper bound



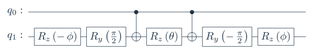
PROJECTIVE E=2 / L=5 / score ≈ 18.006 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

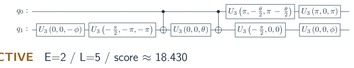
cx_rz_ry

EXACT PHASE E=2 / L=5 / score ≈ 18.410 upper bound

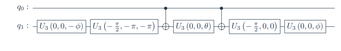


cx_u

EXACT PHASE E=2 / L=7 / score ≈ 22.842 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 18.430 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

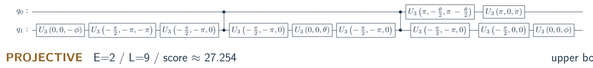
cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 31.622 upper bound



cz_u

EXACT PHASE E=2 / L=11 / score ≈ 31.666 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 27.254 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=11 / score ≈ 34.317 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 29.445 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=11 / score ≈ 34.325 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 29.453 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=15 / score ≈ 51.141 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.269 upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 82.341 upper bound



PROJECTIVE E=12 / L=13 / score ≈ 77.469 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=8 / score ≈ 24.810 upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.810 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.218 upper bound

PROJECTIVE E=2 / L=15 / score ≈ 39.218 upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 41.214 upper bound



PROJECTIVE E=2 / L=15 / score ≈ 39.214 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 50.239 upper bound

PROJECTIVE E=2 / L=18 / score ≈ 49.839 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 49.435 upper bound



PROJECTIVE E=2 / L=18 / score ≈ 49.035 upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=20 / score ≈ 55.035 upper bound



PROJECTIVE E=2 / L=20 / score ≈ 54.635 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=19 / score ≈ 50.876 upper bound

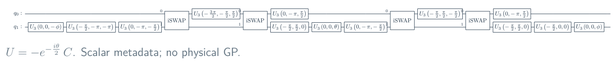


iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.278 upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 93.098 upper bound



PROJECTIVE E=2 / L=14 / score ≈ 38.840 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=16 / score ≈ 47.250 upper bound



PROJECTIVE E=2 / L=14 / score ≈ 39.244 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.218 upper bound



PROJECTIVE E=2 / L=15 / score ≈ 39.218 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.290 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 44.878 upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 44.878 upper bound



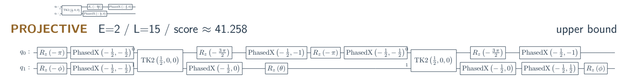
$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=35 / score ≈ 98.142 upper bound



PROJECTIVE E=2 / L=15 / score ≈ 41.258 upper bound



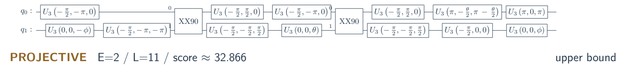
$U = -i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=13 / score ≈ 37.278 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 32.866 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=23 / score ≈ 61.252 upper bound



PROJECTIVE E=2 / L=19 / score ≈ 49.242 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 32.870 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 28.458 upper bound



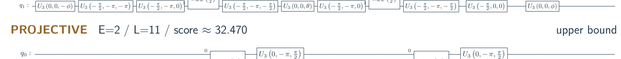
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 38.888 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 32.470 upper bound



$U = ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

CRX (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=13 / score ≈ 32.842 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=12 / score ≈ 44.072 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

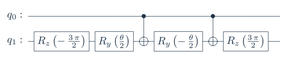
cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 29.212 — upper bound



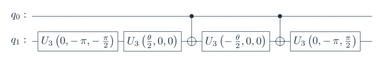
cx_rz_ry

EXACT PHASE E=2 / L=4 / score ≈ 16.808 — upper bound



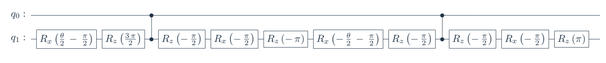
cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound



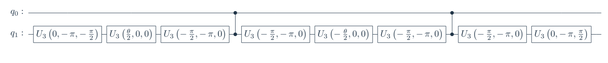
cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.620 — upper bound



cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — upper bound



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.509 — upper bound



google_fsim

EXACT PHASE E=2 / L=8 / score ≈ 27.517 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=12 / score ≈ 44.333 — upper bound

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 75.533 — upper bound

ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — upper bound

ibm_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 46.020 — upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=18 / score ≈ 45.216 — upper bound

ionq_aria

EXACT PHASE E=2 / L=22 / score ≈ 58.717 — upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.317 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=22 / score ≈ 57.913 — upper bound

PROJECTIVE E=2 / L=22 / score ≈ 57.513 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 23.279 — conditional minimum

iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.256 — upper bound

iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 94.159 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 39.901 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=5 / score ≈ 15.879 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 46.020 — upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.272 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — upper bound

PROJECTIVE E=4 / L=12 / score ≈ 43.272 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=34 / score ≈ 97.201 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.317 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — upper bound

xx_rz_ry

EXACT PHASE E=1 / L=15 / score ≈ 36.032 — conditional minimum

zx_u

EXACT PHASE E=1 / L=5 / score ≈ 15.632 — conditional minimum

zz_u

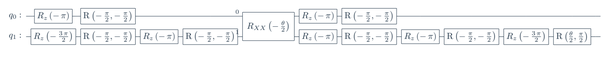
EXACT PHASE E=1 / L=3 / score ≈ 11.220 — conditional minimum

CRY (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

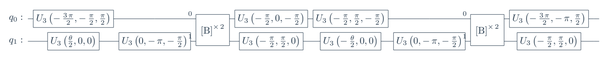
aqt_rxx

EXACT PHASE E=1 / L=14 / score ≈ 35.444 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=10 / score ≈ 39.660 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

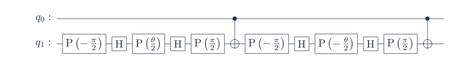
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

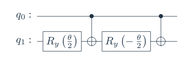
cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 29.212 — upper bound



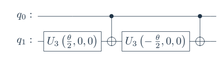
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — upper bound



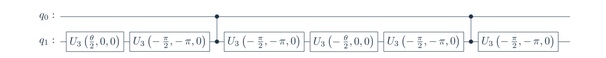
cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.022 — upper bound



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — upper bound



google_cz

EXACT PHASE E=2 / L=6 / score ≈ 23.097 — upper bound



google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 23.105 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=10 / score ≈ 39.921 — upper bound

google_syc

EXACT PHASE E=12 / L=10 / score ≈ 71.121 — upper bound

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.408 — upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 45.616 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 40.812 — upper bound

ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 49.509 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 49.109 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 48.705 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 48.305 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=12 / score ≈ 31.887 — conditional minimum

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.440 — upper bound

iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.755 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.497 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=6 / score ≈ 18.481 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 45.616 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=10 / score ≈ 38.860 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — upper bound

PROJECTIVE E=4 / L=10 / score ≈ 38.860 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=32 / score ≈ 92.797 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.913 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — upper bound

xx_rz_ry

EXACT PHASE E=1 / L=14 / score ≈ 34.030 — conditional minimum

zx_u

EXACT PHASE E=1 / L=7 / score ≈ 19.644 — conditional minimum

zz_u

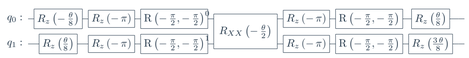
EXACT PHASE E=1 / L=5 / score ≈ 15.232 — conditional minimum

CRZ (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

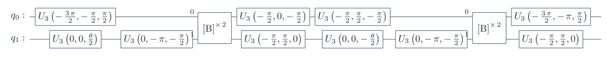
aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 32.234 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=10 / score ≈ 39.660 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

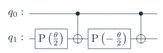
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

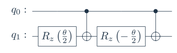
cx_h_phase

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



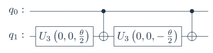
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



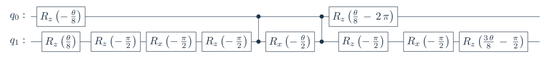
cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — upper bound



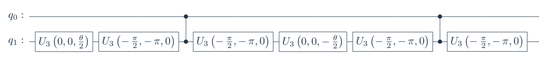
cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 31.620 — upper bound



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — upper bound



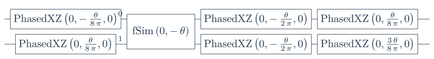
google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.697 — upper bound



google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 20.822 — conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=10 / score ≈ 39.521 — upper bound

google_syc

EXACT PHASE E=12 / L=10 / score ≈ 70.721 — upper bound

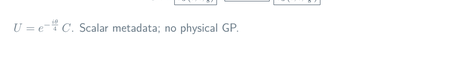
ibm_cx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — upper bound



PROJECTIVE E=2 / L=11 / score ≈ 31.212 — upper bound

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.612 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=14 / score ≈ 41.101 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.701 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=14 / score ≈ 40.297 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 39.897 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=2 / score ≈ 9.667 — conditional minimum

iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.656 — upper bound

iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.860 — conditional minimum

PROJECTIVE E=2 / L=8 / score ≈ 27.048 — conditional minimum

$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 82.210 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 27.952 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 14.940 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 31.212 — upper bound

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=10 / score ≈ 38.860 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — upper bound

PROJECTIVE E=4 / L=10 / score ≈ 38.860 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 14.944 — conditional minimum

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — upper bound

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 31.426 — conditional minimum

zx_u

EXACT PHASE E=1 / L=8 / score ≈ 23.850 — conditional minimum

PROJECTIVE E=1 / L=6 / score ≈ 19.038 — conditional minimum

$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 19.438 — conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 14.626 — conditional minimum

$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

CS

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

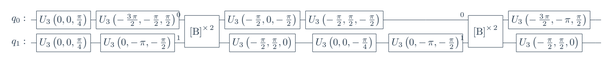
PROJECTIVE E=1 / L=10 / score ≈ 26.830 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 42.466 upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=3 / score ≈ 14.000 proved minimum



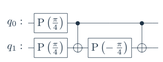
clifford_t_cz

EXACT PHASE E=2 / L=7 / score ≈ 22.000 proved minimum



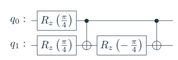
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.206 upper bound



cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 15.206 upper bound



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 proved minimum



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 26.216 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.042 proved minimum



Exact definitions: η_{24}, η_2 in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 20.030 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

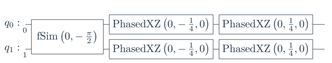
google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.030 proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.828 proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 41.666 upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.866 upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.206 upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.014 upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

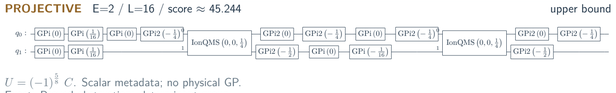
PROJECTIVE E=2 / L=6 / score ≈ 21.208 proved minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

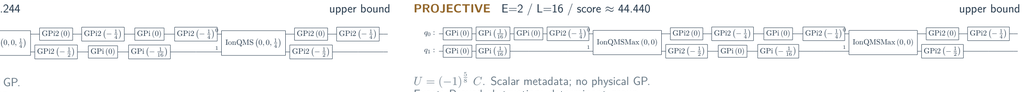
PROJECTIVE E=2 / L=16 / score ≈ 45.244 upper bound



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.440 upper bound



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.210 proved minimum



$U = \eta_{24}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.644 proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

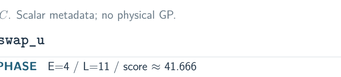
iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved minimum



Exact definitions: η_{24}, η_2 in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.220 proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 9.206 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

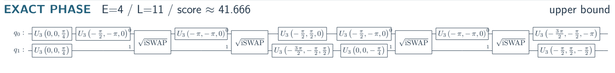
PROJECTIVE E=2 / L=12 / score ≈ 34.014 upper bound



$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

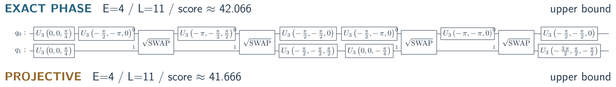
sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.666 upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.066 upper bound



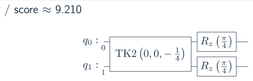
PROJECTIVE E=4 / L=11 / score ≈ 41.666 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.210 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

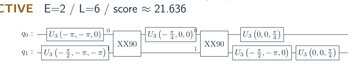
xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved minimum



Exact definitions: η_{24}, η_{24} in the expression glossary.

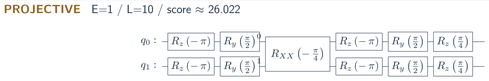
PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved minimum



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

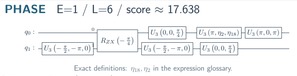
PROJECTIVE E=1 / L=10 / score ≈ 26.022 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.638 proved minimum



Exact definitions: η_{24}, η_2 in the expression glossary.

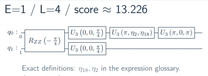
PROJECTIVE E=1 / L=4 / score ≈ 13.626 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.226 proved minimum



Exact definitions: η_{24}, η_2 in the expression glossary.

PROJECTIVE E=1 / L=2 / score ≈ 9.214 proved minimum



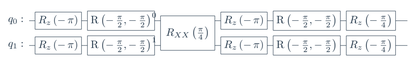
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

CSdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

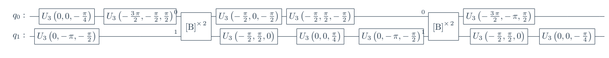
PROJECTIVE E=1 / L=10 / score ≈ 26.830 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 42.466 upper bound

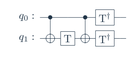


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=3 / score ≈ 14.000 proved minimum



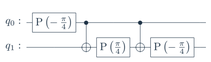
clifford_t_cz

EXACT PHASE E=2 / L=7 / score ≈ 22.000 proved minimum



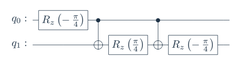
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.206 upper bound



cx_rz_ry

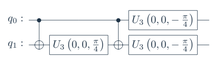
PROJECTIVE E=2 / L=3 / score ≈ 15.206 upper bound



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 proved minimum



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 26.216 proved minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.042 proved minimum



Exact definitions: η_{14}, η_{8} in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 20.030 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

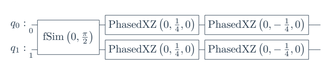
google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.030 proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.828 proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 41.666 upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.866 upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.206 upper bound

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.014 proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

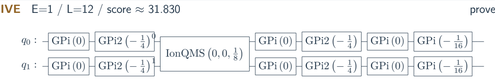
PROJECTIVE E=2 / L=6 / score ≈ 21.208 proved minimum



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

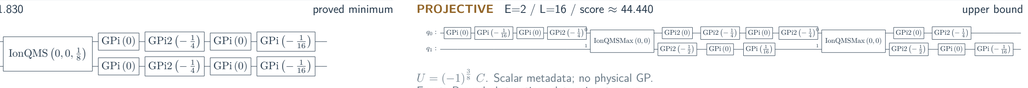
PROJECTIVE E=1 / L=12 / score ≈ 31.830 proved minimum



$U = \eta_{27}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.440 upper bound



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.210 proved minimum



$U = \eta_{27}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.644 proved minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved minimum



Exact definitions: η_{11}, η_{6} in the expression glossary.

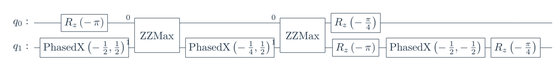
PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

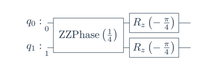
PROJECTIVE E=2 / L=7 / score ≈ 24.220 proved minimum



$U = -(-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

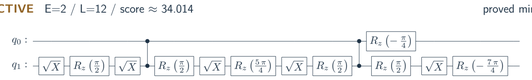
PROJECTIVE E=1 / L=2 / score ≈ 9.206 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

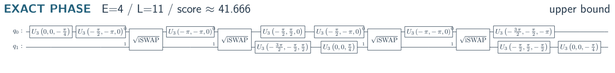
PROJECTIVE E=2 / L=12 / score ≈ 34.014 proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

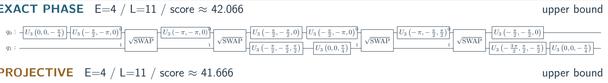
sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.666 upper bound

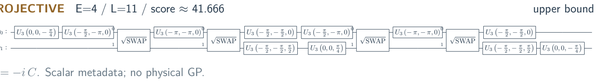


sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.066 upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.666 upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.210 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved minimum



Exact definitions: η_{11}, η_{6} in the expression glossary.

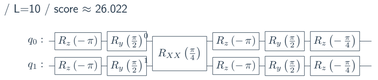
PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

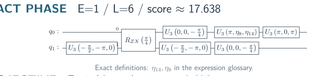
PROJECTIVE E=1 / L=10 / score ≈ 26.022 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

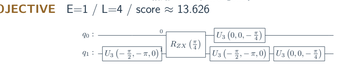
zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.638 proved minimum



Exact definitions: η_{11}, η_{6} in the expression glossary.

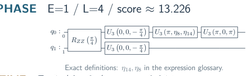
PROJECTIVE E=1 / L=4 / score ≈ 13.626 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

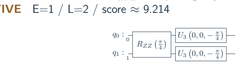
zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.226 proved minimum



Exact definitions: η_{11}, η_{6} in the expression glossary.

PROJECTIVE E=1 / L=2 / score ≈ 9.214 proved minimum



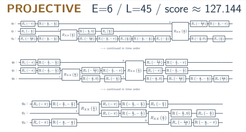
$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

CSWAP

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

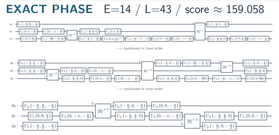
PROJECTIVE E=6 / L=45 / score ≈ 127.144 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=14 / L=43 / score ≈ 159.058 — upper bound

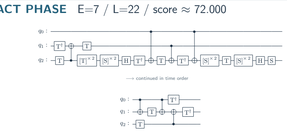


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

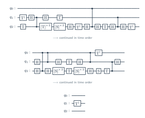
clifford_t_cx

EXACT PHASE E=7 / L=22 / score ≈ 72.000 — upper bound



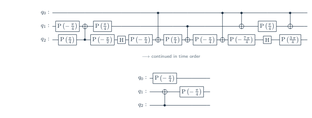
clifford_t_cz

EXACT PHASE E=7 / L=34 / score ≈ 96.000 — upper bound



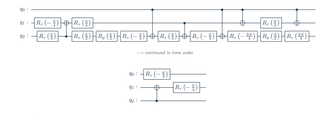
cx_h_phase

EXACT PHASE E=7 / L=14 / score ≈ 60.624 — upper bound



cx_rz_ry

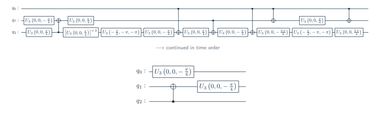
PROJECTIVE E=7 / L=14 / score ≈ 61.028 — upper bound



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=7 / L=15 / score ≈ 63.690 — upper bound



cz_rz_rx

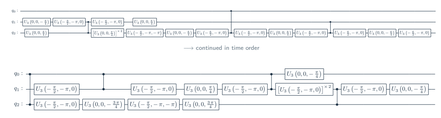
PROJECTIVE E=7 / L=45 / score ≈ 129.090 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

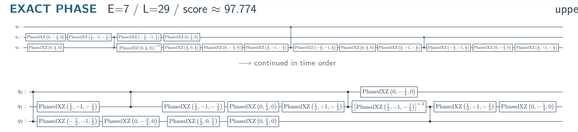
cz_u

EXACT PHASE E=7 / L=29 / score ≈ 94.574 — upper bound



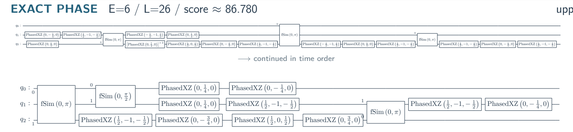
google_cz

EXACT PHASE E=7 / L=29 / score ≈ 97.774 — upper bound



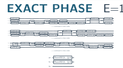
google_fsim

EXACT PHASE E=6 / L=26 / score ≈ 86.780 — upper bound



google_sqrt_iswap

EXACT PHASE E=14 / L=43 / score ≈ 157.258 — upper bound



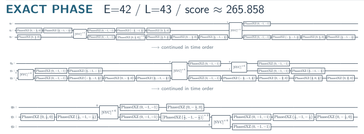
PROJECTIVE E=14 / L=43 / score ≈ 156.658 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

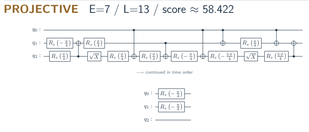
google_syc

EXACT PHASE E=42 / L=43 / score ≈ 265.858 — upper bound



ibm_cx

PROJECTIVE E=7 / L=13 / score ≈ 58.422 — upper bound



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

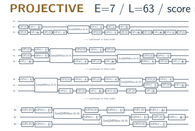
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria_full

PROJECTIVE E=7 / L=63 / score ≈ 172.354 — upper bound



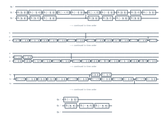
$U = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iqm_cz

PROJECTIVE E=7 / L=69 / score ≈ 188.276 — upper bound



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=40 / score ≈ 141.240 — upper bound



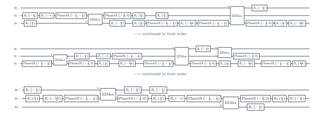
PROJECTIVE E=12 / L=38 / score ≈ 137.228 — upper bound



$U = \eta_{81}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

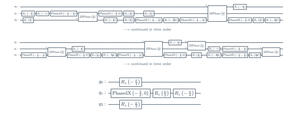
PROJECTIVE E=7 / L=45 / score ≈ 130.522 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 105.506 — upper bound



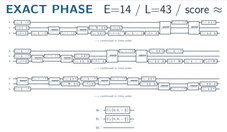
$U = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

sqrt_iswap_u

EXACT PHASE E=14 / L=43 / score ≈ 156.258 — upper bound

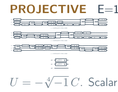


sqrt_swap_u

EXACT PHASE E=14 / L=43 / score ≈ 157.058 — upper bound



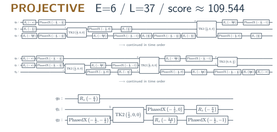
PROJECTIVE E=14 / L=43 / score ≈ 156.258 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

tket_tk2

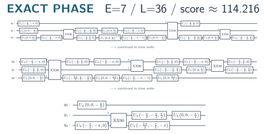
PROJECTIVE E=6 / L=37 / score ≈ 109.544 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

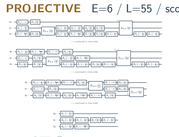
xx90_u

EXACT PHASE E=7 / L=36 / score ≈ 114.216 — upper bound



xx_rz_ry

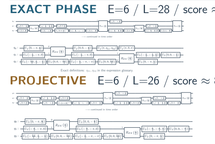
PROJECTIVE E=6 / L=55 / score ≈ 146.922 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=28 / score ≈ 91.180 — upper bound

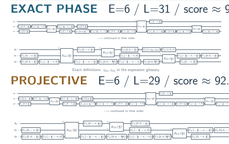


PROJECTIVE E=6 / L=26 / score ≈ 87.168 — upper bound

$U = \eta_{13}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=31 / score ≈ 96.798 — upper bound



PROJECTIVE E=6 / L=29 / score ≈ 92.786 — upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

CSX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

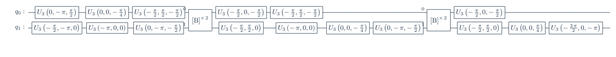
PROJECTIVE E=1 / L=14 / score ≈ 35.644 — proved minimum



$U = \eta_{24}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 53.096 — upper bound

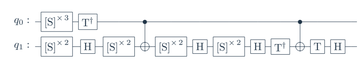


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=18 / score ≈ 44.000 — upper bound



clifford_t_cz

EXACT PHASE E=2 / L=22 / score ≈ 52.000 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 48.000 — upper bound



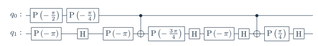
$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 33.416 — upper bound



PROJECTIVE E=2 / L=11 / score ≈ 31.414 — upper bound



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=8 / score ≈ 25.816 — upper bound



$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

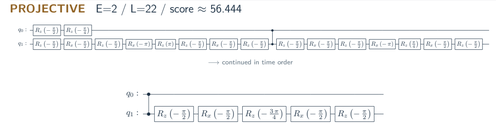
cx_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — upper bound



cz_rz_rx

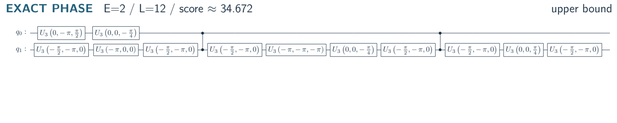
PROJECTIVE E=2 / L=22 / score ≈ 56.444 — upper bound



$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.672 — upper bound



google_cz

EXACT PHASE E=2 / L=12 / score ≈ 35.872 — upper bound



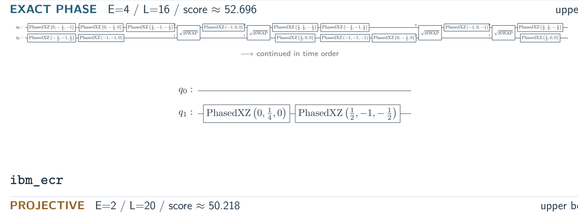
google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 35.880 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 52.696 — upper bound



google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.896 — upper bound



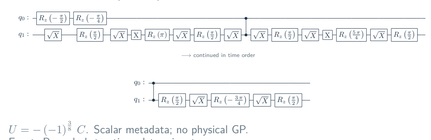
ibm_cx

PROJECTIVE E=2 / L=12 / score ≈ 33.814 — upper bound

$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

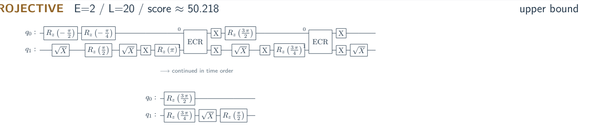
PROJECTIVE E=2 / L=22 / score ≈ 54.622 — upper bound



$U = -(-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=20 / score ≈ 50.218 — upper bound



$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

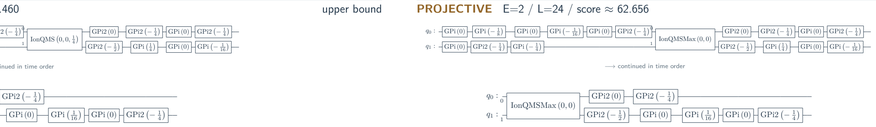
PROJECTIVE E=2 / L=24 / score ≈ 63.460 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

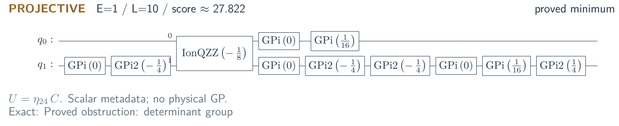
PROJECTIVE E=2 / L=24 / score ≈ 62.656 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

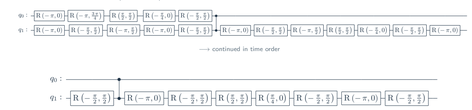
PROJECTIVE E=1 / L=10 / score ≈ 27.822 — proved minimum



$U = \eta_{24}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

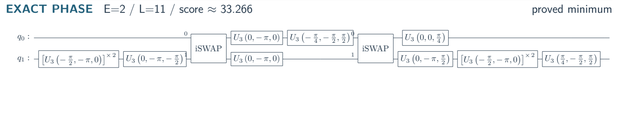
PROJECTIVE E=2 / L=25 / score ≈ 64.900 — upper bound



$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

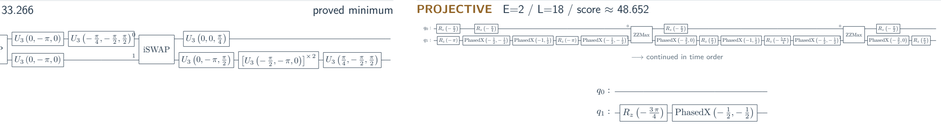
iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — proved minimum



quantinuum_zzmax

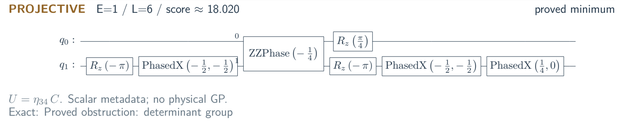
PROJECTIVE E=2 / L=18 / score ≈ 48.652 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

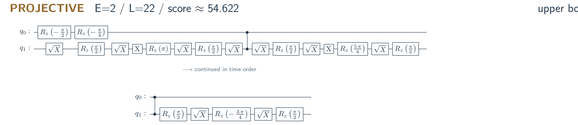
PROJECTIVE E=1 / L=6 / score ≈ 18.020 — proved minimum



$U = \eta_{24}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

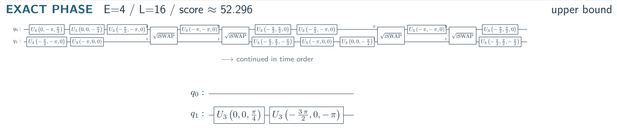
PROJECTIVE E=2 / L=22 / score ≈ 54.622 — upper bound



$U = -(-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

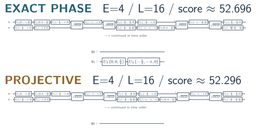
sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.296 — upper bound

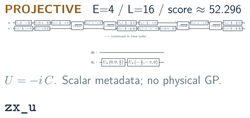


sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.696 — upper bound



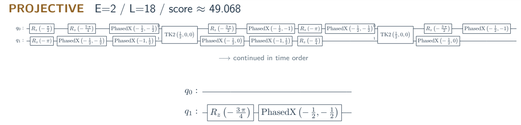
PROJECTIVE E=4 / L=16 / score ≈ 52.296 — upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

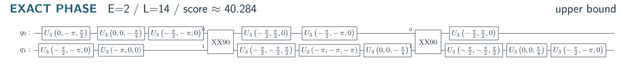
PROJECTIVE E=2 / L=18 / score ≈ 49.068 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=14 / score ≈ 40.284 — upper bound



xx_rz_ry

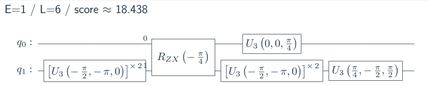
PROJECTIVE E=1 / L=16 / score ≈ 38.834 — proved minimum



$U = \eta_{24}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

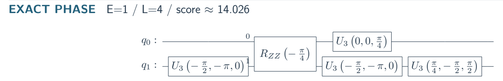
zx_u

EXACT PHASE E=1 / L=6 / score ≈ 18.438 — proved minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 14.026 — proved minimum

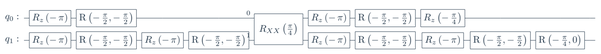


CSXdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=14 / score ≈ 35.644 — proved minimum



$U = \eta_{235}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=13 / score ≈ 46.878 — upper bound

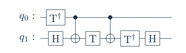


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=5 / score ≈ 18.000 — proved minimum



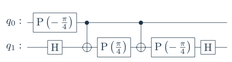
clifford_t_cz

EXACT PHASE E=2 / L=9 / score ≈ 26.000 — proved minimum



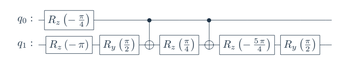
cx_h_phase

EXACT PHASE E=2 / L=5 / score ≈ 19.206 — proved minimum



cx_rz_ry

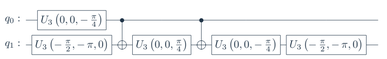
PROJECTIVE E=2 / L=6 / score ≈ 21.612 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

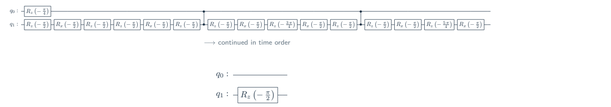
cx_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved minimum



cz_rz_rx

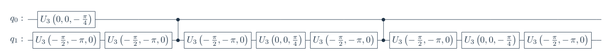
PROJECTIVE E=2 / L=17 / score ≈ 46.034 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

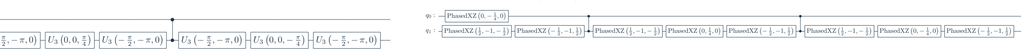
cz_u

EXACT PHASE E=2 / L=9 / score ≈ 28.454 — proved minimum



google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.654 — proved minimum



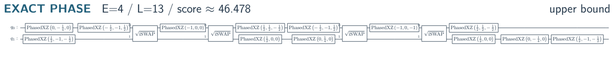
google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 29.662 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=13 / score ≈ 46.478 — upper bound



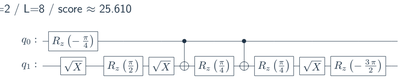
google_syc

EXACT PHASE E=12 / L=13 / score ≈ 77.678 — upper bound



ibm_cx

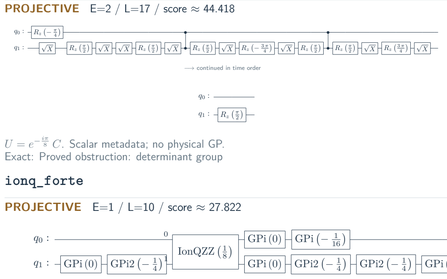
PROJECTIVE E=2 / L=8 / score ≈ 25.610 — proved minimum



$U = e^{-\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

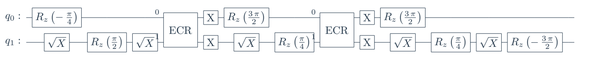
PROJECTIVE E=2 / L=17 / score ≈ 44.418 — proved minimum



$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

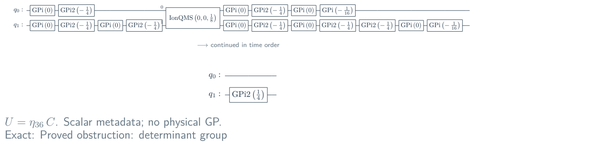
PROJECTIVE E=2 / L=16 / score ≈ 42.014 — proved minimum



$U = e^{-\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

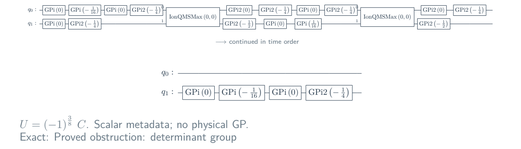
PROJECTIVE E=1 / L=18 / score ≈ 45.442 — proved minimum



$U = \eta_{236}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

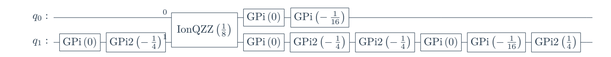
PROJECTIVE E=2 / L=20 / score ≈ 53.248 — upper bound



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

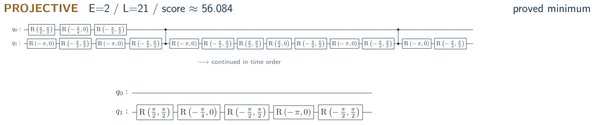
PROJECTIVE E=1 / L=10 / score ≈ 27.822 — proved minimum



$U = \eta_{236}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=21 / score ≈ 56.084 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

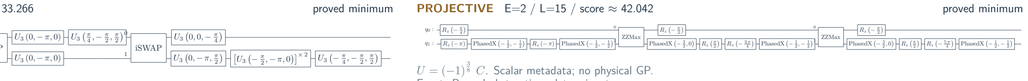
iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — proved minimum



quantinuum_zzmax

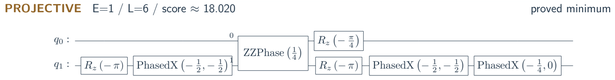
PROJECTIVE E=2 / L=15 / score ≈ 42.042 — proved minimum



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

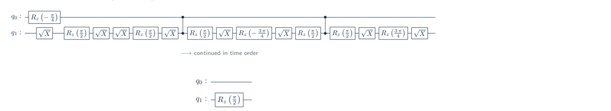
PROJECTIVE E=1 / L=6 / score ≈ 18.020 — proved minimum



$U = \eta_{235}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

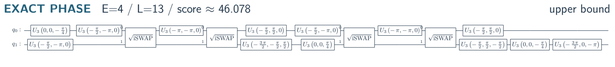
PROJECTIVE E=2 / L=17 / score ≈ 44.418 — proved minimum



$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

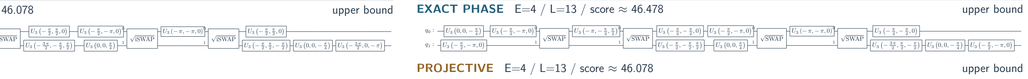
sqrt_iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 46.078 — upper bound

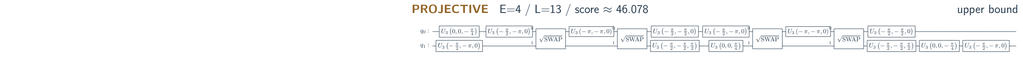


sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 46.478 — upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.078 — upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

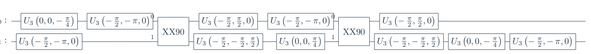
PROJECTIVE E=2 / L=15 / score ≈ 42.458 — upper bound



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 34.066 — proved minimum



xx_rz_ry

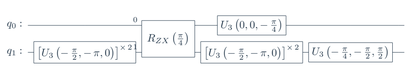
PROJECTIVE E=1 / L=16 / score ≈ 38.834 — proved minimum



$U = \eta_{235}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

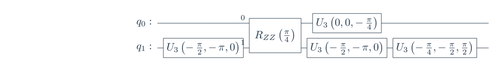
zx_u

EXACT PHASE E=1 / L=6 / score ≈ 18.438 — proved minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 14.026 — proved minimum



CT

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

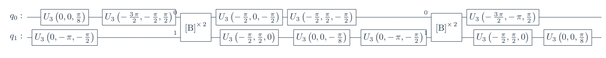
PROJECTIVE E=1 / L=10 / score ≈ 27.430 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 43.066 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.806 — proved minimum



cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 15.806 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

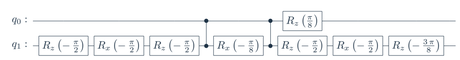
cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.818 — proved minimum



cz_rz_rx

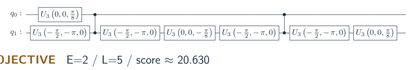
PROJECTIVE E=2 / L=8 / score ≈ 26.816 — proved minimum



$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — upper bound



PROJECTIVE E=2 / L=5 / score ≈ 20.630 — proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

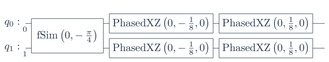
google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.630 — proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.828 — proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 42.266 — upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.466 — upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.806 — proved minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

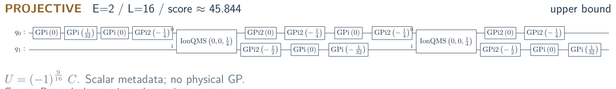
PROJECTIVE E=2 / L=6 / score ≈ 21.808 — proved minimum



$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

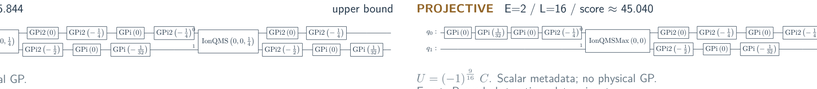
PROJECTIVE E=2 / L=16 / score ≈ 45.844 — upper bound



$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

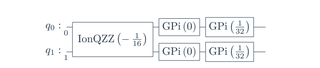
PROJECTIVE E=2 / L=16 / score ≈ 45.040 — upper bound



$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

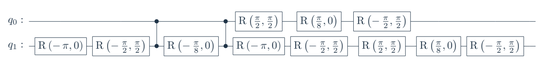
PROJECTIVE E=1 / L=4 / score ≈ 14.810 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

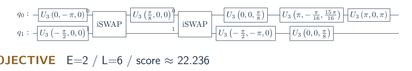
PROJECTIVE E=2 / L=11 / score ≈ 34.244 — proved minimum



$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.848 — proved minimum



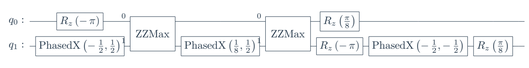
PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.820 — proved minimum



$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

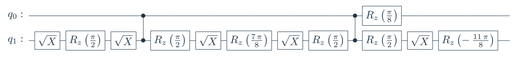
PROJECTIVE E=1 / L=2 / score ≈ 9.806 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

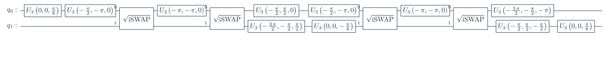
PROJECTIVE E=2 / L=12 / score ≈ 34.614 — proved minimum



$U = -(-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.266 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.666 — upper bound



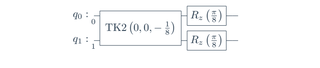
PROJECTIVE E=4 / L=11 / score ≈ 42.266 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

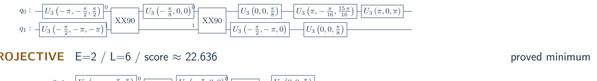
PROJECTIVE E=1 / L=2 / score ≈ 9.810 — proved minimum



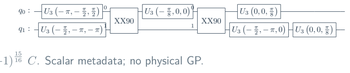
$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 28.248 — proved minimum



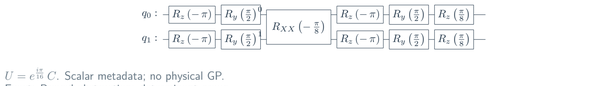
PROJECTIVE E=2 / L=6 / score ≈ 22.636 — proved minimum



$U = -(-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 26.622 — proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 19.838 — proved minimum



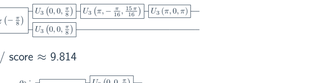
PROJECTIVE E=1 / L=4 / score ≈ 14.226 — proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 15.426 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 9.814 — proved minimum



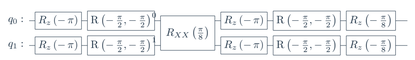
$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

CTdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

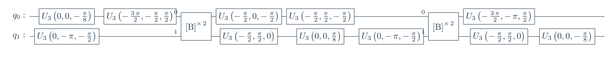
PROJECTIVE E=1 / L=10 / score ≈ 27.430 proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 43.066 upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

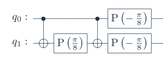
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

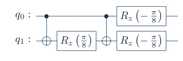
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.806 proved minimum



cx_rz_ry

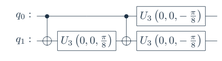
PROJECTIVE E=2 / L=3 / score ≈ 15.806 proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.818 proved minimum



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 26.816 proved minimum



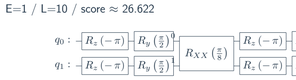
$U = (-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.642 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 20.630 proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.630 proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.828 proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 42.266 upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.466 upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.806 proved minimum

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 proved minimum



$U = -(-1)^{\frac{11}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=11 / score ≈ 32.210 upper bound



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=1 / L=12 / score ≈ 32.430 proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 45.440 upper bound



$U = (-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

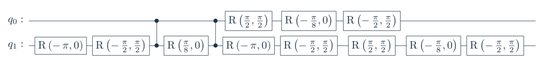
PROJECTIVE E=1 / L=4 / score ≈ 14.810 proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

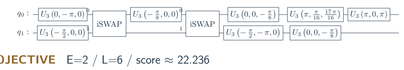
PROJECTIVE E=2 / L=11 / score ≈ 34.244 proved minimum



$U = (-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.848 proved minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.236 proved minimum



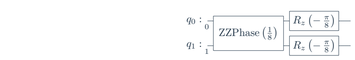
$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

Exact: Proved obstruction: determinant group
Projective: Open: construction not found

quantinuum_zzphase

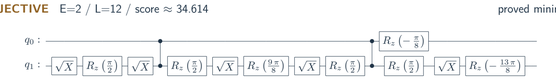
PROJECTIVE E=1 / L=2 / score ≈ 9.806 proved minimum



$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

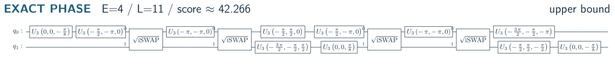
PROJECTIVE E=2 / L=12 / score ≈ 34.614 proved minimum



$U = -(-1)^{\frac{11}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

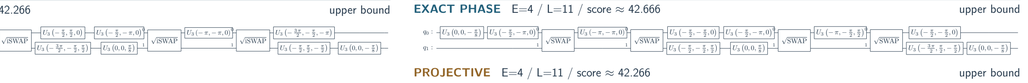
sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.266 upper bound

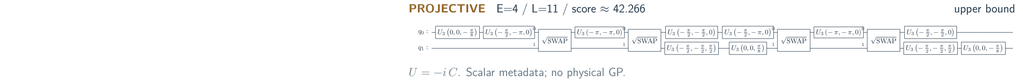


sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.666 upper bound



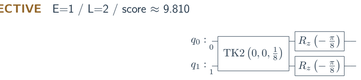
PROJECTIVE E=4 / L=11 / score ≈ 42.266 upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

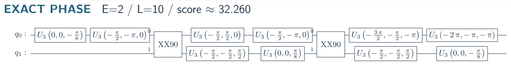
PROJECTIVE E=1 / L=2 / score ≈ 9.810 proved minimum



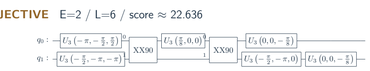
$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 32.260 upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.636 proved minimum



$U = \sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 26.622 proved minimum

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 19.838 proved minimum



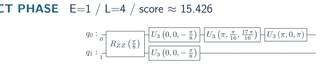
PROJECTIVE E=1 / L=4 / score ≈ 14.226 proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 15.426 proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 9.814 proved minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

CU $(\theta, \phi, lbda, \gamma)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=16 / score ≈ 43.852 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=14 / score ≈ 48.284 upper bound



$U = \eta_{39}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 33.616 upper bound



cx_rz_ry

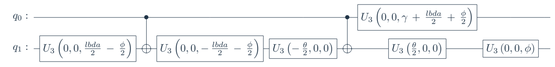
PROJECTIVE E=2 / L=6 / score ≈ 21.012 upper bound



$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.036 upper bound



cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 38.628 upper bound



$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 29.860 upper bound



google_cz

EXACT PHASE E=2 / L=10 / score ≈ 33.042 upper bound



google_fsim

EXACT PHASE E=2 / L=10 / score ≈ 33.050 upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=14 / score ≈ 49.866 upper bound



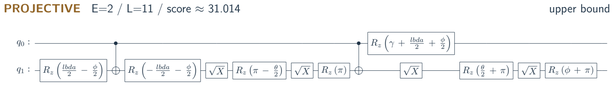
google_syc

EXACT PHASE E=12 / L=14 / score ≈ 81.066 upper bound



ibm_cx

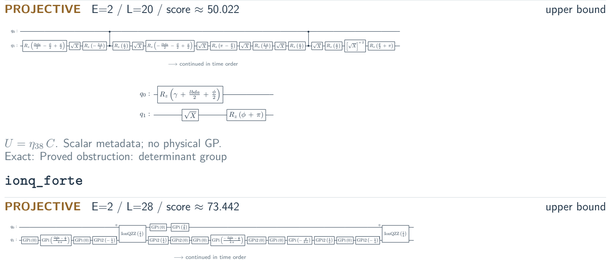
PROJECTIVE E=2 / L=11 / score ≈ 31.014 upper bound



$U = -e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

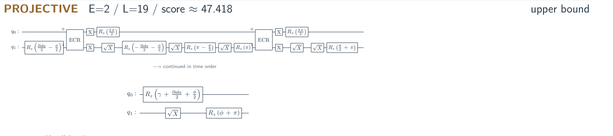
PROJECTIVE E=2 / L=20 / score ≈ 50.022 upper bound



$U = \eta_{38}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

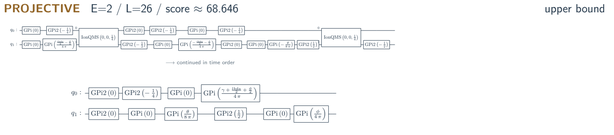
PROJECTIVE E=2 / L=19 / score ≈ 47.418 upper bound



$U = -e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

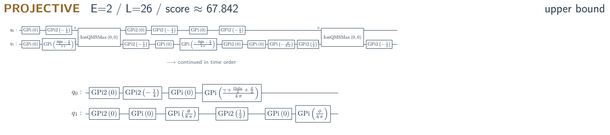
PROJECTIVE E=2 / L=26 / score ≈ 68.646 upper bound



$U = \eta_{39}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

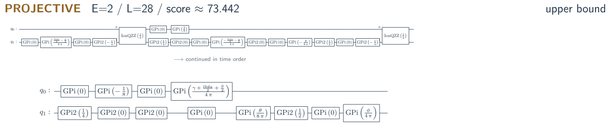
PROJECTIVE E=2 / L=26 / score ≈ 67.842 upper bound



$U = \eta_{39}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

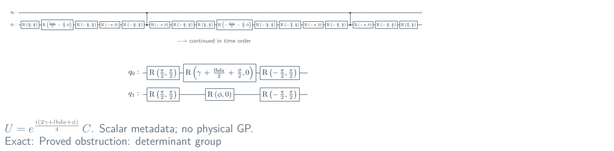
PROJECTIVE E=2 / L=28 / score ≈ 73.442 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=22 / score ≈ 58.288 upper bound



$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

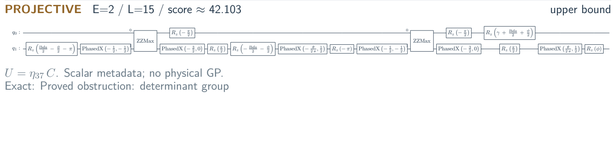
iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 upper bound



quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 42.103 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

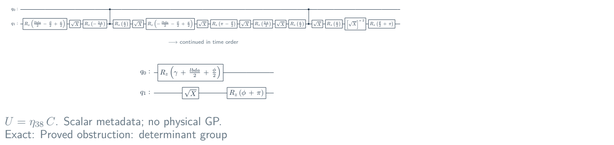
PROJECTIVE E=2 / L=15 / score ≈ 42.507 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

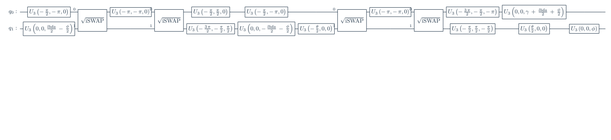
PROJECTIVE E=2 / L=20 / score ≈ 50.022 upper bound



$U = \eta_{38}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

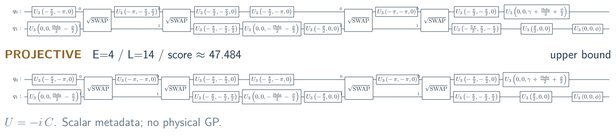
sqrt_iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.484 upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 upper bound



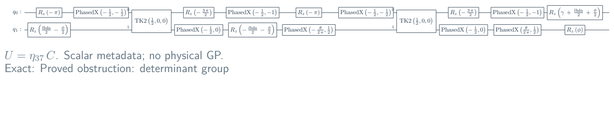
PROJECTIVE E=4 / L=14 / score ≈ 47.484 upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

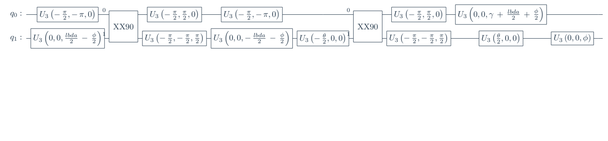
PROJECTIVE E=2 / L=16 / score ≈ 44.521 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

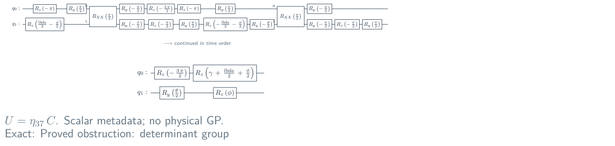
xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 35.472 upper bound



xx_rz_ry

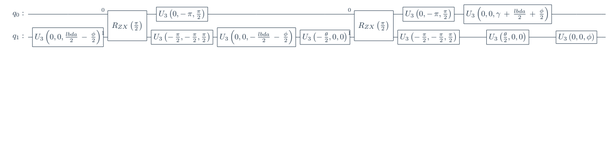
PROJECTIVE E=2 / L=20 / score ≈ 51.844 upper bound



$U = \eta_{37}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

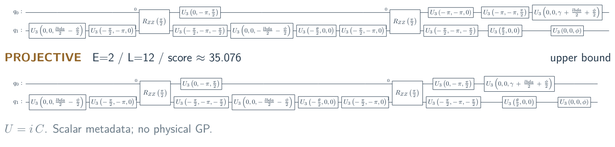
zx_u

EXACT PHASE E=2 / L=10 / score ≈ 31.064 upper bound



zz_u

EXACT PHASE E=2 / L=13 / score ≈ 37.082 upper bound



PROJECTIVE E=2 / L=12 / score ≈ 35.076 upper bound

$U = i\, C$. Scalar metadata; no physical GP.

CU1 (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

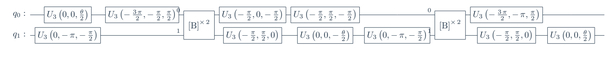
PROJECTIVE E=1 / L=10 / score ≈ 26.230 conditional minimum



$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.866 upper bound



$U = ie^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

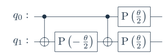
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

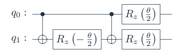
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 upper bound



cx_rz_ry

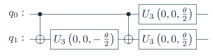
PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 upper bound



cz_rz_rx

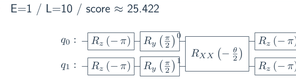
PROJECTIVE E=2 / L=8 / score ≈ 25.616 upper bound

$U = -e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.430 upper bound



$U = e^{-i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.421 upper bound



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 conditional minimum

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 42.057 upper bound



google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.257 upper bound

ibm_cx

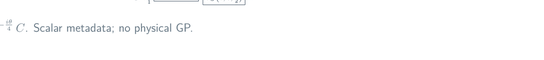
PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound



$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound



$U = -(-1)^{\frac{3}{4}}\,e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 upper bound



$U = ie^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

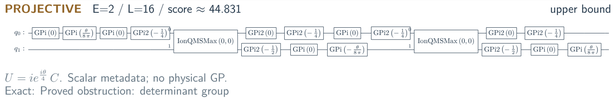
PROJECTIVE E=2 / L=16 / score ≈ 45.635 upper bound



$U = ie^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

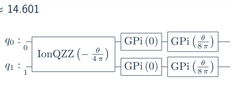
PROJECTIVE E=2 / L=16 / score ≈ 44.831 upper bound



$U = ie^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.601 conditional minimum



$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 upper bound

$U = -e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 conditional minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.036 conditional minimum

$U = e^{-i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.950 upper bound

$U = -ie^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.936 conditional minimum

$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound

$U = -(-1)^{\frac{3}{4}}\,e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.066 upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.940 conditional minimum

$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.036 upper bound

$U = ie^{-i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 conditional minimum

$U = e^{i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.026 conditional minimum

$U = e^{-i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.614 conditional minimum

$U = e^{-i\frac{\theta}{4}}\,C$. Scalar metadata; no physical GP.

CU3 $(\theta, \phi, lbda)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

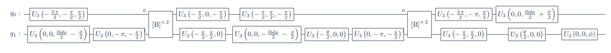
PROJECTIVE E=2 / L=16 / score ≈ 43.852 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=14 / score ≈ 48.284 upper bound



$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

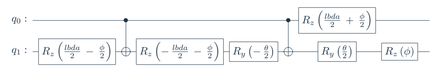
EXACT PHASE E=2 / L=12 / score ≈ 33.616 upper bound



$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_rz_ry

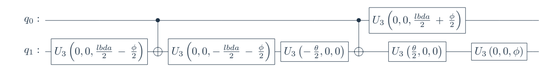
PROJECTIVE E=2 / L=6 / score ≈ 21.012 upper bound



$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.036 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 38.628 upper bound



$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 29.860 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_cz

EXACT PHASE E=2 / L=10 / score ≈ 33.042 upper bound



google_fsim

EXACT PHASE E=2 / L=10 / score ≈ 33.050 upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=14 / score ≈ 49.866 upper bound

google_syc

EXACT PHASE E=12 / L=14 / score ≈ 81.066 upper bound



ibm_cx

PROJECTIVE E=2 / L=11 / score ≈ 31.014 upper bound

$U = -e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

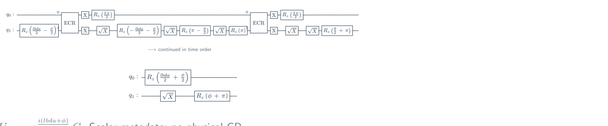
PROJECTIVE E=2 / L=20 / score ≈ 50.022 upper bound



$U = \eta_{41}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

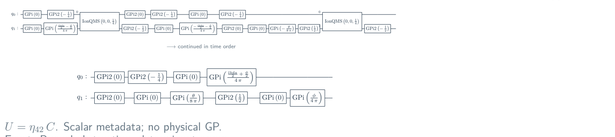
PROJECTIVE E=2 / L=19 / score ≈ 47.418 upper bound



$U = -e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

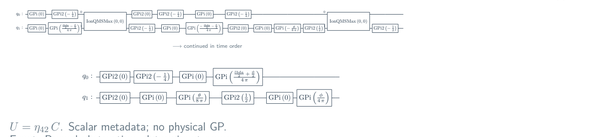
PROJECTIVE E=2 / L=26 / score ≈ 68.646 upper bound



$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

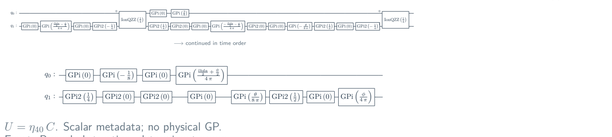
PROJECTIVE E=2 / L=26 / score ≈ 67.842 upper bound



$U = \eta_{12}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

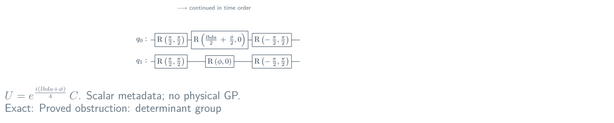
PROJECTIVE E=2 / L=28 / score ≈ 73.442 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=22 / score ≈ 58.288 upper bound



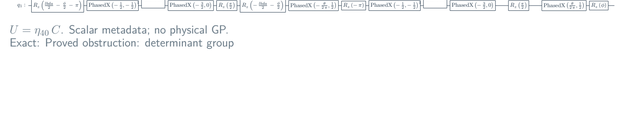
$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 upper bound

quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 42.103 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=15 / score ≈ 42.507 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=20 / score ≈ 50.022 upper bound



$U = \eta_{41}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

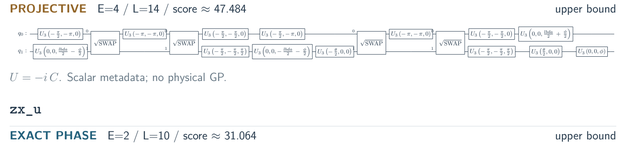
sqrt_iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.484 upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 upper bound



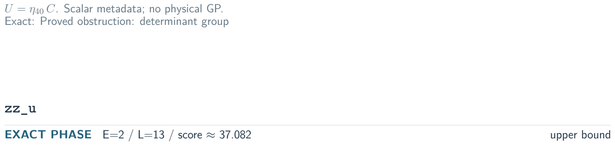
PROJECTIVE E=4 / L=14 / score ≈ 47.484 upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=16 / score ≈ 44.521 upper bound



$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 35.472 upper bound

xx_rz_ry

PROJECTIVE E=2 / L=20 / score ≈ 51.844 upper bound

$U = \eta_{40}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=10 / score ≈ 31.064 upper bound

zz_u

EXACT PHASE E=2 / L=13 / score ≈ 37.082 upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.076 upper bound

$U = i\, C$. Scalar metadata; no physical GP.

CX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=5 / score ≈ 15.218 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — upper bound



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



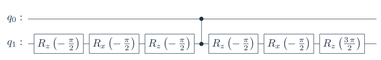
cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.212 — proved minimum



cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved minimum



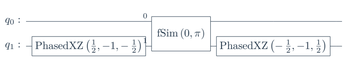
google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — upper bound

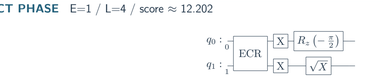


google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 17.824 — upper bound



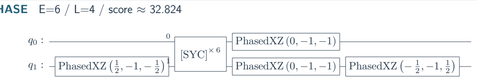
PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — upper bound



ibm_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



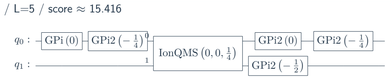
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved minimum



ionq_aria

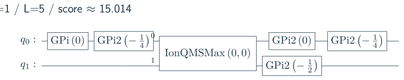
PROJECTIVE E=1 / L=5 / score ≈ 15.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

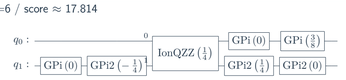
PROJECTIVE E=1 / L=5 / score ≈ 15.014 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

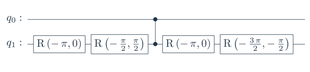
PROJECTIVE E=1 / L=6 / score ≈ 17.814 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

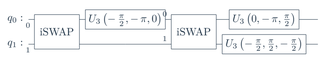
iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved minimum



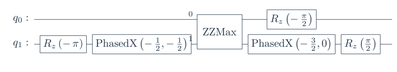
iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved minimum



quantinuum_zzmax

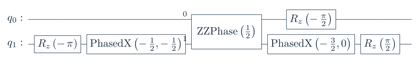
PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

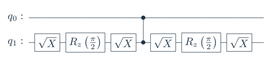
PROJECTIVE E=1 / L=5 / score ≈ 15.216 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

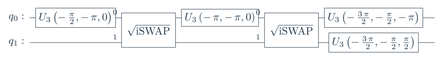
rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



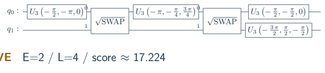
sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — upper bound

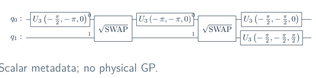


sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — upper bound



PROJECTIVE E=2 / L=4 / score ≈ 17.224 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

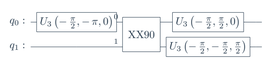
PROJECTIVE E=1 / L=5 / score ≈ 15.222 — upper bound



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

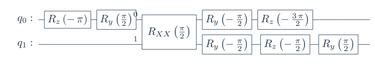
xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved minimum



xx_rz_ry

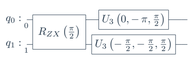
PROJECTIVE E=1 / L=7 / score ≈ 19.416 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

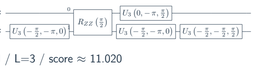
zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.014 — upper bound

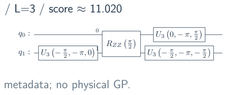


zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — upper bound



PROJECTIVE E=1 / L=3 / score ≈ 11.020 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

CXPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

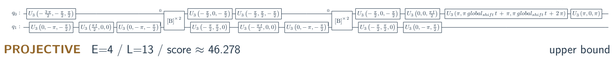
PROJECTIVE E=2 / L=15 / score ≈ 41.850 upper bound



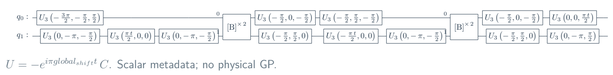
$U = ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 50.290 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.278 upper bound



$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=18 / score ≈ 45.420 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 31.414 upper bound



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

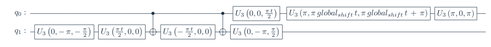
PROJECTIVE E=2 / L=5 / score ≈ 19.010 upper bound



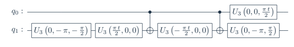
$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 upper bound



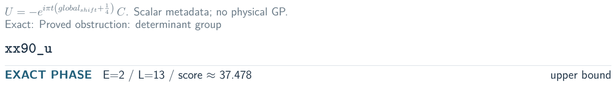
PROJECTIVE E=2 / L=5 / score ≈ 19.030 upper bound



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=2 / L=11 / score ≈ 31.822 upper bound



$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=11 / score ≈ 31.866 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 27.854 upper bound



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=11 / score ≈ 33.066 upper bound



PROJECTIVE E=2 / L=9 / score ≈ 29.054 upper bound



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=11 / score ≈ 33.074 upper bound

PROJECTIVE E=2 / L=9 / score ≈ 29.062 upper bound

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=15 / score ≈ 49.890 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.878 upper bound

$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.090 upper bound



PROJECTIVE E=12 / L=13 / score ≈ 77.078 upper bound

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

ibm_cx

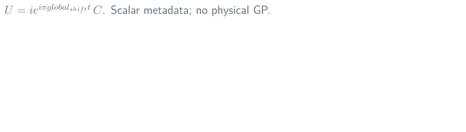
PROJECTIVE E=2 / L=11 / score ≈ 31.014 upper bound



$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=19 / score ≈ 48.222 upper bound



$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=19 / score ≈ 47.418 upper bound

$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=24 / score ≈ 62.260 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=24 / score ≈ 61.456 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=26 / score ≈ 67.056 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=17 / score ≈ 46.268 upper bound

$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.890 upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.878 upper bound

$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 41.442 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=15 / score ≈ 41.846 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=19 / score ≈ 48.222 upper bound

$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.490 upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.478 upper bound

$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.890 upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.478 upper bound

$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=15 / score ≈ 41.858 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=13 / score ≈ 37.478 upper bound

PROJECTIVE E=2 / L=11 / score ≈ 33.466 upper bound

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=19 / score ≈ 49.842 upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 33.070 upper bound

PROJECTIVE E=2 / L=9 / score ≈ 29.058 upper bound

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 39.088 upper bound

PROJECTIVE E=2 / L=11 / score ≈ 33.070 upper bound

$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

CY

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

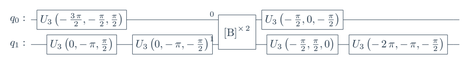
PROJECTIVE E=1 / L=7 / score ≈ 19.622 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

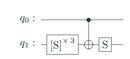
berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — upper bound



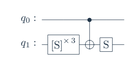
clifford_only

EXACT PHASE E=1 / L=4 / score ≈ 12.000 — proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=4 / score ≈ 12.000 — proved minimum



clifford_t_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.000 — proved minimum

cx_h_phase

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved minimum

cx_rz_ry

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved minimum

cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved minimum

cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 16.812 — proved minimum

PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved minimum



google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.228 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 22.236 — upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.236 — upper bound

ibm_cx

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved minimum

ibm_cz

EXACT PHASE E=1 / L=5 / score ≈ 14.404 — proved minimum



ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved minimum

ionq_aria

PROJECTIVE E=1 / L=9 / score ≈ 24.624 — upper bound

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=9 / score ≈ 24.222 — upper bound

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=10 / score ≈ 27.022 — upper bound

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=8 / score ≈ 21.632 — proved minimum

iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved minimum

quantinuum_zzmax

PROJECTIVE E=1 / L=6 / score ≈ 17.416 — proved minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=6 / score ≈ 17.618 — upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=5 / score ≈ 14.404 — proved minimum

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — upper bound

sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — upper bound

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=7 / score ≈ 19.626 — upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved minimum

xx_rz_ry

PROJECTIVE E=1 / L=9 / score ≈ 23.820 — upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — upper bound

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — upper bound

PROJECTIVE E=1 / L=5 / score ≈ 15.432 — upper bound

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

CZ

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

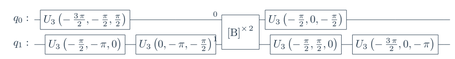
PROJECTIVE E=1 / L=9 / score ≈ 24.030 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

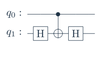
berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — upper bound



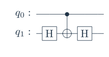
clifford_only

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



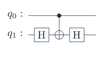
clifford_t_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



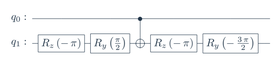
cx_h_phase

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved minimum



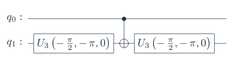
cx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.408 — proved minimum



cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved minimum



cz_rz_rx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



cz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



google_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.628 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.036 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

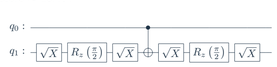
google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.636 — upper bound



ibm_cx

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved minimum



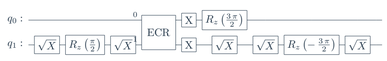
ibm_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



ibm_ecr

EXACT PHASE E=1 / L=10 / score ≈ 24.606 — proved minimum



ionq_aria

PROJECTIVE E=1 / L=9 / score ≈ 24.224 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=9 / score ≈ 23.822 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=10 / score ≈ 26.622 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



iswap_u

EXACT PHASE E=2 / L=5 / score ≈ 19.430 — proved minimum



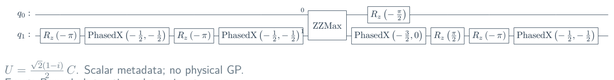
PROJECTIVE E=2 / L=4 / score ≈ 17.024 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

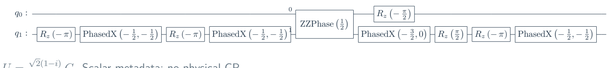
PROJECTIVE E=1 / L=9 / score ≈ 23.826 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=9 / score ≈ 24.028 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

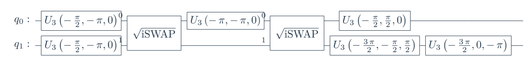
rigetti_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound



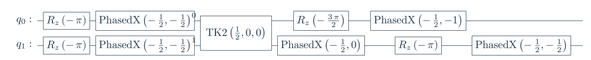
PROJECTIVE E=2 / L=6 / score ≈ 21.636 — upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

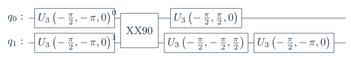
PROJECTIVE E=1 / L=9 / score ≈ 24.034 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved minimum



xx_rz_ry

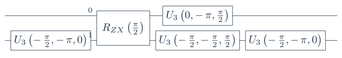
PROJECTIVE E=1 / L=11 / score ≈ 27.824 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

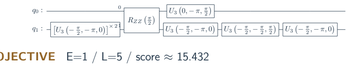
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — upper bound



zz_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — upper bound



PROJECTIVE E=1 / L=5 / score ≈ 15.432 — upper bound



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

CZPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

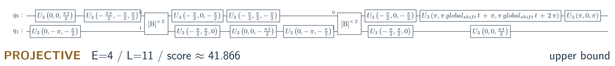
PROJECTIVE E=1 / L=10 / score ≈ 26.230 — conditional minimum



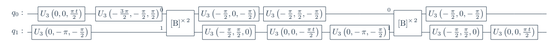
$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=13 / score ≈ 45.878 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.866 — upper bound



$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.606 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — upper bound



$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.618 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cz_rz_rx

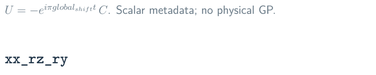
PROJECTIVE E=2 / L=8 / score ≈ 25.616 — upper bound



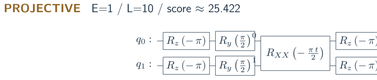
$U = -e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.242 — upper bound



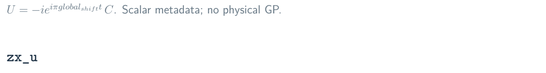
PROJECTIVE E=2 / L=5 / score ≈ 19.430 — upper bound



$U = e^{\frac{i\pi\left(4 global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — upper bound



PROJECTIVE E=2 / L=5 / score ≈ 19.430 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 16.840 — conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 12.828 — conditional minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=13 / score ≈ 45.078 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.066 — upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 76.278 — upper bound

PROJECTIVE E=12 / L=11 / score ≈ 72.266 — upper bound

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — upper bound



$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — upper bound



$U = \eta_{43} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 — upper bound



$U = ie^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 44.644 — upper bound



$U = ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

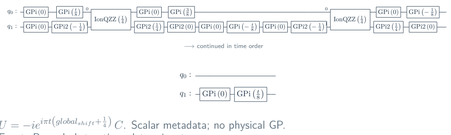
PROJECTIVE E=2 / L=16 / score ≈ 43.840 — upper bound



$U = ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=18 / score ≈ 49.440 — upper bound



$U = -ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 — upper bound

$U = -e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — conditional minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — conditional minimum

$U = e^{\frac{i\pi\left(4 global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.620 — upper bound

$U = -ie^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.606 — conditional minimum

$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — upper bound

$U = \eta_{43} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.078 — upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 — upper bound

$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.478 — upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 — upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.610 — conditional minimum

$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 — upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — upper bound

$U = ie^{\frac{i\pi\left(4 global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 — conditional minimum

$U = e^{\frac{i\pi\left(4 global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.026 — conditional minimum

$U = e^{\frac{i\pi\left(4 global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.614 — conditional minimum

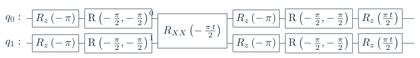
$U = e^{\frac{i\pi\left(4 global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

CZPowGate (t)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

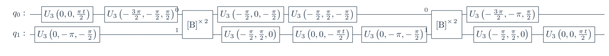
PROJECTIVE E=1 / L=10 / score ≈ 26.230 conditional minimum



$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.866 upper bound



$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 upper bound



cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 upper bound

cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 25.616 upper bound



$U = -e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.430 upper bound



$U = e^{\frac{-i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 19.430 upper bound



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.828 conditional minimum

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 41.066 upper bound

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.266 upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 upper bound

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound

$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 upper bound

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 44.644 upper bound

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 43.840 upper bound

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 13.610 conditional minimum

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 upper bound

$U = -e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 conditional minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.036 conditional minimum

$U = e^{\frac{-i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.620 upper bound

$U = -ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.606 conditional minimum

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 upper bound

$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.066 upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.610 conditional minimum

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.036 upper bound

$U = ie^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 conditional minimum

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.026 conditional minimum

$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.614 conditional minimum

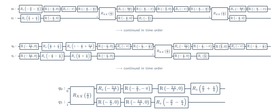
$U = e^{\frac{-i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

CZS (θ, ϕ, γ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

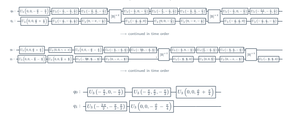
PROJECTIVE E=4 / L=40 / score ≈ 106.532 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=8 / L=37 / score ≈ 122.222 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

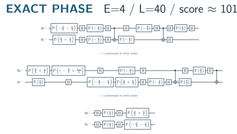
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

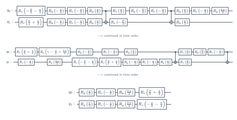
cx_h_phase

EXACT PHASE E=4 / L=40 / score ≈ 101.640 upper bound



cx_rz_ry

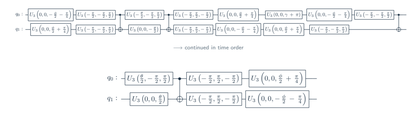
PROJECTIVE E=4 / L=40 / score ≈ 105.680 upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

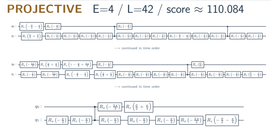
cx_u

EXACT PHASE E=4 / L=21 / score ≈ 67.726 upper bound



cz_rz_rx

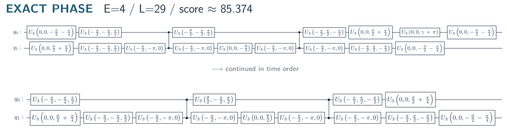
PROJECTIVE E=4 / L=42 / score ≈ 110.084 upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

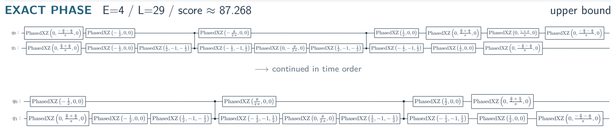
cz_u

EXACT PHASE E=4 / L=29 / score ≈ 85.374 upper bound



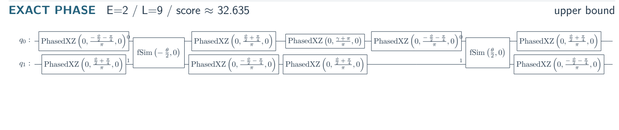
google_cz

EXACT PHASE E=4 / L=29 / score ≈ 87.268 upper bound



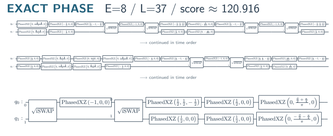
google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 32.635 upper bound



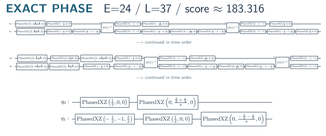
google_sqrt_iswap

EXACT PHASE E=8 / L=37 / score ≈ 120.916 upper bound



google_syc

EXACT PHASE E=24 / L=37 / score ≈ 183.316 upper bound



ibm_cx

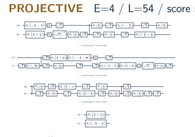
PROJECTIVE E=4 / L=32 / score ≈ 84.832 upper bound



$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=4 / L=54 / score ≈ 130.448 upper bound



$U = -e^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

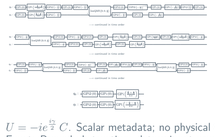
PROJECTIVE E=4 / L=44 / score ≈ 108.832 upper bound



$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

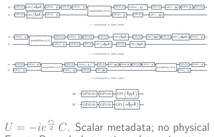
PROJECTIVE E=4 / L=58 / score ≈ 153.634 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=4 / L=58 / score ≈ 152.026 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

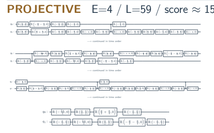
PROJECTIVE E=4 / L=62 / score ≈ 163.226 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

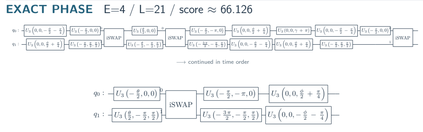
PROJECTIVE E=4 / L=59 / score ≈ 151.836 upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

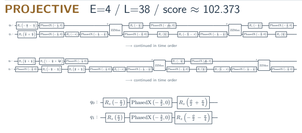
iswap_u

EXACT PHASE E=4 / L=21 / score ≈ 66.126 upper bound



quantinuum_zzmax

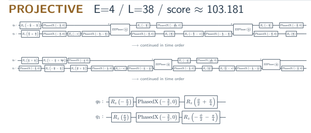
PROJECTIVE E=4 / L=38 / score ≈ 102.373 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

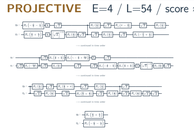
PROJECTIVE E=4 / L=38 / score ≈ 103.181 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

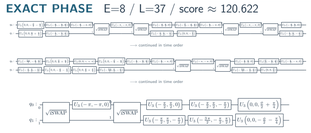
PROJECTIVE E=4 / L=54 / score ≈ 130.448 upper bound



$U = -e^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

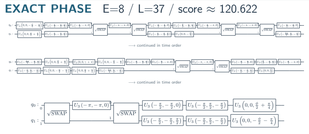
sqrt_iswap_u

EXACT PHASE E=8 / L=37 / score ≈ 120.622 upper bound



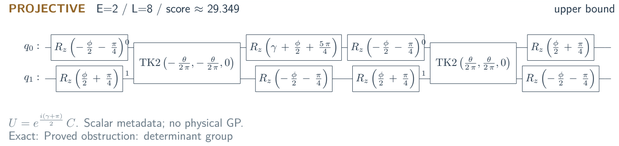
sqrt_swap_u

EXACT PHASE E=8 / L=37 / score ≈ 120.622 upper bound



tket_tk2

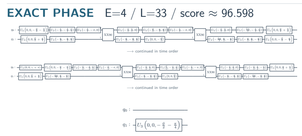
PROJECTIVE E=2 / L=8 / score ≈ 29.349 upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

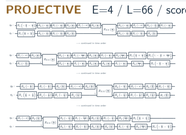
xx90_u

EXACT PHASE E=4 / L=33 / score ≈ 96.598 upper bound



xx_rz_ry

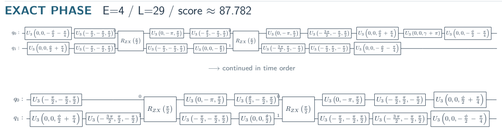
PROJECTIVE E=4 / L=66 / score ≈ 162.540 upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

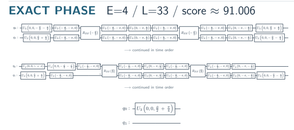
zx_u

EXACT PHASE E=4 / L=29 / score ≈ 87.782 upper bound



zz_u

EXACT PHASE E=4 / L=33 / score ≈ 91.006 upper bound

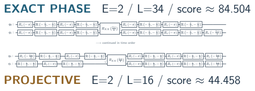


DB

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 84.504 — proved minimum



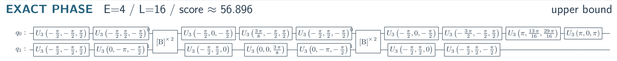
PROJECTIVE E=2 / L=16 / score ≈ 44.458 — upper bound



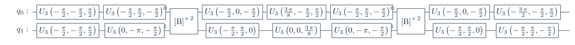
$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 56.896 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 51.284 — upper bound



$U = (-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

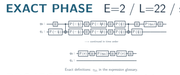
Exact: Proved obstruction: outside clifford t number field
Projective: Proved obstruction: outside clifford t number field

clifford_t_cz

Exact: Proved obstruction: outside clifford t number field
Projective: Proved obstruction: outside clifford t number field

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.424 — proved minimum



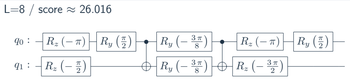
PROJECTIVE E=2 / L=14 / score ≈ 38.416 — proved minimum



$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

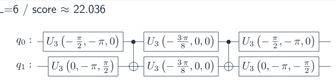
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 26.016 — proved minimum



cx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved minimum



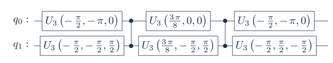
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 35.224 — proved minimum



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 23.236 — proved minimum



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.309 — proved minimum



Exact definitions: η_{44}, η_{45} in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.636 — proved minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.604 — proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 53.157 — upper bound

PROJECTIVE E=4 / L=14 / score ≈ 48.484 — upper bound

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 84.357 — upper bound



Exact definitions: η_{44}, η_{45} in the expression glossary.

PROJECTIVE E=12 / L=14 / score ≈ 79.684 — upper bound

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.612 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 33.608 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.014 — proved minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.820 — proved minimum



PROJECTIVE E=2 / L=14 / score ≈ 38.416 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.452 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 54.052 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.648 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.248 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 59.248 — upper bound



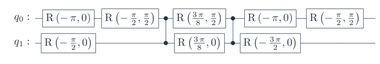
PROJECTIVE E=2 / L=22 / score ≈ 58.848 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

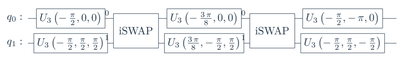
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.632 — proved minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.236 — proved minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.886 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 26.628 — proved minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.256 — proved minimum



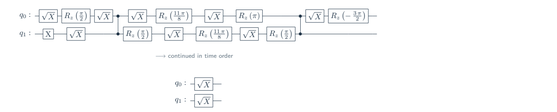
PROJECTIVE E=2 / L=15 / score ≈ 42.452 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

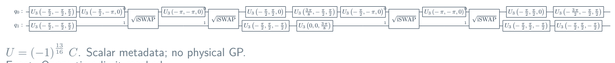
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.014 — proved minimum



sqrt_iswap_u

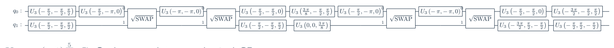
PROJECTIVE E=4 / L=14 / score ≈ 50.484 — upper bound



$U = (-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

sqrt_swap_u

PROJECTIVE E=4 / L=14 / score ≈ 50.484 — upper bound



$U = -(-1)^{\frac{5}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

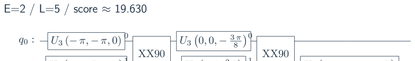
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.206 — proved minimum



xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved minimum



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 81.272 — proved minimum



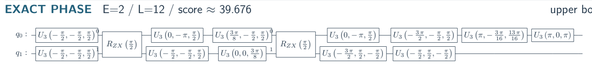
PROJECTIVE E=2 / L=29 / score ≈ 72.462 — upper bound



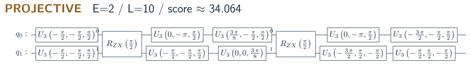
$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 39.676 — upper bound



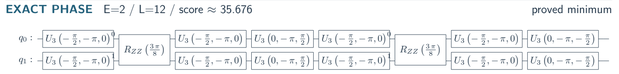
PROJECTIVE E=2 / L=10 / score ≈ 34.064 — upper bound



$U = -(-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.676 — proved minimum



DCX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=12 / score ≈ 38.446 — upper bound



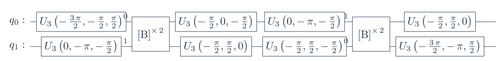
PROJECTIVE E=2 / L=10 / score ≈ 30.436 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=8 / score ≈ 35.248 — upper bound



clifford_only

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



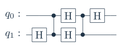
clifford_t_cx

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



clifford_t_cz

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved minimum



cx_h_phase

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



cx_rz_ry

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



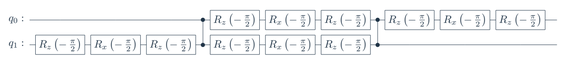
cx_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



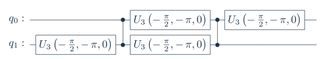
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — proved minimum



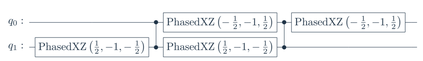
cz_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved minimum



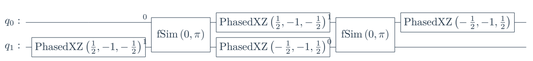
google_cz

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved minimum



google_fsim

EXACT PHASE E=2 / L=4 / score ≈ 17.632 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=8 / score ≈ 34.448 — upper bound

google_syc

EXACT PHASE E=12 / L=8 / score ≈ 65.648 — upper bound



ibm_cx

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum

ibm_cz

EXACT PHASE E=2 / L=12 / score ≈ 32.808 — proved minimum



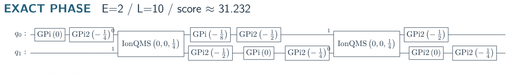
ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.404 — proved minimum



ionq_aria

EXACT PHASE E=2 / L=10 / score ≈ 31.232 — upper bound



PROJECTIVE E=2 / L=10 / score ≈ 30.832 — upper bound



$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=10 / score ≈ 30.428 — upper bound



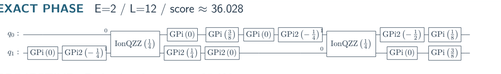
PROJECTIVE E=2 / L=10 / score ≈ 30.028 — upper bound



$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=12 / score ≈ 36.028 — upper bound



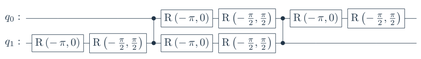
PROJECTIVE E=2 / L=12 / score ≈ 35.628 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.632 — proved minimum



iswap_u

EXACT PHASE E=1 / L=3 / score ≈ 10.818 — proved minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=25 / score ≈ 75.478 — upper bound



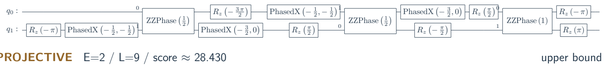
PROJECTIVE E=2 / L=6 / score ≈ 21.220 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=11 / score ≈ 36.436 — upper bound



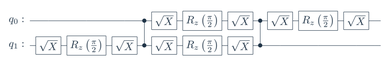
PROJECTIVE E=2 / L=9 / score ≈ 28.430 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

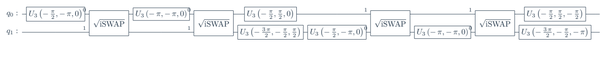
rigetti_cz

EXACT PHASE E=2 / L=12 / score ≈ 32.808 — proved minimum



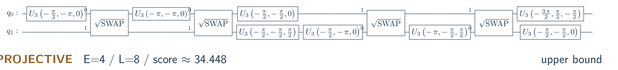
sqrt_iswap_u

EXACT PHASE E=4 / L=8 / score ≈ 34.448 — upper bound

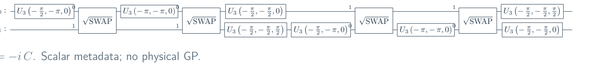


sqrt_swap_u

EXACT PHASE E=4 / L=8 / score ≈ 34.848 — upper bound



PROJECTIVE E=4 / L=8 / score ≈ 34.448 — upper bound



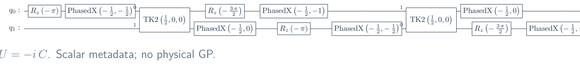
$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=30 / score ≈ 87.328 — upper bound



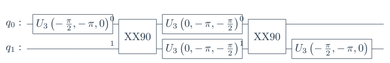
PROJECTIVE E=2 / L=10 / score ≈ 30.444 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved minimum



xx_rz_ry

EXACT PHASE E=3 / L=18 / score ≈ 50.842 — upper bound



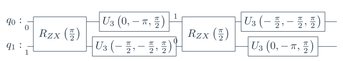
PROJECTIVE E=2 / L=14 / score ≈ 38.832 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

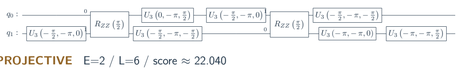
zx_u

EXACT PHASE E=2 / L=4 / score ≈ 18.028 — upper bound



zz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.046 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.040 — upper bound



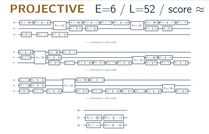
$U = iC$. Scalar metadata; no physical GP.

Deutsch (θ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

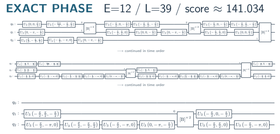
PROJECTIVE E=6 / L=52 / score ≈ 140.576 — upper bound



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=39 / score ≈ 141.034 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

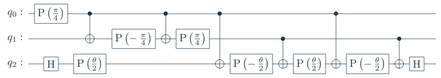
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

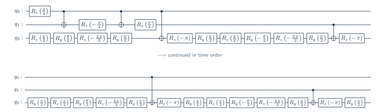
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.014 — upper bound



cx_rz_ry

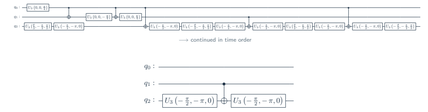
PROJECTIVE E=6 / L=27 / score ≈ 83.254 — upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

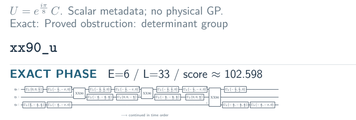
cx_u

EXACT PHASE E=6 / L=15 / score ≈ 59.290 — upper bound



cz_rz_rx

PROJECTIVE E=6 / L=16 / score ≈ 59.832 — upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

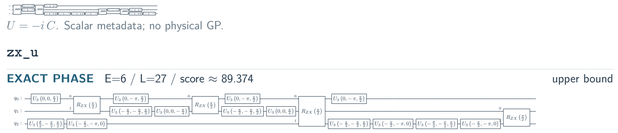
cz_u

EXACT PHASE E=6 / L=11 / score ≈ 50.466 — upper bound



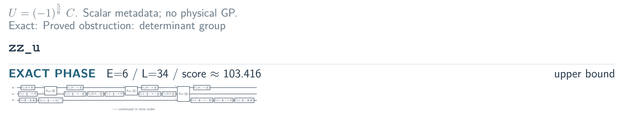
google_cz

EXACT PHASE E=6 / L=11 / score ≈ 50.987 — upper bound



google_fsim

EXACT PHASE E=6 / L=27 / score ≈ 89.507 — upper bound

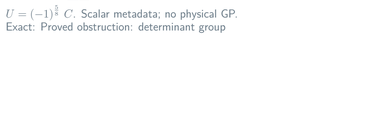


google_sqrt_iswap

EXACT PHASE E=12 / L=39 / score ≈ 139.955 — upper bound

google_syc

EXACT PHASE E=36 / L=39 / score ≈ 233.555 — upper bound



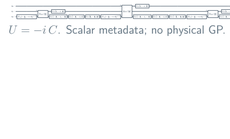
ibm_cx

PROJECTIVE E=6 / L=40 / score ≈ 107.846 — upper bound

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=32 / score ≈ 92.640 — upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=64 / score ≈ 157.058 — upper bound

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=68 / score ≈ 181.093 — upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=68 / score ≈ 178.681 — upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=74 / score ≈ 195.481 — upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=21 / score ≈ 72.084 — upper bound

$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=39 / score ≈ 139.834 — upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=52 / score ≈ 140.673 — upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=52 / score ≈ 141.885 — upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=32 / score ≈ 92.640 — upper bound

$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=39 / score ≈ 138.634 — upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=39 / score ≈ 139.034 — upper bound

PROJECTIVE E=12 / L=39 / score ≈ 138.634 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=52 / score ≈ 141.921 — upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=33 / score ≈ 102.598 — upper bound

xx_rz_ry

PROJECTIVE E=6 / L=68 / score ≈ 173.748 — upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=27 / score ≈ 89.374 — upper bound

zz_u

EXACT PHASE E=6 / L=34 / score ≈ 103.416 — upper bound

PROJECTIVE E=6 / L=33 / score ≈ 101.410 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

Diagonal

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

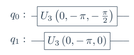
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cx_rz_ry

EXACT PHASE E=3 / L=12 / score ≈ 38.424 upper bound



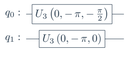
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



cz_rz_rx

EXACT PHASE E=3 / L=24 / score ≈ 64.448 upper bound



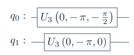
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

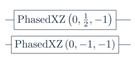


google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

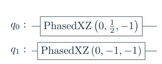
google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



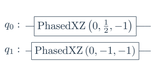
google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



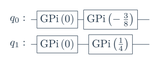
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 9.008 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

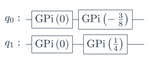
PROJECTIVE E=0 / L=4 / score ≈ 9.008 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 9.008 proved minimum



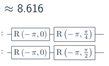
$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=3 / L=22 / score ≈ 60.488 upper bound



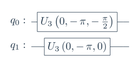
PROJECTIVE E=0 / L=4 / score ≈ 8.616 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

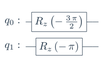
iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



quantinuum_zzmax

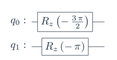
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



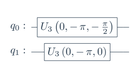
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

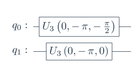
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



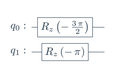
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



tket_tk2

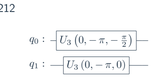
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

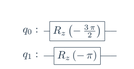
xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



xx_rz_ry

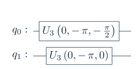
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

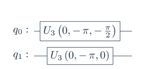
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



ECR

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=9 / score ≈ 27.836 — upper bound



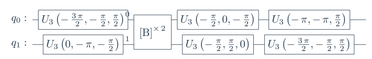
PROJECTIVE E=1 / L=7 / score ≈ 19.826 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound



clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=9 / score ≈ 22.000 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=11 / score ≈ 26.000 — proved minimum



PROJECTIVE E=1 / L=10 / score ≈ 24.000 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=7 / score ≈ 18.608 — proved minimum



PROJECTIVE E=1 / L=7 / score ≈ 18.406 — proved minimum



$U = \sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=16 / score ≈ 51.032 — upper bound



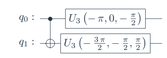
PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved minimum



cz_rz_rx

EXACT PHASE E=4 / L=31 / score ≈ 83.862 — upper bound



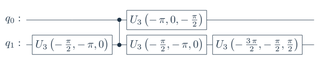
PROJECTIVE E=1 / L=9 / score ≈ 23.618 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved minimum



google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.224 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.428 — upper bound



PROJECTIVE E=1 / L=4 / score ≈ 13.228 — upper bound



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

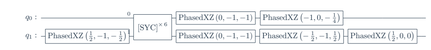
google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — upper bound

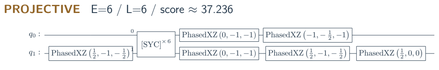


google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.436 — upper bound



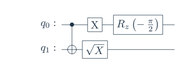
PROJECTIVE E=6 / L=6 / score ≈ 37.236 — upper bound



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

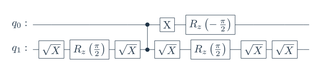
ibm_cx

EXACT PHASE E=1 / L=3 / score ≈ 10.202 — proved minimum



ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved minimum



ibm_ecr

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



ionq_aria

EXACT PHASE E=1 / L=7 / score ≈ 20.420 — upper bound



PROJECTIVE E=1 / L=7 / score ≈ 20.020 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=1 / L=7 / score ≈ 20.018 — upper bound



PROJECTIVE E=1 / L=7 / score ≈ 19.618 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 22.818 — upper bound



PROJECTIVE E=1 / L=8 / score ≈ 22.418 — upper bound



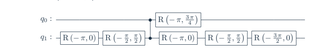
$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=4 / L=24 / score ≈ 69.296 — upper bound



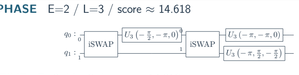
PROJECTIVE E=1 / L=6 / score ≈ 17.424 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 — proved minimum



PROJECTIVE E=2 / L=3 / score ≈ 14.218 — proved minimum



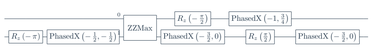
$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=26 / score ≈ 73.880 — upper bound



PROJECTIVE E=1 / L=7 / score ≈ 19.622 — proved minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=9 / score ≈ 27.830 — upper bound



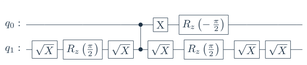
PROJECTIVE E=1 / L=7 / score ≈ 19.824 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

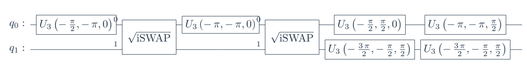
rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved minimum



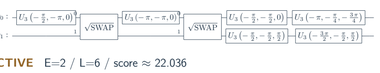
sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — upper bound



sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.036 — upper bound



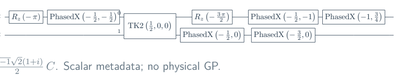
$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=4 / L=27 / score ≈ 76.714 — upper bound



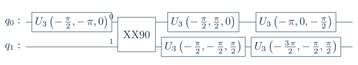
PROJECTIVE E=1 / L=7 / score ≈ 19.830 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

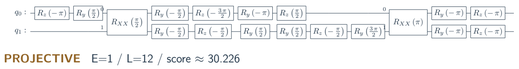
xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 — proved minimum

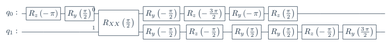


xx_rz_ry

EXACT PHASE E=2 / L=16 / score ≈ 42.236 — upper bound



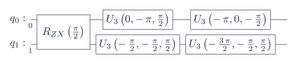
PROJECTIVE E=1 / L=12 / score ≈ 30.226 — upper bound



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

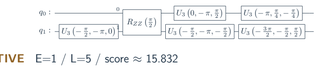
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.826 — upper bound

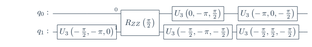


zz_u

EXACT PHASE E=1 / L=5 / score ≈ 16.432 — upper bound



PROJECTIVE E=1 / L=5 / score ≈ 15.832 — upper bound



$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.

fFredkin

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=17 / L=162 / score ≈ 434.912 — upper bound



PROJECTIVE E=5 / L=49 / score ≈ 131.362 — upper bound



$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=10 / L=42 / score ≈ 143.052 — upper bound



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=5 / L=50 / score ≈ 120.000 — upper bound



PROJECTIVE E=5 / L=42 / score ≈ 104.000 — upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=5 / L=52 / score ≈ 124.000 — upper bound



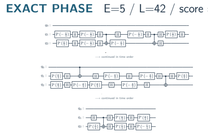
PROJECTIVE E=5 / L=48 / score ≈ 116.000 — upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

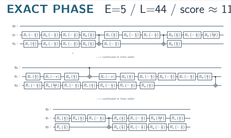
cx_h_phase

EXACT PHASE E=5 / L=42 / score ≈ 110.440 — upper bound



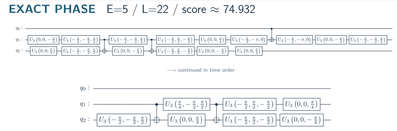
cx_rz_ry

EXACT PHASE E=5 / L=44 / score ≈ 118.888 — upper bound



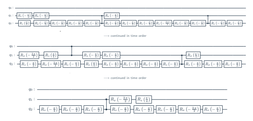
cx_u

EXACT PHASE E=5 / L=22 / score ≈ 74.932 — upper bound



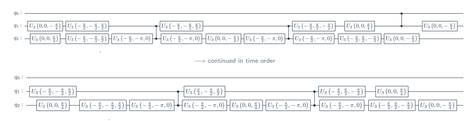
cz_rz_rx

EXACT PHASE E=5 / L=42 / score ≈ 114.884 — upper bound



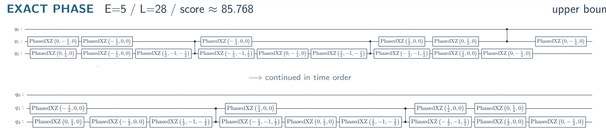
cz_u

EXACT PHASE E=5 / L=28 / score ≈ 88.168 — upper bound



google_cz

EXACT PHASE E=5 / L=28 / score ≈ 85.768 — upper bound



google_fsim

EXACT PHASE E=3 / L=12 / score ≈ 41.684 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — upper bound

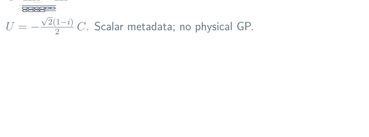


PROJECTIVE E=4 / L=16 / score ≈ 53.296 — upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=30 / L=42 / score ≈ 215.452 — upper bound



ibm_cx

EXACT PHASE E=5 / L=38 / score ≈ 102.036 — upper bound

PROJECTIVE E=5 / L=37 / score ≈ 100.036 — upper bound

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=5 / L=54 / score ≈ 135.652 — upper bound



PROJECTIVE E=5 / L=54 / score ≈ 135.248 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=5 / L=54 / score ≈ 134.238 — upper bound



PROJECTIVE E=5 / L=53 / score ≈ 132.238 — upper bound



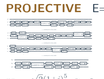
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=17 / L=161 / score ≈ 440.224 — upper bound



PROJECTIVE E=5 / L=65 / score ≈ 169.960 — upper bound



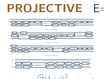
$U = \frac{\sqrt{2}(1+i)}{8}\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=17 / L=161 / score ≈ 433.390 — upper bound



PROJECTIVE E=5 / L=65 / score ≈ 167.950 — upper bound



$U = \frac{\sqrt{2}(1+i)}{8}\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=17 / L=178 / score ≈ 480.990 — upper bound



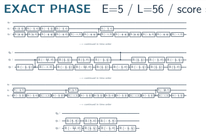
PROJECTIVE E=5 / L=70 / score ≈ 181.950 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

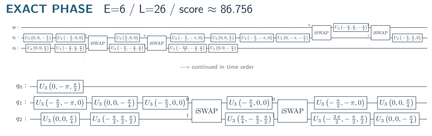
iqm_cz

EXACT PHASE E=5 / L=56 / score ≈ 149.824 — upper bound



iswap_u

EXACT PHASE E=6 / L=26 / score ≈ 86.756 — upper bound



quantinuum_zzmax

EXACT PHASE E=17 / L=117 / score ≈ 333.334 — upper bound



PROJECTIVE E=5 / L=47 / score ≈ 126.338 — upper bound



$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=17 / L=120 / score ≈ 343.970 — upper bound



PROJECTIVE E=5 / L=47 / score ≈ 127.348 — upper bound



$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=5 / L=54 / score ≈ 135.652 — upper bound



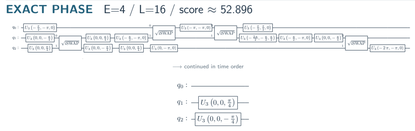
PROJECTIVE E=5 / L=54 / score ≈ 135.248 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

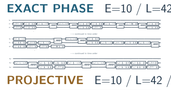
sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.896 — upper bound

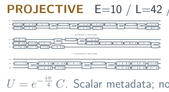


sqrt_swap_u

EXACT PHASE E=10 / L=42 / score ≈ 141.852 — upper bound



PROJECTIVE E=10 / L=42 / score ≈ 141.052 — upper bound



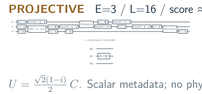
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=15 / L=96 / score ≈ 280.772 — upper bound



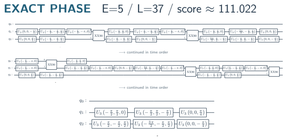
PROJECTIVE E=3 / L=16 / score ≈ 50.860 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=5 / L=37 / score ≈ 111.022 — upper bound



xx_rz_ry

EXACT PHASE E=17 / L=186 / score ≈ 477.406 — upper bound



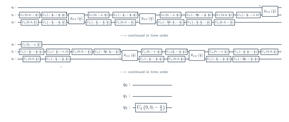
PROJECTIVE E=5 / L=77 / score ≈ 191.164 — upper bound



$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=5 / L=32 / score ≈ 100.002 — upper bound

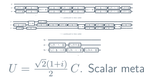


zz_u

EXACT PHASE E=5 / L=38 / score ≈ 107.638 — upper bound



PROJECTIVE E=5 / L=37 / score ≈ 105.232 — upper bound



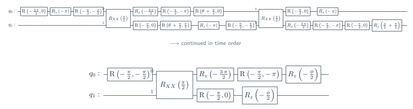
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

fSim (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

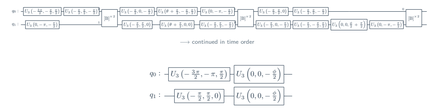
PROJECTIVE E=3 / L=22 / score ≈ 61.276 upper bound



$U = \eta_{18}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=19 / score ≈ 69.514 upper bound



$U = \eta_{19}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

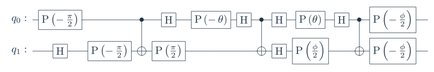
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

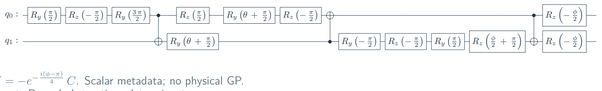
cx_h_phase

EXACT PHASE E=3 / L=14 / score ≈ 41.216 upper bound



cx_rz_ry

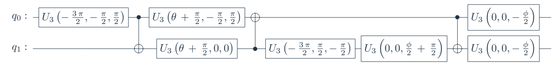
PROJECTIVE E=3 / L=13 / score ≈ 40.626 upper bound



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

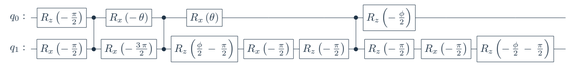
cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.642 upper bound



cz_rz_rx

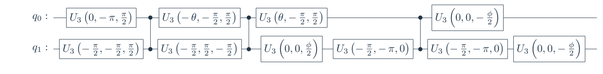
PROJECTIVE E=3 / L=12 / score ≈ 38.024 upper bound



$U = -(-1)^{\frac{3}{4}} e^{-\frac{i\phi}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

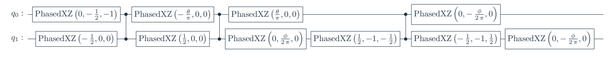
cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 upper bound



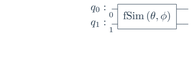
google_cz

EXACT PHASE E=3 / L=10 / score ≈ 35.711 upper bound



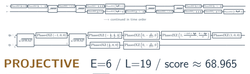
google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.004 upper bound

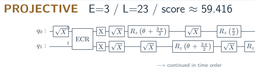


google_sqrt_iswap

EXACT PHASE E=6 / L=21 / score ≈ 74.238 upper bound



PROJECTIVE E=6 / L=19 / score ≈ 68.965 upper bound



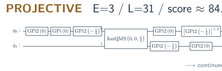
$U = -\frac{\sqrt{2}(1-i)e^{-i\theta}}{2}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=21 / score ≈ 120.438 upper bound



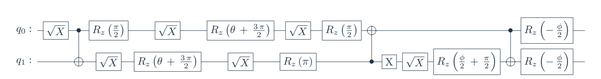
PROJECTIVE E=18 / L=19 / score ≈ 115.765 upper bound



$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=3 / L=15 / score ≈ 43.416 upper bound



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

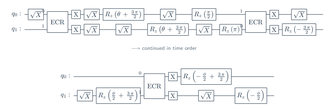
PROJECTIVE E=3 / L=21 / score ≈ 56.024 upper bound



$U = \sqrt[4]{-1}e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

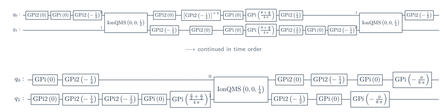
PROJECTIVE E=3 / L=23 / score ≈ 59.416 upper bound



$U = -ie^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

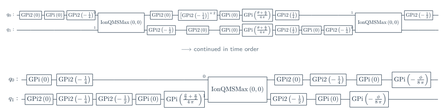
PROJECTIVE E=3 / L=31 / score ≈ 84.131 upper bound



$U = \eta_{19}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

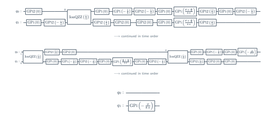
PROJECTIVE E=3 / L=31 / score ≈ 82.925 upper bound



$U = \eta_{19}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

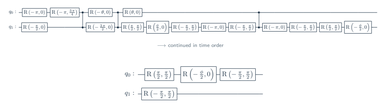
PROJECTIVE E=3 / L=34 / score ≈ 91.325 upper bound



$U = \eta_{19}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

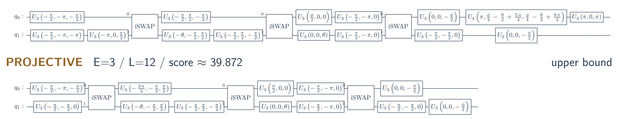
PROJECTIVE E=3 / L=19 / score ≈ 54.676 upper bound



$U = -(-1)^{\frac{3}{4}} e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=15 / score ≈ 46.690 upper bound



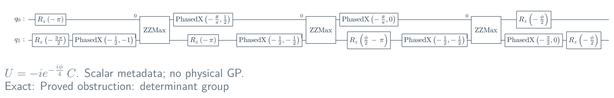
PROJECTIVE E=3 / L=12 / score ≈ 39.872 upper bound



$U = \eta_{50}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

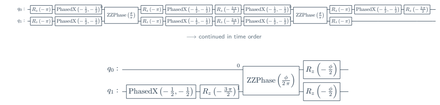
PROJECTIVE E=3 / L=12 / score ≈ 38.897 upper bound



$U = -ie^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

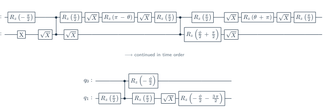
PROJECTIVE E=3 / L=20 / score ≈ 57.653 upper bound



$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

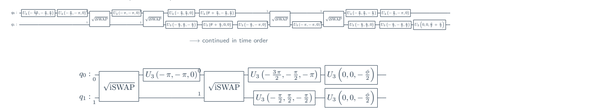
PROJECTIVE E=3 / L=21 / score ≈ 56.024 upper bound



$U = \sqrt[4]{-1}e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.314 upper bound



sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 69.114 upper bound



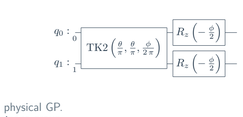
PROJECTIVE E=6 / L=19 / score ≈ 68.314 upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.601 conditional minimum



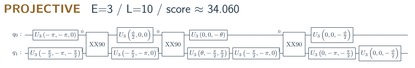
$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 39.272 upper bound



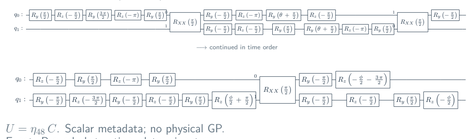
PROJECTIVE E=3 / L=10 / score ≈ 34.060 upper bound



$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=32 / score ≈ 82.270 upper bound



$U = \eta_{48}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 upper bound

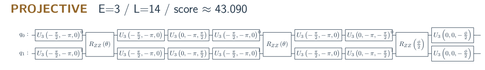


zz_u

EXACT PHASE E=3 / L=16 / score ≈ 47.902 upper bound



PROJECTIVE E=3 / L=14 / score ≈ 43.090 upper bound



$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

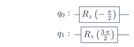
FSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=18 / score ≈ 48.062 — upper bound



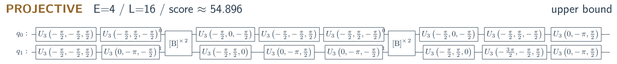


berkeley_b

EXACT PHASE E=4 / L=17 / score ≈ 56.902 — upper bound



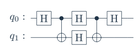
PROJECTIVE E=4 / L=16 / score ≈ 54.896 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved minimum



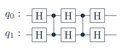
clifford_t_cx

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved minimum



clifford_t_cz

EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved minimum



cx_h_phase

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved minimum



cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 24.816 — proved minimum



cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved minimum



cz_rz_rx

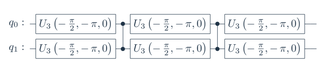
EXACT PHASE E=2 / L=18 / score ≈ 47.636 — proved minimum





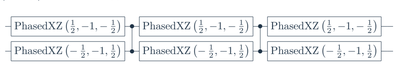
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — proved minimum



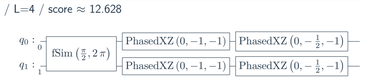
google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — proved minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved minimum



google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.296 — upper bound



ibm_cx

EXACT PHASE E=2 / L=12 / score ≈ 32.808 — proved minimum

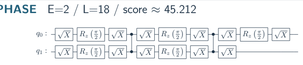


PROJECTIVE E=2 / L=11 / score ≈ 30.606 — upper bound

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

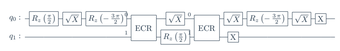
ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved minimum

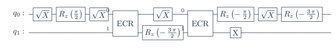


ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — proved minimum



PROJECTIVE E=2 / L=9 / score ≈ 26.808 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=21 / score ≈ 55.254 — upper bound





ionq_aria_full

EXACT PHASE E=2 / L=21 / score ≈ 54.450 — upper bound

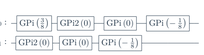




ionq_forte

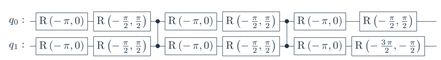
EXACT PHASE E=2 / L=23 / score ≈ 60.050 — upper bound





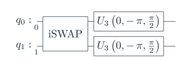
iqm_cz

EXACT PHASE E=2 / L=12 / score ≈ 34.448 — proved minimum



iswap_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — upper bound



quantinuum_zzmax

EXACT PHASE E=2 / L=17 / score ≈ 45.652 — upper bound



PROJECTIVE E=2 / L=10 / score ≈ 30.032 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=17 / score ≈ 46.056 — upper bound





rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved minimum





sqrt_iswap_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved minimum

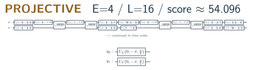


sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — upper bound



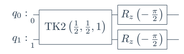
PROJECTIVE E=4 / L=16 / score ≈ 54.096 — upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

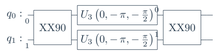
tket_tk2

EXACT PHASE E=1 / L=2 / score ≈ 8.810 — proved minimum



xx90_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved minimum



xx_rz_ry

EXACT PHASE E=2 / L=32 / score ≈ 78.468 — upper bound



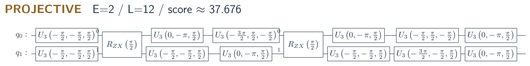


xx_u

EXACT PHASE E=2 / L=13 / score ≈ 39.682 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 37.676 — upper bound



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=15 / score ≈ 43.694 — upper bound



PROJECTIVE E=2 / L=14 / score ≈ 41.688 — upper bound



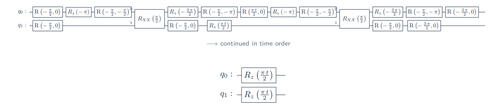
$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

FSWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=18 / score ≈ 48.062 — conditional minimum



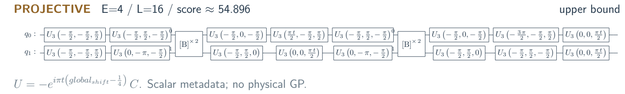
$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=18 / score ≈ 59.708 — upper bound



PROJECTIVE E=4 / L=16 / score ≈ 54.896 — upper bound



$U = -e^{i\pi t\left(global_{shift}-\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

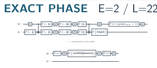
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 53.824 — conditional minimum



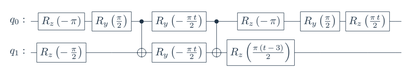
PROJECTIVE E=2 / L=15 / score ≈ 39.818 — conditional minimum



$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

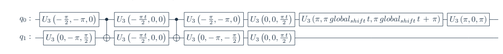
PROJECTIVE E=2 / L=9 / score ≈ 27.418 — conditional minimum



$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cx_u

EXACT PHASE E=2 / L=10 / score ≈ 29.660 — conditional minimum



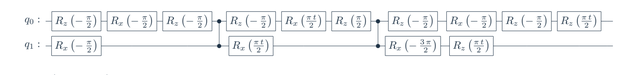
PROJECTIVE E=2 / L=8 / score ≈ 25.648 — conditional minimum



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

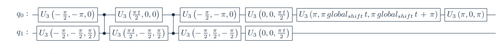
PROJECTIVE E=2 / L=14 / score ≈ 38.828 — conditional minimum



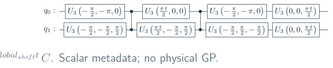
$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 30.860 — conditional minimum



PROJECTIVE E=2 / L=8 / score ≈ 26.848 — conditional minimum



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — conditional minimum



PROJECTIVE E=2 / L=8 / score ≈ 26.248 — conditional minimum



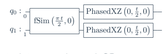
$U = e^{\frac{i\pi t\left(2global_{shift}-1\right)}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — conditional minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.616 — conditional minimum



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=18 / score ≈ 56.308 — upper bound



PROJECTIVE E=4 / L=16 / score ≈ 52.096 — upper bound

$U = \eta_{52}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=18 / score ≈ 87.508 — upper bound



PROJECTIVE E=12 / L=16 / score ≈ 83.296 — upper bound

$U = e^{\frac{i\pi t\left(2global_{shift}-1\right)}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

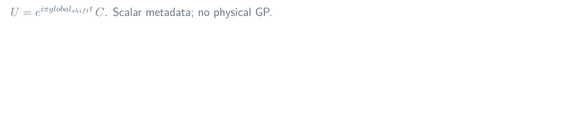
PROJECTIVE E=2 / L=14 / score ≈ 37.212 — conditional minimum



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

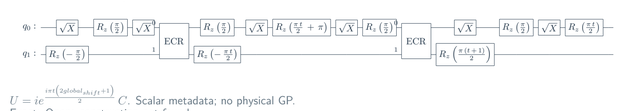
PROJECTIVE E=2 / L=20 / score ≈ 49.618 — conditional minimum



$U = -e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

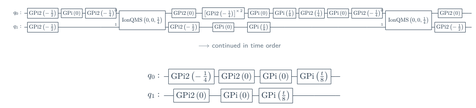
PROJECTIVE E=2 / L=15 / score ≈ 39.818 — conditional minimum



$U = ie^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

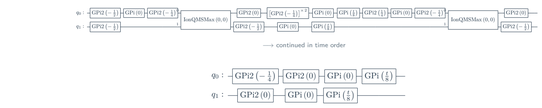
PROJECTIVE E=2 / L=24 / score ≈ 62.460 — conditional minimum



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

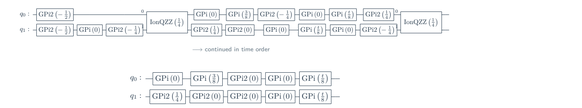
PROJECTIVE E=2 / L=24 / score ≈ 61.656 — conditional minimum



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

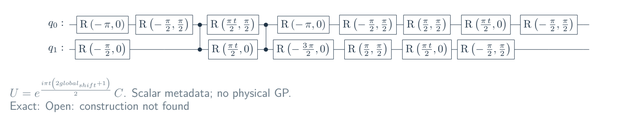
PROJECTIVE E=2 / L=26 / score ≈ 67.256 — conditional minimum



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

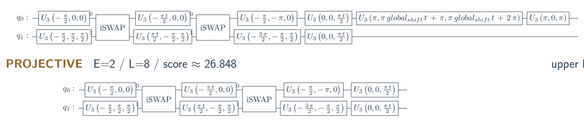
PROJECTIVE E=2 / L=14 / score ≈ 39.856 — conditional minimum



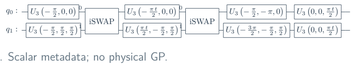
$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.860 — upper bound



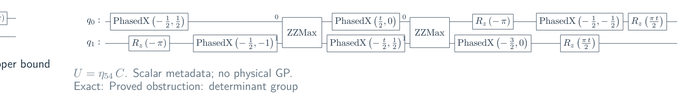
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — upper bound



$U = \eta_{53}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

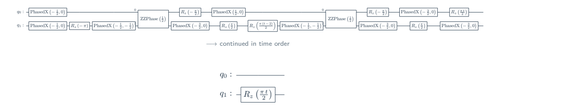
PROJECTIVE E=2 / L=10 / score ≈ 30.232 — conditional minimum



$U = \eta_{54}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

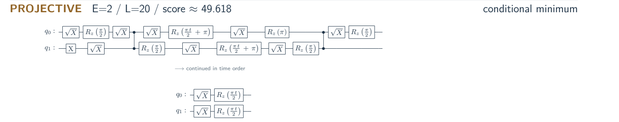
PROJECTIVE E=2 / L=17 / score ≈ 46.056 — conditional minimum



$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=20 / score ≈ 49.618 — conditional minimum



$U = -e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=4 / L=18 / score ≈ 58.908 — upper bound



PROJECTIVE E=4 / L=16 / score ≈ 54.096 — upper bound



$U = -e^{i\pi t\left(global_{shift}-\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=18 / score ≈ 59.308 — upper bound



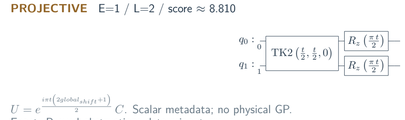
PROJECTIVE E=4 / L=16 / score ≈ 54.096 — upper bound



$U = -ie^{i\pi t\left(global_{shift}-\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

tket_tk2

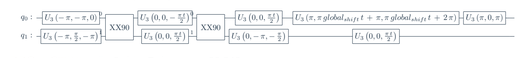
PROJECTIVE E=1 / L=2 / score ≈ 8.810 — conditional minimum



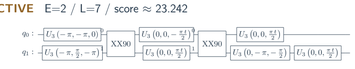
$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=9 / score ≈ 27.254 — conditional minimum



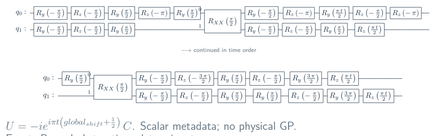
PROJECTIVE E=2 / L=7 / score ≈ 23.242 — conditional minimum



$U = \eta_{55}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=31 / score ≈ 76.066 — conditional minimum



$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=14 / score ≈ 42.488 — conditional minimum



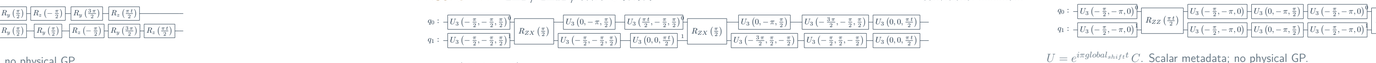
PROJECTIVE E=2 / L=12 / score ≈ 37.676 — conditional minimum

$U = e^{i\pi t\left(global_{shift}-\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=16 / score ≈ 43.300 — conditional minimum

PROJECTIVE E=2 / L=14 / score ≈ 39.288 — conditional minimum



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

Givens (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.710 — conditional minimum



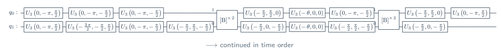
PROJECTIVE E=2 / L=21 / score ≈ 54.866 — upper bound

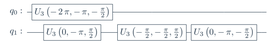


$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=20 / score ≈ 62.120 — upper bound





clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 58.628 — conditional minimum



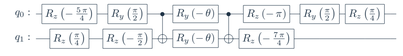
PROJECTIVE E=2 / L=17 / score ≈ 44.622 — conditional minimum



$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

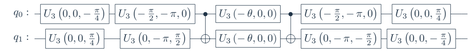
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 30.220 — conditional minimum



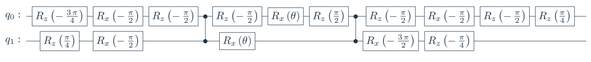
cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — conditional minimum



cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.430 — conditional minimum



cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.660 — conditional minimum

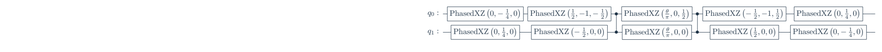


google_cz

EXACT PHASE E=2 / L=12 / score ≈ 36.393 — conditional minimum



PROJECTIVE E=2 / L=10 / score ≈ 31.721 — conditional minimum



$U = e^{-i\theta}\,C$. Scalar metadata; no physical GP.

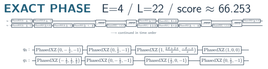
google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.628 — conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=22 / score ≈ 66.253 — upper bound



PROJECTIVE E=4 / L=20 / score ≈ 61.581 — upper bound



$U = -ie^{i\theta}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=22 / score ≈ 97.453 — upper bound



PROJECTIVE E=12 / L=20 / score ≈ 92.781 — upper bound



$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — upper bound



PROJECTIVE E=2 / L=18 / score ≈ 45.624 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

PROJECTIVE E=2 / L=21 / score ≈ 52.422 — conditional minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 42.824 — conditional minimum



PROJECTIVE E=2 / L=16 / score ≈ 42.622 — conditional minimum



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.145 — upper bound



PROJECTIVE E=2 / L=36 / score ≈ 89.745 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 89.341 — upper bound



PROJECTIVE E=2 / L=36 / score ≈ 88.941 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=38 / score ≈ 94.941 — upper bound



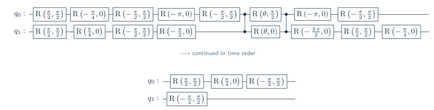
PROJECTIVE E=2 / L=38 / score ≈ 94.541 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

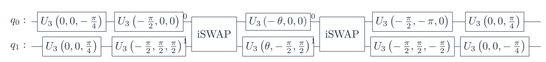
iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 54.280 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.660 — upper bound

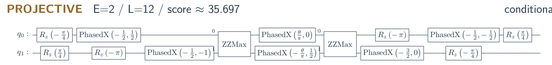


quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.955 — upper bound



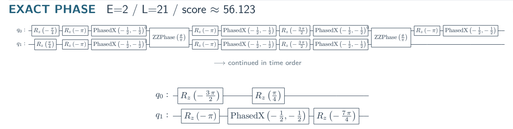
PROJECTIVE E=2 / L=12 / score ≈ 35.697 — conditional minimum



$U = \frac{\sqrt[3]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

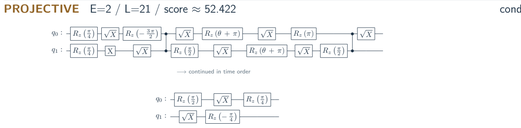
quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 56.123 — conditional minimum



rigetti_cz

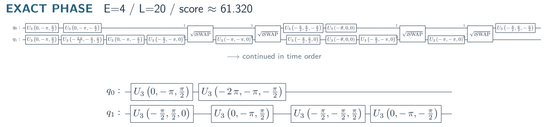
PROJECTIVE E=2 / L=21 / score ≈ 52.422 — conditional minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

sqrt_iswap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.320 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.720 — upper bound



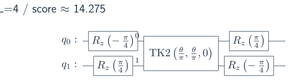
PROJECTIVE E=4 / L=20 / score ≈ 61.320 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

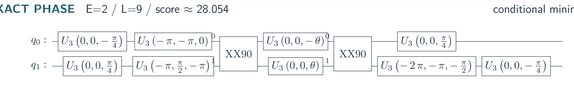
tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 14.275 — conditional minimum



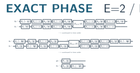
xx90_u

EXACT PHASE E=2 / L=9 / score ≈ 28.054 — conditional minimum



xx_rz_ry

EXACT PHASE E=2 / L=37 / score ≈ 87.478 — conditional minimum



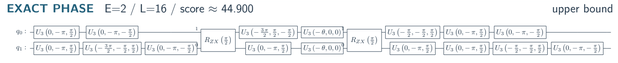
PROJECTIVE E=2 / L=29 / score ≈ 71.662 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

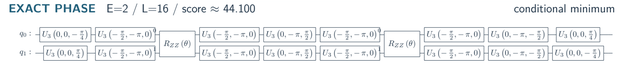
zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — upper bound



zz_u

EXACT PHASE E=2 / L=16 / score ≈ 44.100 — conditional minimum



GP

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

clifford_only

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum



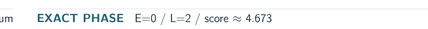
PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.673 proved minimum



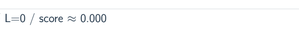
PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.673 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.673 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.673 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

GPi (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

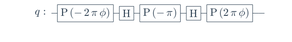
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.006 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum



ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

zz_u

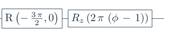
EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

GPi2 (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

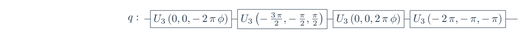
aqt_rxx

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved minimum



berkeley_b

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.812 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=5 / score ≈ 10.610 proved minimum



cx_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

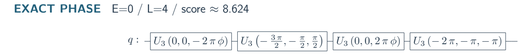
cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum



cz_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

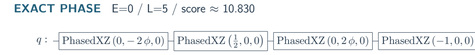


$U = -1\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

iqm_cz

EXACT PHASE E=0 / L=8 / score ≈ 17.832 proved minimum



PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved minimum



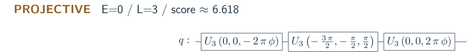
$U = -1\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved minimum

xx90_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=5 / score ≈ 10.610 proved minimum

zx_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

H

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



cx_rz_ry

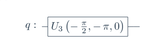
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

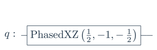
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

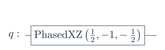


$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



H_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



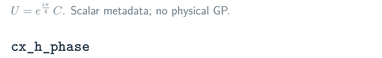
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

H_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

H_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

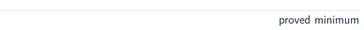
PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

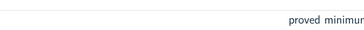


$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

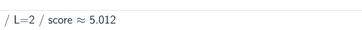


$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

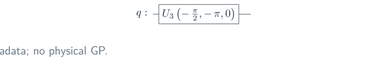


PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

H_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.008 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum



ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

H_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=15 / score ≈ 30.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

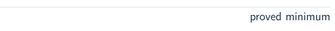
ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

H_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

H_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved minimum



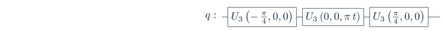
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=16 / score ≈ 33.216 proved minimum



PROJECTIVE E=0 / L=9 / score ≈ 19.210 proved minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

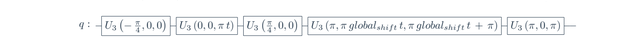
PROJECTIVE E=0 / L=3 / score ≈ 6.806 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=5 / score ≈ 11.210 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

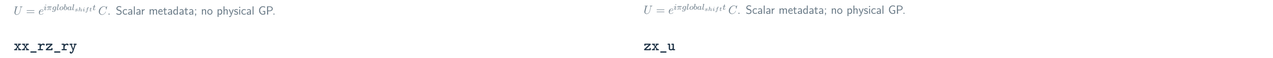
PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved minimum

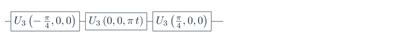
PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

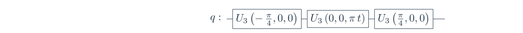
PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved minimum



$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved minimum



$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=0 / L=5 / score ≈ 12.020 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.806 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

HS

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

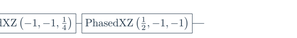
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



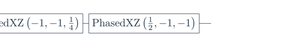
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

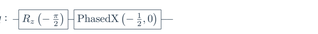
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



HS_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HS_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HS_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HS_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum



ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum



ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

HS_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

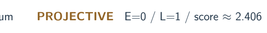


PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HS_q6

1 qubits / 32 preserved native bases / alphabetical panel order

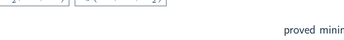
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum



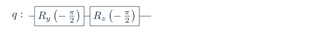
PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



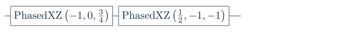
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

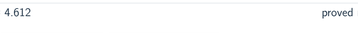
PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



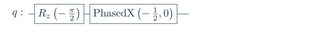
$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HS_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



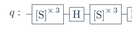
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



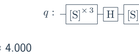
PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

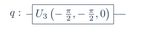
PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

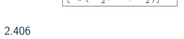
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

HSdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



cx_rz_ry

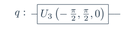
PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

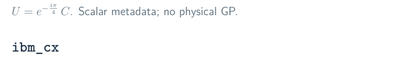
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

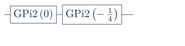
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



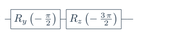
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



HSdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HSdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



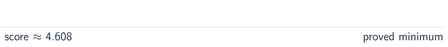
PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

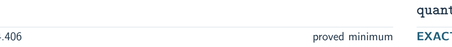
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

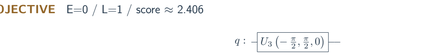
EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

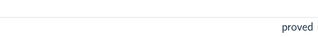
EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

HSdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HSdg_q7

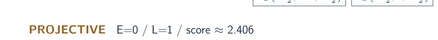
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

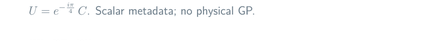
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

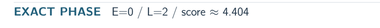
EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



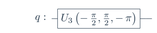
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

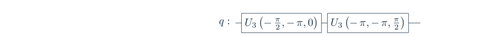
PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



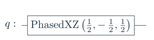
$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



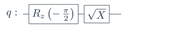
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

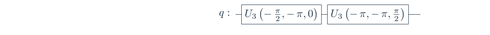
PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum



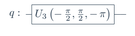
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

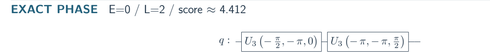
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

HSX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



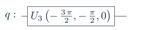
$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

HSX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HSX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

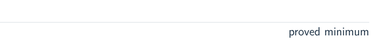
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



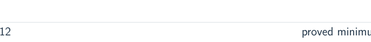
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HSX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

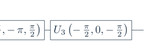


$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

ibm_cx

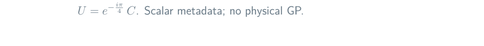
EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



cx_rz_ry

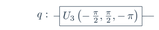
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



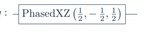
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

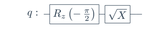
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



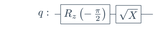
$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

HSX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

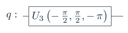


berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



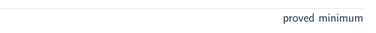
quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

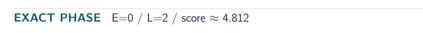


PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

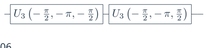


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

HSY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

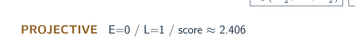
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

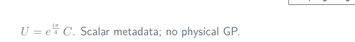
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

HSY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

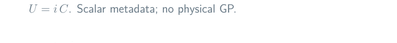
$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HSY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



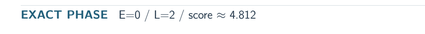
PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum



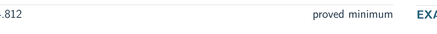
PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

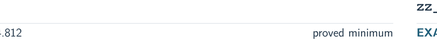


PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

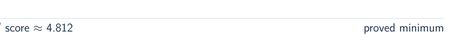
quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



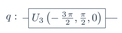
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

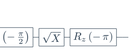
quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



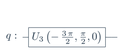
$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



HSY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

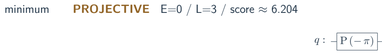
$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

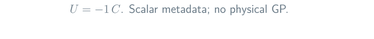


$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

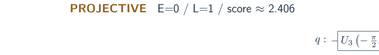
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

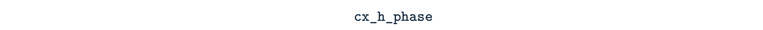
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

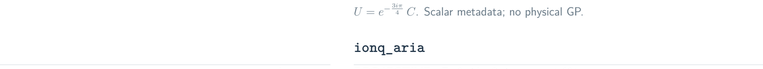
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

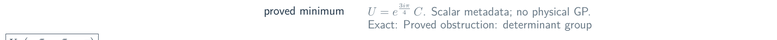
quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HSY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

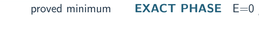
EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

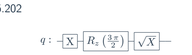
EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



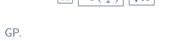
quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum



cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

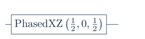


google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

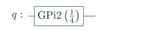
ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



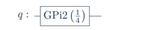
ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



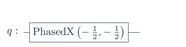
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



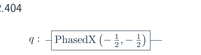
quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

HX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



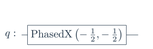
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



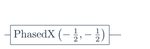
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

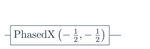
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

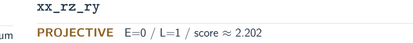
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

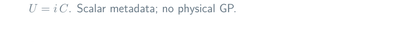
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



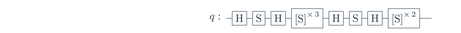
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.006 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

HX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



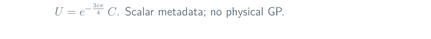
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

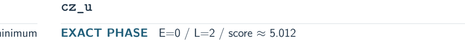
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

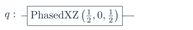


$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

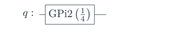
ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum



ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum



ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

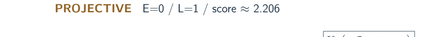
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

HX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

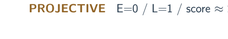
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

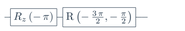
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



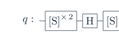
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



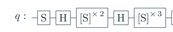
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

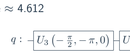


$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

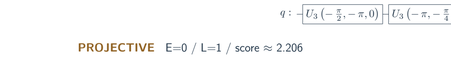
$U = -i\,C$. Scalar metadata; no physical GP.

HY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

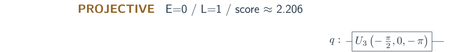
EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



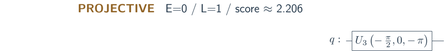
google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

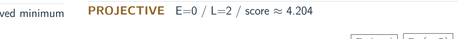
PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

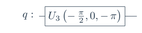
PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum



cx_rz_ry

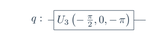
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

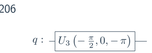
rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



HY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

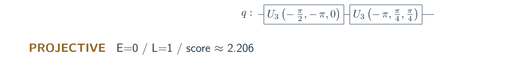
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

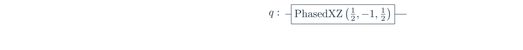
PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

HY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



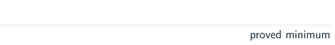
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

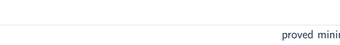


PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum

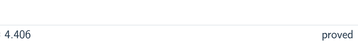


PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

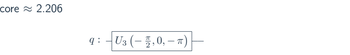


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

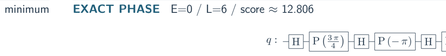


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

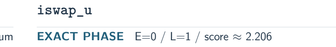
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

HY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



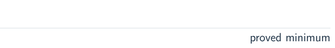
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



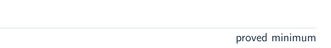
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum



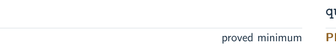
PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

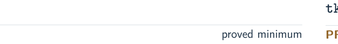
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



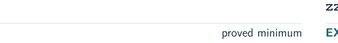
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

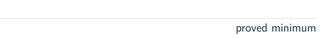
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

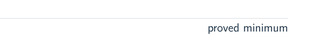
EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HZ

1 qubits / 32 preserved native bases / alphabetical panel order

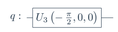
aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



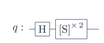
clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



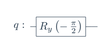
cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum



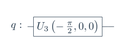
cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

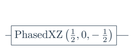
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



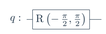
ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



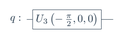
iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-i\frac{\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

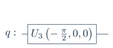
EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

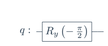
xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



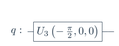
zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



HZ_q1

1 qubits / 32 preserved native bases / alphabetical panel order

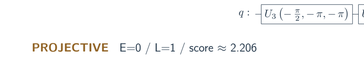
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

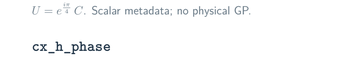
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

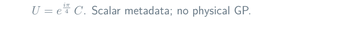
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

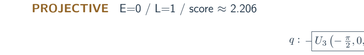
EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

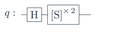
$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



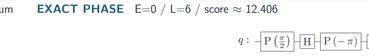
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



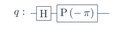
$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

HZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



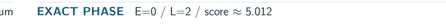
$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.006 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

HZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

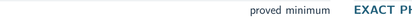
EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

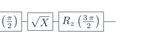
EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

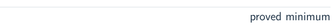
iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

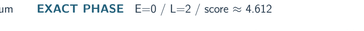
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

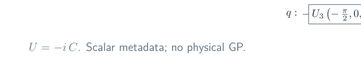


PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

HZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

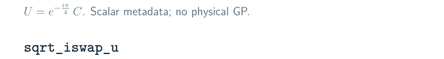
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

berkeley_b

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

clifford_only

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

clifford_t_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

clifford_t_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

cx_h_phase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

cx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

cx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

cz_rz_rx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

cz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

google_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

google_fsim

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

google_sqrt_iswap

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

google_syc

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ibm_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ibm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ibm_ecr

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ionq_aria

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ionq_aria_full

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

ionq_forte

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

iqm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

quantinuum_zzmax

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

quantinuum_zzphase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

rigetti_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

sqrt_iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

sqrt_swap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

tket_tk2

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

xx90_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

xx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

zx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

zz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

q : —

l_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

l_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = i\,C.$ Scalar metadata; no physical GP.

l_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved minimum

$q:$ —[S]$^{\times 3}$—[H]—[S]—[H]—[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[H]—[S]—[H]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

$q:$ —[S]—[H]—[S]$^{\times 2}$—[H]—[S]—[T]—[H]—[S]$^{\times 2}$—[H]—[T]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

$q:$ —[S]—[H]—[S]$^{\times 2}$—[H]—[S]—[T]—[H]—[S]$^{\times 2}$—[H]—[T]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved minimum

$q:$ —[H]—[P$(-\pi)$]—[H]—[P$(\frac{3\pi}{4})$]—[H]—[P$(-\pi)$]—[H]—[P$(\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,-\frac{1}{2},-\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,-\frac{1}{2},-\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,-\frac{1}{2},-\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,-\frac{1}{2},-\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,-\frac{\pi}{4},\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

l_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

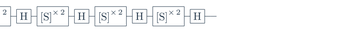


PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

I_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

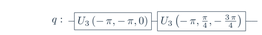
PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=16 / score ≈ 32.000 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

l_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

l_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

$q:$ —[$|\mathrm{S}|^{\times 3}$]—[H]—[$|\mathrm{S}|^{\times 3}$]—[H]—[$|\mathrm{S}|^{\times 3}$]—[H]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

$q:$ —[H]—[$|\mathrm{S}|^{\times 2}$]—[H]—[$\mathrm{T}^\dagger$]—[H]—[$|\mathrm{S}|^{\times 2}$]—[H]—[$\mathrm{T}^\dagger$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

$q:$ —[H]—[$|\mathrm{S}|^{\times 2}$]—[H]—[$\mathrm{T}^\dagger$]—[H]—[$|\mathrm{S}|^{\times 2}$]—[H]—[$\mathrm{T}^\dagger$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[H]—[$\mathrm{P}(-\frac{\pi}{2})$]—[H]—[$\mathrm{P}(-\frac{\pi}{2})$]—[H]—[$\mathrm{P}(-\frac{\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[$\mathrm{PhasedXZ}(-1,0,0)$]—[$\mathrm{PhasedXZ}(-1,-\frac{1}{2},\frac{1}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[$\mathrm{PhasedXZ}(-1,0,0)$]—[$\mathrm{PhasedXZ}(-1,-\frac{1}{2},\frac{1}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[$\mathrm{PhasedXZ}(-1,0,0)$]—[$\mathrm{PhasedXZ}(-1,-\frac{1}{2},\frac{1}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

$q:$ —[$\mathrm{PhasedXZ}(-1,0,0)$]—[$\mathrm{PhasedXZ}(-1,-\frac{1}{2},\frac{1}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{3\pi}{4},-\frac{\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Identity

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



berkeley_b

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



clifford_only

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



cx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



cx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



cz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



google_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



google_fsim

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



google_syc

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ibm_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ibm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ionq_aria

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



ionq_forte

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



iqm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



tket_tk2

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



xx90_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



xx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



zx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum



zz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved minimum

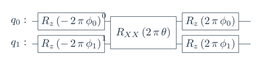


IonQMS (ϕ_0, ϕ_1, θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=4 / score ≈ 12.010 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=19 / score ≈ 58.114 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound



$U = -e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=14 / score ≈ 36.016 — upper bound



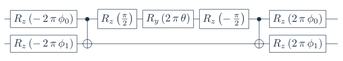
PROJECTIVE E=2 / L=7 / score ≈ 22.010 — upper bound



$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

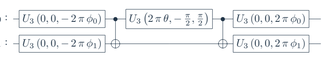
cx_rz_ry

EXACT PHASE E=2 / L=7 / score ≈ 22.414 — upper bound



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — upper bound



cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.220 — upper bound



cz_u

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — upper bound



google_cz

EXACT PHASE E=2 / L=9 / score ≈ 26.854 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.842 — upper bound



$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

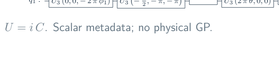
google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.044 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=19 / score ≈ 58.114 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound

$U = -ie^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 89.314 — upper bound

PROJECTIVE E=12 / L=17 / score ≈ 85.302 — upper bound

$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 34.418 — upper bound

PROJECTIVE E=2 / L=9 / score ≈ 26.414 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=14 / score ≈ 37.220 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 37.018 — upper bound

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=12 / score ≈ 32.016 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.012 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — upper bound

ionq_aria_full

EXACT PHASE E=2 / L=28 / score ≈ 69.064 — upper bound

PROJECTIVE E=2 / L=28 / score ≈ 68.664 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=30 / score ≈ 74.664 — upper bound

PROJECTIVE E=2 / L=30 / score ≈ 74.264 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=17 / score ≈ 46.068 — upper bound

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 24.448 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 79.280 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 25.022 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=12 / score ≈ 29.634 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=14 / score ≈ 37.220 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 37.018 — upper bound

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=19 / score ≈ 57.314 — upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.302 — upper bound

$U = -e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=19 / score ≈ 57.714 — upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.302 — upper bound

$U = -ie^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 12.014 — conditional minimum

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 24.848 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.448 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.010 — conditional minimum

zx_u

EXACT PHASE E=1 / L=10 / score ≈ 25.262 — conditional minimum

zz_u

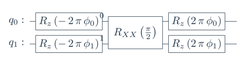
EXACT PHASE E=1 / L=8 / score ≈ 20.850 — conditional minimum

IonQMSMax (ϕ_0, ϕ_1)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=4 / score ≈ 12.210 — proved minimum



berkeley_b

EXACT PHASE E=4 / L=18 / score ≈ 56.308 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.302 — upper bound



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=7 / score ≈ 22.412 — upper bound

PROJECTIVE E=2 / L=7 / score ≈ 22.210 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=7 / score ≈ 22.614 — upper bound



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — upper bound

cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.420 — upper bound



cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — upper bound

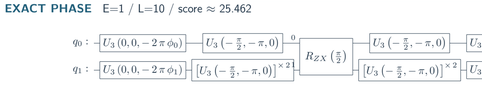


google_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.448 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 23.042 — upper bound



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.844 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=17 / score ≈ 54.902 — upper bound

PROJECTIVE E=4 / L=17 / score ≈ 54.302 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=18 / score ≈ 87.708 — upper bound

PROJECTIVE E=12 / L=17 / score ≈ 85.502 — upper bound

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.614 — upper bound

PROJECTIVE E=2 / L=5 / score ≈ 18.008 — upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — upper bound

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.210 — upper bound

PROJECTIVE E=2 / L=7 / score ≈ 22.412 — upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=28 / score ≈ 70.268 — upper bound

PROJECTIVE E=2 / L=28 / score ≈ 69.868 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=1 / L=0 / score ≈ 4.004 — upper bound

ionq_forte

EXACT PHASE E=2 / L=30 / score ≈ 75.064 — upper bound

PROJECTIVE E=2 / L=30 / score ≈ 74.664 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=17 / score ≈ 46.268 — upper bound

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 24.648 — proved minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 79.480 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 25.222 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=12 / score ≈ 29.834 — proved minimum

rigetti_cz

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — upper bound

sqrt_iswap_u

EXACT PHASE E=4 / L=17 / score ≈ 54.302 — upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.502 — upper bound

$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=17 / score ≈ 54.302 — upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.502 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 12.214 — proved minimum

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 25.048 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.648 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.210 — proved minimum

zx_u

EXACT PHASE E=1 / L=10 / score ≈ 25.462 — proved minimum

zz_u

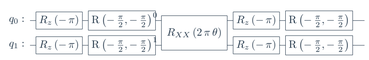
EXACT PHASE E=1 / L=8 / score ≈ 21.050 — proved minimum

IonQZZ (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — conditional minimum

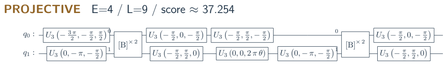


berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 37.254 — upper bound



$U = -e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=9 / score ≈ 26.010 — upper bound



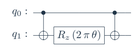
PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound



$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

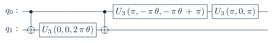
cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.018 — upper bound



PROJECTIVE E=2 / L=1 / score ≈ 10.006 — upper bound



$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — upper bound



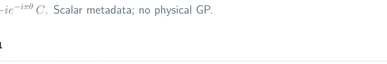
cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — upper bound



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 — upper bound



$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.028 — conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.016 — conditional minimum



$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 40.466 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 71.666 — upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — upper bound

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=12 / score ≈ 35.436 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.036 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 34.632 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 34.232 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — upper bound

iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — upper bound

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.274 — upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.466 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — conditional minimum

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — conditional minimum

zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — conditional minimum

zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — conditional minimum

iSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — proved minimum



PROJECTIVE E=2 / L=16 / score ≈ 43.658 — upper bound



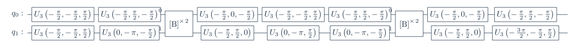
$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 52.490 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 50.484 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved minimum



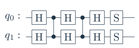
clifford_t_cx

EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved minimum



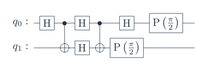
clifford_t_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.000 — proved minimum



cx_h_phase

EXACT PHASE E=2 / L=6 / score ≈ 20.404 — proved minimum



cx_rz_ry

EXACT PHASE E=2 / L=16 / score ≈ 43.232 — proved minimum



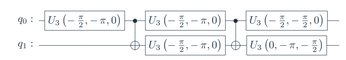
PROJECTIVE E=2 / L=9 / score ≈ 27.218 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved minimum



cz_rz_rx

EXACT PHASE E=2 / L=17 / score ≈ 45.234 — proved minimum



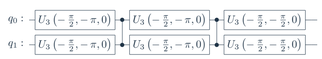
PROJECTIVE E=2 / L=16 / score ≈ 43.232 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved minimum



google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved minimum



google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — proved minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum

google_syc

EXACT PHASE E=12 / L=14 / score ≈ 78.884 — upper bound



ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 34.808 — proved minimum



PROJECTIVE E=2 / L=9 / score ≈ 26.202 — proved minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

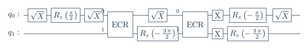
EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved minimum

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.810 — proved minimum



PROJECTIVE E=2 / L=10 / score ≈ 28.808 — proved minimum



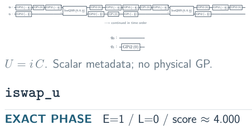
$U = \eta_{56}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.446 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 46.046 — upper bound



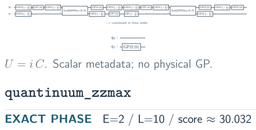
$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.642 — upper bound



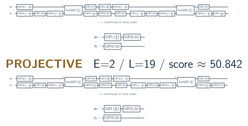
PROJECTIVE E=2 / L=17 / score ≈ 45.242 — upper bound



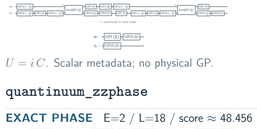
$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.242 — upper bound



PROJECTIVE E=2 / L=19 / score ≈ 50.842 — upper bound



$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.060 — proved minimum



PROJECTIVE E=2 / L=12 / score ≈ 34.848 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



quantinuum_zzmax

EXACT PHASE E=2 / L=10 / score ≈ 30.032 — proved minimum

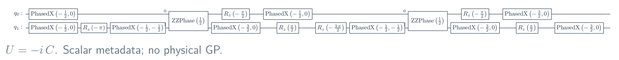


quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.456 — proved minimum



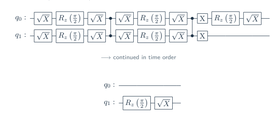
PROJECTIVE E=2 / L=15 / score ≈ 41.652 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved minimum

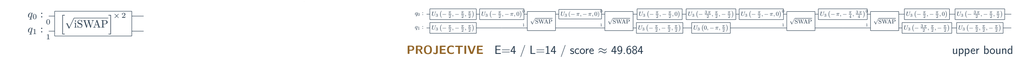


sqrt_iswap_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum

sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 50.484 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 49.684 — upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.406 — proved minimum



xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved minimum



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — proved minimum

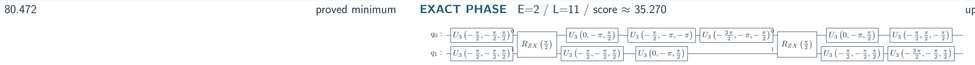
PROJECTIVE E=2 / L=30 / score ≈ 74.064 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 35.270 — upper bound



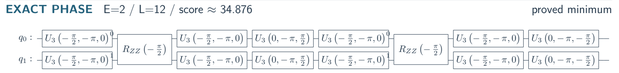
PROJECTIVE E=2 / L=10 / score ≈ 33.264 — upper bound



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — proved minimum



iSWAP_pow (α)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — conditional minimum



PROJECTIVE E=2 / L=16 / score ≈ 43.658 — upper bound



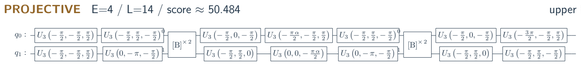
$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 55.296 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 50.484 — upper bound



$U = -e^{\frac{i\pi\alpha}{4}}C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.024 — conditional minimum



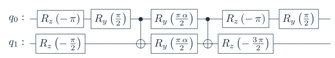
PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = e^{-\frac{i\pi\alpha}{2}}C$. Scalar metadata; no physical GP.

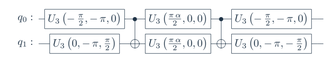
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — conditional minimum



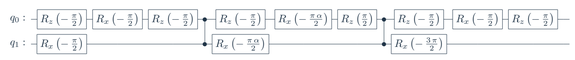
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — conditional minimum



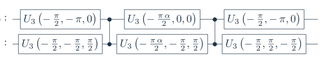
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — conditional minimum



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — conditional minimum



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.048 — conditional minimum



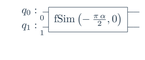
PROJECTIVE E=2 / L=6 / score ≈ 21.836 — conditional minimum



$U = e^{\frac{i\pi\alpha}{2}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 51.896 — upper bound

PROJECTIVE E=4 / L=14 / score ≈ 47.684 — upper bound

$U = -ie^{\frac{i\pi\alpha}{2}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.096 — upper bound



PROJECTIVE E=12 / L=14 / score ≈ 78.884 — upper bound

$U = e^{\frac{i\pi\alpha}{2}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 34.812 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 32.808 — upper bound

$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — conditional minimum



PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 53.652 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.252 — upper bound



$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 52.848 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 52.448 — upper bound



$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.448 — upper bound



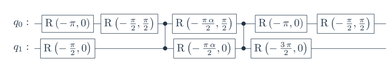
PROJECTIVE E=2 / L=22 / score ≈ 58.048 — upper bound



$U = iC$. Scalar metadata; no physical GP.

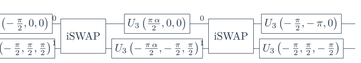
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound

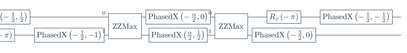


quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.086 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 25.828 — conditional minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.456 — conditional minimum



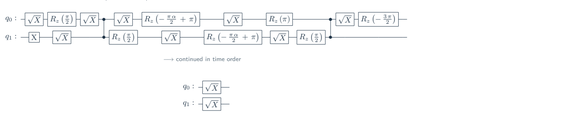
PROJECTIVE E=2 / L=15 / score ≈ 41.652 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum

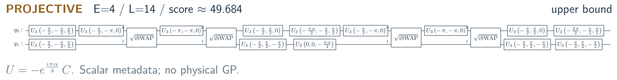


sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.496 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 49.684 — upper bound



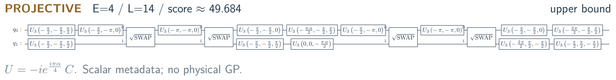
$U = -e^{\frac{i\pi\alpha}{4}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 49.684 — upper bound



$U = -ie^{\frac{i\pi\alpha}{4}}C$. Scalar metadata; no physical GP.

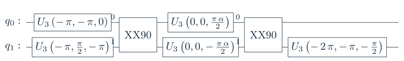
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.406 — conditional minimum



xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — conditional minimum



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — conditional minimum



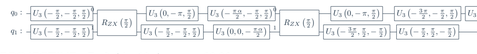
PROJECTIVE E=2 / L=29 / score ≈ 71.662 — upper bound



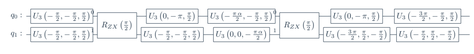
$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 38.076 — upper bound



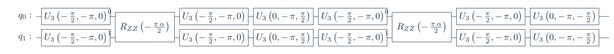
PROJECTIVE E=2 / L=10 / score ≈ 33.264 — upper bound



$U = e^{\frac{i\pi\alpha}{4}}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — conditional minimum



ISWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

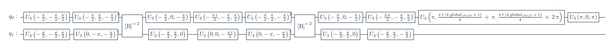
PROJECTIVE E=2 / L=16 / score ≈ 43.658 — upper bound



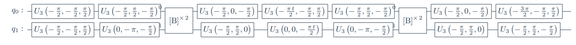
$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 55.296 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 50.484 — upper bound



$U = -e^{i\pi\left(global_{shift}+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

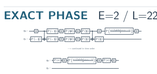
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

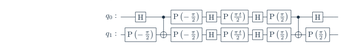
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.024 — conditional minimum



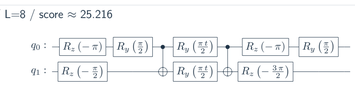
PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = e^{\frac{i\pi\left(2global_{shift}-1\right)}{2}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

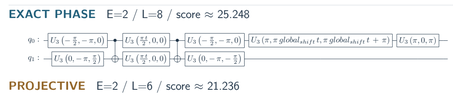
PROJECTIVE E=2 / L=8 / score ≈ 25.216 — conditional minimum



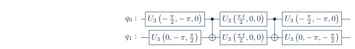
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=8 / score ≈ 25.248 — conditional minimum



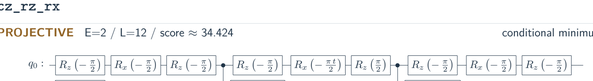
PROJECTIVE E=2 / L=6 / score ≈ 21.236 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=2 / L=12 / score ≈ 34.424 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — conditional minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.436 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.048 — conditional minimum



PROJECTIVE E=2 / L=6 / score ≈ 21.836 — conditional minimum



$U = e^{\frac{i\pi\left(2global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.216 — conditional minimum



PROJECTIVE E=1 / L=0 / score ≈ 4.204 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 51.896 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 47.684 — upper bound

$U = \eta_{58}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.096 — upper bound



PROJECTIVE E=12 / L=14 / score ≈ 78.884 — upper bound

$U = e^{\frac{i\pi\left(2global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

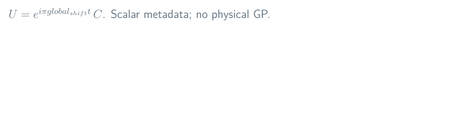
PROJECTIVE E=2 / L=12 / score ≈ 32.808 — upper bound



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=18 / score ≈ 45.214 — conditional minimum



$U = -e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=20 / score ≈ 53.252 — upper bound

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 52.448 — upper bound

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=22 / score ≈ 58.048 — upper bound

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=8 / score ≈ 25.832 — conditional minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — upper bound

PROJECTIVE E=2 / L=6 / score ≈ 22.436 — upper bound

$U = \eta_{53}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=8 / score ≈ 25.828 — conditional minimum

$U = \eta_{59}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=15 / score ≈ 41.652 — upper bound

$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=18 / score ≈ 45.214 — conditional minimum

$U = -e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.496 — upper bound

PROJECTIVE E=4 / L=14 / score ≈ 49.684 — upper bound

$U = -e^{i\pi\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — upper bound

PROJECTIVE E=4 / L=14 / score ≈ 49.684 — upper bound

$U = -ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.406 — conditional minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — conditional minimum

PROJECTIVE E=2 / L=5 / score ≈ 18.830 — conditional minimum

$U = \eta_{55}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=29 / score ≈ 71.662 — upper bound

$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 38.076 — upper bound

PROJECTIVE E=2 / L=10 / score ≈ 33.264 — upper bound

$U = e^{i\pi\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 38.888 — conditional minimum

PROJECTIVE E=2 / L=12 / score ≈ 34.876 — conditional minimum

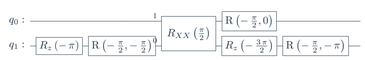
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

LinearFunction

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

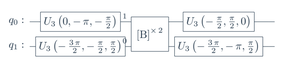
PROJECTIVE E=1 / L=5 / score ≈ 15.218 proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 proved minimum



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum



clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum



clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 proved minimum

cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum

cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum

cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum

cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.212 proved minimum

cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 proved minimum

google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 proved minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 17.824 proved minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 proved minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 upper bound

ibm_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 proved minimum

ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 proved minimum



ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 proved minimum

ionq_aria

PROJECTIVE E=1 / L=5 / score ≈ 15.416 proved minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 proved minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=6 / score ≈ 17.814 proved minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 proved minimum

iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 proved minimum

quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 proved minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=5 / score ≈ 15.216 proved minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 proved minimum

sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 proved minimum

sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 proved minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=5 / score ≈ 15.222 proved minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 proved minimum

xx_rz_ry

PROJECTIVE E=1 / L=7 / score ≈ 19.416 proved minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.014 proved minimum

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 proved minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 proved minimum

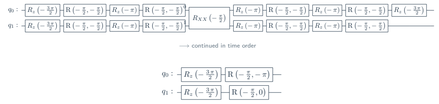
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

Magic

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

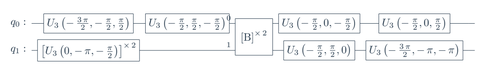
PROJECTIVE E=1 / L=21 / score ≈ 50.864 — proved minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 27.048 — proved minimum



clifford_only

EXACT PHASE E=1 / L=13 / score ≈ 30.000 — proved minimum



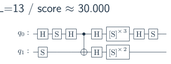
PROJECTIVE E=1 / L=11 / score ≈ 26.000 — proved minimum



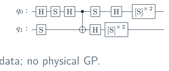
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=13 / score ≈ 30.000 — proved minimum



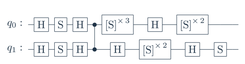
PROJECTIVE E=1 / L=11 / score ≈ 26.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

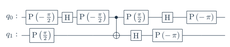
clifford_t_cz

EXACT PHASE E=1 / L=17 / score ≈ 38.000 — proved minimum



cx_h_phase

EXACT PHASE E=1 / L=9 / score ≈ 22.812 — proved minimum



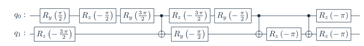
PROJECTIVE E=1 / L=9 / score ≈ 22.610 — proved minimum



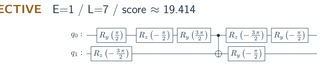
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=3 / L=10 / score ≈ 33.420 — upper bound



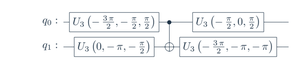
PROJECTIVE E=1 / L=7 / score ≈ 19.414 — proved minimum



$U = \eta_1\, C$. Scalar metadata; no physical GP.

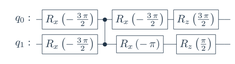
cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved minimum



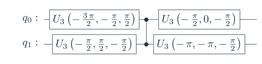
cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.012 — proved minimum



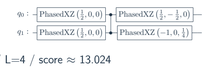
cz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.824 — proved minimum



google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.024 — proved minimum



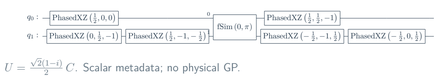
$U = \frac{\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=7 / score ≈ 20.246 — proved minimum



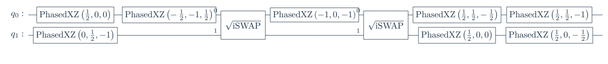
PROJECTIVE E=1 / L=6 / score ≈ 18.040 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

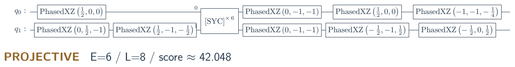
google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — proved minimum

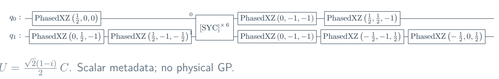


google_syc

EXACT PHASE E=6 / L=9 / score ≈ 44.254 — upper bound



PROJECTIVE E=6 / L=8 / score ≈ 42.048 — upper bound



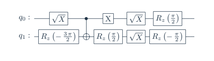
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.808 — proved minimum



$U = \frac{\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved minimum



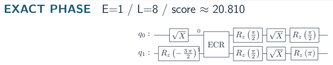
PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved minimum



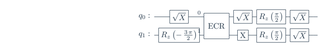
$U = \frac{\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=8 / score ≈ 20.810 — proved minimum



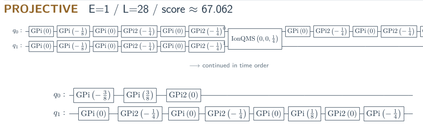
PROJECTIVE E=1 / L=8 / score ≈ 20.606 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=1 / L=28 / score ≈ 67.062 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

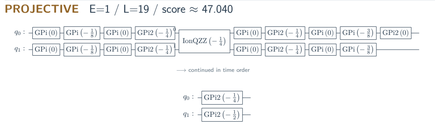
PROJECTIVE E=1 / L=11 / score ≈ 28.626 — proved minimum



$U = \sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=19 / score ≈ 47.040 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

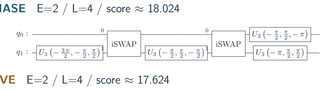
iqm_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.420 — proved minimum



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — proved minimum



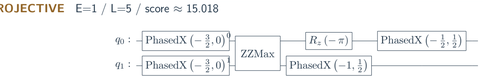
PROJECTIVE E=2 / L=4 / score ≈ 17.624 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.018 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=13 / score ≈ 33.240 — proved minimum



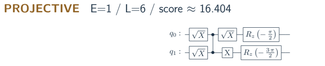
$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved minimum



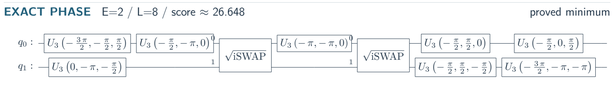
PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved minimum



$U = \frac{\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

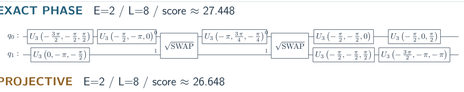
sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 26.648 — proved minimum



sqrt_swap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.448 — proved minimum



PROJECTIVE E=2 / L=8 / score ≈ 26.648 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=10 / score ≈ 26.438 — proved minimum



$U = \sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

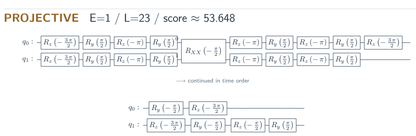
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved minimum



xx_rz_ry

PROJECTIVE E=1 / L=23 / score ≈ 53.648 — proved minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=12 / score ≈ 31.274 — proved minimum



zz_u

EXACT PHASE E=1 / L=10 / score ≈ 26.862 — proved minimum

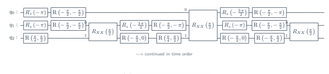


Margolus

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

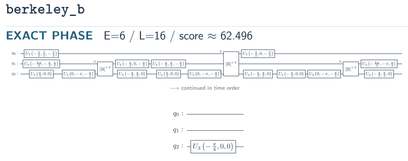
PROJECTIVE E=3 / L=19 / score ≈ 56.070 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

berkeley_b

EXACT PHASE E=6 / L=16 / score ≈ 62.496 — upper bound

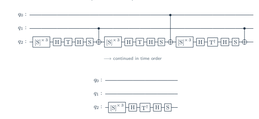


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

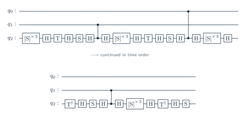
clifford_t_cx

EXACT PHASE E=3 / L=28 / score ≈ 68.000 — proved minimum



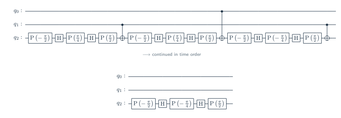
clifford_t_cz

EXACT PHASE E=3 / L=34 / score ≈ 80.000 — proved minimum



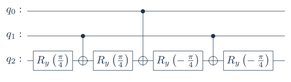
cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 55.224 — proved minimum



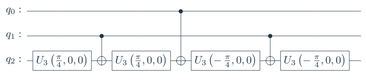
cx_rz_ry

EXACT PHASE E=3 / L=4 / score ≈ 21.608 — proved minimum



cx_u

EXACT PHASE E=3 / L=4 / score ≈ 21.624 — proved minimum



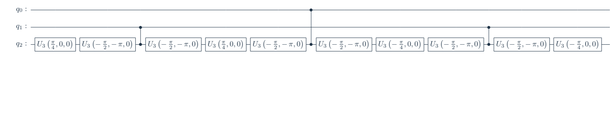
cz_rz_rx

EXACT PHASE E=3 / L=18 / score ≈ 51.836 — proved minimum



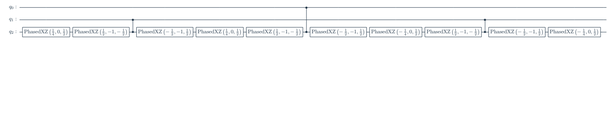
cz_u

EXACT PHASE E=3 / L=10 / score ≈ 34.860 — proved minimum



google_cz

EXACT PHASE E=3 / L=10 / score ≈ 36.860 — proved minimum



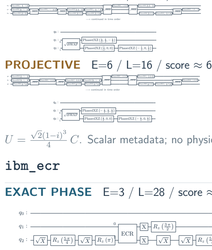
google_fsim

EXACT PHASE E=3 / L=10 / score ≈ 36.872 — upper bound



google_sqrt_iswap

EXACT PHASE E=6 / L=16 / score ≈ 62.696 — upper bound



PROJECTIVE E=6 / L=16 / score ≈ 62.096 — upper bound

$U = \frac{\sqrt{2}(1-i)^2}{4} C$. Scalar metadata; no physical GP.

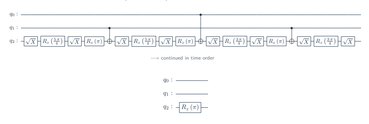
google_syc

EXACT PHASE E=18 / L=16 / score ≈ 108.896 — upper bound



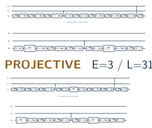
ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.616 — proved minimum



ibm_cz

EXACT PHASE E=3 / L=31 / score ≈ 77.630 — proved minimum

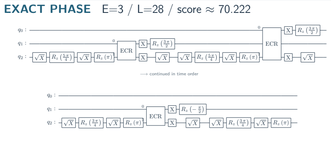


PROJECTIVE E=3 / L=31 / score ≈ 77.428 — proved minimum

$U = -(-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

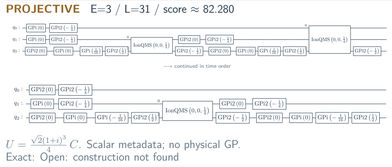
ibm_ecr

EXACT PHASE E=3 / L=28 / score ≈ 70.222 — proved minimum



ionq_aria

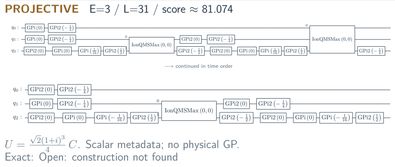
PROJECTIVE E=3 / L=31 / score ≈ 82.280 — upper bound



$U = \frac{\sqrt{2}(1+i)^2}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

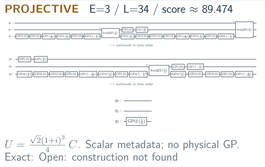
PROJECTIVE E=3 / L=31 / score ≈ 81.074 — upper bound



$U = \frac{\sqrt{2}(1+i)^2}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=3 / L=34 / score ≈ 89.474 — upper bound



$U = \frac{\sqrt{2}(1+i)^2}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

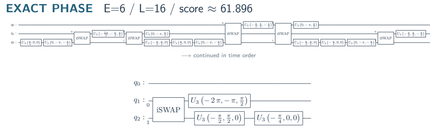
iqm_cz

EXACT PHASE E=3 / L=16 / score ≈ 48.864 — proved minimum



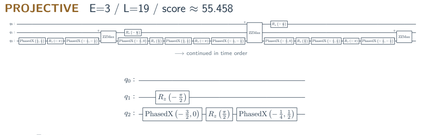
iswap_u

EXACT PHASE E=6 / L=16 / score ≈ 61.896 — upper bound



quantinuum_zzmax

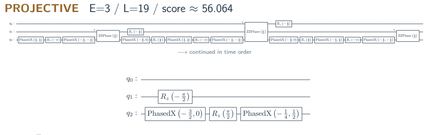
PROJECTIVE E=3 / L=19 / score ≈ 55.458 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

quantinuum_zzphase

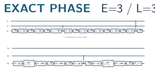
PROJECTIVE E=3 / L=19 / score ≈ 56.064 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

rigetti_cz

EXACT PHASE E=3 / L=31 / score ≈ 77.630 — proved minimum



PROJECTIVE E=3 / L=31 / score ≈ 77.428 — proved minimum



$U = -(-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=6 / L=16 / score ≈ 62.096 — upper bound



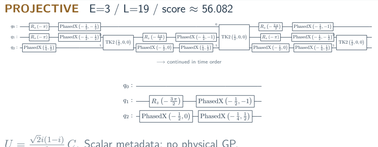
PROJECTIVE E=6 / L=16 / score ≈ 61.296 — upper bound



$U = -\sqrt[4]{-1} C$. Scalar metadata; no physical GP.

tket_tk2

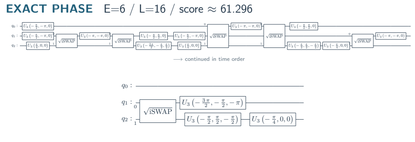
PROJECTIVE E=3 / L=19 / score ≈ 56.082 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

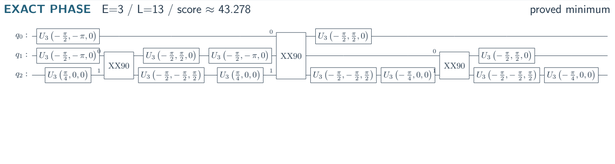
sqrt_iswap_u

EXACT PHASE E=6 / L=16 / score ≈ 61.296 — upper bound



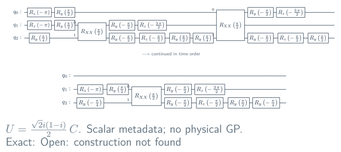
xx90_u

EXACT PHASE E=3 / L=13 / score ≈ 43.278 — proved minimum



xx_rz_ry

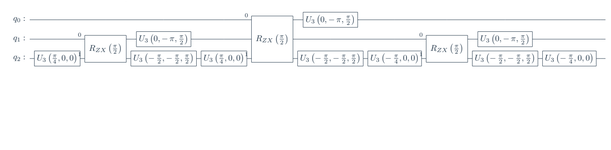
PROJECTIVE E=3 / L=25 / score ≈ 67.856 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

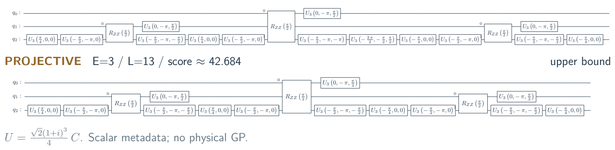
zx_u

EXACT PHASE E=3 / L=10 / score ≈ 36.666 — upper bound



zz_u

EXACT PHASE E=3 / L=14 / score ≈ 45.090 — upper bound



PROJECTIVE E=3 / L=13 / score ≈ 42.684 — upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

MCMT

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=24 / score ≈ 66.486 — upper bound



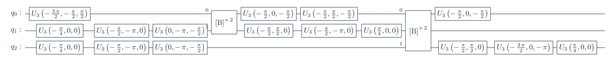
PROJECTIVE E=2 / L=22 / score ≈ 58.476 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 53.696 — upper bound

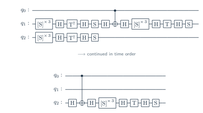


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

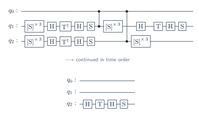
clifford_t_cx

EXACT PHASE E=2 / L=32 / score ≈ 72.000 — proved minimum



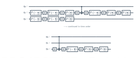
clifford_t_cz

EXACT PHASE E=2 / L=28 / score ≈ 64.000 — proved minimum



cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 59.224 — proved minimum



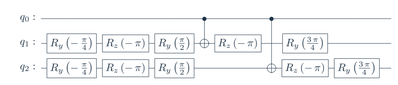
PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

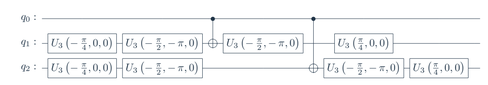
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 30.020 — proved minimum



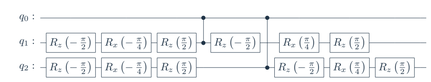
cx_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — proved minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 35.224 — proved minimum



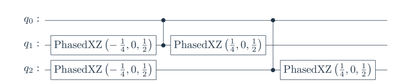
cz_u

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved minimum



google_cz

EXACT PHASE E=2 / L=4 / score ≈ 18.424 — proved minimum



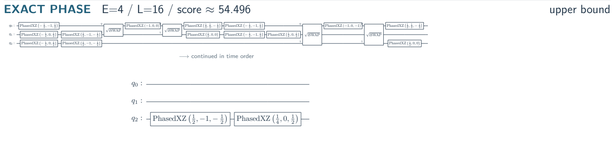
google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 37.680 — proved minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 54.496 — upper bound



google_syc

EXACT PHASE E=12 / L=16 / score ≈ 85.696 — upper bound



ibm_cx

EXACT PHASE E=2 / L=26 / score ≈ 63.228 — proved minimum



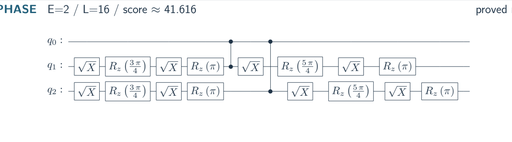
PROJECTIVE E=2 / L=26 / score ≈ 62.824 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

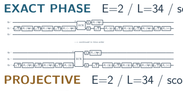
ibm_cz

EXACT PHASE E=2 / L=16 / score ≈ 41.616 — proved minimum



ibm_ecr

EXACT PHASE E=2 / L=34 / score ≈ 79.632 — proved minimum



PROJECTIVE E=2 / L=34 / score ≈ 79.228 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=34 / score ≈ 84.880 — proved minimum



PROJECTIVE E=2 / L=34 / score ≈ 84.480 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=34 / score ≈ 84.076 — proved minimum



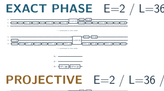
PROJECTIVE E=2 / L=34 / score ≈ 83.676 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=36 / score ≈ 89.676 — proved minimum



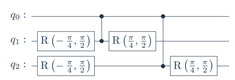
PROJECTIVE E=2 / L=36 / score ≈ 89.276 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

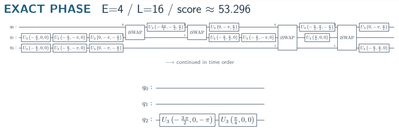
iqm_cz

EXACT PHASE E=2 / L=4 / score ≈ 18.416 — proved minimum



iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.296 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=41 / score ≈ 112.326 — upper bound



PROJECTIVE E=2 / L=22 / score ≈ 58.068 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=24 / score ≈ 66.478 — upper bound



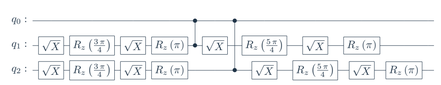
PROJECTIVE E=2 / L=22 / score ≈ 58.472 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=16 / score ≈ 41.616 — proved minimum



sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.896 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.296 — upper bound



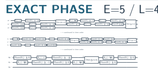
PROJECTIVE E=4 / L=16 / score ≈ 52.896 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=42 / score ≈ 115.368 — upper bound



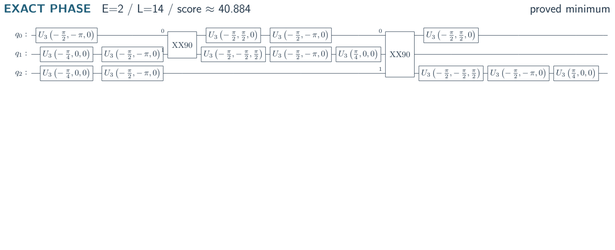
PROJECTIVE E=2 / L=22 / score ≈ 58.484 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

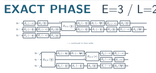
xx90_u

EXACT PHASE E=2 / L=14 / score ≈ 40.884 — proved minimum



xx_rz_ry

EXACT PHASE E=3 / L=28 / score ≈ 72.862 — upper bound



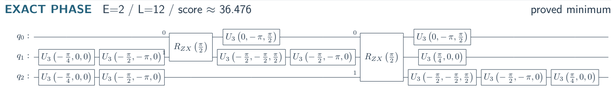
PROJECTIVE E=2 / L=24 / score ≈ 60.852 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

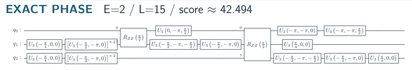
zx_u

EXACT PHASE E=2 / L=12 / score ≈ 36.476 — proved minimum

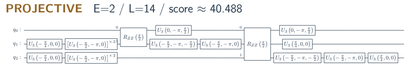


zz_u

EXACT PHASE E=2 / L=15 / score ≈ 42.494 — proved minimum



PROJECTIVE E=2 / L=14 / score ≈ 40.488 — proved minimum



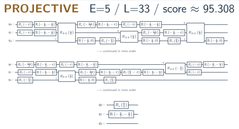
$U = i\,C$. Scalar metadata; no physical GP.

MCPhase

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=5 / L=33 / score ≈ 95.308 — upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

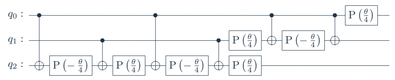
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

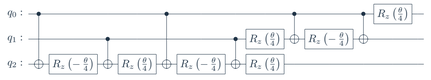
cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — upper bound



cx_rz_ry

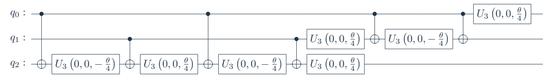
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

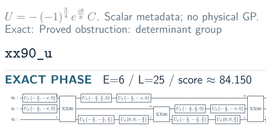
cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — upper bound



cz_rz_rx

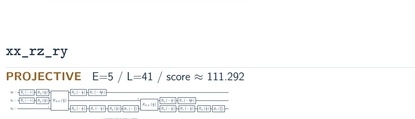
PROJECTIVE E=6 / L=33 / score ≈ 98.066 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

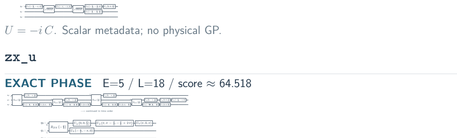
cz_u

EXACT PHASE E=6 / L=19 / score ≈ 67.314 — upper bound



google_cz

EXACT PHASE E=6 / L=19 / score ≈ 72.026 — upper bound



google_fsim

EXACT PHASE E=5 / L=16 / score ≈ 61.358 — upper bound

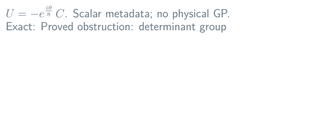


google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 122.498 — upper bound

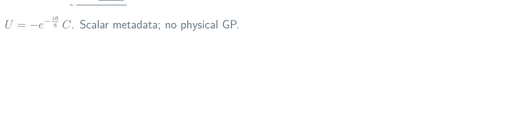
google_syc

EXACT PHASE E=36 / L=31 / score ≈ 216.098 — upper bound



ibm_cx

PROJECTIVE E=6 / L=7 / score ≈ 40.814 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

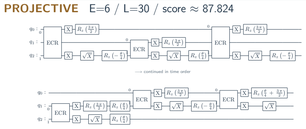
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

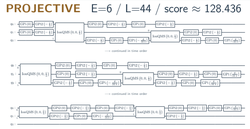
PROJECTIVE E=6 / L=30 / score ≈ 87.824 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

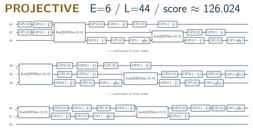
PROJECTIVE E=6 / L=44 / score ≈ 128.436 — upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

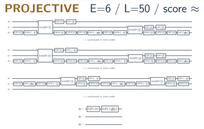
PROJECTIVE E=6 / L=44 / score ≈ 126.024 — upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

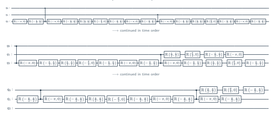
PROJECTIVE E=6 / L=50 / score ≈ 142.824 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

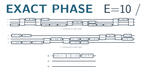
PROJECTIVE E=6 / L=45 / score ≈ 127.380 — upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=28 / score ≈ 106.168 — upper bound



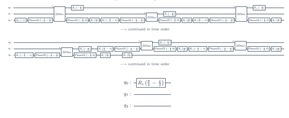
PROJECTIVE E=10 / L=26 / score ≈ 100.956 — upper bound



$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

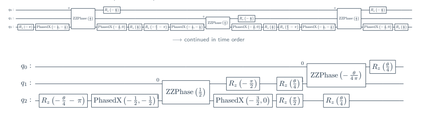
PROJECTIVE E=6 / L=31 / score ≈ 94.686 — upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

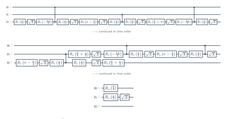
PROJECTIVE E=5 / L=23 / score ≈ 74.002 — upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

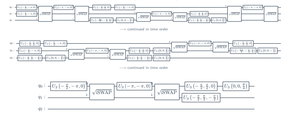
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

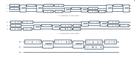
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — upper bound



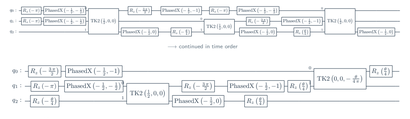
PROJECTIVE E=12 / L=31 / score ≈ 120.186 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

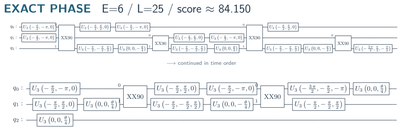
PROJECTIVE E=5 / L=26 / score ≈ 80.036 — upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

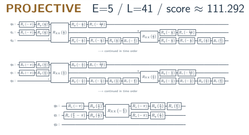
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — upper bound



xx_rz_ry

PROJECTIVE E=5 / L=41 / score ≈ 111.292 — upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=18 / score ≈ 64.518 — upper bound



PROJECTIVE E=5 / L=16 / score ≈ 59.306 — upper bound



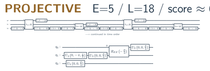
$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=20 / score ≈ 68.130 — upper bound



PROJECTIVE E=5 / L=18 / score ≈ 62.918 — upper bound



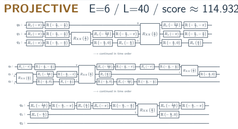
$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

MCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

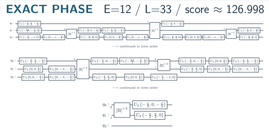
PROJECTIVE E=6 / L=40 / score ≈ 114.932 upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 upper bound

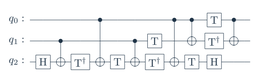


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

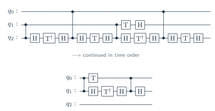
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 upper bound



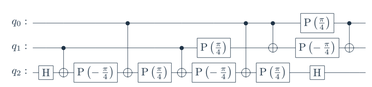
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 upper bound



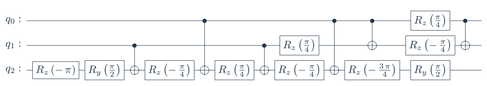
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 upper bound



cx_rz_ry

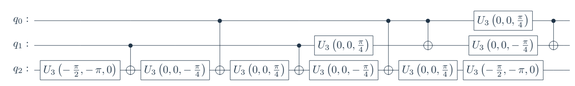
PROJECTIVE E=6 / L=10 / score ≈ 47.220 upper bound



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

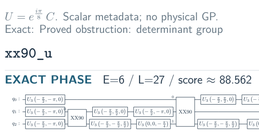
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 45.254 upper bound



cz_rz_rx

PROJECTIVE E=6 / L=40 / score ≈ 113.280 upper bound



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.726 upper bound



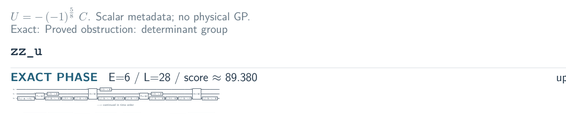
google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 upper bound



google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 74.550 upper bound

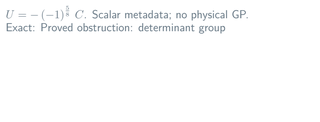


google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 upper bound

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 upper bound



ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 upper bound

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 upper bound



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=35 / score ≈ 98.228 upper bound

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 134.932 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 132.520 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 upper bound

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.798 upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 103.498 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 104.710 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 upper bound

PROJECTIVE E=12 / L=33 / score ≈ 124.598 upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=40 / score ≈ 114.956 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 upper bound

xx_rz_ry

PROJECTIVE E=6 / L=51 / score ≈ 137.514 upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 75.338 upper bound

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 89.380 upper bound

PROJECTIVE E=6 / L=27 / score ≈ 87.374 upper bound

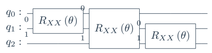
$U = -i\,C$. Scalar metadata; no physical GP.

MS

3 qubits / 32 preserved native bases / alphabetical panel order

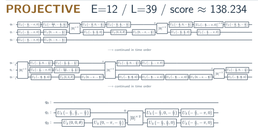
aqt_rxx

EXACT PHASE E=3 / L=0 / score ≈ 12.006 — upper bound



berkeley_b

PROJECTIVE E=12 / L=39 / score ≈ 138.234 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

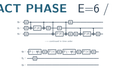
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=21 / score ≈ 66.414 — upper bound



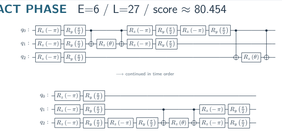
PROJECTIVE E=6 / L=13 / score ≈ 50.006 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=6 / L=27 / score ≈ 80.454 — upper bound

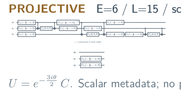


cx_u

EXACT PHASE E=6 / L=17 / score ≈ 60.902 — upper bound



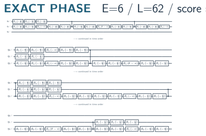
PROJECTIVE E=6 / L=15 / score ≈ 56.490 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

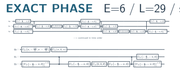
cz_rz_rx

EXACT PHASE E=6 / L=62 / score ≈ 158.524 — upper bound

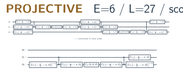


cz_u

EXACT PHASE E=6 / L=29 / score ≈ 87.374 — upper bound



PROJECTIVE E=6 / L=27 / score ≈ 82.962 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

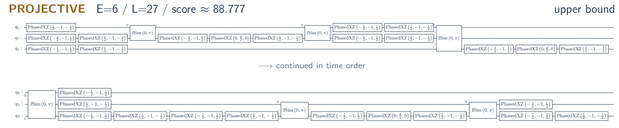
PROJECTIVE E=6 / L=27 / score ≈ 88.753 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_fsim

PROJECTIVE E=6 / L=27 / score ≈ 88.777 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_sqrt_iswap

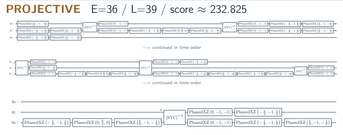
PROJECTIVE E=12 / L=39 / score ≈ 139.225 — upper bound



$U = ie^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

google_syc

PROJECTIVE E=36 / L=39 / score ≈ 232.825 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_cx

EXACT PHASE E=6 / L=38 / score ≈ 102.836 — upper bound



PROJECTIVE E=6 / L=38 / score ≈ 102.432 — upper bound



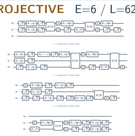
$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

ibm_ecr

PROJECTIVE E=6 / L=62 / score ≈ 151.644 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ionq_aria

EXACT PHASE E=6 / L=60 / score ≈ 159.947 — upper bound



PROJECTIVE E=6 / L=60 / score ≈ 159.547 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=6 / L=60 / score ≈ 157.535 — upper bound



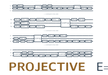
PROJECTIVE E=6 / L=60 / score ≈ 157.135 — upper bound



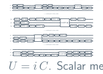
$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=6 / L=66 / score ≈ 174.335 — upper bound



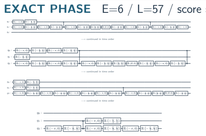
PROJECTIVE E=6 / L=66 / score ≈ 173.935 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

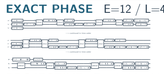
iqm_cz

EXACT PHASE E=6 / L=57 / score ≈ 150.228 — upper bound



iswap_u

EXACT PHASE E=12 / L=41 / score ≈ 141.446 — upper bound



PROJECTIVE E=12 / L=39 / score ≈ 137.034 — upper bound



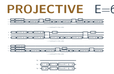
$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=9 / L=73 / score ≈ 197.214 — upper bound



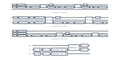
PROJECTIVE E=6 / L=54 / score ≈ 142.956 — upper bound



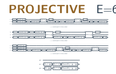
$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=7 / L=56 / score ≈ 152.174 — upper bound



PROJECTIVE E=6 / L=54 / score ≈ 144.168 — upper bound



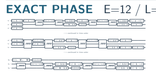
$U = i\,C$. Scalar metadata; no physical GP.

rigetti_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

sqrt_iswap_u

EXACT PHASE E=12 / L=41 / score ≈ 140.246 — upper bound



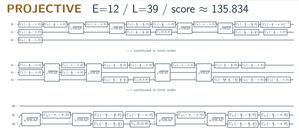
PROJECTIVE E=12 / L=39 / score ≈ 135.834 — upper bound



$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

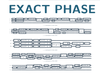
PROJECTIVE E=12 / L=39 / score ≈ 135.834 — upper bound



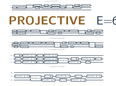
$U = -ie^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

tket_tk2

EXACT PHASE E=9 / L=77 / score ≈ 207.094 — upper bound



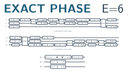
PROJECTIVE E=6 / L=57 / score ≈ 150.210 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=6 / L=35 / score ≈ 104.210 — upper bound



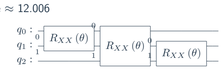
PROJECTIVE E=6 / L=33 / score ≈ 99.798 — upper bound



$U = -e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=0 / score ≈ 12.006 — upper bound



zx_u

EXACT PHASE E=6 / L=29 / score ≈ 90.986 — upper bound



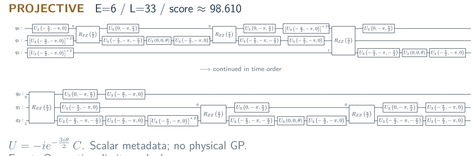
PROJECTIVE E=6 / L=27 / score ≈ 86.574 — upper bound



$U = -e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

zz_u

PROJECTIVE E=6 / L=33 / score ≈ 98.610 — upper bound



$U = -ie^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

P (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

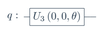
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



cx_rz_ry

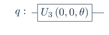
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

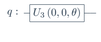
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum



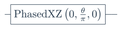
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum



ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

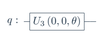
PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

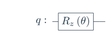
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

Pauli

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=5 / score ≈ 14.220 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.210 — upper bound



$U = iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — upper bound



PROJECTIVE E=0 / L=12 / score ≈ 24.000 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — upper bound



PROJECTIVE E=0 / L=12 / score ≈ 24.000 — upper bound



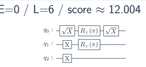
$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — upper bound



PROJECTIVE E=0 / L=12 / score ≈ 24.000 — upper bound

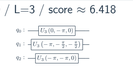


$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — upper bound

PROJECTIVE E=0 / L=8 / score ≈ 16.008 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=7 / score ≈ 22.014 — upper bound



PROJECTIVE E=0 / L=4 / score ≈ 8.008 — upper bound



$U = iC$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=14 / score ≈ 37.628 — upper bound



PROJECTIVE E=0 / L=4 / score ≈ 8.008 — upper bound



$U = iC$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound



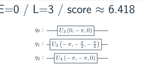
$U = -iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.672 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.416 — upper bound

$U = iC$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=3 / L=22 / score ≈ 60.468 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — upper bound

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=5 / score ≈ 14.216 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — upper bound

$U = iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=3 / L=23 / score ≈ 63.094 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — upper bound

$U = iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.018 — upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — upper bound

$U = iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

PauliRotation

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=4 / L=29 / score ≈ 80.098 — upper bound



PROJECTIVE E=3 / L=27 / score ≈ 72.088 — upper bound



$U = iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=8 / L=25 / score ≈ 90.150 — upper bound



PROJECTIVE E=8 / L=23 / score ≈ 85.738 — upper bound



$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

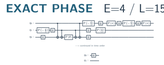
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=4 / L=15 / score ≈ 46.814 — upper bound



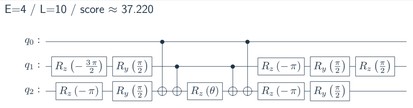
PROJECTIVE E=4 / L=7 / score ≈ 30.406 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

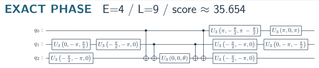
cx_rz_ry

EXACT PHASE E=4 / L=10 / score ≈ 37.220 — upper bound

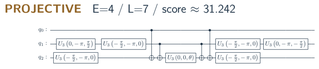


cx_u

EXACT PHASE E=4 / L=9 / score ≈ 35.654 — upper bound



PROJECTIVE E=4 / L=7 / score ≈ 31.242 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=30 / score ≈ 80.860 — upper bound

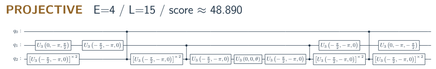


cz_u

EXACT PHASE E=4 / L=17 / score ≈ 53.302 — upper bound



PROJECTIVE E=4 / L=15 / score ≈ 48.890 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=4 / L=17 / score ≈ 56.493 — upper bound



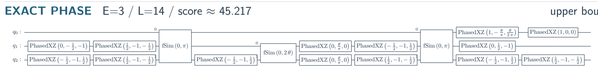
PROJECTIVE E=4 / L=15 / score ≈ 51.620 — upper bound



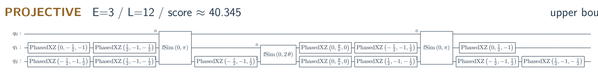
$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=3 / L=14 / score ≈ 45.217 — upper bound



PROJECTIVE E=3 / L=12 / score ≈ 40.345 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=8 / L=25 / score ≈ 90.141 — upper bound



PROJECTIVE E=8 / L=23 / score ≈ 85.268 — upper bound



$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=24 / L=25 / score ≈ 152.541 — upper bound



PROJECTIVE E=24 / L=23 / score ≈ 147.668 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=4 / L=12 / score ≈ 41.216 — upper bound



PROJECTIVE E=4 / L=12 / score ≈ 40.812 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=4 / L=30 / score ≈ 78.436 — upper bound



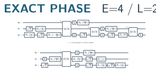
PROJECTIVE E=4 / L=30 / score ≈ 78.434 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=4 / L=28 / score ≈ 74.024 — upper bound



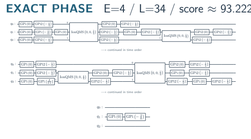
PROJECTIVE E=4 / L=28 / score ≈ 73.620 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

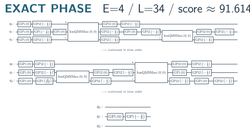
ionq_aria

EXACT PHASE E=4 / L=34 / score ≈ 93.222 — upper bound



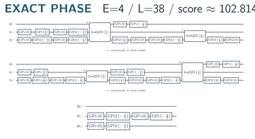
ionq_aria_full

EXACT PHASE E=4 / L=34 / score ≈ 91.614 — upper bound



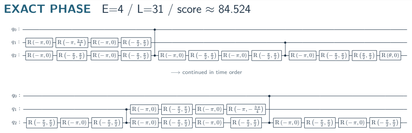
ionq_forte

EXACT PHASE E=4 / L=38 / score ≈ 102.814 — upper bound



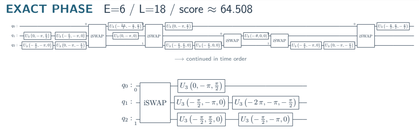
iqm_cz

EXACT PHASE E=4 / L=31 / score ≈ 84.524 — upper bound



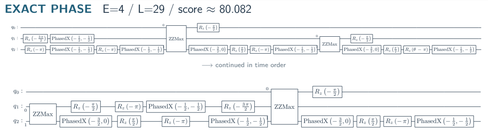
iswap_u

EXACT PHASE E=6 / L=18 / score ≈ 64.508 — upper bound



quantinuum_zzmax

EXACT PHASE E=4 / L=29 / score ≈ 80.082 — upper bound

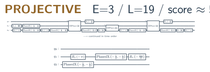


quantinuum_zzphase

EXACT PHASE E=4 / L=21 / score ≈ 62.796 — upper bound



PROJECTIVE E=3 / L=19 / score ≈ 54.790 — upper bound



$U = iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=4 / L=30 / score ≈ 78.436 — upper bound



PROJECTIVE E=4 / L=30 / score ≈ 78.434 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=8 / L=25 / score ≈ 88.550 — upper bound



PROJECTIVE E=8 / L=23 / score ≈ 84.138 — upper bound



$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=8 / L=25 / score ≈ 88.550 — upper bound



PROJECTIVE E=8 / L=23 / score ≈ 84.138 — upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=4 / L=30 / score ≈ 82.916 — upper bound



PROJECTIVE E=3 / L=19 / score ≈ 54.806 — upper bound



$U = iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=4 / L=21 / score ≈ 64.526 — upper bound



PROJECTIVE E=4 / L=19 / score ≈ 60.114 — upper bound



$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=4 / L=35 / score ≈ 90.878 — upper bound



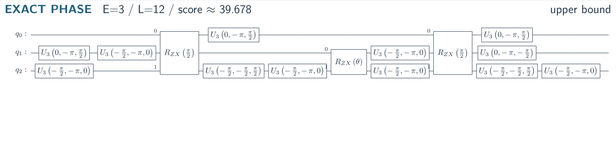
PROJECTIVE E=3 / L=31 / score ≈ 78.868 — upper bound



$U = iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=12 / score ≈ 39.678 — upper bound



zz_u

EXACT PHASE E=3 / L=13 / score ≈ 41.284 — upper bound



PROJECTIVE E=3 / L=12 / score ≈ 39.278 — upper bound



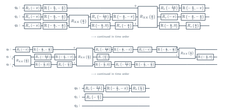
$U = iC$. Scalar metadata; no physical GP.

Peres

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

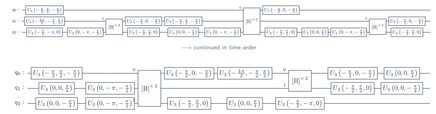
PROJECTIVE E=5 / L=35 / score ≈ 99.714 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=10 / L=29 / score ≈ 109.374 — upper bound

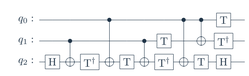


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

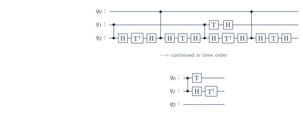
clifford_t_cx

EXACT PHASE E=5 / L=9 / score ≈ 38.000 — proved minimum



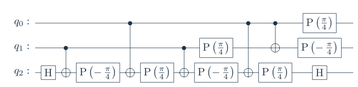
clifford_t_cz

EXACT PHASE E=5 / L=17 / score ≈ 54.000 — proved minimum



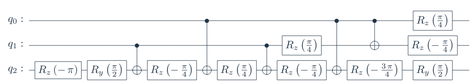
cx_h_phase

EXACT PHASE E=5 / L=9 / score ≈ 40.814 — proved minimum



cx_rz_ry

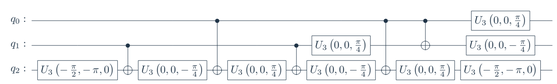
PROJECTIVE E=5 / L=10 / score ≈ 43.220 — proved minimum



$U = -\sqrt[8]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

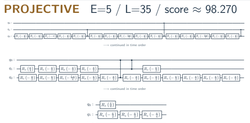
cx_u

EXACT PHASE E=5 / L=9 / score ≈ 41.254 — proved minimum



cz_rz_rx

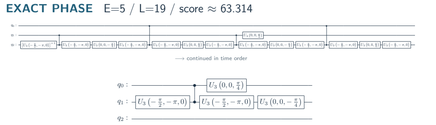
PROJECTIVE E=5 / L=35 / score ≈ 98.270 — proved minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

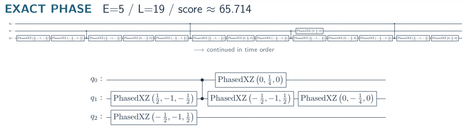
cz_u

EXACT PHASE E=5 / L=19 / score ≈ 63.314 — proved minimum



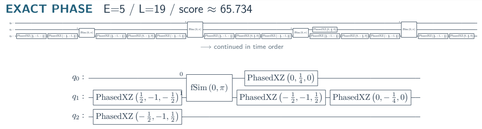
google_cz

EXACT PHASE E=5 / L=19 / score ≈ 65.714 — proved minimum



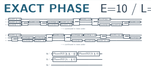
google_fsim

EXACT PHASE E=5 / L=19 / score ≈ 65.734 — upper bound



google_sqrt_iswap

EXACT PHASE E=10 / L=29 / score ≈ 108.374 — upper bound



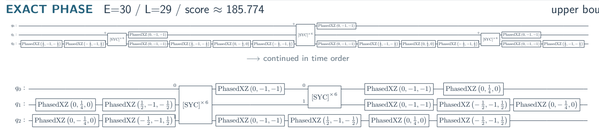
PROJECTIVE E=10 / L=29 / score ≈ 107.774 — upper bound



$U = \frac{\sqrt{2}(1-i)^5}{8}\, C$. Scalar metadata; no physical GP.

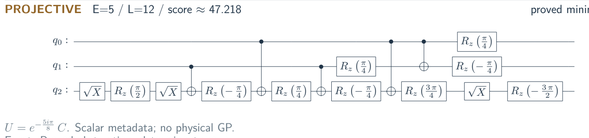
google_syc

EXACT PHASE E=30 / L=29 / score ≈ 185.774 — upper bound



ibm_cx
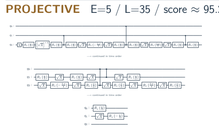
PROJECTIVE E=5 / L=12 / score ≈ 47.218 — proved minimum



$U = e^{-\frac{3i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

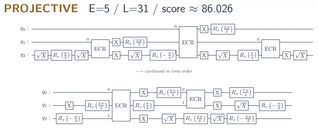
PROJECTIVE E=5 / L=35 / score ≈ 95.238 — proved minimum



$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

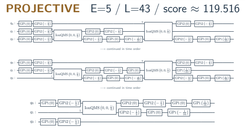
PROJECTIVE E=5 / L=31 / score ≈ 86.026 — proved minimum



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

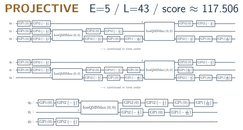
PROJECTIVE E=5 / L=43 / score ≈ 119.516 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

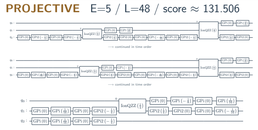
PROJECTIVE E=5 / L=43 / score ≈ 117.506 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

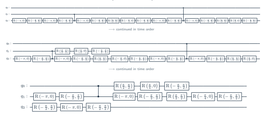
PROJECTIVE E=5 / L=48 / score ≈ 131.506 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

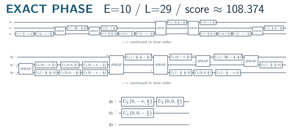
PROJECTIVE E=5 / L=45 / score ≈ 123.380 — proved minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

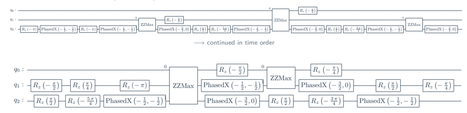
iswap_u

EXACT PHASE E=10 / L=29 / score ≈ 108.374 — upper bound



quantinuum_zzmax

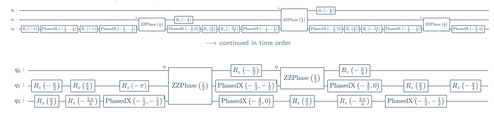
PROJECTIVE E=5 / L=31 / score ≈ 90.486 — proved minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

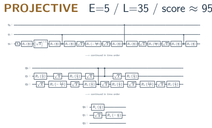
PROJECTIVE E=5 / L=31 / score ≈ 91.496 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=5 / L=35 / score ≈ 95.238 — proved minimum



$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=10 / L=29 / score ≈ 107.374 — upper bound



sqrt_swap_u

EXACT PHASE E=10 / L=29 / score ≈ 108.174 — upper bound



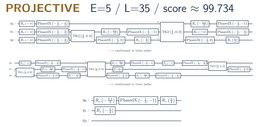
PROJECTIVE E=10 / L=29 / score ≈ 107.374 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

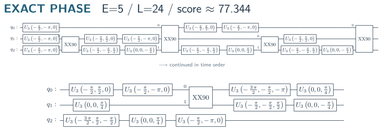
PROJECTIVE E=5 / L=35 / score ≈ 99.734 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

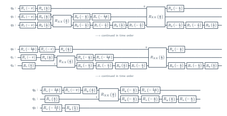
xx90_u

EXACT PHASE E=5 / L=24 / score ≈ 77.344 — proved minimum



xx_rz_ry

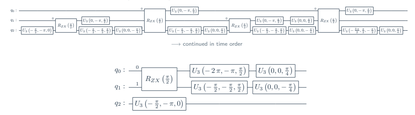
PROJECTIVE E=5 / L=44 / score ≈ 118.098 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=19 / score ≈ 66.324 — upper bound

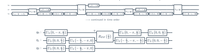


zz_u

EXACT PHASE E=5 / L=25 / score ≈ 78.760 — upper bound



PROJECTIVE E=5 / L=24 / score ≈ 76.354 — upper bound



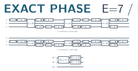
$U = \frac{\sqrt{2}(1+i)^5}{8}\, C$. Scalar metadata; no physical GP.

Permutation

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=7 / L=32 / score ≈ 99.318 — upper bound



PROJECTIVE E=6 / L=30 / score ≈ 91.308 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=12 / L=24 / score ≈ 105.744 — upper bound



clifford_only

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



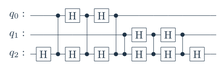
clifford_t_cx

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



clifford_t_cz

EXACT PHASE E=6 / L=10 / score ≈ 44.000 — upper bound



cx_h_phase

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



cx_rz_ry

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



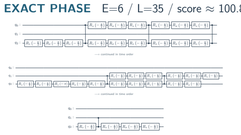
cx_u

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



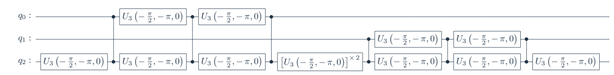
cz_rz_rx

EXACT PHASE E=6 / L=35 / score ≈ 100.870 — upper bound



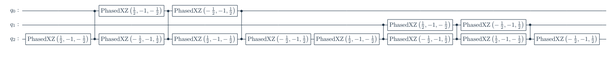
cz_u

EXACT PHASE E=6 / L=12 / score ≈ 50.472 — upper bound



google_cz

EXACT PHASE E=6 / L=12 / score ≈ 52.872 — upper bound



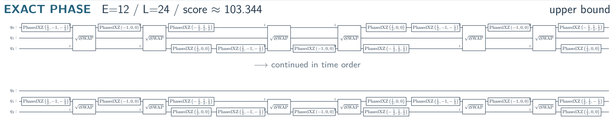
google_fsim

EXACT PHASE E=6 / L=12 / score ≈ 52.896 — upper bound



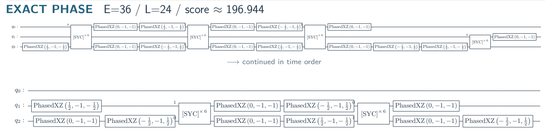
google_sqrt_iswap

EXACT PHASE E=12 / L=24 / score ≈ 103.344 — upper bound



google_syc

EXACT PHASE E=36 / L=24 / score ≈ 196.944 — upper bound



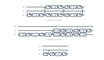
ibm_cx

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — upper bound



ibm_cz

EXACT PHASE E=6 / L=35 / score ≈ 96.830 — upper bound



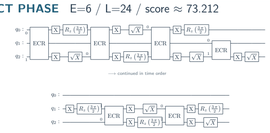
PROJECTIVE E=6 / L=35 / score ≈ 96.426 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=6 / L=24 / score ≈ 73.212 — upper bound



ionq_aria

EXACT PHASE E=6 / L=30 / score ≈ 92.896 — upper bound



PROJECTIVE E=6 / L=30 / score ≈ 92.496 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=6 / L=30 / score ≈ 90.484 — upper bound



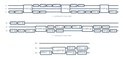
PROJECTIVE E=6 / L=30 / score ≈ 90.084 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=6 / L=36 / score ≈ 107.284 — upper bound



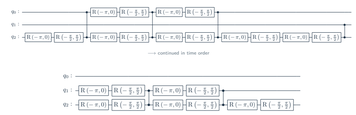
PROJECTIVE E=6 / L=36 / score ≈ 106.884 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

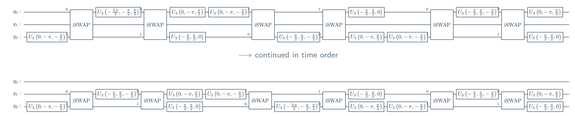
iqm_cz

EXACT PHASE E=6 / L=24 / score ≈ 76.896 — upper bound



iswap_u

EXACT PHASE E=12 / L=24 / score ≈ 104.544 — upper bound



quantinuum_zzmax

EXACT PHASE E=9 / L=45 / score ≈ 136.334 — upper bound



PROJECTIVE E=6 / L=26 / score ≈ 82.076 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=7 / L=28 / score ≈ 91.294 — upper bound



PROJECTIVE E=6 / L=26 / score ≈ 83.288 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=6 / L=35 / score ≈ 96.830 — upper bound



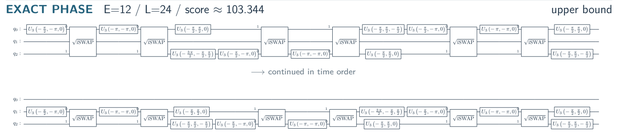
PROJECTIVE E=6 / L=35 / score ≈ 96.426 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

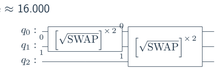
sqrt_iswap_u

EXACT PHASE E=12 / L=24 / score ≈ 103.344 — upper bound



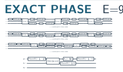
sqrt_swap_u

EXACT PHASE E=4 / L=0 / score ≈ 16.000 — upper bound



tket_tk2

EXACT PHASE E=9 / L=50 / score ≈ 148.216 — upper bound



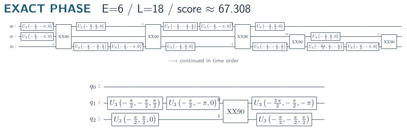
PROJECTIVE E=6 / L=30 / score ≈ 91.332 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

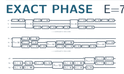
xx90_u

EXACT PHASE E=6 / L=18 / score ≈ 67.308 — upper bound



xx_rz_ry

EXACT PHASE E=7 / L=46 / score ≈ 128.506 — upper bound



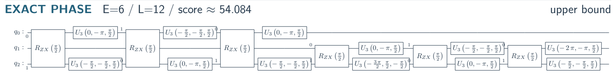
PROJECTIVE E=6 / L=42 / score ≈ 116.496 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=6 / L=12 / score ≈ 54.084 — upper bound



zz_u

EXACT PHASE E=6 / L=19 / score ≈ 68.126 — upper bound



PROJECTIVE E=6 / L=18 / score ≈ 66.120 — upper bound



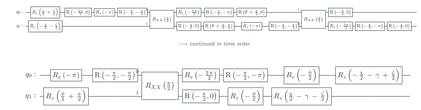
$U = -i\,C$. Scalar metadata; no physical GP.

PhasedFSim $(\theta, \zeta, \chi, \gamma, \phi)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

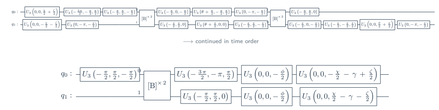
PROJECTIVE E=3 / L=26 / score ≈ 70.084 — upper bound



$U = \eta_{60}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=23 / score ≈ 78.338 — upper bound



$U = \eta_{60}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

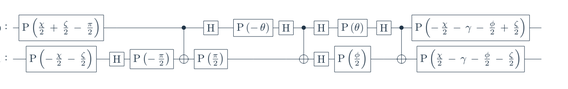
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

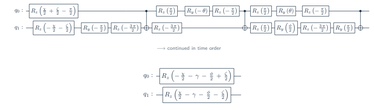
EXACT PHASE E=3 / L=15 / score ≈ 43.418 — upper bound



$U = \eta_{50}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

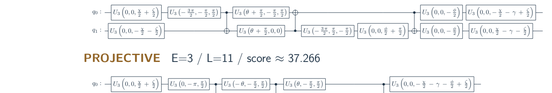
PROJECTIVE E=3 / L=17 / score ≈ 49.034 — upper bound



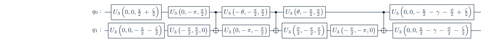
$U = \eta_{61}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=11 / score ≈ 37.466 — upper bound



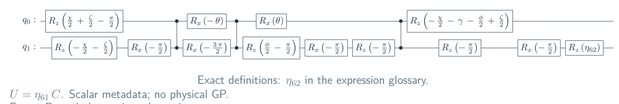
PROJECTIVE E=3 / L=11 / score ≈ 37.266 — upper bound



$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=3 / L=13 / score ≈ 40.226 — upper bound

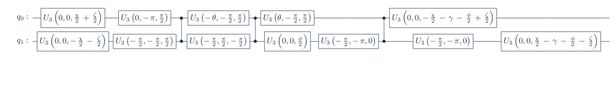


Exact definitions: η_{62} in the expression glossary.

$U = \eta_{61}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

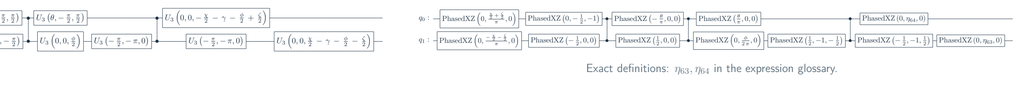
cz_u

EXACT PHASE E=3 / L=12 / score ≈ 39.672 — upper bound



google_cz

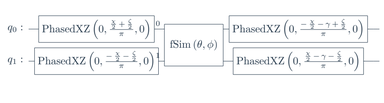
EXACT PHASE E=3 / L=12 / score ≈ 40.784 — upper bound



Exact definitions: η_{63}, η_{64} in the expression glossary.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — upper bound



google_sqrt_iswap

EXACT PHASE E=6 / L=25 / score ≈ 84.383 — upper bound



PROJECTIVE E=6 / L=23 / score ≈ 79.111 — upper bound

$U = -\frac{\sqrt{2}(1-i)e^{-i\theta}}{2}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=25 / score ≈ 130.583 — upper bound



PROJECTIVE E=18 / L=23 / score ≈ 125.911 — upper bound

$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

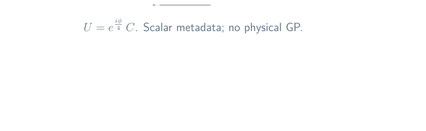
ibm_cx

PROJECTIVE E=3 / L=19 / score ≈ 52.224 — upper bound

$U = -e^{-\frac{i(\phi_3+\phi_4-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — upper bound



Exact definitions: η_{65} in the expression glossary.

$U = \sqrt[4]{-1}e^{-\frac{i(\phi_3+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=27 / score ≈ 68.224 — upper bound

$U = -ie^{-\frac{i(4+\phi_3+\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=39 / score ≈ 103.869 — upper bound

$U = \eta_{66}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=39 / score ≈ 102.663 — upper bound

$U = \eta_{66}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=42 / score ≈ 111.063 — upper bound

$U = \eta_{66}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=27 / score ≈ 68.224 — upper bound

$U = -ie^{-\frac{i(4+\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=25 / score ≈ 68.700 — upper bound

$U = \eta_{61}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=17 / score ≈ 51.102 — upper bound

PROJECTIVE E=3 / L=14 / score ≈ 44.284 — upper bound

$U = \eta_{50}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=13 / score ≈ 41.299 — upper bound

$U = -ie^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=3 / L=22 / score ≈ 62.057 — upper bound

$U = e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — upper bound

Exact definitions: η_{65} in the expression glossary.

$U = \sqrt[4]{-1}e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.138 — upper bound

sqrt_swap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.938 — upper bound

PROJECTIVE E=6 / L=23 / score ≈ 77.138 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=4 / score ≈ 14.005 — conditional minimum

$U = e^{-\frac{i(4+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=14 / score ≈ 43.684 — upper bound

PROJECTIVE E=3 / L=12 / score ≈ 38.472 — upper bound

$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=36 / score ≈ 91.078 — upper bound

$U = \eta_{60}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=17 / score ≈ 52.508 — upper bound

zz_u

EXACT PHASE E=3 / L=18 / score ≈ 52.314 — upper bound

PROJECTIVE E=3 / L=16 / score ≈ 47.502 — upper bound

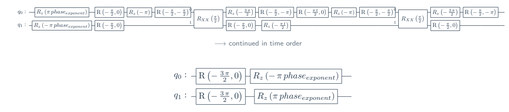
$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

PhasedISWAPPow $(phase_{exponent}, t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=20 / score ≈ 51.666 — conditional minimum



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cz_rz_rx

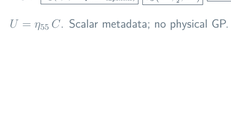
PROJECTIVE E=2 / L=15 / score ≈ 40.430 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_sqrt_iswap

EXACT PHASE E=4 / L=20 / score ≈ 59.920 — upper bound



PROJECTIVE E=4 / L=18 / score ≈ 55.708 — upper bound

$U = \eta_{58}\, C$. Scalar metadata; no physical GP.

ibm_ecr

PROJECTIVE E=2 / L=16 / score ≈ 41.822 — conditional minimum

$U = -\left(-1\right)^{\frac{3}{4}} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=20 / score ≈ 53.080 — conditional minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=21 / score ≈ 51.422 — conditional minimum

$U = \sqrt[4]{-1} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 30.866 — conditional minimum

PROJECTIVE E=2 / L=9 / score ≈ 26.854 — conditional minimum

$U = \eta_{55}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=20 / score ≈ 63.320 — upper bound



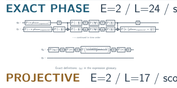
PROJECTIVE E=4 / L=18 / score ≈ 58.508 — upper bound



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 58.028 — conditional minimum



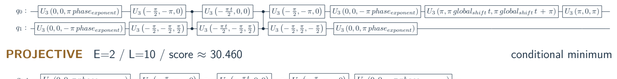
PROJECTIVE E=2 / L=17 / score ≈ 43.622 — conditional minimum



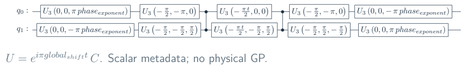
$U = e^{\frac{i\pi t\left(2 global_{shift}-1\right)}{2}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — conditional minimum



PROJECTIVE E=2 / L=10 / score ≈ 30.460 — conditional minimum



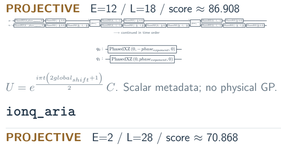
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=20 / score ≈ 91.120 — upper bound



PROJECTIVE E=12 / L=18 / score ≈ 86.908 — upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

ionq_aria

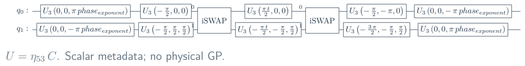
PROJECTIVE E=2 / L=28 / score ≈ 70.868 — conditional minimum

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — upper bound

PROJECTIVE E=2 / L=10 / score ≈ 30.460 — upper bound



$U = \eta_{53}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

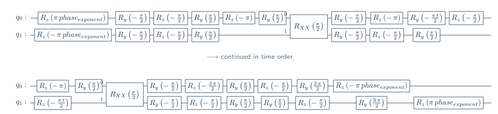
EXACT PHASE E=4 / L=20 / score ≈ 62.520 — upper bound

PROJECTIVE E=4 / L=18 / score ≈ 57.708 — upper bound

$U = -e^{i\pi t\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=33 / score ≈ 79.670 — conditional minimum



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_rz_ry

PROJECTIVE E=2 / L=10 / score ≈ 29.220 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_cz

EXACT PHASE E=2 / L=12 / score ≈ 34.072 — conditional minimum



PROJECTIVE E=2 / L=10 / score ≈ 29.860 — conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=16 / score ≈ 40.816 — conditional minimum



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

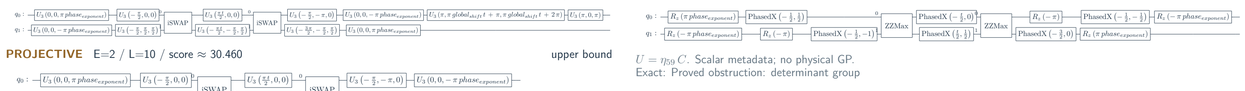
PROJECTIVE E=2 / L=28 / score ≈ 70.064 — conditional minimum



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzmax

PROJECTIVE E=2 / L=12 / score ≈ 33.836 — conditional minimum



$U = \eta_{59}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 62.920 — upper bound



PROJECTIVE E=4 / L=18 / score ≈ 57.708 — upper bound



$U = -ie^{i\pi t\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 46.100 — conditional minimum

PROJECTIVE E=2 / L=14 / score ≈ 41.288 — conditional minimum

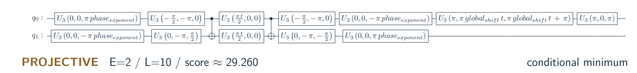
$U = e^{i\pi t\left(global_{shift}+\frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

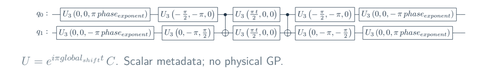
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_u

EXACT PHASE E=2 / L=12 / score ≈ 33.272 — conditional minimum



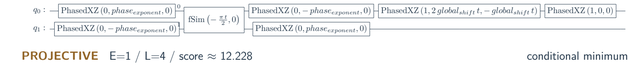
PROJECTIVE E=2 / L=10 / score ≈ 29.260 — conditional minimum



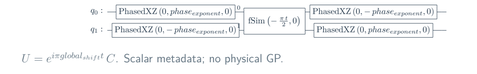
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 16.240 — conditional minimum



PROJECTIVE E=1 / L=4 / score ≈ 12.228 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

ibm_cz

PROJECTIVE E=2 / L=21 / score ≈ 51.422 — conditional minimum



$U = \sqrt[4]{-1} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

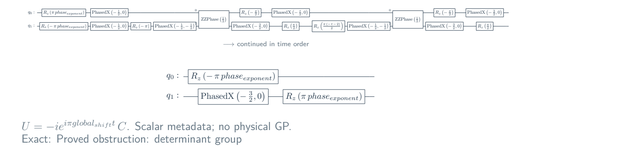
PROJECTIVE E=2 / L=30 / score ≈ 75.664 — conditional minimum



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

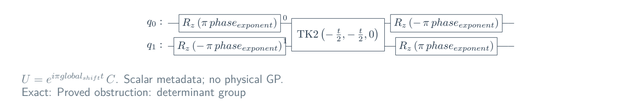
PROJECTIVE E=2 / L=19 / score ≈ 49.660 — conditional minimum



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

tket_tk2

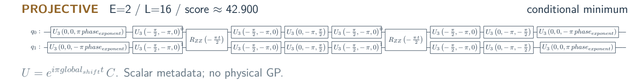
PROJECTIVE E=1 / L=4 / score ≈ 12.414 — conditional minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zz_u

EXACT PHASE E=2 / L=18 / score ≈ 46.912 — conditional minimum



PROJECTIVE E=2 / L=16 / score ≈ 42.900 — conditional minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

PhasedX (α, β)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=3 / score ≈ 6.008 proved minimum



berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

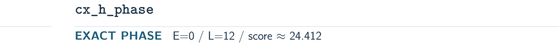
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

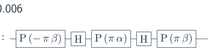
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = e^{-\frac{i\,\pi\,\alpha}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum



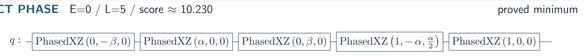
cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

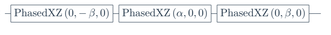


google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.230 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum



$U = e^{-\frac{i\,\pi\,\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.230 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum



$U = e^{-\frac{i\,\pi\,\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.230 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum

$U = e^{-\frac{i\,\pi\,\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.230 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum

$U = e^{-\frac{i\,\pi\,\alpha}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=8 / score ≈ 18.016 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=8 / score ≈ 18.016 proved minimum

ionq_forte

EXACT PHASE E=0 / L=8 / score ≈ 18.016 proved minimum

iqm_cz

EXACT PHASE E=0 / L=7 / score ≈ 15.628 proved minimum

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PhasedXPow $(t, phase_{exponent}, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.012 proved minimum

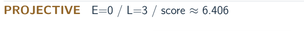
PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

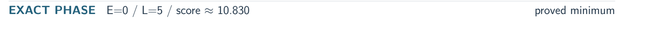
PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum



$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum



$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved minimum



$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

PhasedXZ $(x_{exponent}, z_{exponent}, axis_{phaseexponent})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum



$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.006 proved minimum



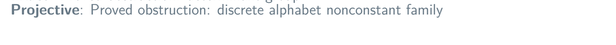
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum

Exact definitions: η_{68} in the expression glossary.
$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum



$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum



$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



Exact definitions: η_{69} in the expression glossary.
$U = \eta_{70}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



Exact definitions: η_{69} in the expression glossary.
$U = \eta_{70}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

Exact definitions: η_{69} in the expression glossary.
$U = \eta_{70}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

Exact definitions: η_{69} in the expression glossary.
$U = \eta_{70}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum

Exact definitions: η_{68} in the expression glossary.
$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

QFT

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

berkeley_b

Exact: Open: time limit reached
Projective: Open: time limit reached

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

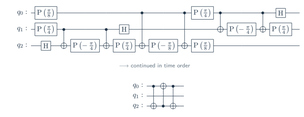
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

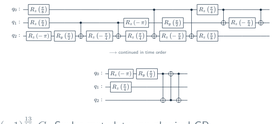
cx_h_phase

EXACT PHASE E=9 / L=12 / score ≈ 64.218 — upper bound



cx_rz_ry

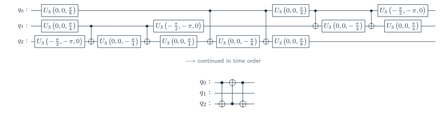
PROJECTIVE E=9 / L=15 / score ≈ 70.830 — upper bound



$U = (-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

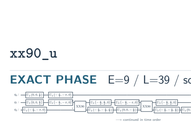
cx_u

EXACT PHASE E=9 / L=12 / score ≈ 64.872 — upper bound



cz_rz_rx

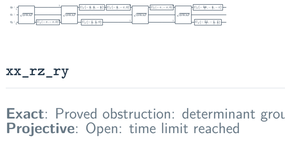
PROJECTIVE E=9 / L=63 / score ≈ 177.126 — upper bound



$U = -(-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=9 / L=30 / score ≈ 104.580 — upper bound



google_cz

EXACT PHASE E=9 / L=31 / score ≈ 111.186 — upper bound

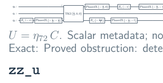


PROJECTIVE E=9 / L=30 / score ≈ 108.780 — upper bound

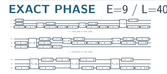
$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=9 / L=31 / score ≈ 111.222 — upper bound



PROJECTIVE E=9 / L=30 / score ≈ 108.816 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=18 / L=48 / score ≈ 184.488 — upper bound

google_syc

EXACT PHASE E=54 / L=49 / score ≈ 327.294 — upper bound

PROJECTIVE E=54 / L=48 / score ≈ 324.888 — upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=9 / L=18 / score ≈ 76.824 — upper bound

$U = e^{\frac{9i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ibm_ecr

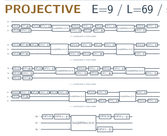
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

ionq_aria_full

PROJECTIVE E=9 / L=69 / score ≈ 192.174 — upper bound



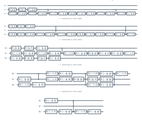
$U = \eta_{71}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

iqm_cz

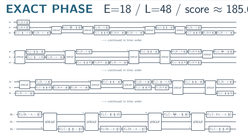
PROJECTIVE E=9 / L=69 / score ≈ 194.076 — upper bound



$U = -(-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

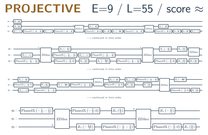
iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 185.688 — upper bound



quantinuum_zzmax

PROJECTIVE E=9 / L=55 / score ≈ 160.552 — upper bound



$U = \eta_{72}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

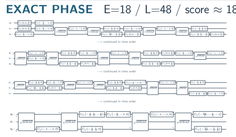
Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

rigetti_cz

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

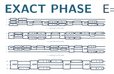
sqrt_iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 183.888 — upper bound

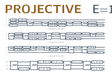


sqrt_swap_u

EXACT PHASE E=18 / L=48 / score ≈ 184.688 — upper bound



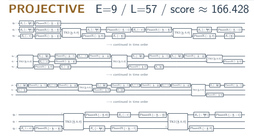
PROJECTIVE E=18 / L=48 / score ≈ 183.888 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

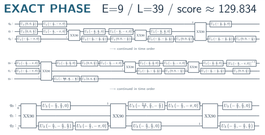
PROJECTIVE E=9 / L=57 / score ≈ 166.428 — upper bound



$U = \eta_{72}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=9 / L=39 / score ≈ 129.834 — upper bound

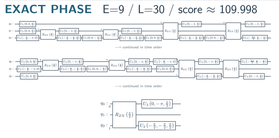


xx_rz_ry

Exact: Proved obstruction: determinant group
Projective: Open: time limit reached

zx_u

EXACT PHASE E=9 / L=30 / score ≈ 109.998 — upper bound



zz_u

EXACT PHASE E=9 / L=40 / score ≈ 130.458 — upper bound

PROJECTIVE E=9 / L=39 / score ≈ 128.052 — upper bound



$U = \frac{\sqrt{2}(1+i)^9}{32}\, C$. Scalar metadata; no physical GP.

QFTdg

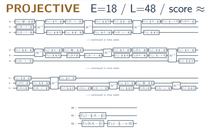
3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Open: time limit reached
Projective: Open: time limit reached

berkeley_b

PROJECTIVE E=18 / L=48 / score ≈ 187.488 upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

clifford_only

Exact: Open: time limit reached
Projective: Proved obstruction: not a clifford

clifford_t_cx

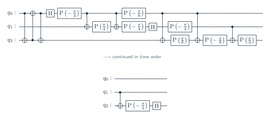
Exact: Open: time limit reached
Projective: Open: time limit reached

clifford_t_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

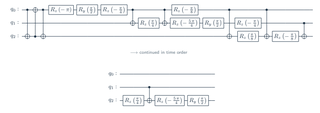
cx_h_phase

EXACT PHASE E=9 / L=12 / score ≈ 64.218 upper bound



cx_rz_ry

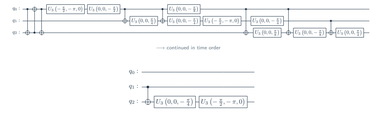
PROJECTIVE E=9 / L=13 / score ≈ 66.826 upper bound



$U = (-1)^{\frac{3}{16}} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

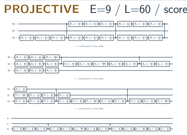
cx_u

EXACT PHASE E=9 / L=12 / score ≈ 64.872 upper bound



cz_rz_rx

PROJECTIVE E=9 / L=60 / score ≈ 170.520 upper bound



$U = -(-1)^{\frac{3}{16}} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

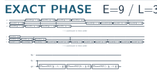
cz_u

EXACT PHASE E=9 / L=30 / score ≈ 104.580 upper bound



google_cz

EXACT PHASE E=9 / L=31 / score ≈ 111.186 upper bound



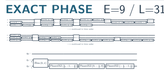
PROJECTIVE E=9 / L=30 / score ≈ 108.780 upper bound



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=9 / L=31 / score ≈ 111.222 upper bound



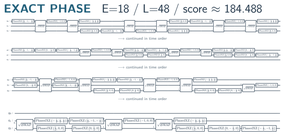
PROJECTIVE E=9 / L=30 / score ≈ 108.816 upper bound



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

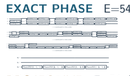
google_sqrt_iswap

EXACT PHASE E=18 / L=48 / score ≈ 184.488 upper bound



google_syc

EXACT PHASE E=54 / L=49 / score ≈ 327.294 upper bound



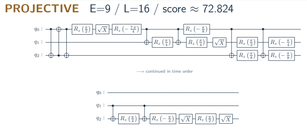
PROJECTIVE E=54 / L=48 / score ≈ 324.888 upper bound



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=9 / L=16 / score ≈ 72.824 upper bound



$U = -(-1)^{\frac{3}{16}} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ibm_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

ibm_ecr

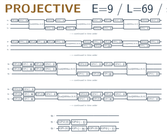
Exact: Open: time limit reached
Projective: Open: time limit reached

ionq_aria

Exact: Open: time limit reached
Projective: Open: time limit reached

ionq_aria_full

PROJECTIVE E=9 / L=69 / score ≈ 192.174 upper bound



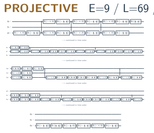
$U = \eta_{73} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ionq_forte

Exact: Open: time limit reached
Projective: Open: time limit reached

iqm_cz

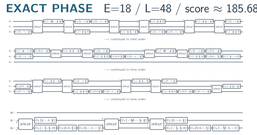
PROJECTIVE E=9 / L=69 / score ≈ 194.076 upper bound



$U = -(-1)^{\frac{3}{16}} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

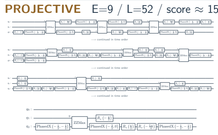
iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 185.688 upper bound



quantinuum_zzmax

PROJECTIVE E=9 / L=52 / score ≈ 154.546 upper bound



$U = \eta_{74} C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

quantinuum_zzphase

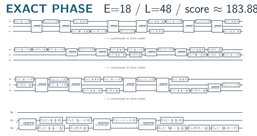
Exact: Open: time limit reached
Projective: Open: time limit reached

rigetti_cz

Exact: Open: time limit reached
Projective: Open: time limit reached

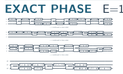
sqrt_iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 183.888 upper bound



sqrt_swap_u

EXACT PHASE E=18 / L=48 / score ≈ 184.688 upper bound



PROJECTIVE E=18 / L=48 / score ≈ 183.888 upper bound



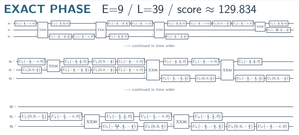
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

Exact: Open: time limit reached
Projective: Open: time limit reached

xx90_u

EXACT PHASE E=9 / L=39 / score ≈ 129.834 upper bound

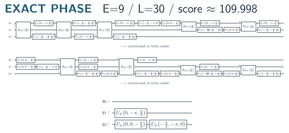


xx_rz_ry

Exact: Open: time limit reached
Projective: Open: time limit reached

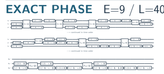
zx_u

EXACT PHASE E=9 / L=30 / score ≈ 109.998 upper bound

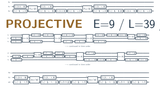


zz_u

EXACT PHASE E=9 / L=40 / score ≈ 130.458 upper bound



PROJECTIVE E=9 / L=39 / score ≈ 128.052 upper bound



$U = \frac{\sqrt{2}(1+i)^9}{32} C$. Scalar metadata; no physical GP.

R (θ, ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum



berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved minimum

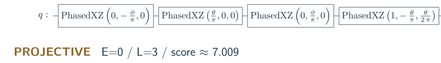
PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved minimum



ionq_forte

EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved minimum



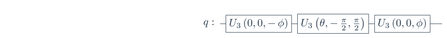
iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum



iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

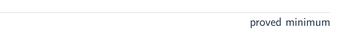


tket_tk2

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum



zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

RC3X

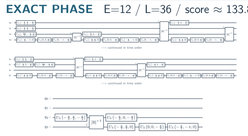
4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Open: time limit reached
Projective: Open: time limit reached

berkeley_b

EXACT PHASE E=12 / L=36 / score ≈ 133.816 — upper bound

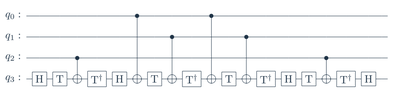


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

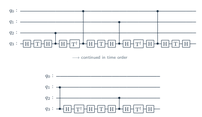
clifford_t_cx

EXACT PHASE E=6 / L=12 / score ≈ 48.000 — upper bound



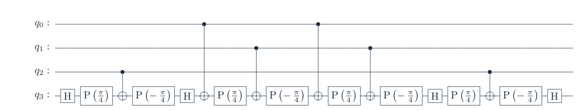
clifford_t_cz

EXACT PHASE E=6 / L=22 / score ≈ 68.000 — upper bound
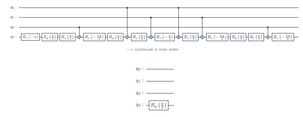


cx_h_phase

EXACT PHASE E=6 / L=12 / score ≈ 51.216 — upper bound



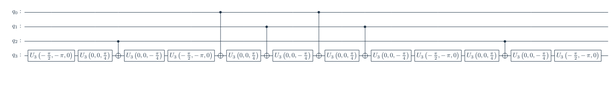
cx_rz_ry

EXACT PHASE E=6 / L=13 / score ≈ 54.026 — upper bound



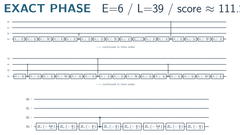
cx_u

EXACT PHASE E=6 / L=12 / score ≈ 52.072 — upper bound



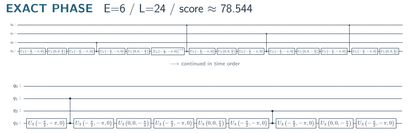
cz_rz_rx

EXACT PHASE E=6 / L=39 / score ≈ 111.278 — upper bound



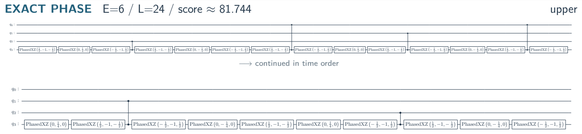
cz_u

EXACT PHASE E=6 / L=24 / score ≈ 78.544 — upper bound



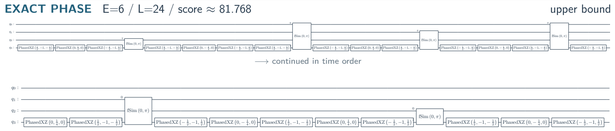
google_cz

EXACT PHASE E=6 / L=24 / score ≈ 81.744 — upper bound



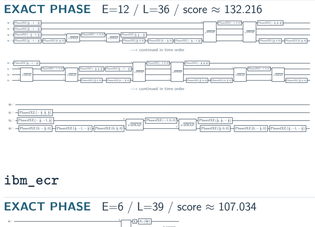
google_fsim

EXACT PHASE E=6 / L=24 / score ≈ 81.768 — upper bound



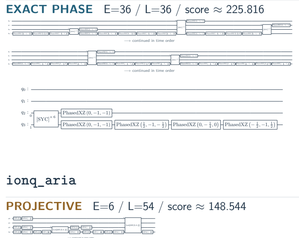
google_sqrt_iswap

EXACT PHASE E=12 / L=36 / score ≈ 132.216 — upper bound



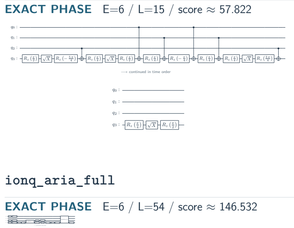
google_syc

EXACT PHASE E=36 / L=36 / score ≈ 225.816 — upper bound



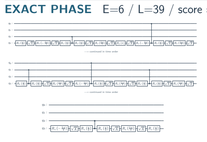
ibm_cx

EXACT PHASE E=6 / L=15 / score ≈ 57.822 — upper bound



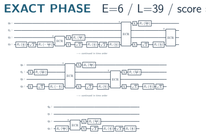
ibm_cz

EXACT PHASE E=6 / L=39 / score ≈ 108.046 — upper bound



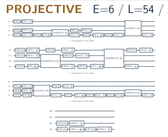
ibm_ecr

EXACT PHASE E=6 / L=39 / score ≈ 107.034 — upper bound



ionq_aria

PROJECTIVE E=6 / L=54 / score ≈ 148.544 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Open: time limit reached

ionq_aria_full

EXACT PHASE E=6 / L=54 / score ≈ 146.532 — upper bound



PROJECTIVE E=6 / L=54 / score ≈ 146.132 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=6 / L=60 / score ≈ 163.332 — upper bound



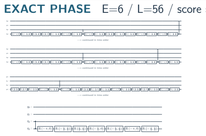
PROJECTIVE E=6 / L=60 / score ≈ 162.932 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

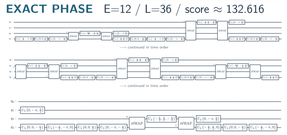
iqm_cz

EXACT PHASE E=6 / L=56 / score ≈ 152.224 — upper bound



iswap_u

EXACT PHASE E=12 / L=36 / score ≈ 132.616 — upper bound



quantinuum_zzmax

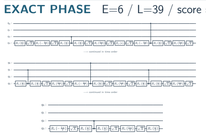
Exact: Open: time limit reached
Projective: Open: construction not found

quantinuum_zzphase

Exact: Open: time limit reached
Projective: Open: construction not found

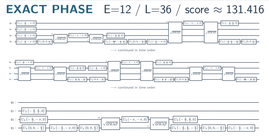
rigetti_cz

EXACT PHASE E=6 / L=39 / score ≈ 108.046 — upper bound
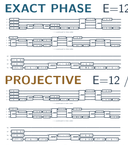


sqrt_iswap_u

EXACT PHASE E=12 / L=36 / score ≈ 131.416 — upper bound

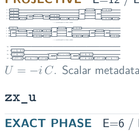


sqrt_swap_u

EXACT PHASE E=12 / L=36 / score ≈ 131.816 — upper bound



PROJECTIVE E=12 / L=36 / score ≈ 131.416 — upper bound



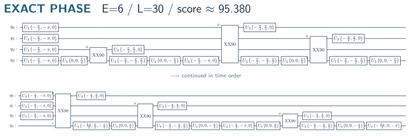
$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

Exact: Open: time limit reached
Projective: Open: time limit reached

xx90_u

EXACT PHASE E=6 / L=30 / score ≈ 95.380 — upper bound

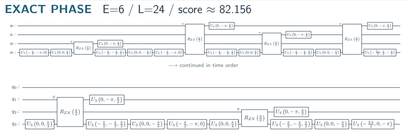


xx_rz_ry

Exact: Open: time limit reached
Projective: Open: time limit reached
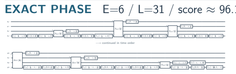
zx_u

EXACT PHASE E=6 / L=24 / score ≈ 82.156 — upper bound



zz_u

EXACT PHASE E=6 / L=31 / score ≈ 96.198 — upper bound



PROJECTIVE E=6 / L=30 / score ≈ 94.192 — upper bound



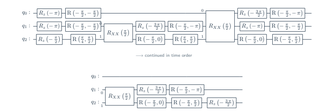
$U = -i\,C$. Scalar metadata; no physical GP.

RCCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

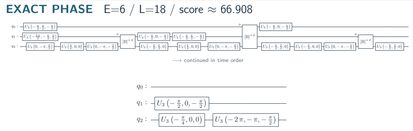
PROJECTIVE E=3 / L=21 / score ≈ 60.474 — upper bound



$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

berkeley_b

EXACT PHASE E=6 / L=18 / score ≈ 66.908 — upper bound

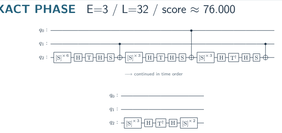


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

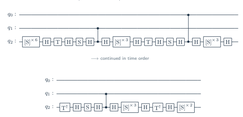
clifford_t_cx

EXACT PHASE E=3 / L=32 / score ≈ 76.000 — proved minimum



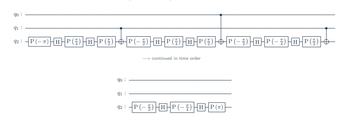
clifford_t_cz

EXACT PHASE E=3 / L=38 / score ≈ 88.000 — proved minimum



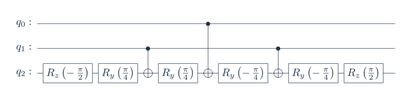
cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 54.824 — proved minimum



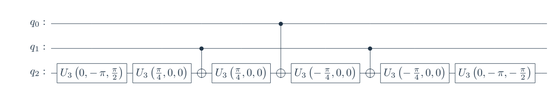
cx_rz_ry

EXACT PHASE E=3 / L=6 / score ≈ 26.012 — proved minimum



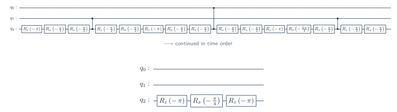
cx_u

EXACT PHASE E=3 / L=6 / score ≈ 26.036 — proved minimum



cz_rz_rx

EXACT PHASE E=3 / L=18 / score ≈ 51.436 — proved minimum



cz_u

EXACT PHASE E=3 / L=12 / score ≈ 39.272 — proved minimum



google_cz

EXACT PHASE E=3 / L=12 / score ≈ 41.272 — proved minimum



google_fsim

EXACT PHASE E=3 / L=12 / score ≈ 41.284 — upper bound



google_sqrt_iswap

EXACT PHASE E=6 / L=18 / score ≈ 67.108 — upper bound



PROJECTIVE E=6 / L=18 / score ≈ 66.508 — upper bound



$U = \frac{\sqrt{2}(1-i)^3}{4} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=18 / score ≈ 113.308 — upper bound



ibm_cx

EXACT PHASE E=3 / L=17 / score ≈ 48.018 — proved minimum



ibm_cz

EXACT PHASE E=3 / L=32 / score ≈ 80.032 — proved minimum



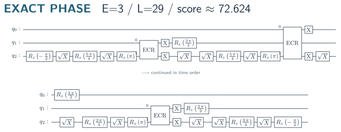
PROJECTIVE E=3 / L=32 / score ≈ 79.830 — proved minimum



$U = -(-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

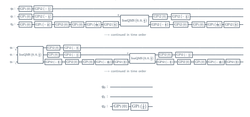
ibm_ecr

EXACT PHASE E=3 / L=29 / score ≈ 72.624 — proved minimum



ionq_aria

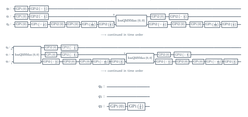
PROJECTIVE E=3 / L=35 / score ≈ 91.488 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

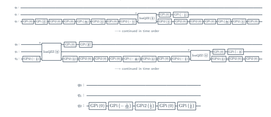
PROJECTIVE E=3 / L=35 / score ≈ 90.282 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

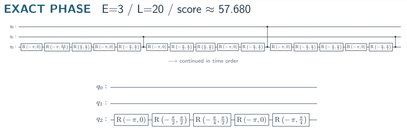
PROJECTIVE E=3 / L=38 / score ≈ 98.682 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

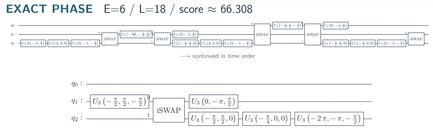
iqm_cz

EXACT PHASE E=3 / L=20 / score ≈ 57.680 — proved minimum



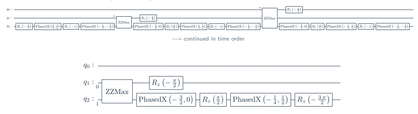
iswap_u

EXACT PHASE E=6 / L=18 / score ≈ 66.308 — upper bound



quantinuum_zzmax

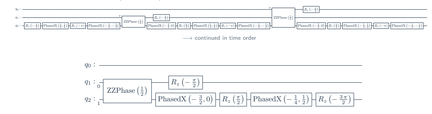
PROJECTIVE E=3 / L=21 / score ≈ 59.862 — proved minimum



$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

quantinuum_zzphase

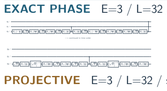
PROJECTIVE E=3 / L=21 / score ≈ 60.468 — upper bound



$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

rigetti_cz

EXACT PHASE E=3 / L=32 / score ≈ 80.032 — proved minimum



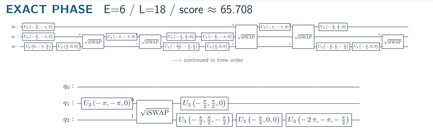
PROJECTIVE E=3 / L=32 / score ≈ 79.830 — proved minimum



$U = -(-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=6 / L=18 / score ≈ 65.708 — upper bound



sqrt_swap_u

EXACT PHASE E=6 / L=18 / score ≈ 66.508 — upper bound



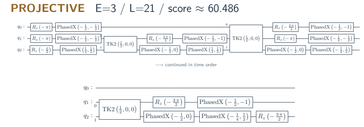
PROJECTIVE E=6 / L=18 / score ≈ 65.708 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

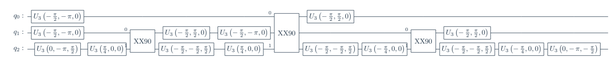
PROJECTIVE E=3 / L=21 / score ≈ 60.486 — upper bound



$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

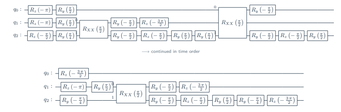
xx90_u

EXACT PHASE E=3 / L=15 / score ≈ 47.690 — proved minimum



xx_rz_ry

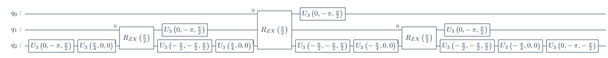
PROJECTIVE E=3 / L=27 / score ≈ 72.260 — upper bound



$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

zx_u

EXACT PHASE E=3 / L=12 / score ≈ 41.078 — upper bound

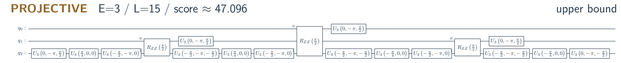


zz_u

EXACT PHASE E=3 / L=16 / score ≈ 49.502 — upper bound



PROJECTIVE E=3 / L=15 / score ≈ 47.096 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

RV (vx, vy, vz)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

Exact: Open: construction not found
Projective: Open: construction not found

berkeley_b

Exact: Open: construction not found
Projective: Open: construction not found

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

Exact: Open: construction not found
Projective: Open: construction not found

cx_rz_ry

Exact: Open: construction not found
Projective: Open: construction not found

cx_u

Exact: Open: construction not found
Projective: Open: construction not found

cz_rz_rx

Exact: Open: construction not found
Projective: Open: construction not found

cz_u

Exact: Open: construction not found
Projective: Open: construction not found

google_cz

Exact: Open: construction not found
Projective: Open: construction not found

google_fsim

Exact: Open: construction not found
Projective: Open: construction not found

google_sqrt_iswap

Exact: Open: construction not found
Projective: Open: construction not found

google_syc

Exact: Open: construction not found
Projective: Open: construction not found

ibm_cx

Exact: Open: construction not found
Projective: Open: construction not found

ibm_cz

Exact: Open: construction not found
Projective: Open: construction not found

ibm_ecr

Exact: Open: construction not found
Projective: Open: construction not found

ionq_aria

Exact: Open: construction not found
Projective: Open: construction not found

ionq_aria_full

Exact: Open: construction not found
Projective: Open: construction not found

ionq_forte

Exact: Open: construction not found
Projective: Open: construction not found

iqm_cz

Exact: Open: construction not found
Projective: Open: construction not found

iswap_u

Exact: Open: construction not found
Projective: Open: construction not found

quantinuum_zzmax

Exact: Open: construction not found
Projective: Open: construction not found

quantinuum_zzphase

Exact: Open: construction not found
Projective: Open: construction not found

rigetti_cz

Exact: Open: construction not found
Projective: Open: construction not found

sqrt_iswap_u

Exact: Open: construction not found
Projective: Open: construction not found

sqrt_swap_u

Exact: Open: construction not found
Projective: Open: construction not found

tket_tk2

Exact: Open: construction not found
Projective: Open: construction not found

xx90_u

Exact: Open: construction not found
Projective: Open: construction not found

xx_rz_ry

Exact: Open: construction not found
Projective: Open: construction not found

zx_u

Exact: Open: construction not found
Projective: Open: construction not found

zz_u

Exact: Open: construction not found
Projective: Open: construction not found

Rx (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

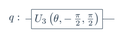
aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.410 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

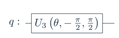
cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum



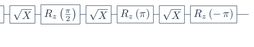
PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved minimum



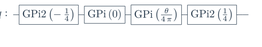
ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved minimum



ionq_forte

EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved minimum



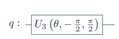
iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved minimum



xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



Rxmy (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — conditional minimum



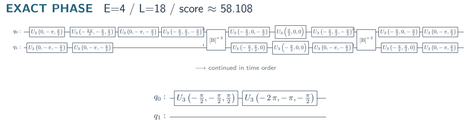
PROJECTIVE E=2 / L=19 / score ≈ 50.862 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=18 / score ≈ 58.108 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

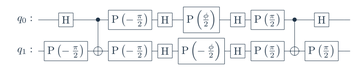
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

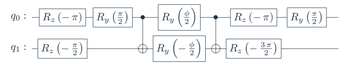
cx_h_phase

EXACT PHASE E=2 / L=14 / score ≈ 37.616 — conditional minimum



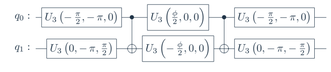
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — conditional minimum



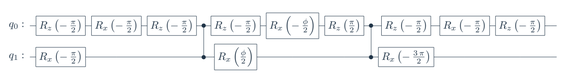
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — conditional minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — conditional minimum



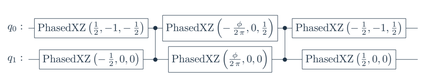
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — conditional minimum



google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.497 — conditional minimum



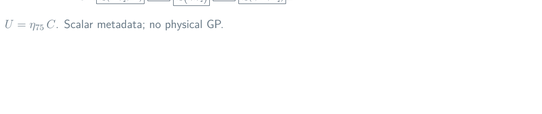
google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.220 — upper bound



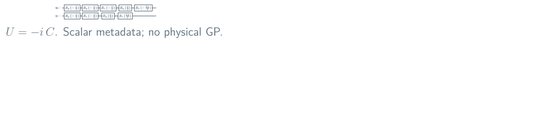
google_sqrt_iswap

EXACT PHASE E=4 / L=18 / score ≈ 57.569 — upper bound



google_syc

EXACT PHASE E=12 / L=18 / score ≈ 88.769 — upper bound



ibm_cx

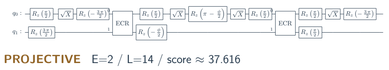
EXACT PHASE E=2 / L=16 / score ≈ 41.216 — conditional minimum

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — conditional minimum



PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=32 / score ≈ 81.337 — upper bound



PROJECTIVE E=2 / L=32 / score ≈ 80.937 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=32 / score ≈ 80.533 — upper bound



PROJECTIVE E=2 / L=32 / score ≈ 80.133 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=34 / score ≈ 86.133 — upper bound



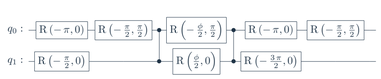
PROJECTIVE E=2 / L=34 / score ≈ 85.733 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

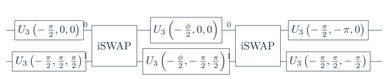
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.747 — upper bound



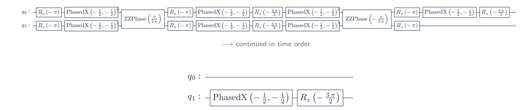
PROJECTIVE E=2 / L=8 / score ≈ 26.489 — conditional minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

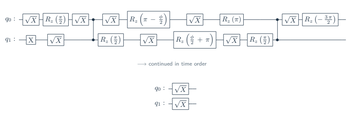
quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.117 — conditional minimum



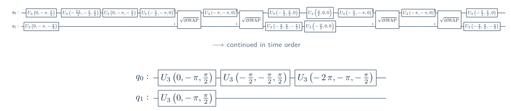
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum



sqrt_iswap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.308 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.708 — upper bound



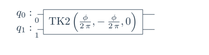
PROJECTIVE E=4 / L=18 / score ≈ 57.308 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.067 — conditional minimum



xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — conditional minimum

PROJECTIVE E=2 / L=5 / score ≈ 18.830 — conditional minimum

$U = \eta_{75}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — conditional minimum

PROJECTIVE E=2 / L=25 / score ≈ 63.254 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=14 / score ≈ 40.888 — upper bound

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — conditional minimum

Rxx (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=0 / score ≈ 4.002 upper bound



berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 50.490 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.078 upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 30.410 upper bound



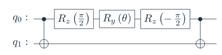
PROJECTIVE E=2 / L=3 / score ≈ 14.002 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

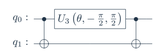
cx_rz_ry

EXACT PHASE E=2 / L=3 / score ≈ 14.406 upper bound



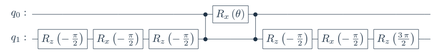
cx_u

EXACT PHASE E=2 / L=1 / score ≈ 10.406 upper bound



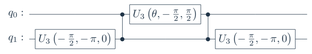
cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 upper bound



cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 upper bound



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.021 upper bound



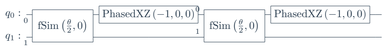
PROJECTIVE E=2 / L=3 / score ≈ 15.148 upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.420 upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=15 / score ≈ 51.281 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.408 upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 82.481 upper bound



PROJECTIVE E=12 / L=13 / score ≈ 77.608 upper bound



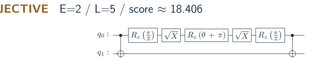
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 18.406 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 30.810 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.004 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 53.582 upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.182 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 52.778 upper bound



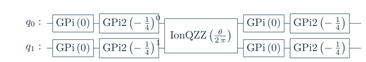
PROJECTIVE E=2 / L=20 / score ≈ 52.378 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

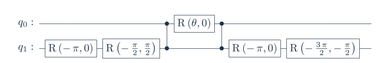
ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 22.148 conditional minimum



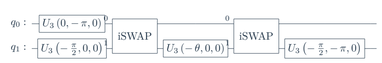
iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 upper bound



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 conditional minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.604 upper bound



PROJECTIVE E=2 / L=5 / score ≈ 19.346 upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

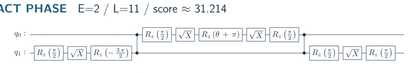
quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.956 conditional minimum

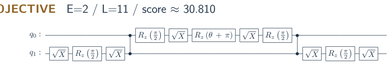


rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 upper bound



PROJECTIVE E=2 / L=11 / score ≈ 30.810 upper bound



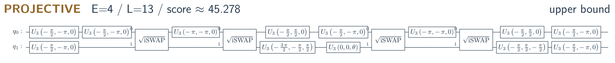
$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.278 upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 50.090 upper bound



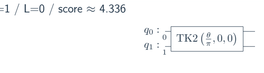
PROJECTIVE E=4 / L=13 / score ≈ 45.278 upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

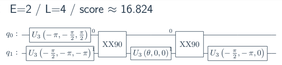
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 conditional minimum

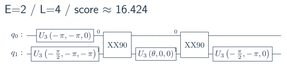


xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 upper bound



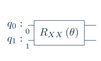
PROJECTIVE E=2 / L=4 / score ≈ 16.424 upper bound



$U = i\,C$. Scalar metadata; no physical GP.

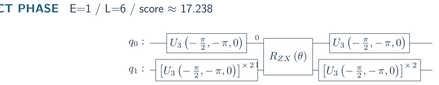
xx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.002 upper bound



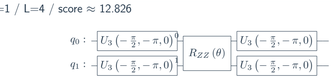
zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.238 conditional minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 conditional minimum

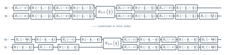


Rxy (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — conditional minimum



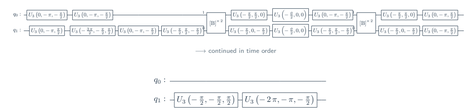
PROJECTIVE E=2 / L=19 / score ≈ 50.862 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=18 / score ≈ 58.108 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

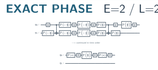
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

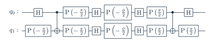
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.024 — conditional minimum



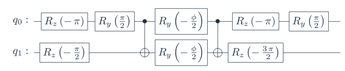
PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = e^{\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

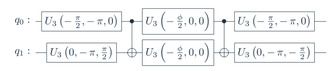
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — conditional minimum



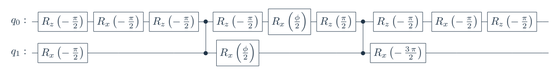
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — conditional minimum



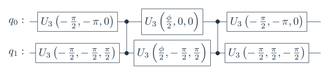
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — conditional minimum



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — conditional minimum



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.369 — conditional minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.497 — conditional minimum



$U = e^{-\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — conditional minimum



google_sqrt_iswap

EXACT PHASE E=4 / L=20 / score ≈ 62.441 — upper bound

PROJECTIVE E=4 / L=18 / score ≈ 57.569 — upper bound

$U = -ie^{\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=20 / score ≈ 93.641 — upper bound



PROJECTIVE E=12 / L=18 / score ≈ 88.769 — upper bound

$U = e^{\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=16 / score ≈ 41.216 — conditional minimum

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — conditional minimum



PROJECTIVE E=2 / L=14 / score ≈ 37.616 — conditional minimum



$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=32 / score ≈ 81.337 — upper bound



PROJECTIVE E=2 / L=32 / score ≈ 80.937 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=32 / score ≈ 80.533 — upper bound



PROJECTIVE E=2 / L=32 / score ≈ 80.133 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=34 / score ≈ 86.133 — upper bound



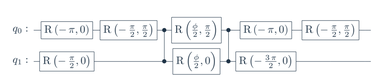
PROJECTIVE E=2 / L=34 / score ≈ 85.733 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

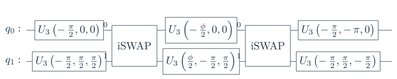
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — upper bound

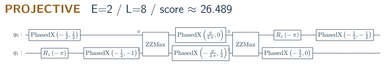


quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.747 — upper bound



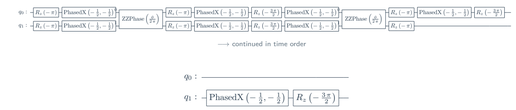
PROJECTIVE E=2 / L=8 / score ≈ 26.489 — conditional minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}C$. Scalar metadata; no physical GP.

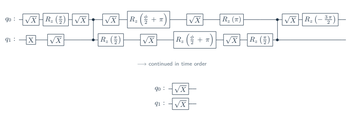
quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.117 — conditional minimum



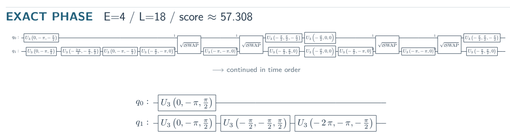
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — conditional minimum



sqrt_iswap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.308 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.708 — upper bound



PROJECTIVE E=4 / L=18 / score ≈ 57.308 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.067 — conditional minimum



xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — conditional minimum



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — conditional minimum



PROJECTIVE E=2 / L=27 / score ≈ 67.658 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

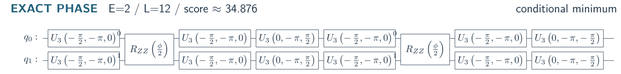
zx_u

EXACT PHASE E=2 / L=14 / score ≈ 40.888 — upper bound



zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — conditional minimum

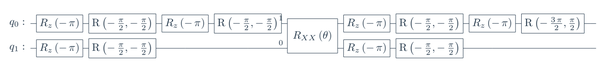


Rxz (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 30.438 — conditional minimum

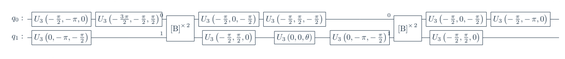


berkeley_b

EXACT PHASE E=4 / L=13 / score ≈ 46.078 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.666 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

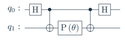
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 30.410 — upper bound



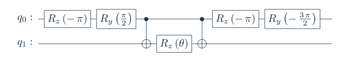
PROJECTIVE E=2 / L=3 / score ≈ 14.002 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

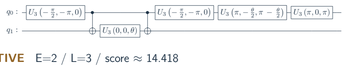
cx_rz_ry

EXACT PHASE E=2 / L=5 / score ≈ 18.410 — upper bound



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.418 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=17 / score ≈ 45.234 — upper bound



cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — upper bound

PROJECTIVE E=2 / L=7 / score ≈ 23.242 — upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.645 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 24.772 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 29.653 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 24.780 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=13 / score ≈ 46.469 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.596 — upper bound

$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 77.669 — upper bound

PROJECTIVE E=12 / L=11 / score ≈ 72.796 — upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=7 / score ≈ 22.406 — upper bound

ibm_cz

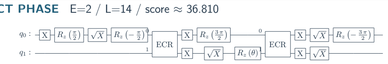
EXACT PHASE E=2 / L=17 / score ≈ 43.618 — upper bound

PROJECTIVE E=2 / L=17 / score ≈ 43.214 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 36.810 — upper bound



PROJECTIVE E=2 / L=13 / score ≈ 34.608 — upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=16 / score ≈ 44.774 — upper bound



PROJECTIVE E=2 / L=16 / score ≈ 44.374 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=16 / score ≈ 43.970 — upper bound



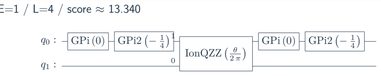
PROJECTIVE E=2 / L=16 / score ≈ 43.570 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=4 / score ≈ 13.340 — conditional minimum



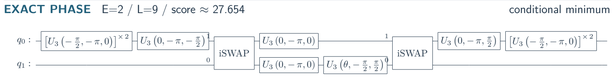
iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.260 — upper bound



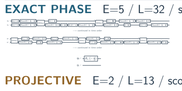
iswap_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — conditional minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=32 / score ≈ 91.096 — upper bound



PROJECTIVE E=2 / L=13 / score ≈ 36.838 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 13.144 — conditional minimum



rigetti_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.618 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 43.214 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 40.866 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.678 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 40.866 — upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=35 / score ≈ 98.142 — upper bound



PROJECTIVE E=2 / L=15 / score ≈ 41.258 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — upper bound



PROJECTIVE E=2 / L=9 / score ≈ 28.854 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

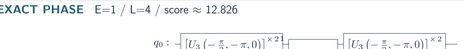
xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 29.226 — conditional minimum



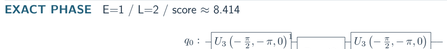
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 — conditional minimum



zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — conditional minimum



Ry (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

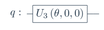
aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_cz

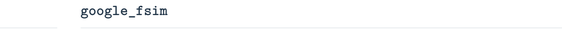
EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved minimum

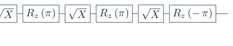


PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved minimum



ionq_forte

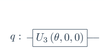
EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

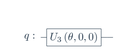
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved minimum

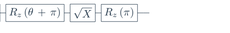


quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

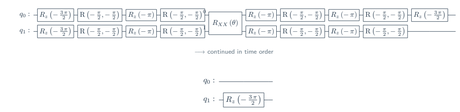
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

Ryy (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

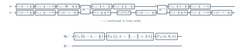
aqt_rxx

EXACT PHASE E=1 / L=18 / score ≈ 44.054 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=19 / score ≈ 59.314 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.902 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

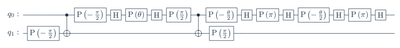
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=15 / score ≈ 39.218 — upper bound



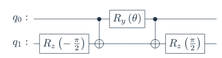
PROJECTIVE E=2 / L=7 / score ≈ 22.810 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=3 / score ≈ 14.406 — upper bound



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — upper bound



cz_rz_rx

EXACT PHASE E=2 / L=5 / score ≈ 18.810 — upper bound



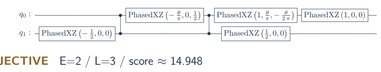
cz_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 — upper bound



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 19.821 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.948 — upper bound



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.420 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=19 / score ≈ 60.105 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 55.232 — upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 91.305 — upper bound



PROJECTIVE E=12 / L=17 / score ≈ 86.432 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 20.408 — upper bound



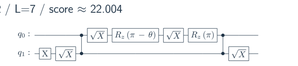
$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.004 — upper bound



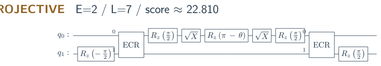
$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.810 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=28 / score ≈ 71.998 — upper bound



PROJECTIVE E=2 / L=28 / score ≈ 71.598 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=28 / score ≈ 71.194 — upper bound



PROJECTIVE E=2 / L=28 / score ≈ 70.794 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

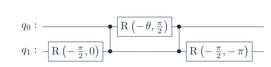
ionq_forte

EXACT PHASE E=1 / L=16 / score ≈ 40.564 — conditional minimum



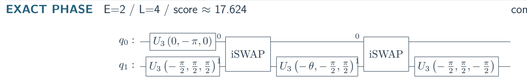
iqm_cz

EXACT PHASE E=2 / L=3 / score ≈ 14.612 — upper bound



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — conditional minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.004 — upper bound



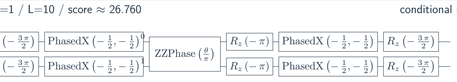
PROJECTIVE E=2 / L=5 / score ≈ 18.746 — upper bound



$U = \frac{\sqrt{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=10 / score ≈ 26.760 — conditional minimum



rigetti_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.004 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=19 / score ≈ 58.514 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=19 / score ≈ 58.914 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

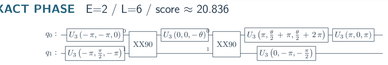
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 — conditional minimum

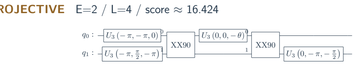


xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — upper bound



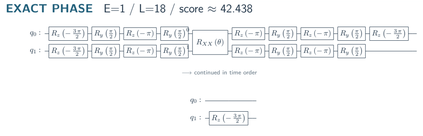
PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound



$U = \eta_{76}\,C$. Scalar metadata; no physical GP.

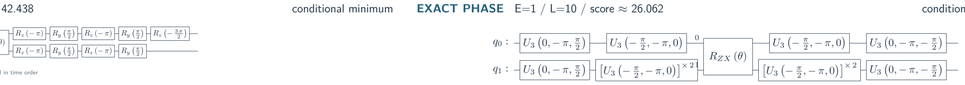
xx_rz_ry

EXACT PHASE E=1 / L=18 / score ≈ 42.438 — conditional minimum



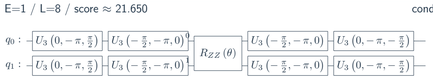
zx_u

EXACT PHASE E=1 / L=10 / score ≈ 26.062 — conditional minimum



zz_u

EXACT PHASE E=1 / L=8 / score ≈ 21.650 — conditional minimum



Rz (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

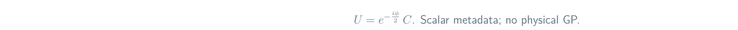
EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved minimum

iqm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.812 proved minimum



iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

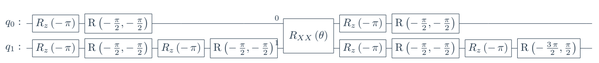
$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

Rzx (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 30.438 — conditional minimum

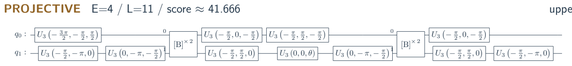


berkeley_b

EXACT PHASE E=4 / L=13 / score ≈ 46.078 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.666 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 30.410 — upper bound



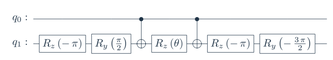
PROJECTIVE E=2 / L=3 / score ≈ 14.002 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=5 / score ≈ 18.410 — upper bound



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.418 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 40.430 — upper bound



cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — upper bound

PROJECTIVE E=2 / L=7 / score ≈ 23.242 — upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.645 — upper bound



PROJECTIVE E=2 / L=7 / score ≈ 24.772 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 29.653 — upper bound



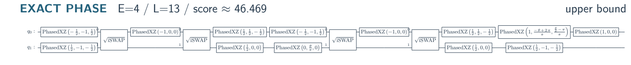
PROJECTIVE E=2 / L=7 / score ≈ 24.780 — upper bound



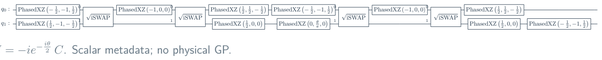
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=13 / score ≈ 46.469 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.596 — upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 77.669 — upper bound



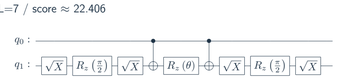
PROJECTIVE E=12 / L=11 / score ≈ 72.796 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=7 / score ≈ 22.406 — upper bound



ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.216 — upper bound



ibm_ecr

EXACT PHASE E=2 / L=15 / score ≈ 38.810 — upper bound



ionq_aria

EXACT PHASE E=2 / L=16 / score ≈ 44.774 — upper bound



PROJECTIVE E=2 / L=16 / score ≈ 44.374 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=16 / score ≈ 43.970 — upper bound



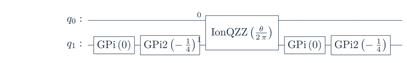
PROJECTIVE E=2 / L=16 / score ≈ 43.570 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=4 / score ≈ 13.340 — conditional minimum



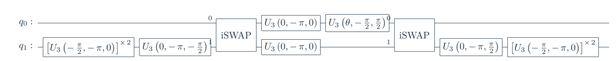
iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.260 — upper bound



iswap_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — conditional minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 93.098 — upper bound



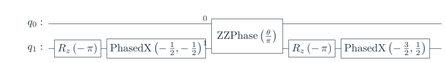
PROJECTIVE E=2 / L=14 / score ≈ 38.840 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 13.144 — conditional minimum



rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.216 — upper bound



sqrt_iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 40.866 — upper bound



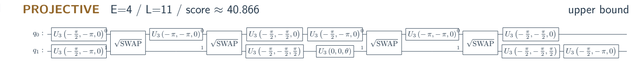
$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.678 — upper bound



PROJECTIVE E=4 / L=11 / score ≈ 40.866 — upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=35 / score ≈ 98.142 — upper bound



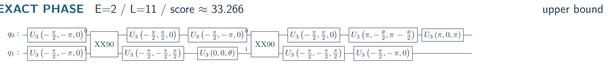
PROJECTIVE E=2 / L=15 / score ≈ 41.258 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — upper bound



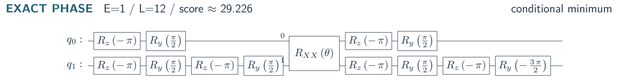
PROJECTIVE E=2 / L=9 / score ≈ 28.854 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

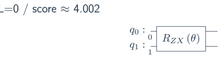
xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 29.226 — conditional minimum



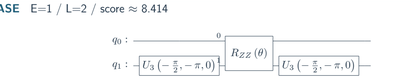
zx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — upper bound



zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — conditional minimum

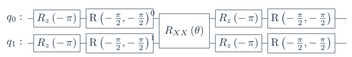


Rzz (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

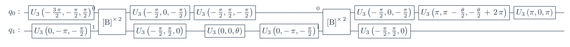
aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — conditional minimum

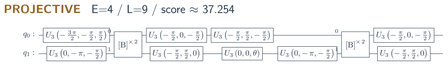


berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.666 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 37.254 — upper bound



$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — upper bound



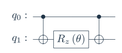
PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — upper bound



PROJECTIVE E=2 / L=1 / score ≈ 10.006 — upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — upper bound



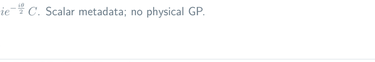
cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — upper bound



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.021 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 15.148 — upper bound



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.549 — conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.677 — conditional minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 41.657 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 36.784 — upper bound

$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.857 — upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.984 — upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=12 / score ≈ 35.966 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.566 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 35.162 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 34.762 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=0 / score ≈ 4.532 — conditional minimum

iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — upper bound

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.604 — upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.346 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.332 — conditional minimum

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 — conditional minimum

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — conditional minimum

zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — conditional minimum

zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — upper bound

S

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



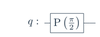
clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

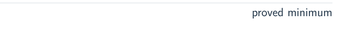
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum



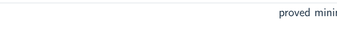
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



S_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

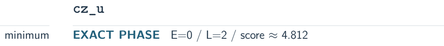
$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

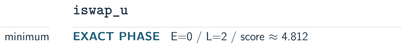
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



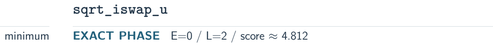
PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

S_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

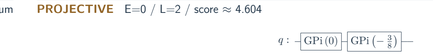
$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

S_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

S_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

S_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

S_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

S_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg

1 qubits / 32 preserved native bases / alphabetical panel order

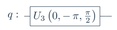
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

Sdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

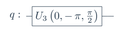


berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

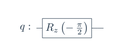
EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

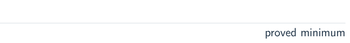
$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

Sdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

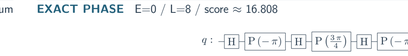
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

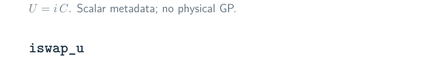
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

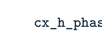
berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

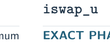
iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

Sdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=15 / score ≈ 30.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

Sdg_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

SH

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

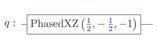
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

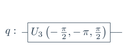
PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

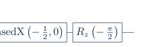
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

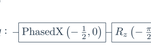
PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

SH_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



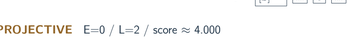
PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SH_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

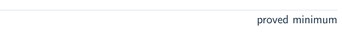
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

SH_q3

1 qubits / 32 preserved native bases / alphabetical panel order

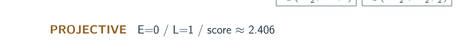
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



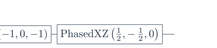
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

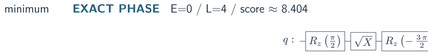
$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

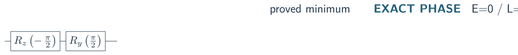
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SH_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

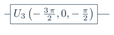
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

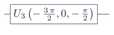


$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

SH_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

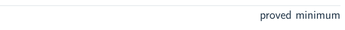
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SH_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SH_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



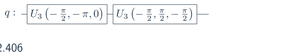
PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



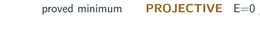
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



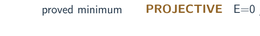
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

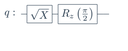
PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ionq_aria

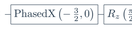
PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

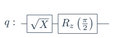
PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

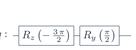
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

SHX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

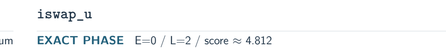
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

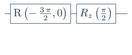
$U = i\,C$. Scalar metadata; no physical GP.

SHX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

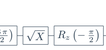
$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

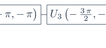
EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



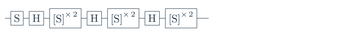
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

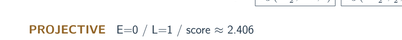
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

SHX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

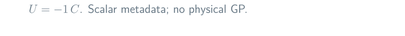
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

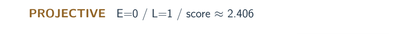
EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

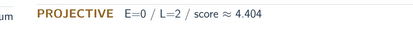


$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SHX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



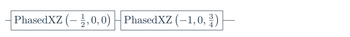
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



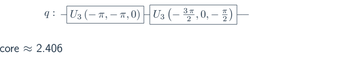
PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SHY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

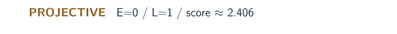
PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



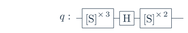
clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum



cx_rz_ry

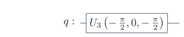
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

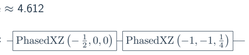
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



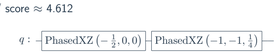
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

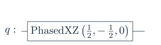


$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



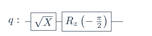
$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

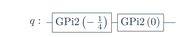
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

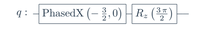
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



quantinuum_zzmax

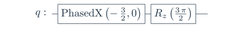
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



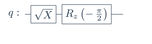
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



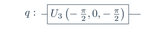
PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

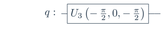
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



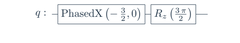
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



tket_tk2

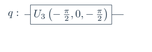
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

SHY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

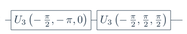
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



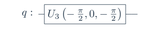
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

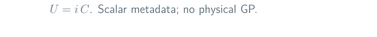
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

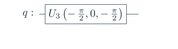
ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

SHY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



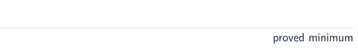
PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

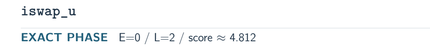


PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum



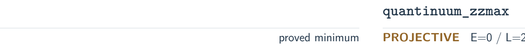
PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

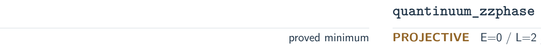
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

SHY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

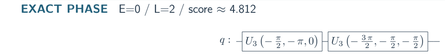
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



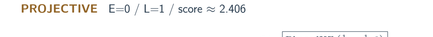
$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



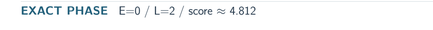
PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

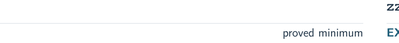
xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHZ

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

SHZ_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

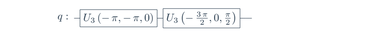
PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

SHZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

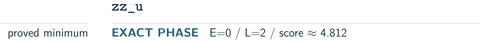


$U = i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

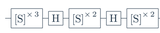
berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

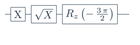
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

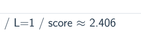
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

SHZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

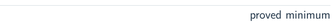
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SHZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SiSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 84.104 — proved minimum



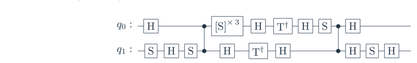
PROJECTIVE E=2 / L=16 / score ≈ 44.058 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

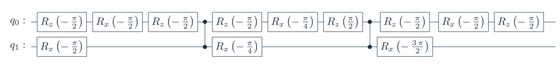
clifford_t_cz

EXACT PHASE E=2 / L=18 / score ≈ 44.000 — proved minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.824 — proved minimum



google_sqrt_iswap

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.420 — proved minimum



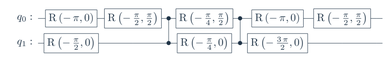
PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

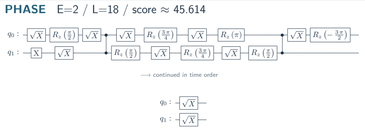
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.232 — proved minimum



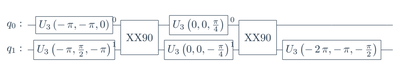
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved minimum



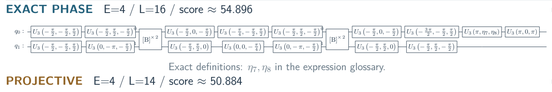
xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved minimum



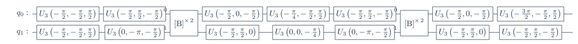
berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — upper bound



Exact definitions: η_7, η_8 in the expression glossary.

PROJECTIVE E=4 / L=14 / score ≈ 50.884 — upper bound



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=16 / score ≈ 41.814 — proved minimum



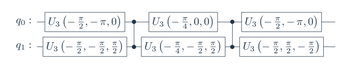
PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved minimum



google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.690 — upper bound



PROJECTIVE E=12 / L=14 / score ≈ 79.284 — upper bound

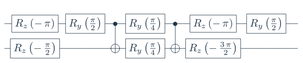
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.616 — proved minimum

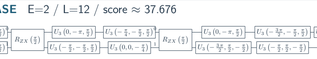


google_cz

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — proved minimum



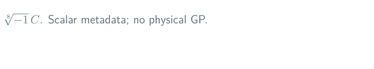
PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.212 — proved minimum

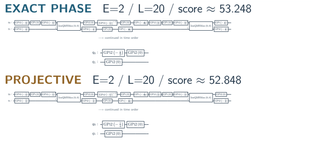


PROJECTIVE E=2 / L=12 / score ≈ 33.208 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.248 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 52.848 — upper bound

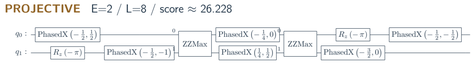
$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.486 — upper bound



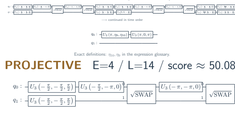
PROJECTIVE E=2 / L=8 / score ≈ 26.228 — proved minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.096 — upper bound



Exact definitions: η_{13}, η_9 in the expression glossary.

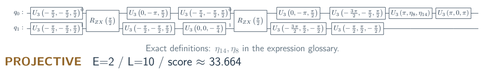
PROJECTIVE E=4 / L=14 / score ≈ 50.084 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

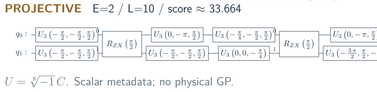
zx_u

EXACT PHASE E=2 / L=12 / score ≈ 37.676 — upper bound



Exact definitions: η_{14}, η_8 in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 33.664 — upper bound



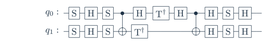
$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=2 / L=18 / score ≈ 44.000 — proved minimum



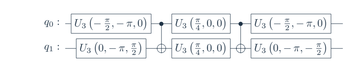
PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved minimum



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

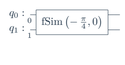
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved minimum



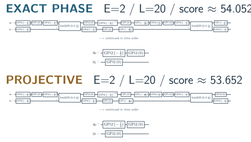
google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.404 — proved minimum



ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.052 — upper bound

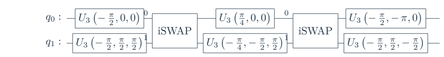


PROJECTIVE E=2 / L=20 / score ≈ 53.652 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved minimum



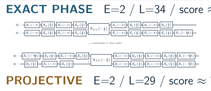
sqrt_iswap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.872 — proved minimum



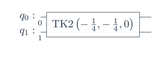
PROJECTIVE E=2 / L=29 / score ≈ 72.062 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

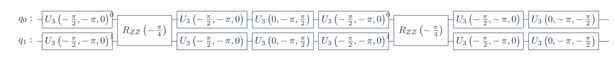
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.806 — proved minimum



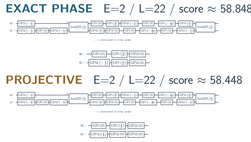
zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.276 — proved minimum



ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.848 — upper bound



PROJECTIVE E=2 / L=22 / score ≈ 58.448 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.856 — proved minimum



PROJECTIVE E=2 / L=15 / score ≈ 42.052 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.806 — proved minimum

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.276 — proved minimum

SiSWAPdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 84.104 — proved minimum



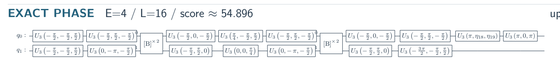
PROJECTIVE E=2 / L=16 / score ≈ 44.058 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

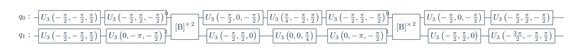
berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — upper bound



Exact definitions: η_{18}, η_{19} in the expression glossary.

PROJECTIVE E=4 / L=14 / score ≈ 50.884 — upper bound



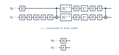
$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.

clifford_only

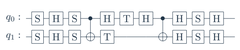
Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=22 / score ≈ 52.000 — proved minimum



PROJECTIVE E=2 / L=16 / score ≈ 40.000 — upper bound

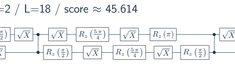


$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=2 / L=24 / score ≈ 56.000 — proved minimum

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=16 / score ≈ 41.814 — upper bound

PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.616 — proved minimum



cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.824 — proved minimum



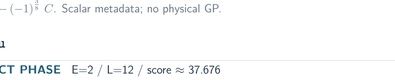
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved minimum



google_cz

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — proved minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.404 — proved minimum

google_sqrt_iswap

EXACT PHASE E=3 / L=2 / score ≈ 16.012 — upper bound

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.690 — upper bound

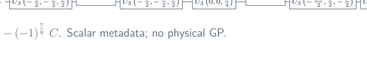


PROJECTIVE E=12 / L=14 / score ≈ 79.284 — upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.212 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 33.208 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved minimum

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.420 — proved minimum

PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.052 — upper bound

PROJECTIVE E=2 / L=20 / score ≈ 53.652 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.248 — upper bound

PROJECTIVE E=2 / L=20 / score ≈ 52.848 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.848 — upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.448 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.232 — proved minimum

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.486 — upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.228 — proved minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.856 — proved minimum

PROJECTIVE E=2 / L=16 / score ≈ 44.054 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved minimum

sqrt_iswap_u

EXACT PHASE E=3 / L=2 / score ≈ 16.012 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.096 — upper bound

Exact definitions: η_{11}, η_{12} in the expression glossary.

PROJECTIVE E=4 / L=14 / score ≈ 50.084 — upper bound

$U = -(-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.806 — proved minimum

xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved minimum

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.872 — proved minimum

PROJECTIVE E=2 / L=29 / score ≈ 72.462 — upper bound

$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 37.676 — upper bound

Exact definitions: η_{18}, η_2 in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 33.664 — upper bound

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.276 — proved minimum

SqrtH

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

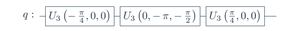
PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=15 / score ≈ 30.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=15 / score ≈ 30.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 19.410 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=5 / score ≈ 11.410 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

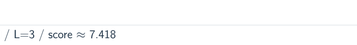


google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum

ibm_cx

EXACT PHASE E=0 / L=15 / score ≈ 31.614 proved minimum



PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



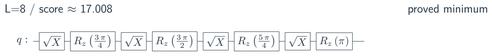
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=15 / score ≈ 31.614 proved minimum



PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=15 / score ≈ 31.614 proved minimum



PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=4 / score ≈ 9.616 proved minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=15 / score ≈ 31.614 proved minimum

PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

SqrtHdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

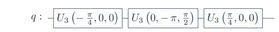
PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=17 / score ≈ 34.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=17 / score ≈ 34.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 19.410 proved minimum



cx_rz_ry

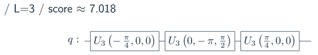
PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=5 / score ≈ 11.410 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum



google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved minimum

ibm_cx

EXACT PHASE E=0 / L=14 / score ≈ 29.614 proved minimum



PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=14 / score ≈ 29.614 proved minimum



PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum

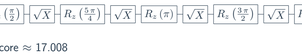


$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=14 / score ≈ 29.614 proved minimum

PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=4 / score ≈ 9.616 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=14 / score ≈ 29.614 proved minimum

PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

zz_u

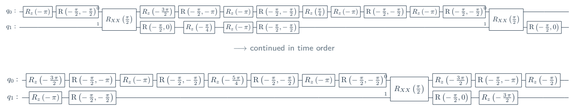
EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved minimum

SSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

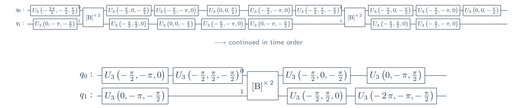
PROJECTIVE E=3 / L=31 / score ≈ 81.698 — upper bound



$U = \eta_{31}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=23 / score ≈ 77.738 — upper bound

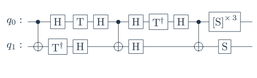


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

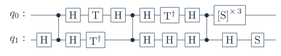
clifford_t_cx

EXACT PHASE E=3 / L=13 / score ≈ 38.000 — proved minimum



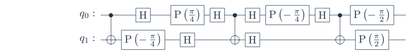
clifford_t_cz

EXACT PHASE E=3 / L=17 / score ≈ 46.000 — proved minimum



cx_h_phase

EXACT PHASE E=3 / L=11 / score ≈ 35.610 — proved minimum



cx_rz_ry

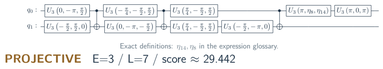
PROJECTIVE E=3 / L=14 / score ≈ 43.428 — proved minimum



$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

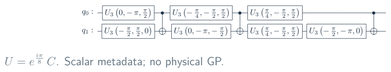
cx_u

EXACT PHASE E=3 / L=9 / score ≈ 33.454 — proved minimum



Exact definitions: η_{14}, η_8 in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 29.442 — proved minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

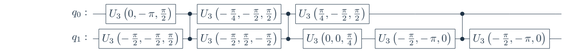
PROJECTIVE E=3 / L=11 / score ≈ 36.822 — proved minimum



$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

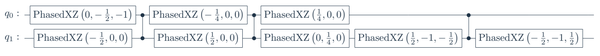
cz_u

EXACT PHASE E=3 / L=8 / score ≈ 31.848 — proved minimum



google_cz

EXACT PHASE E=3 / L=8 / score ≈ 30.648 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 9.416 — proved minimum



google_sqrt_iswap

EXACT PHASE E=6 / L=23 / score ≈ 78.338 — upper bound



PROJECTIVE E=6 / L=23 / score ≈ 77.738 — upper bound

$U = \frac{\sqrt{2}(1-i)}{4}^3\, C$. Scalar metadata; no physical GP.

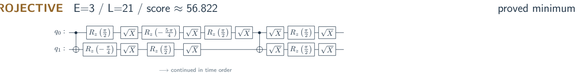
google_syc

EXACT PHASE E=18 / L=23 / score ≈ 124.538 — upper bound



ibm_cx

PROJECTIVE E=3 / L=21 / score ≈ 56.822 — proved minimum



$U = -(-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

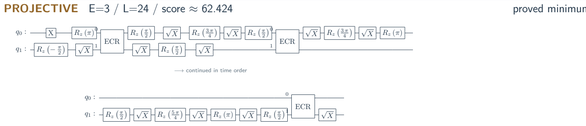
PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved minimum



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

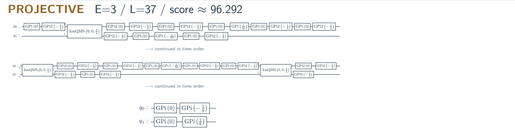
PROJECTIVE E=3 / L=24 / score ≈ 62.424 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=37 / score ≈ 96.292 — upper bound



$U = \eta_{78}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

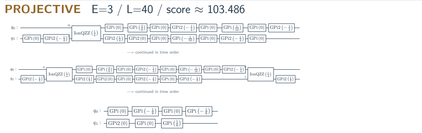
PROJECTIVE E=3 / L=37 / score ≈ 95.086 — upper bound



$U = \eta_{78}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=40 / score ≈ 103.486 — upper bound



$U = \eta_{78}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.652 — proved minimum



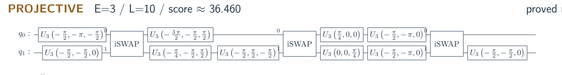
$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=11 / score ≈ 38.466 — proved minimum



PROJECTIVE E=3 / L=10 / score ≈ 36.460 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

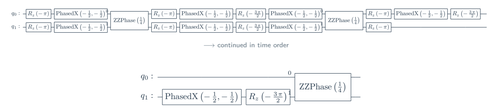
PROJECTIVE E=3 / L=10 / score ≈ 34.832 — proved minimum



$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

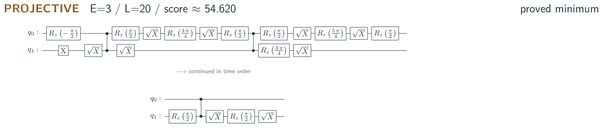
PROJECTIVE E=3 / L=18 / score ≈ 53.258 — upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

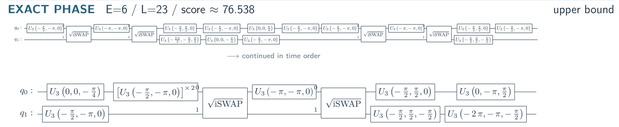
PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved minimum



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=23 / score ≈ 76.538 — upper bound



sqrt_swap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



tket_tk2

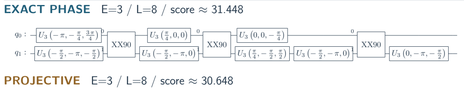
PROJECTIVE E=1 / L=0 / score ≈ 5.206 — proved minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=8 / score ≈ 31.448 — proved minimum



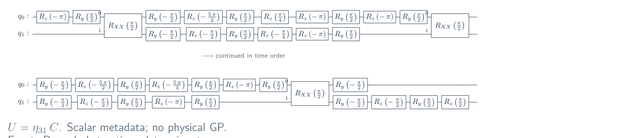
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

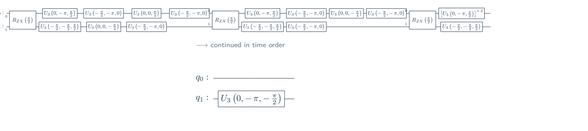
PROJECTIVE E=3 / L=33 / score ≈ 84.672 — upper bound



$U = \eta_{31}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=17 / score ≈ 51.908 — upper bound



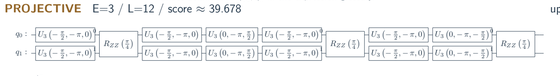
zz_u

EXACT PHASE E=3 / L=14 / score ≈ 43.690 — upper bound



Exact definitions: η_{14}, η_8 in the expression glossary.

PROJECTIVE E=3 / L=12 / score ≈ 39.678 — upper bound



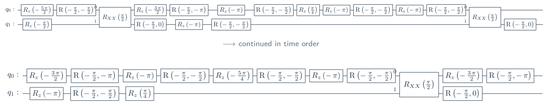
$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.

SSWAPdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

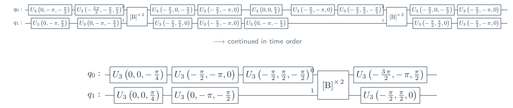
PROJECTIVE E=3 / L=30 / score ≈ 79.696 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=23 / score ≈ 77.738 — upper bound

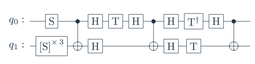


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

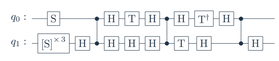
clifford_t_cx

EXACT PHASE E=3 / L=13 / score ≈ 38.000 — upper bound



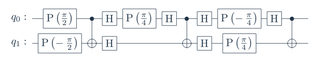
clifford_t_cz

EXACT PHASE E=3 / L=17 / score ≈ 46.000 — upper bound



cx_h_phase

EXACT PHASE E=3 / L=11 / score ≈ 35.610 — upper bound



cx_rz_ry

PROJECTIVE E=3 / L=14 / score ≈ 43.428 — proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=9 / score ≈ 33.454 — proved minimum



Exact definitions: η_{1k}, η_j in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 29.442 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

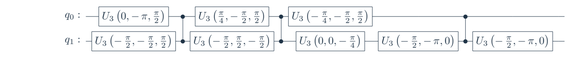
PROJECTIVE E=3 / L=11 / score ≈ 36.822 — proved minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

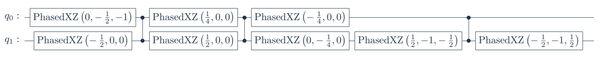
cz_u

EXACT PHASE E=3 / L=8 / score ≈ 31.848 — proved minimum



google_cz

EXACT PHASE E=3 / L=8 / score ≈ 30.648 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 9.416 — proved minimum



google_sqrt_iswap

EXACT PHASE E=6 / L=23 / score ≈ 78.338 — upper bound



PROJECTIVE E=6 / L=23 / score ≈ 77.738 — upper bound



$U = \frac{\sqrt{2}(1-i)^2}{4}\,C$. Scalar metadata; no physical GP.

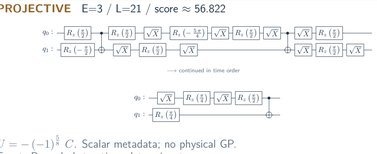
google_syc

EXACT PHASE E=18 / L=23 / score ≈ 124.538 — upper bound



ibm_cx

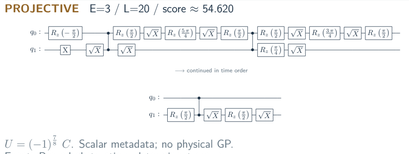
PROJECTIVE E=3 / L=21 / score ≈ 56.822 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

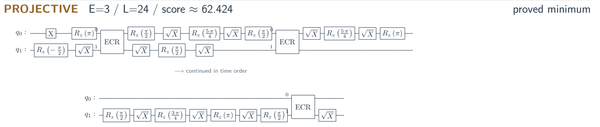
PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

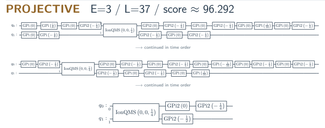
PROJECTIVE E=3 / L=24 / score ≈ 62.424 — proved minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=37 / score ≈ 96.292 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

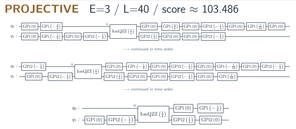
PROJECTIVE E=3 / L=37 / score ≈ 95.086 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=40 / score ≈ 103.486 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.652 — proved minimum



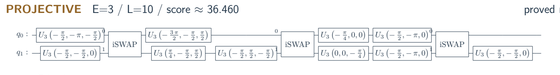
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=11 / score ≈ 38.466 — proved minimum



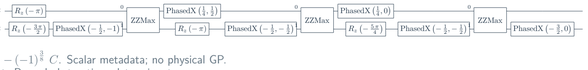
PROJECTIVE E=3 / L=10 / score ≈ 36.460 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

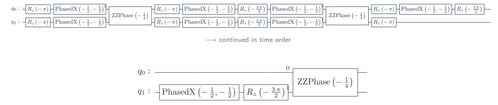
PROJECTIVE E=3 / L=10 / score ≈ 34.832 — proved minimum



$U = -(-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

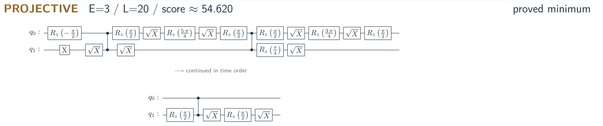
PROJECTIVE E=3 / L=18 / score ≈ 53.258 — upper bound



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

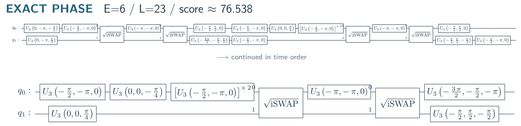
PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=23 / score ≈ 76.538 — upper bound



sqrt_swap_u

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — upper bound



tket_tk2

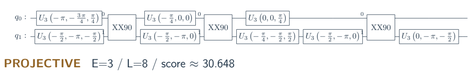
PROJECTIVE E=1 / L=0 / score ≈ 5.206 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=8 / score ≈ 31.448 — proved minimum



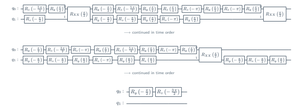
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — proved minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

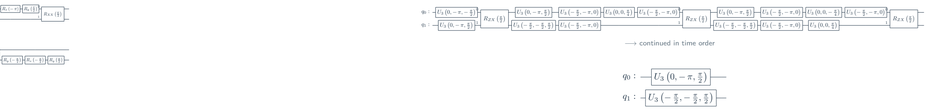
PROJECTIVE E=3 / L=35 / score ≈ 89.076 — upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

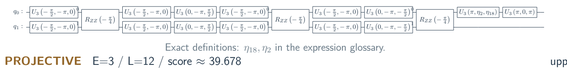
zx_u

EXACT PHASE E=3 / L=17 / score ≈ 51.908 — upper bound



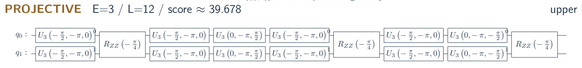
zz_u

EXACT PHASE E=3 / L=14 / score ≈ 43.690 — upper bound



Exact definitions: η_{1k}, η_2 in the expression glossary.

PROJECTIVE E=3 / L=12 / score ≈ 39.678 — upper bound



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

SWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

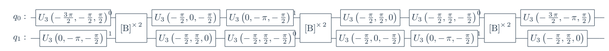
PROJECTIVE E=3 / L=15 / score ≈ 45.654 — upper bound



$U = \frac{\sqrt{2}i(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=12 / score ≈ 52.872 — upper bound



clifford_only

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



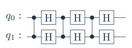
clifford_t_cx

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



clifford_t_cz

EXACT PHASE E=3 / L=6 / score ≈ 24.000 — proved minimum



cx_h_phase

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



cx_rz_ry

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



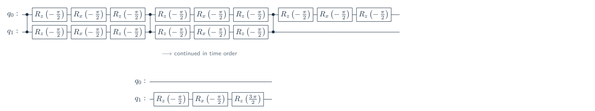
cx_u

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



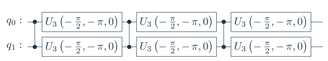
cz_rz_rx

EXACT PHASE E=3 / L=18 / score ≈ 51.636 — proved minimum



cz_u

EXACT PHASE E=3 / L=6 / score ≈ 25.236 — proved minimum



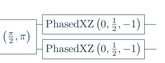
google_cz

EXACT PHASE E=3 / L=6 / score ≈ 26.436 — proved minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.616 — proved minimum



google_sqrt_iswap

EXACT PHASE E=6 / L=12 / score ≈ 52.272 — upper bound



PROJECTIVE E=6 / L=12 / score ≈ 51.672 — upper bound



$U = \frac{\sqrt{2}(1-i)^3}{4} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=12 / score ≈ 98.472 — upper bound



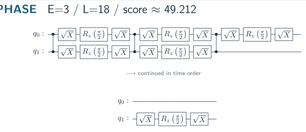
ibm_cx

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved minimum



ibm_cz

EXACT PHASE E=3 / L=18 / score ≈ 49.212 — proved minimum



ibm_ecr

EXACT PHASE E=3 / L=12 / score ≈ 36.606 — proved minimum



PROJECTIVE E=3 / L=11 / score ≈ 34.810 — proved minimum



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=3 / L=15 / score ≈ 46.248 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

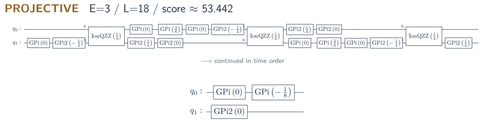
PROJECTIVE E=3 / L=15 / score ≈ 45.042 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

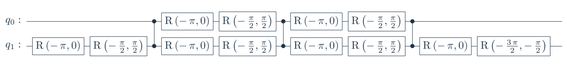
PROJECTIVE E=3 / L=18 / score ≈ 53.442 — upper bound



$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=3 / L=12 / score ≈ 38.448 — proved minimum



iswap_u

EXACT PHASE E=3 / L=5 / score ≈ 23.430 — proved minimum



PROJECTIVE E=3 / L=4 / score ≈ 20.624 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

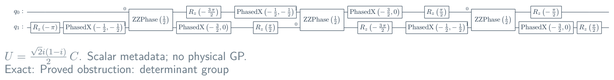
PROJECTIVE E=3 / L=10 / score ≈ 34.032 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

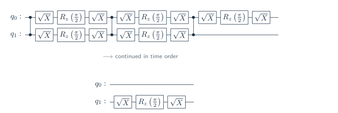
PROJECTIVE E=3 / L=13 / score ≈ 41.644 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=3 / L=18 / score ≈ 49.212 — proved minimum



sqrt_iswap_u

EXACT PHASE E=6 / L=12 / score ≈ 51.672 — upper bound



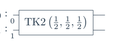
sqrt_swap_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved minimum



tket_tk2

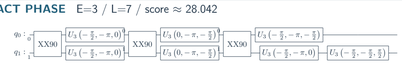
PROJECTIVE E=1 / L=0 / score ≈ 4.606 — proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=7 / score ≈ 28.042 — proved minimum



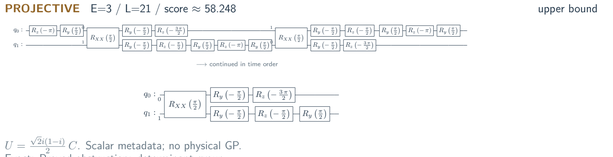
PROJECTIVE E=3 / L=6 / score ≈ 25.636 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.

xx_rz_ry

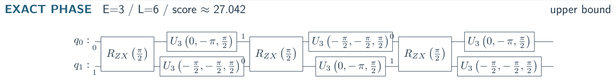
PROJECTIVE E=3 / L=21 / score ≈ 58.248 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

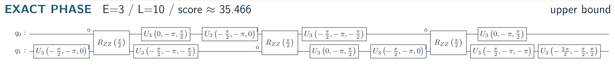
zx_u

EXACT PHASE E=3 / L=6 / score ≈ 27.042 — upper bound

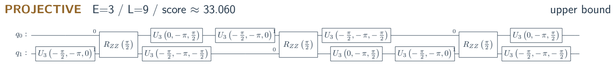


zz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.466 — upper bound



PROJECTIVE E=3 / L=9 / score ≈ 33.060 — upper bound



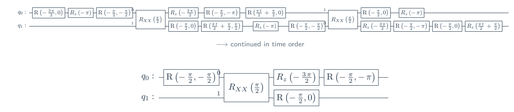
$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

SWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

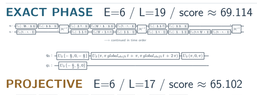
PROJECTIVE E=3 / L=20 / score ≈ 56.872 — upper bound



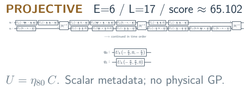
$U = \eta_{79}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 65.102 — upper bound



$U = \eta_{80}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

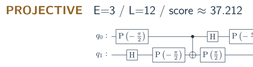
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 53.220 — upper bound



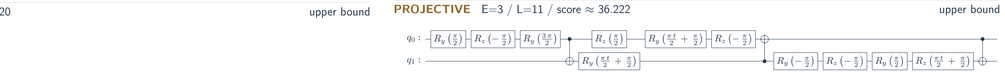
PROJECTIVE E=3 / L=12 / score ≈ 37.212 — upper bound



$U = e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

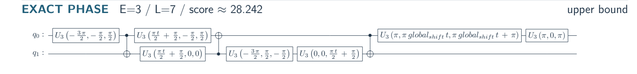
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — upper bound



$U = \eta_{81}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — upper bound



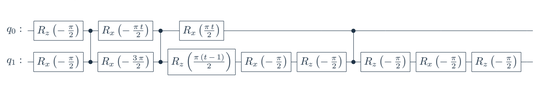
PROJECTIVE E=3 / L=5 / score ≈ 24.230 — upper bound



$U = e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

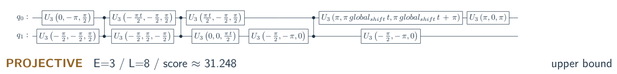
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — upper bound



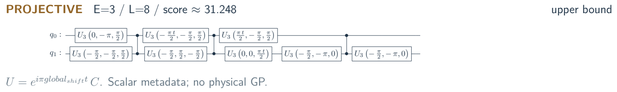
$U = \eta_{43}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 — upper bound



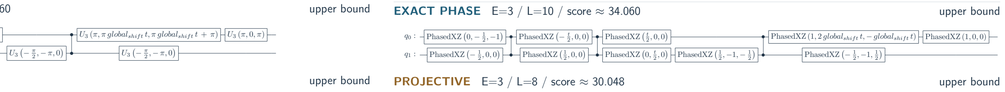
PROJECTIVE E=3 / L=8 / score ≈ 31.248 — upper bound



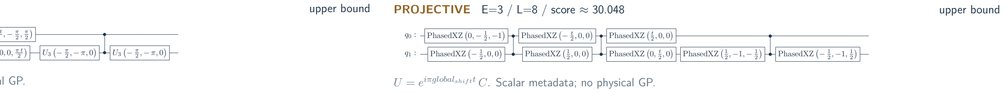
$U = e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 34.060 — upper bound



PROJECTIVE E=3 / L=8 / score ≈ 30.048 — upper bound



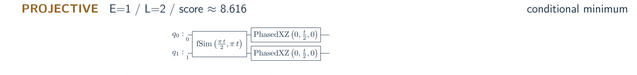
$U = e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — conditional minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.616 — conditional minimum



$U = e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=6 / L=19 / score ≈ 67.514 — upper bound



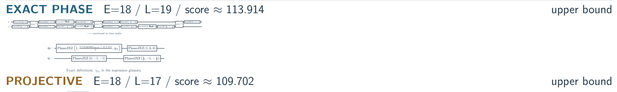
PROJECTIVE E=6 / L=17 / score ≈ 62.902 — upper bound



$U = \eta_{82}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 113.914 — upper bound



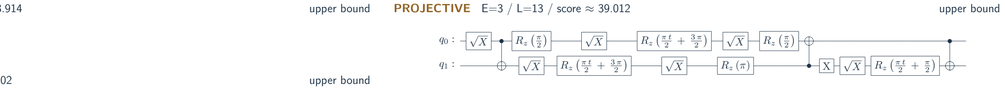
PROJECTIVE E=18 / L=17 / score ≈ 109.702 — upper bound



$U = \eta_{83}\, C$. Scalar metadata; no physical GP.

ibm_cx

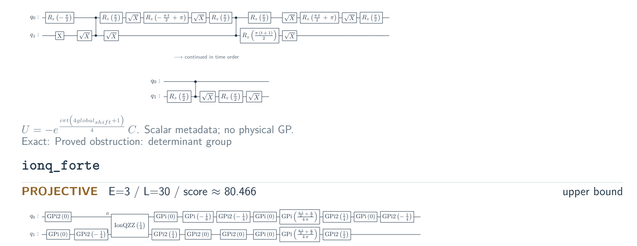
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — upper bound



$U = \eta_{81}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

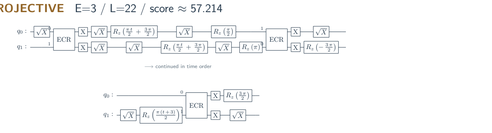
PROJECTIVE E=3 / L=20 / score ≈ 54.020 — upper bound



$U = -e^{\frac{i\pi\left(4global_{shift}t+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

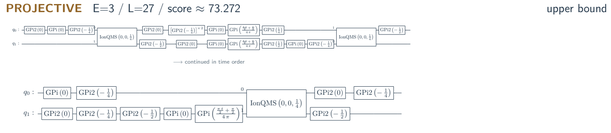
PROJECTIVE E=3 / L=22 / score ≈ 57.214 — upper bound



$U = \eta_{84}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

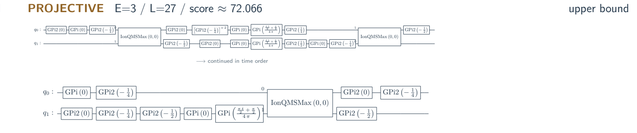
PROJECTIVE E=3 / L=27 / score ≈ 73.272 — upper bound



$U = \eta_{85}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

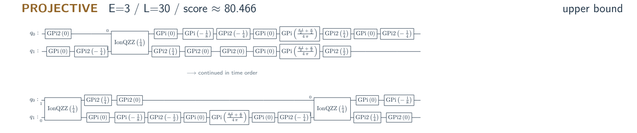
PROJECTIVE E=3 / L=27 / score ≈ 72.066 — upper bound



$U = \eta_{85}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

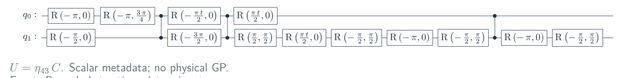
PROJECTIVE E=3 / L=30 / score ≈ 80.466 — upper bound



$U = \eta_{85}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.052 — upper bound



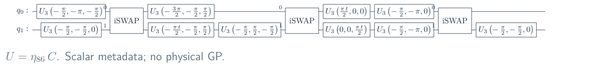
$U = \eta_{43}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.878 — upper bound



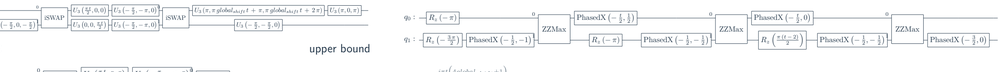
PROJECTIVE E=3 / L=10 / score ≈ 35.860 — upper bound



$U = \eta_{86}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=10 / score ≈ 34.232 — upper bound



$U = -ie^{\frac{i\pi\left(4global_{shift}t+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

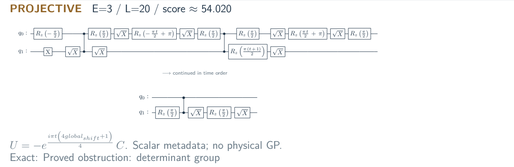
PROJECTIVE E=3 / L=18 / score ≈ 52.658 — upper bound



$U = e^{\frac{i\pi\left(4global_{shift}t+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=20 / score ≈ 54.020 — upper bound



$U = -e^{\frac{i\pi\left(4global_{shift}t+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 67.914 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = \eta_{80}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = -\sqrt[4]{-1}e^{i\pi global_{shift}t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.606 — conditional minimum



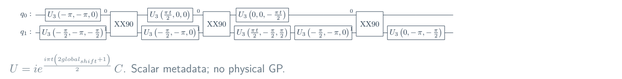
$U = e^{\frac{i\pi\left(4global_{shift}t+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.860 — upper bound



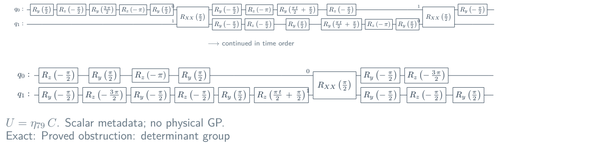
PROJECTIVE E=3 / L=8 / score ≈ 30.048 — upper bound



$U = ie^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

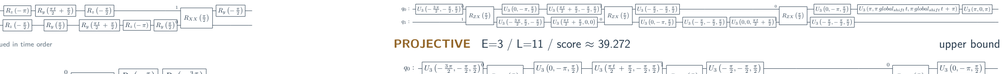
PROJECTIVE E=3 / L=31 / score ≈ 80.068 — upper bound



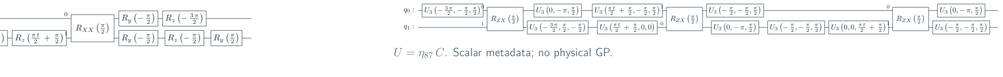
$U = \eta_{79}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — upper bound



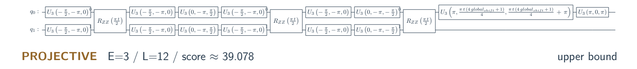
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — upper bound



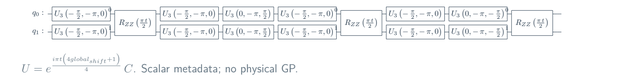
$U = \eta_{87}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=14 / score ≈ 43.890 — upper bound



PROJECTIVE E=3 / L=12 / score ≈ 39.078 — upper bound



$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.

SX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

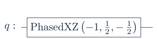
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

Sx

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

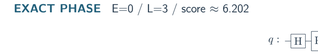
PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum

cx_rz_ry

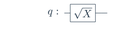
PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

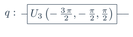
EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

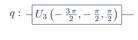


google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

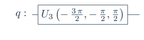
ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

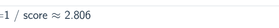
PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

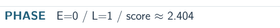


$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

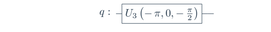


PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

Sx_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

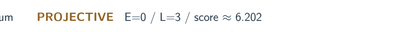
$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

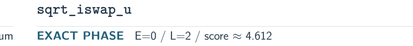
$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

SX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

Sx_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

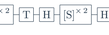
EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

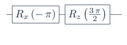
EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

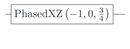
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

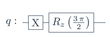
EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum



ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

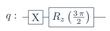
tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

Sx_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



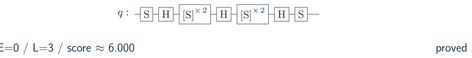
PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

SX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

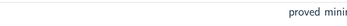
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

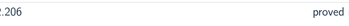


$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

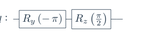
xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



Sx_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved minimum

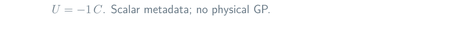


PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

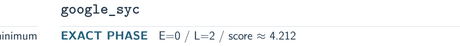
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



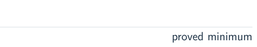
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

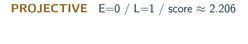
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.204 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

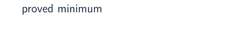
EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

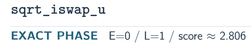
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

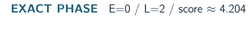


PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

Sx_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

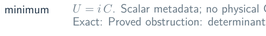
PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

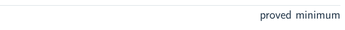
PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

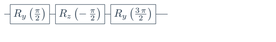
PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

SX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

Sx_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



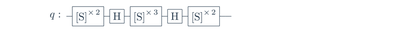
PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



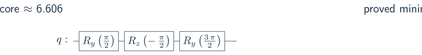
$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

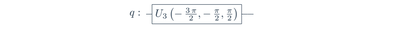


PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{-i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{-i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

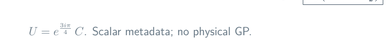
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{-i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sx_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

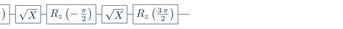
PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

Sxdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

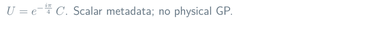


PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



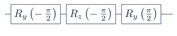
PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

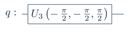
cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

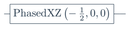


google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



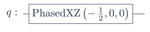
$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum



iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



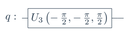
zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

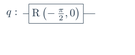


Sxdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



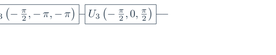
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

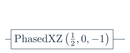
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



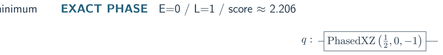
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

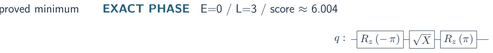
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum



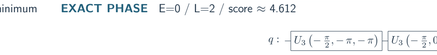
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

Sxdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

Sxdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

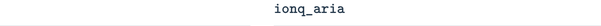
iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

SxY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

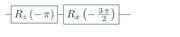


PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum

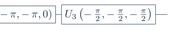
PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

SxY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



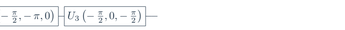
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

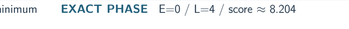


PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

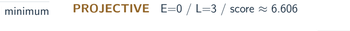
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum



ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

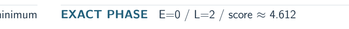
PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

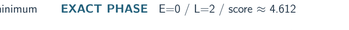
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

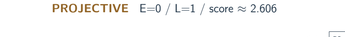


sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



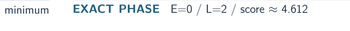
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



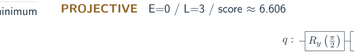
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

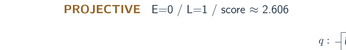
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

SxY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved minimum



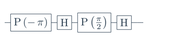
clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum



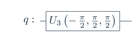
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = iC$. Scalar metadata; no physical GP.

SxY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

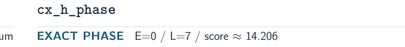


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

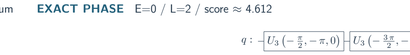
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

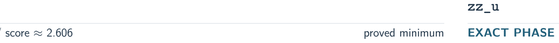
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

SxZ

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

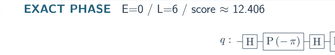
EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



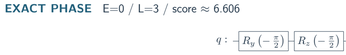
$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



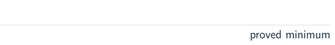
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum



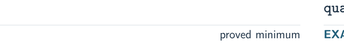
PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

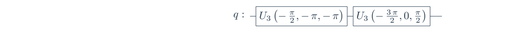
$U = -i\,C$. Scalar metadata; no physical GP.

SxZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

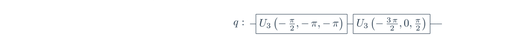
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

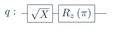
$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

SxZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

SxZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

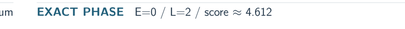
EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

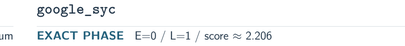
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum



ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum

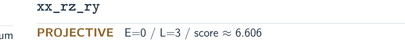


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

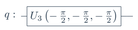
PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

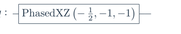


google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



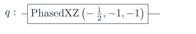
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

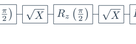
EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 proved minimum

SY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



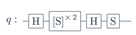
clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



xx_rz_ry

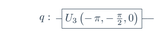
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



Sy

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

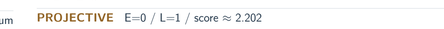
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

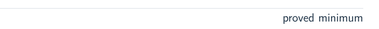


PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

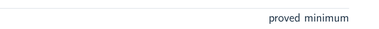
google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



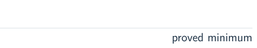
PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.204 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

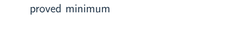
EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

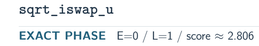
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

SY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum



clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

SY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



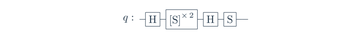
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

SY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

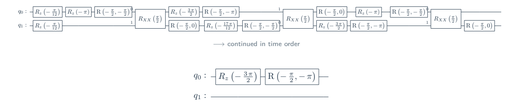
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SYC

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

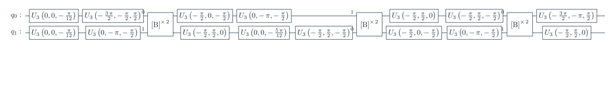
PROJECTIVE E=3 / L=17 / score ≈ 51.809 — upper bound



$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=6 / L=15 / score ≈ 61.041 — upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

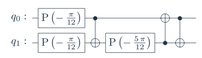
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

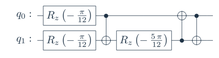
cx_h_phase

EXACT PHASE E=3 / L=3 / score ≈ 20.157 — proved minimum



cx_rz_ry

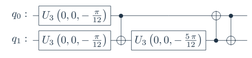
PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved minimum



$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=3 / score ≈ 20.169 — proved minimum



cz_rz_rx

PROJECTIVE E=3 / L=12 / score ≈ 39.975 — proved minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=3 / L=9 / score ≈ 33.405 — proved minimum



google_cz

EXACT PHASE E=3 / L=9 / score ≈ 34.605 — proved minimum



google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.721 — proved minimum

google_sqrt_iswap

EXACT PHASE E=6 / L=15 / score ≈ 60.441 — upper bound

PROJECTIVE E=6 / L=15 / score ≈ 59.841 — upper bound

$U = \frac{\sqrt{2}(1-i)}{4}^3\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum

ibm_cx

PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved minimum

$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved minimum

$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=15 / score ≈ 44.763 — proved minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=25 / score ≈ 70.219 — upper bound

$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=21 / score ≈ 60.405 — upper bound

$U = \eta_{69}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=28 / score ≈ 77.413 — upper bound

$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=19 / score ≈ 56.627 — proved minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=14 / score ≈ 45.835 — proved minimum

Exact definitions: η_{90}, η_{91} in the expression glossary.

PROJECTIVE E=3 / L=12 / score ≈ 41.823 — proved minimum

$U = \eta_{92}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=12 / score ≈ 40.187 — proved minimum

$U = -(-1)^{\frac{11}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=3 / L=16 / score ≈ 49.801 — upper bound

$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved minimum

$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=15 / score ≈ 59.841 — upper bound

sqrt_swap_u

EXACT PHASE E=6 / L=15 / score ≈ 60.641 — upper bound

PROJECTIVE E=6 / L=15 / score ≈ 59.841 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 10.561 — proved minimum

$U = e^{-\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 40.023 — proved minimum

Exact definitions: η_{93}, η_{94} in the expression glossary.

PROJECTIVE E=3 / L=10 / score ≈ 36.011 — proved minimum

$U = -(-1)^{\frac{19}{24}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=23 / score ≈ 64.403 — upper bound

$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=9 / score ≈ 35.211 — upper bound

zz_u

EXACT PHASE E=3 / L=13 / score ≈ 43.635 — upper bound

PROJECTIVE E=3 / L=12 / score ≈ 41.229 — upper bound

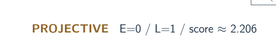
$U = \frac{\sqrt{2}(1+i)}{4}^3\, C$. Scalar metadata; no physical GP.

Sydg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

T

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 7.212 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



tket_tk2

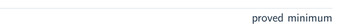
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

Tdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

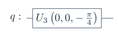
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



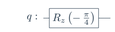
cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

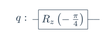
cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



cz_rz_rx

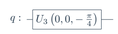
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

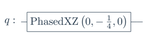
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



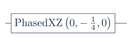
google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



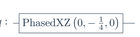
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



ibm_cx

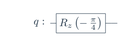
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

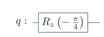
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

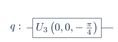
PROJECTIVE E=0 / L=3 / score ≈ 7.212 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

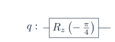
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



quantinuum_zzmax

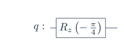
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

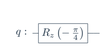
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

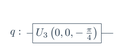
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

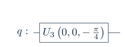
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



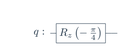
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



tket_tk2

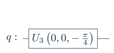
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

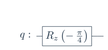
xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

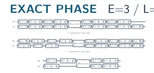


TK2 (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — upper bound



PROJECTIVE E=3 / L=20 / score ≈ 56.872 — upper bound



$U = \eta_{15}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=6 / L=19 / score ≈ 69.514 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 65.102 — upper bound



$U = \eta_{95}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 53.020 — upper bound



PROJECTIVE E=3 / L=12 / score ≈ 36.612 — upper bound

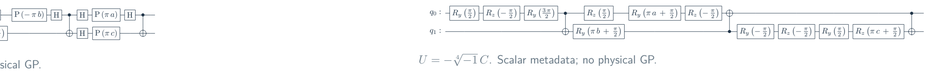
$U = e^{-\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — upper bound



PROJECTIVE E=3 / L=11 / score ≈ 36.222 — upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — upper bound



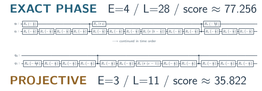
PROJECTIVE E=3 / L=5 / score ≈ 24.230 — upper bound



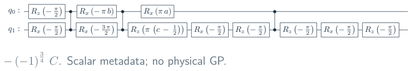
$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — upper bound



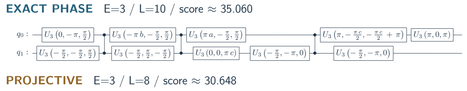
PROJECTIVE E=3 / L=11 / score ≈ 35.822 — upper bound



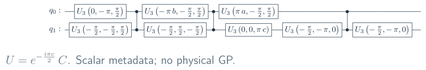
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.060 — upper bound



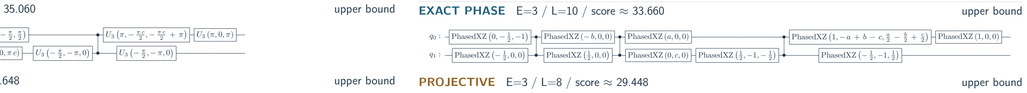
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — upper bound



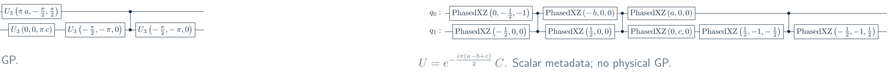
$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 33.660 — upper bound



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — upper bound



$U = e^{\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.644 — upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.432 — upper bound



$U = e^{-\frac{i\phi c}{2}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=6 / L=19 / score ≈ 67.514 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 62.902 — upper bound

$U = \eta_{98}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 113.914 — upper bound

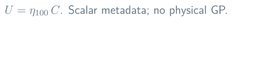


PROJECTIVE E=18 / L=17 / score ≈ 109.702 — upper bound

$U = e^{\frac{i\pi(-a-b-c-1)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — upper bound



PROJECTIVE E=3 / L=13 / score ≈ 39.012 — upper bound

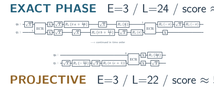
$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — upper bound

ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — upper bound



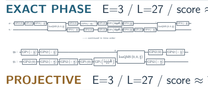
PROJECTIVE E=3 / L=22 / score ≈ 57.214 — upper bound



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 73.472 — upper bound



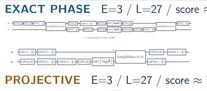
PROJECTIVE E=3 / L=27 / score ≈ 73.272 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 72.266 — upper bound



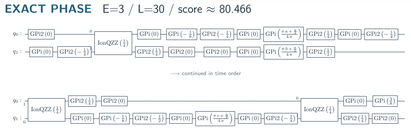
PROJECTIVE E=3 / L=27 / score ≈ 72.066 — upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

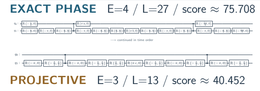
ionq_forte

EXACT PHASE E=3 / L=30 / score ≈ 80.466 — upper bound

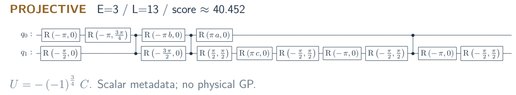


iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — upper bound



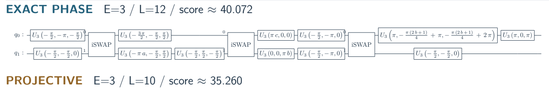
PROJECTIVE E=3 / L=13 / score ≈ 40.452 — upper bound



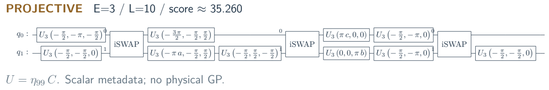
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=12 / score ≈ 40.072 — upper bound



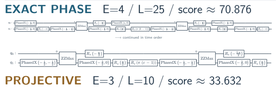
PROJECTIVE E=3 / L=10 / score ≈ 35.260 — upper bound



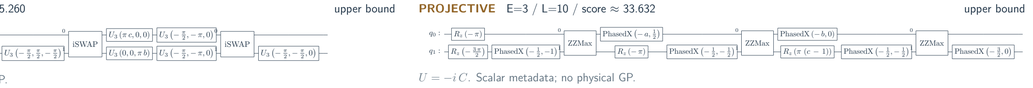
$U = \eta_{99}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 70.876 — upper bound



PROJECTIVE E=3 / L=10 / score ≈ 33.632 — upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

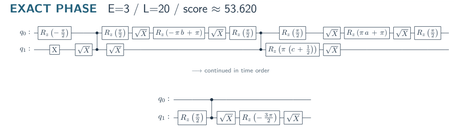
quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 52.058 — upper bound



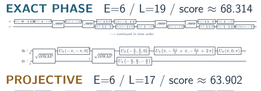
rigetti_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — upper bound



sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.314 — upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



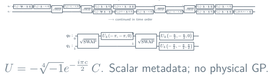
$U = \eta_{95}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — upper bound



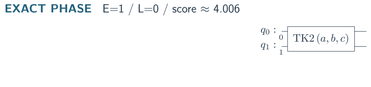
PROJECTIVE E=6 / L=17 / score ≈ 63.902 — upper bound



$U = -\sqrt[4]{-1}e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

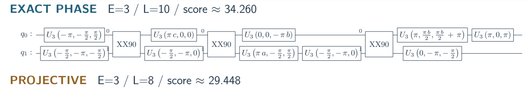
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — upper bound

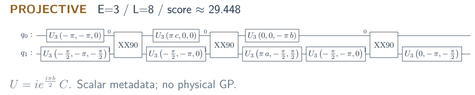


xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.260 — upper bound



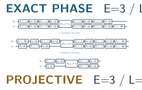
PROJECTIVE E=3 / L=8 / score ≈ 29.448 — upper bound



$U = ie^{\frac{i\pi\lambda}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=41 / score ≈ 98.888 — upper bound



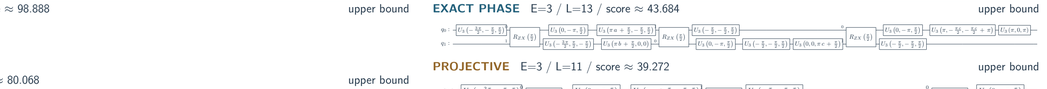
PROJECTIVE E=3 / L=31 / score ≈ 80.068 — upper bound



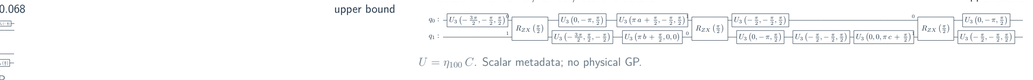
$U = \eta_{15}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 — upper bound



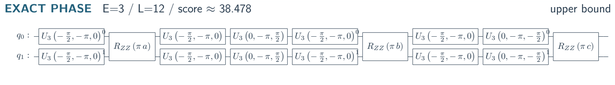
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — upper bound



$U = \eta_{100}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — upper bound

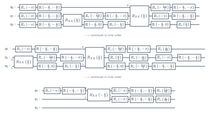


Toffoli

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

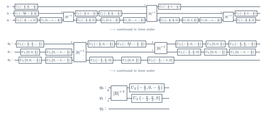
PROJECTIVE E=5 / L=40 / score ≈ 112.926 — upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — upper bound



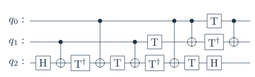
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

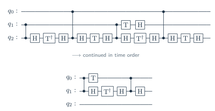
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — proved minimum



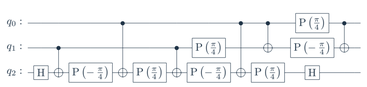
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved minimum



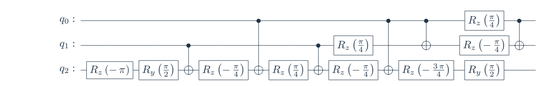
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — proved minimum



cx_rz_ry

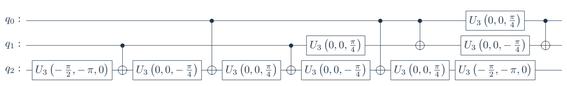
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

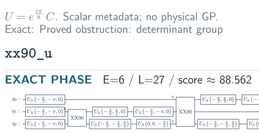
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 45.254 — proved minimum



cz_rz_rx

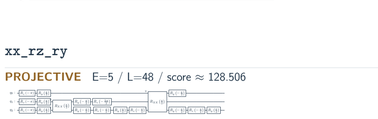
PROJECTIVE E=6 / L=40 / score ≈ 113.280 — proved minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

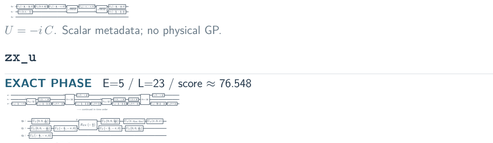
cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.726 — proved minimum



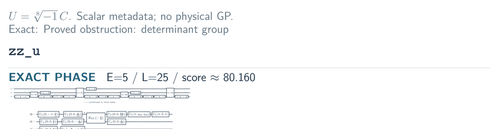
google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 — proved minimum



google_fsim

EXACT PHASE E=5 / L=21 / score ≈ 72.346 — upper bound

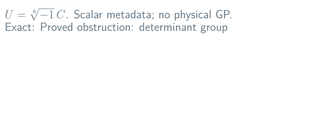


google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — upper bound



ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 — proved minimum



$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=35 / score ≈ 98.228 — proved minimum

$U = e^{-\frac{3\pi i}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 134.932 — upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 132.520 — upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 — upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — proved minimum

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=33 / score ≈ 118.198 — upper bound

PROJECTIVE E=10 / L=31 / score ≈ 114.186 — upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 103.498 — proved minimum

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=5 / L=29 / score ≈ 89.288 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — upper bound

PROJECTIVE E=12 / L=33 / score ≈ 124.598 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=5 / L=32 / score ≈ 95.322 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — proved minimum

xx_rz_ry

PROJECTIVE E=5 / L=48 / score ≈ 128.506 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=23 / score ≈ 76.548 — upper bound

PROJECTIVE E=5 / L=21 / score ≈ 72.536 — upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=25 / score ≈ 80.160 — upper bound

PROJECTIVE E=5 / L=23 / score ≈ 76.148 — upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

U $(\theta, \phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

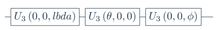
PROJECTIVE E=0 / L=3 / score ≈ 6.208 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



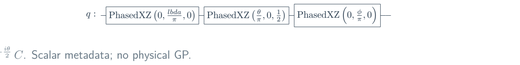
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

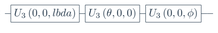
PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



quantinuum_zzmax

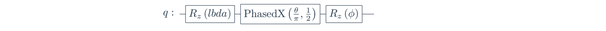
PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

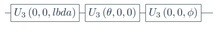
PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

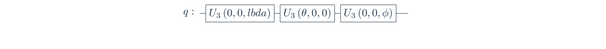
sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

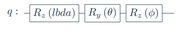
xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



xx_rz_ry

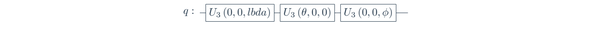
PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



U1 (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved minimum

ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

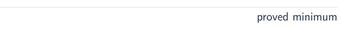
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

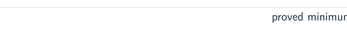
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

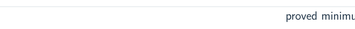
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



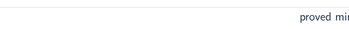
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

U2 $(\phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

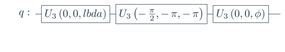
PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

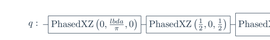
cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved minimum



cx_rz_ry

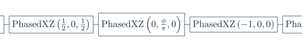
PROJECTIVE E=0 / L=3 / score ≈ 6.206 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 16.028 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.206 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

U3 $(\theta, \phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.208 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_cz

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved minimum



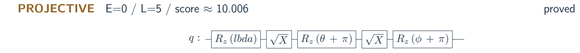
PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved minimum



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

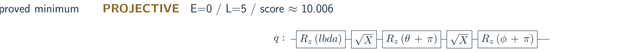
PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum



$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

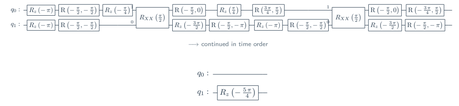
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

UC

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

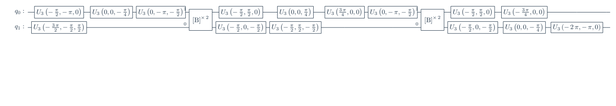
PROJECTIVE E=2 / L=17 / score ≈ 47.256 — upper bound



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 51.490 — upper bound

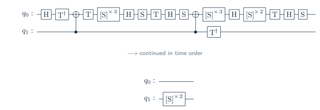


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=23 / score ≈ 54.000 — proved minimum



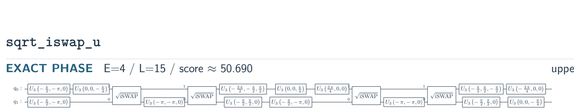
clifford_t_cz

EXACT PHASE E=2 / L=27 / score ≈ 62.000 — proved minimum



cx_h_phase

EXACT PHASE E=2 / L=13 / score ≈ 36.616 — proved minimum



cx_rz_ry

PROJECTIVE E=2 / L=7 / score ≈ 24.214 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

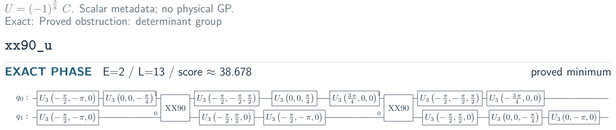
cx_u

EXACT PHASE E=2 / L=7 / score ≈ 24.242 — proved minimum



cz_rz_rx

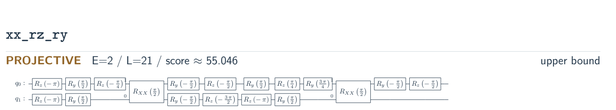
PROJECTIVE E=2 / L=16 / score ≈ 44.032 — proved minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

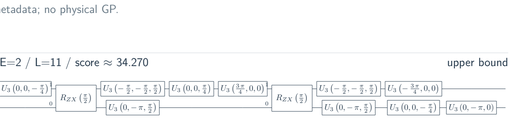
cz_u

EXACT PHASE E=2 / L=11 / score ≈ 33.066 — proved minimum



google_cz

EXACT PHASE E=2 / L=12 / score ≈ 36.872 — proved minimum



PROJECTIVE E=2 / L=11 / score ≈ 34.466 — upper bound

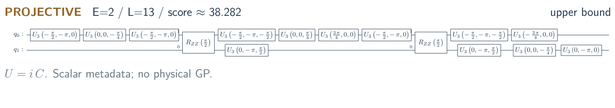
$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 36.880 — upper bound



PROJECTIVE E=2 / L=11 / score ≈ 34.474 — upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=15 / score ≈ 51.890 — upper bound

PROJECTIVE E=4 / L=15 / score ≈ 51.290 — upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 84.896 — upper bound



PROJECTIVE E=12 / L=15 / score ≈ 82.490 — upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=13 / score ≈ 36.216 — proved minimum

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=21 / score ≈ 52.620 — proved minimum

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=28 / score ≈ 72.068 — upper bound

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=28 / score ≈ 71.264 — upper bound

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=30 / score ≈ 76.864 — upper bound

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=23 / score ≈ 61.092 — proved minimum

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 51.090 — upper bound

quantinuum_zzmax

PROJECTIVE E=2 / L=16 / score ≈ 44.846 — proved minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=16 / score ≈ 45.250 — upper bound

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 50.690 — upper bound

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 51.090 — upper bound

PROJECTIVE E=4 / L=15 / score ≈ 50.690 — upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=17 / score ≈ 47.264 — upper bound

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=13 / score ≈ 38.678 — proved minimum

xx_rz_ry

PROJECTIVE E=2 / L=21 / score ≈ 55.046 — upper bound

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 34.270 — upper bound

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 40.288 — upper bound

PROJECTIVE E=2 / L=13 / score ≈ 38.282 — upper bound

$U = i\, C$. Scalar metadata; no physical GP.

UCRX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=22 / score ≈ 60.478 — upper bound



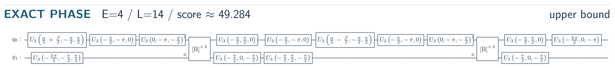
PROJECTIVE E=2 / L=20 / score ≈ 52.468 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=14 / score ≈ 49.284 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 32.812 — upper bound



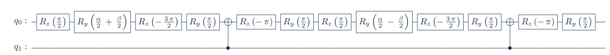
PROJECTIVE E=2 / L=4 / score ≈ 16.404 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

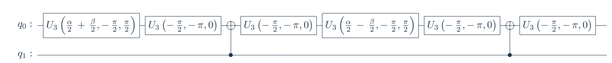
cx_rz_ry

EXACT PHASE E=2 / L=12 / score ≈ 34.024 — upper bound



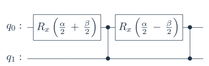
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — upper bound



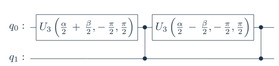
cz_rz_rx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



cz_u

EXACT PHASE E=2 / L=2 / score ≈ 13.212 — upper bound



google_cz

EXACT PHASE E=2 / L=4 / score ≈ 17.945 — upper bound



PROJECTIVE E=2 / L=2 / score ≈ 13.073 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 37.201 — upper bound



PROJECTIVE E=2 / L=10 / score ≈ 32.329 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 54.017 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 49.145 — upper bound



$U = -ie^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 85.217 — upper bound



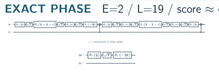
PROJECTIVE E=12 / L=14 / score ≈ 80.345 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.622 — upper bound



PROJECTIVE E=2 / L=19 / score ≈ 47.420 — upper bound



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound



PROJECTIVE E=2 / L=10 / score ≈ 29.212 — upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=27 / score ≈ 64.026 — upper bound



PROJECTIVE E=2 / L=27 / score ≈ 63.824 — upper bound



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=26 / score ≈ 68.325 — upper bound



PROJECTIVE E=2 / L=26 / score ≈ 67.925 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=26 / score ≈ 67.521 — upper bound



PROJECTIVE E=2 / L=26 / score ≈ 67.121 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=28 / score ≈ 73.121 — upper bound



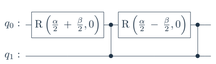
PROJECTIVE E=2 / L=28 / score ≈ 72.721 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=2 / score ≈ 12.408 — upper bound



iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 48.884 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=39 / score ≈ 106.979 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 52.721 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=22 / score ≈ 61.131 — upper bound



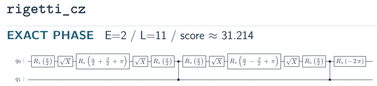
PROJECTIVE E=2 / L=20 / score ≈ 53.125 — upper bound



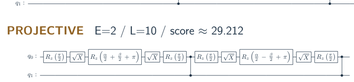
$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound



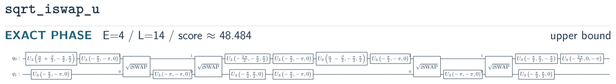
PROJECTIVE E=2 / L=10 / score ≈ 29.212 — upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

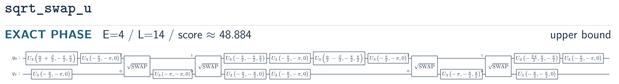
sqrt_iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 48.484 — upper bound

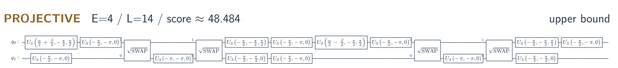


sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 48.884 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 48.484 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=40 / score ≈ 110.021 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.137 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 36.472 — upper bound



xx_rz_ry

EXACT PHASE E=3 / L=30 / score ≈ 76.866 — upper bound



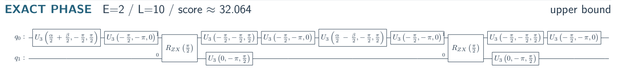
PROJECTIVE E=2 / L=26 / score ≈ 64.856 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

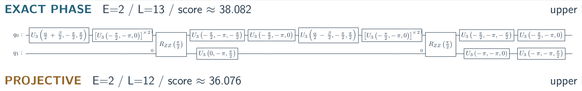
zx_u

EXACT PHASE E=2 / L=10 / score ≈ 32.064 — upper bound

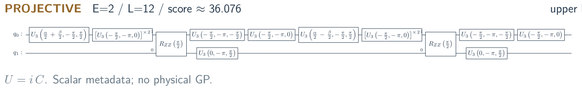


zz_u

EXACT PHASE E=2 / L=13 / score ≈ 38.082 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 36.076 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

UCRY

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=14 / score ≈ 43.254 — upper bound



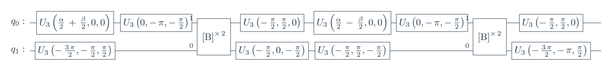
PROJECTIVE E=2 / L=12 / score ≈ 35.244 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=10 / score ≈ 39.660 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=18 / score ≈ 45.620 — upper bound



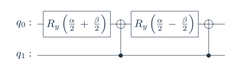
PROJECTIVE E=2 / L=10 / score ≈ 29.212 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

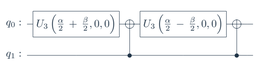
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound



cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — upper bound



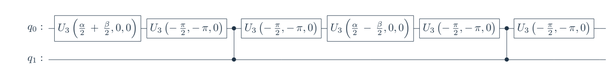
cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.022 — upper bound



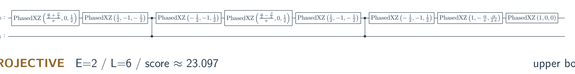
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — upper bound



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.969 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 23.097 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=8 / score ≈ 27.977 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 23.105 — upper bound



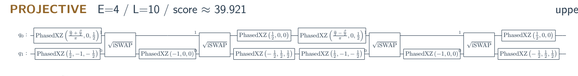
$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=12 / score ≈ 44.793 — upper bound



PROJECTIVE E=4 / L=10 / score ≈ 39.921 — upper bound



$U = -ie^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 75.993 — upper bound



PROJECTIVE E=12 / L=10 / score ≈ 71.121 — upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 24.408 — upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — upper bound



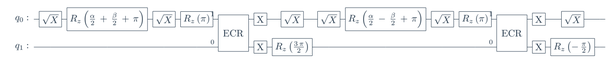
PROJECTIVE E=2 / L=18 / score ≈ 45.616 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 40.812 — upper bound



ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 49.509 — upper bound



PROJECTIVE E=2 / L=18 / score ≈ 49.109 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 48.705 — upper bound



PROJECTIVE E=2 / L=18 / score ≈ 48.305 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=20 / score ≈ 54.305 — upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.905 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

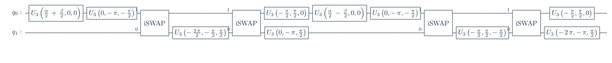
iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.440 — upper bound



iswap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.755 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 35.497 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=14 / score ≈ 43.907 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 35.901 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — upper bound



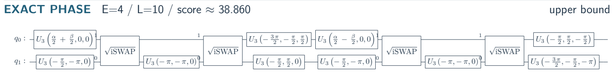
PROJECTIVE E=2 / L=18 / score ≈ 45.616 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=10 / score ≈ 38.860 — upper bound

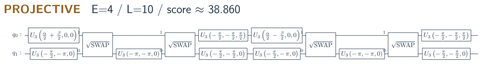


sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — upper bound



PROJECTIVE E=4 / L=10 / score ≈ 38.860 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=32 / score ≈ 92.797 — upper bound



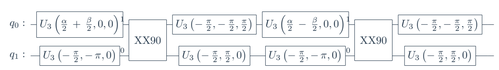
PROJECTIVE E=2 / L=12 / score ≈ 35.913 — upper bound



$U = -iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — upper bound

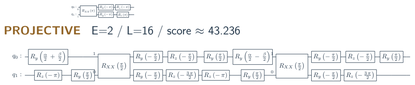


xx_rz_ry

EXACT PHASE E=3 / L=20 / score ≈ 55.246 — upper bound



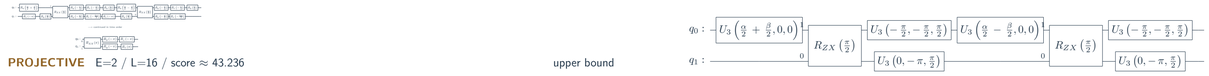
PROJECTIVE E=2 / L=16 / score ≈ 43.236 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

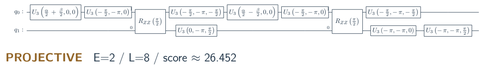
zx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.440 — upper bound

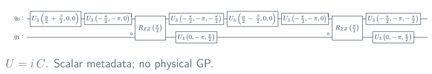


zz_u

EXACT PHASE E=2 / L=9 / score ≈ 28.458 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 26.452 — upper bound



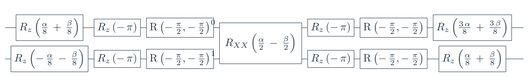
$U = i\,C$. Scalar metadata; no physical GP.

UCRZ

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 32.234 — conditional minimum



berkeley_b

EXACT PHASE E=4 / L=12 / score ≈ 44.072 — upper bound



PROJECTIVE E=4 / L=10 / score ≈ 39.660 — upper bound



$U = -e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

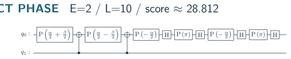
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

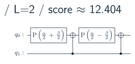
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — upper bound



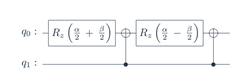
PROJECTIVE E=2 / L=2 / score ≈ 12.404 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound

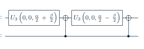


cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound



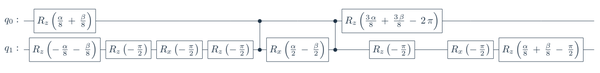
PROJECTIVE E=2 / L=2 / score ≈ 12.412 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 31.620 — upper bound



cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 21.236 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.569 — upper bound



PROJECTIVE E=2 / L=6 / score ≈ 22.697 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=8 / score ≈ 25.694 — conditional minimum

PROJECTIVE E=1 / L=6 / score ≈ 20.822 — conditional minimum

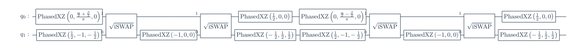
$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=12 / score ≈ 44.393 — upper bound



PROJECTIVE E=4 / L=10 / score ≈ 39.521 — upper bound



$U = -ie^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 75.593 — upper bound



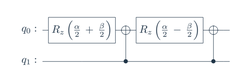
PROJECTIVE E=12 / L=10 / score ≈ 70.721 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — upper bound

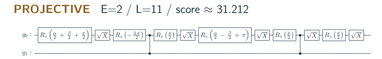


ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — upper bound



PROJECTIVE E=2 / L=11 / score ≈ 31.212 — upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 26.612 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=14 / score ≈ 41.101 — upper bound



PROJECTIVE E=2 / L=14 / score ≈ 40.701 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=14 / score ≈ 40.297 — upper bound



PROJECTIVE E=2 / L=14 / score ≈ 39.897 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=16 / score ≈ 45.897 — upper bound



PROJECTIVE E=2 / L=16 / score ≈ 45.497 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.656 — upper bound

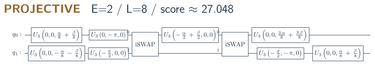


iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.860 — conditional minimum



PROJECTIVE E=2 / L=8 / score ≈ 27.048 — conditional minimum



$U = e^{-\frac{i(\alpha+\beta)}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 82.210 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 27.952 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 14.940 — conditional minimum

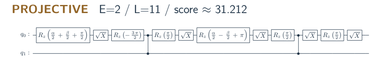


rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — upper bound



PROJECTIVE E=2 / L=11 / score ≈ 31.212 — upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.272 — upper bound



PROJECTIVE E=4 / L=10 / score ≈ 38.860 — upper bound



$U = -e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — upper bound



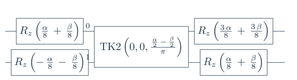
PROJECTIVE E=4 / L=10 / score ≈ 38.860 — upper bound



$U = -ie^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

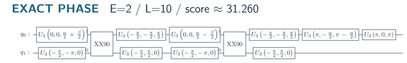
tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 14.944 — conditional minimum



xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — upper bound



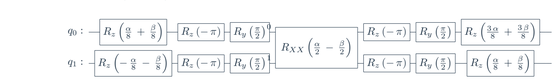
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — upper bound



$U = e^{-\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 31.426 — conditional minimum



zx_u

EXACT PHASE E=1 / L=8 / score ≈ 23.850 — conditional minimum



PROJECTIVE E=1 / L=6 / score ≈ 19.038 — conditional minimum



$U = e^{-\frac{i(\alpha+\beta)}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 19.438 — conditional minimum

PROJECTIVE E=1 / L=4 / score ≈ 14.626 — conditional minimum

$U = e^{-\frac{i(\alpha+\beta)}{4}}\,C$. Scalar metadata; no physical GP.

Unitary

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.904 — proved minimum



PROJECTIVE E=2 / L=16 / score ≈ 43.858 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — upper bound

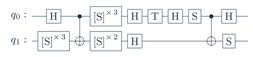


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

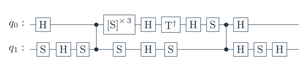
PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

clifford_t_cz

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

cx_h_phase

EXACT PHASE E=2 / L=19 / score ≈ 47.220 — proved minimum



PROJECTIVE E=2 / L=11 / score ≈ 31.212 — proved minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

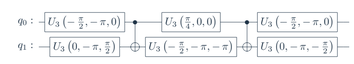
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.416 — proved minimum



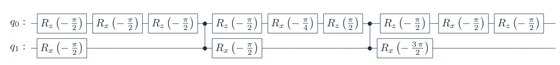
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.436 — proved minimum



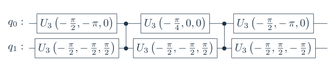
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.624 — proved minimum



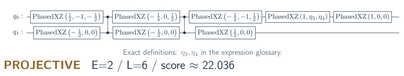
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved minimum



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.709 — proved minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved minimum



$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 13.220 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 52.557 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 47.884 — upper bound



$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.757 — upper bound



Exact definitions: η_3, η_4 in the expression glossary.

PROJECTIVE E=12 / L=14 / score ≈ 79.084 — upper bound



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.008 — upper bound



PROJECTIVE E=2 / L=9 / score ≈ 26.402 — upper bound

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved minimum



PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.220 — proved minimum



PROJECTIVE E=2 / L=14 / score ≈ 37.816 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.646 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 46.246 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.842 — upper bound



PROJECTIVE E=2 / L=17 / score ≈ 45.442 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.442 — upper bound



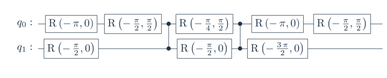
PROJECTIVE E=2 / L=19 / score ≈ 51.042 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

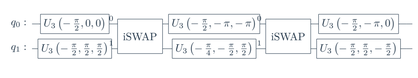
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.032 — proved minimum



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved minimum

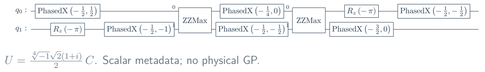


quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.286 — upper bound



PROJECTIVE E=2 / L=8 / score ≈ 26.028 — proved minimum



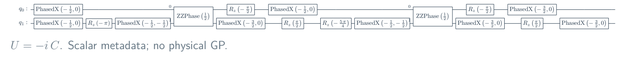
$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.656 — proved minimum



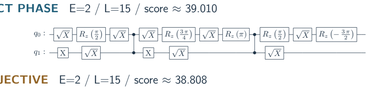
PROJECTIVE E=2 / L=15 / score ≈ 41.852 — upper bound



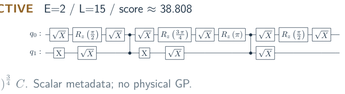
$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved minimum



PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved minimum



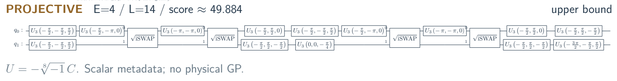
$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — upper bound



PROJECTIVE E=4 / L=14 / score ≈ 49.884 — upper bound



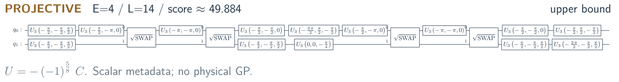
$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — upper bound



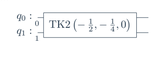
PROJECTIVE E=4 / L=14 / score ≈ 49.884 — upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.606 — proved minimum

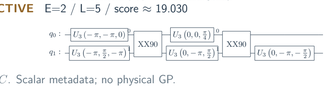


xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — proved minimum



PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved minimum



$U = \eta_{13}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.672 — proved minimum



PROJECTIVE E=2 / L=30 / score ≈ 74.264 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 37.476 — upper bound



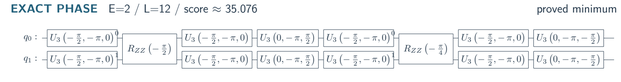
Exact definitions: η_{14}, η_k in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 33.464 — upper bound

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.076 — proved minimum



WalshHadamard

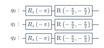
3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.228 — upper bound



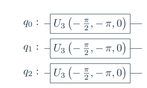
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — upper bound



clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — upper bound



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — upper bound



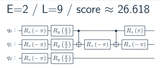
cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — upper bound



cx_rz_ry

EXACT PHASE E=2 / L=9 / score ≈ 26.618 — upper bound



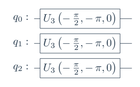
PROJECTIVE E=0 / L=6 / score ≈ 12.612 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound

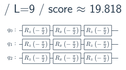


cz_rz_rx

EXACT PHASE E=2 / L=19 / score ≈ 49.438 — upper bound



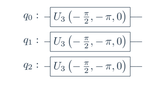
PROJECTIVE E=0 / L=9 / score ≈ 19.818 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

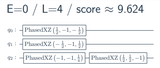
cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound

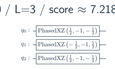


google_cz

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — upper bound



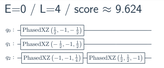
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — upper bound



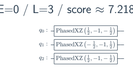
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — upper bound



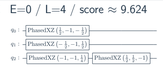
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — upper bound



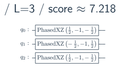
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — upper bound



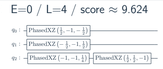
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — upper bound



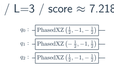
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — upper bound



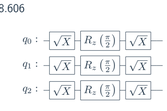
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

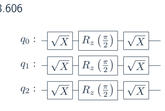
ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — upper bound



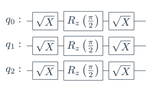
ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — upper bound



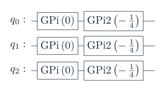
ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — upper bound



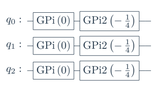
ionq_aria

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — upper bound



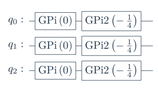
ionq_aria_full

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — upper bound



ionq_forte

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — upper bound

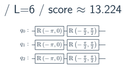


iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 51.480 — upper bound



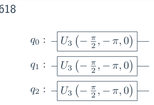
PROJECTIVE E=0 / L=6 / score ≈ 13.224 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound

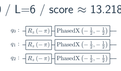


quantinuum_zzmax

EXACT PHASE E=3 / L=25 / score ≈ 67.476 — upper bound



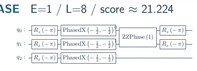
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — upper bound



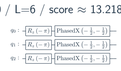
$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.224 — upper bound



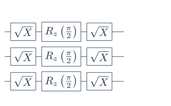
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

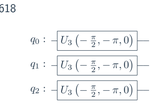
rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — upper bound



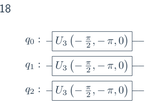
sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound



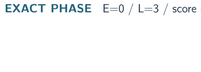
sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound

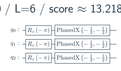


tket_tk2

EXACT PHASE E=3 / L=26 / score ≈ 70.102 — upper bound



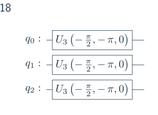
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound

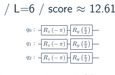


xx_rz_ry

EXACT PHASE E=1 / L=10 / score ≈ 24.622 — upper bound



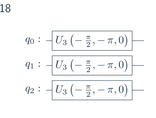
PROJECTIVE E=0 / L=6 / score ≈ 12.612 — upper bound



$U = i\,C$. Scalar metadata; no physical GP.

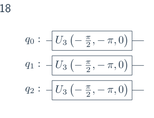
zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound



zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — upper bound



X

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



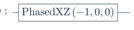
google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



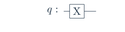
google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved minimum



sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



X_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

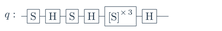


PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved minimum



$U = iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum



$U = iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

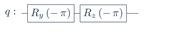
xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = iC$. Scalar metadata; no physical GP.

X_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



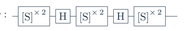
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



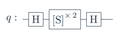
$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.006 — proved minimum



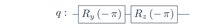
PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

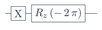
EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

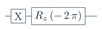
PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

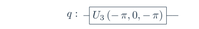
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum



X_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum



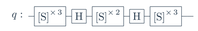
PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

X_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

XPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.010 proved minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved minimum

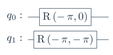
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

XX

2 qubits / 32 preserved native bases / alphabetical panel order

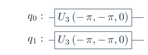
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.008 upper bound



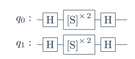
berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 upper bound



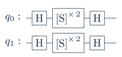
clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 upper bound



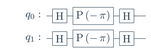
clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 upper bound



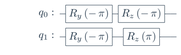
cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.004 upper bound



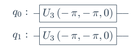
cx_rz_ry

EXACT PHASE E=0 / L=4 / score ≈ 8.008 upper bound



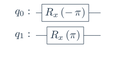
cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



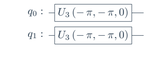
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



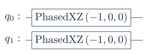
google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 upper bound



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 upper bound



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.000 upper bound



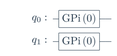
ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



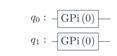
ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



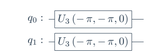
iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.008 upper bound



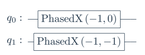
iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



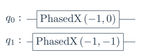
quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.008 upper bound



quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.008 upper bound



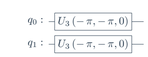
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 upper bound



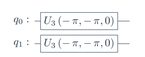
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



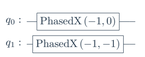
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



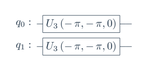
tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.008 upper bound



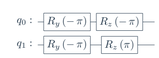
xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



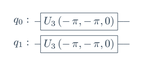
xx_rz_ry

EXACT PHASE E=0 / L=4 / score ≈ 8.008 upper bound



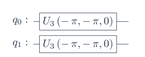
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



XX90

2 qubits / 32 preserved native bases / alphabetical panel order

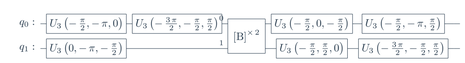
aqt_rxx

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum



berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 24.842 — proved minimum

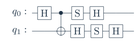


clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved minimum



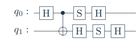
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved minimum



PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved minimum

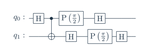
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved minimum



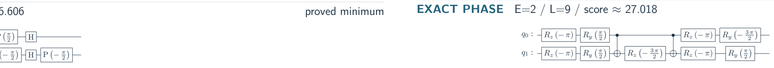
PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=9 / score ≈ 27.018 — upper bound



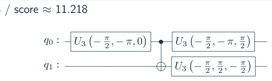
PROJECTIVE E=1 / L=7 / score ≈ 19.214 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=21 / score ≈ 53.842 — upper bound



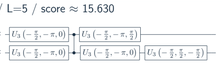
PROJECTIVE E=1 / L=10 / score ≈ 26.020 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved minimum



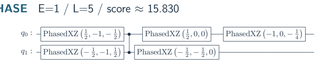
PROJECTIVE E=1 / L=4 / score ≈ 13.224 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.830 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.624 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.820 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=7 / score ≈ 24.242 — proved minimum

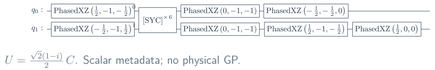


google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — upper bound



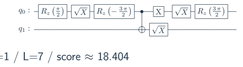
PROJECTIVE E=6 / L=7 / score ≈ 39.842 — upper bound



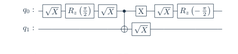
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum



PROJECTIVE E=1 / L=7 / score ≈ 18.404 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=1 / L=12 / score ≈ 29.010 — proved minimum



PROJECTIVE E=1 / L=12 / score ≈ 28.808 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.808 — proved minimum



PROJECTIVE E=1 / L=6 / score ≈ 16.606 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=1 / L=16 / score ≈ 39.638 — proved minimum



ionq_aria_full

EXACT PHASE E=1 / L=10 / score ≈ 26.024 — proved minimum



ionq_forte

EXACT PHASE E=1 / L=11 / score ≈ 28.024 — proved minimum



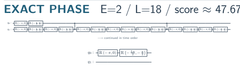
PROJECTIVE E=1 / L=10 / score ≈ 26.022 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=18 / score ≈ 47.672 — upper bound



PROJECTIVE E=1 / L=8 / score ≈ 22.032 — proved minimum



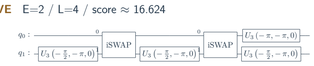
$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved minimum



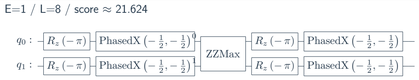
PROJECTIVE E=2 / L=4 / score ≈ 16.624 — proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

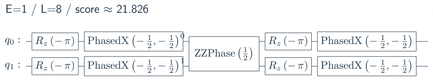
quantinuum_zzmax

EXACT PHASE E=1 / L=8 / score ≈ 21.624 — proved minimum



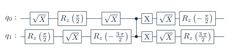
quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved minimum



rigetti_cz

EXACT PHASE E=1 / L=12 / score ≈ 29.010 — proved minimum



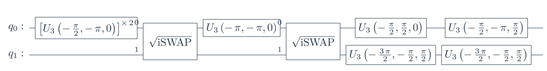
PROJECTIVE E=1 / L=12 / score ≈ 28.808 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved minimum



sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 25.242 — proved minimum



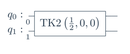
PROJECTIVE E=2 / L=7 / score ≈ 24.442 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved minimum



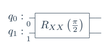
xx90_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



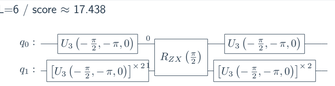
xx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum



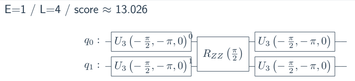
zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.438 — proved minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.026 — proved minimum

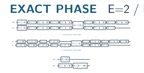


xx_minus_yy (ϕ, β)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.310 — conditional minimum



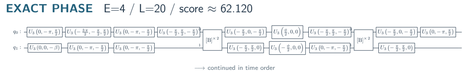
PROJECTIVE E=2 / L=21 / score ≈ 54.866 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=20 / score ≈ 62.120 — upper bound





clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

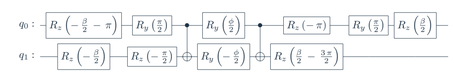
EXACT PHASE E=2 / L=17 / score ≈ 44.222 — conditional minimum





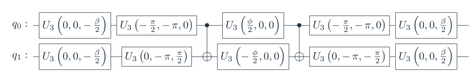
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 29.820 — conditional minimum



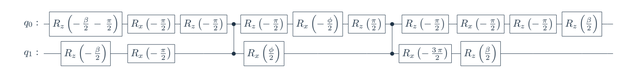
cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — conditional minimum



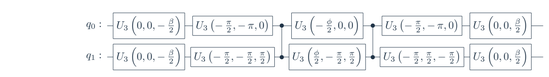
cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.030 — conditional minimum



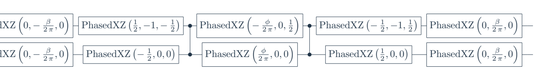
cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — conditional minimum



google_cz

EXACT PHASE E=2 / L=10 / score ≈ 32.642 — conditional minimum



google_fsim

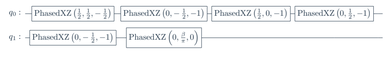
EXACT PHASE E=2 / L=6 / score ≈ 22.365 — upper bound



google_sqrt_iswap

EXACT PHASE E=4 / L=20 / score ≈ 62.241 — upper bound

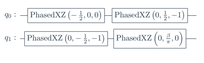




google_syc

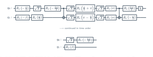
EXACT PHASE E=12 / L=20 / score ≈ 93.441 — upper bound





ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — upper bound



PROJECTIVE E=2 / L=18 / score ≈ 45.822 — conditional minimum



$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — conditional minimum



PROJECTIVE E=2 / L=21 / score ≈ 52.022 — conditional minimum



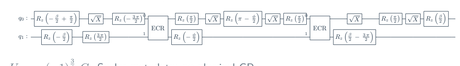
$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 42.424 — conditional minimum



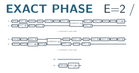
PROJECTIVE E=2 / L=16 / score ≈ 42.222 — conditional minimum



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.805 — upper bound



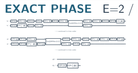
PROJECTIVE E=2 / L=36 / score ≈ 90.405 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 90.001 — upper bound



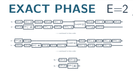
PROJECTIVE E=2 / L=36 / score ≈ 89.601 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=38 / score ≈ 95.601 — upper bound



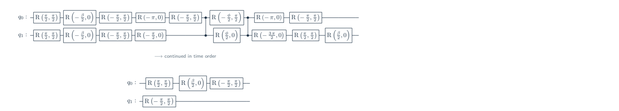
PROJECTIVE E=2 / L=38 / score ≈ 95.201 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 53.880 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.555 — upper bound



PROJECTIVE E=2 / L=12 / score ≈ 35.297 — conditional minimum



$U = \frac{\sqrt[8]{-\mathbb{I}}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 55.723 — conditional minimum



PROJECTIVE E=2 / L=21 / score ≈ 55.123 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — conditional minimum



PROJECTIVE E=2 / L=21 / score ≈ 52.022 — conditional minimum

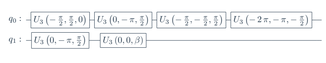


$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.320 — upper bound





sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.720 — upper bound



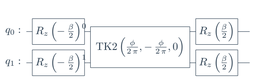
PROJECTIVE E=4 / L=20 / score ≈ 61.320 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 13.875 — conditional minimum



xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — conditional minimum



PROJECTIVE E=2 / L=9 / score ≈ 27.654 — conditional minimum



$U = \eta_{75}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=37 / score ≈ 87.078 — conditional minimum



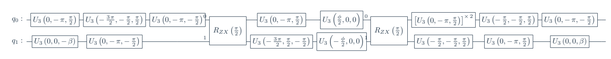
PROJECTIVE E=2 / L=27 / score ≈ 67.258 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

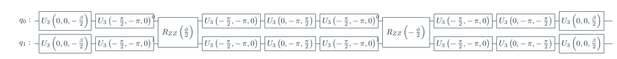
zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — upper bound



zz_u

EXACT PHASE E=2 / L=16 / score ≈ 43.700 — conditional minimum

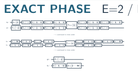


xx_plus_yy (ϕ, β)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.310 — conditional minimum



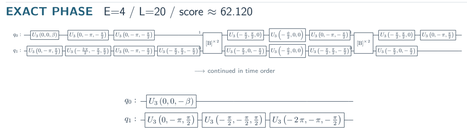
PROJECTIVE E=2 / L=21 / score ≈ 54.866 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=20 / score ≈ 62.120 — upper bound



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

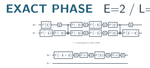
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 58.428 — conditional minimum



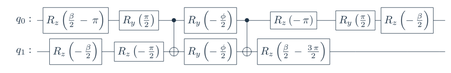
PROJECTIVE E=2 / L=17 / score ≈ 44.222 — conditional minimum



$U = e^{\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

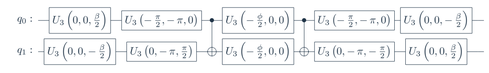
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 29.820 — conditional minimum



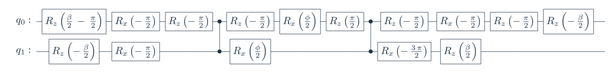
cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — conditional minimum



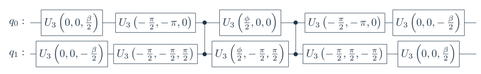
cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.030 — conditional minimum



cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — conditional minimum



google_cz

EXACT PHASE E=2 / L=12 / score ≈ 37.514 — conditional minimum



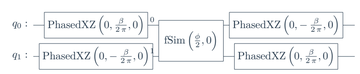
PROJECTIVE E=2 / L=10 / score ≈ 32.642 — conditional minimum



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.349 — conditional minimum



google_sqrt_iswap

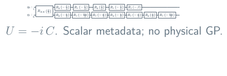
EXACT PHASE E=4 / L=22 / score ≈ 67.114 — upper bound

PROJECTIVE E=4 / L=20 / score ≈ 62.241 — upper bound

$U = -ie^{\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=22 / score ≈ 98.314 — upper bound



PROJECTIVE E=12 / L=20 / score ≈ 93.441 — upper bound

$U = e^{\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — upper bound

PROJECTIVE E=2 / L=18 / score ≈ 45.624 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — conditional minimum

PROJECTIVE E=2 / L=21 / score ≈ 52.022 — conditional minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 42.424 — conditional minimum



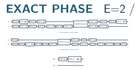
PROJECTIVE E=2 / L=16 / score ≈ 42.222 — conditional minimum



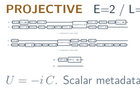
$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.805 — upper bound



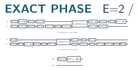
PROJECTIVE E=2 / L=36 / score ≈ 90.405 — upper bound



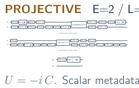
$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 90.001 — upper bound



PROJECTIVE E=2 / L=36 / score ≈ 89.601 — upper bound



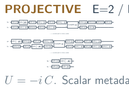
$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=38 / score ≈ 95.601 — upper bound



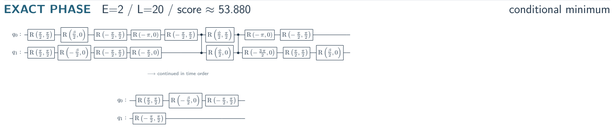
PROJECTIVE E=2 / L=38 / score ≈ 95.201 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

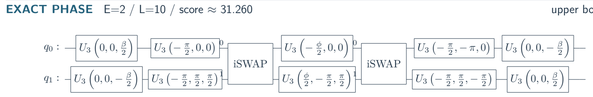
iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 53.880 — conditional minimum



iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.555 — upper bound



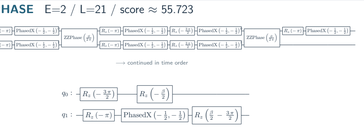
PROJECTIVE E=2 / L=12 / score ≈ 35.297 — conditional minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 55.723 — conditional minimum



rigetti_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — conditional minimum



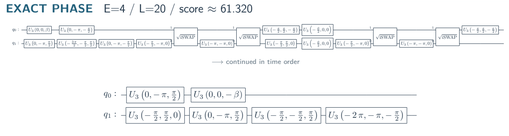
PROJECTIVE E=2 / L=21 / score ≈ 52.022 — conditional minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.320 — upper bound



sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.720 — upper bound



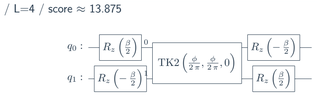
PROJECTIVE E=4 / L=20 / score ≈ 61.320 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

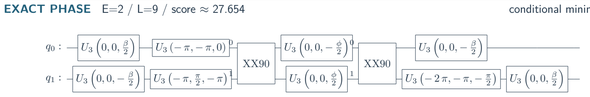
tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 13.875 — conditional minimum



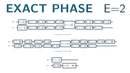
xx90_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — conditional minimum



xx_rz_ry

EXACT PHASE E=2 / L=37 / score ≈ 87.078 — conditional minimum



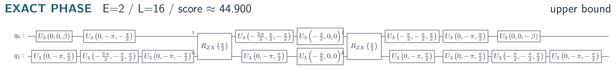
PROJECTIVE E=2 / L=29 / score ≈ 71.662 — upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — upper bound



zz_u

EXACT PHASE E=2 / L=16 / score ≈ 43.700 — conditional minimum



XXPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=0 / score ≈ 4.002 upper bound

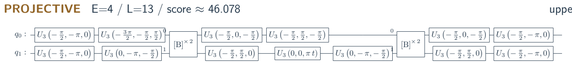
$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

berkeley_b

EXACT PHASE E=4 / L=15 / score ≈ 50.090 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.078 upper bound



$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 30.010 upper bound



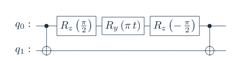
PROJECTIVE E=2 / L=3 / score ≈ 14.002 upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 14.406 upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 upper bound



PROJECTIVE E=2 / L=1 / score ≈ 10.406 upper bound

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

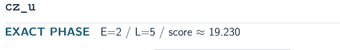
PROJECTIVE E=2 / L=7 / score ≈ 23.214 upper bound



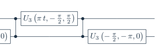
$U = -e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=4 / score ≈ 16.632 upper bound



PROJECTIVE E=2 / L=2 / score ≈ 12.420 upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=15 / score ≈ 50.090 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.078 upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

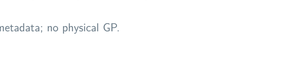
EXACT PHASE E=12 / L=15 / score ≈ 81.290 upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.278 upper bound

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=5 / score ≈ 18.406 upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 upper bound

$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

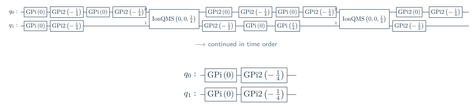
PROJECTIVE E=2 / L=4 / score ≈ 16.004 upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

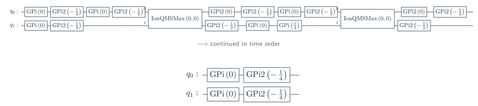
PROJECTIVE E=2 / L=20 / score ≈ 52.852 upper bound



$U = ie^{i\pi t\left(global_{shift} + \frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 52.048 upper bound



$U = ie^{i\pi t\left(global_{shift} + \frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=1 / L=8 / score ≈ 21.818 conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=2 / L=5 / score ≈ 18.820 upper bound



$U = -e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 conditional minimum



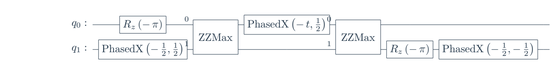
PROJECTIVE E=2 / L=4 / score ≈ 16.424 conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

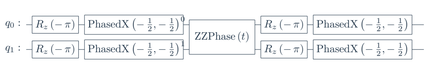
PROJECTIVE E=2 / L=5 / score ≈ 19.016 upper bound



$U = -ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

quantinuum_zzphase

PROJECTIVE E=1 / L=8 / score ≈ 21.626 conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

rigetti_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 upper bound



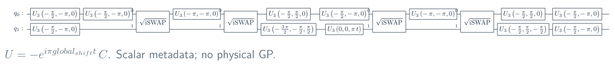
$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.290 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.278 upper bound



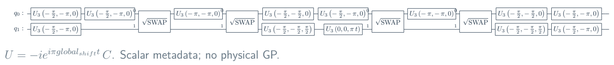
$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 upper bound



PROJECTIVE E=4 / L=13 / score ≈ 45.278 upper bound



$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.006 conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 upper bound



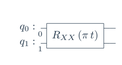
PROJECTIVE E=2 / L=4 / score ≈ 16.424 upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=0 / score ≈ 4.002 upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

zx_u

EXACT PHASE E=1 / L=8 / score ≈ 21.650 conditional minimum



PROJECTIVE E=1 / L=6 / score ≈ 17.238 conditional minimum



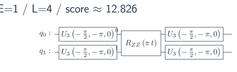
$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 17.238 conditional minimum



PROJECTIVE E=1 / L=4 / score ≈ 12.826 conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

Y

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

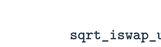
EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -iC$. Scalar metadata; no physical GP.

Y_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum



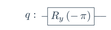
cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum



cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum



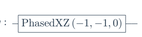
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



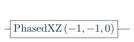
google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



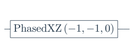
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum



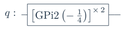
PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum



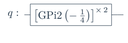
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum



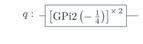
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum



rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

Y_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

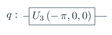
EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

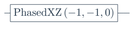
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

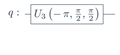
PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



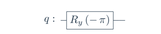
$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



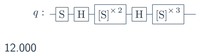
PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

Y_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

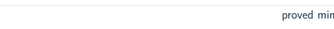
PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

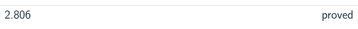


$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved minimum

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

Y_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



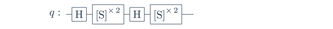
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

YPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 proved minimum



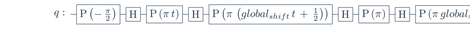
PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved minimum



$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum



$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved minimum

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

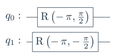
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

YY

2 qubits / 32 preserved native bases / alphabetical panel order

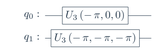
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.408 upper bound



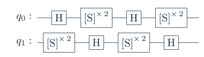
berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



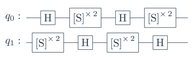
clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 upper bound



clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 upper bound



clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 upper bound

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 upper bound

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

cz_rz_rx

EXACT PHASE E=0 / L=4 / score ≈ 8.008 upper bound

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 upper bound

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 upper bound

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 upper bound

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 upper bound

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 upper bound

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 upper bound

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 upper bound

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.408 upper bound

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.408 upper bound

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 upper bound

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.408 upper bound

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

zz_u

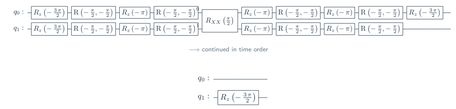
EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

YY90

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=18 / score ≈ 44.254 — proved minimum



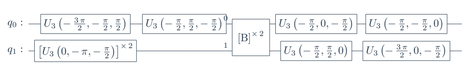
PROJECTIVE E=1 / L=14 / score ≈ 32.000 — proved minimum



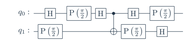
$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 27.248 — proved minimum



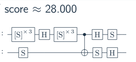
PROJECTIVE E=1 / L=8 / score ≈ 20.808 — proved minimum



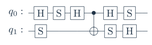
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=19 / score ≈ 42.000 — proved minimum



PROJECTIVE E=1 / L=14 / score ≈ 32.000 — proved minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=8 / score ≈ 21.010 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.808 — proved minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=11 / score ≈ 31.822 — upper bound



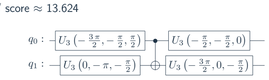
PROJECTIVE E=1 / L=8 / score ≈ 21.616 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.624 — proved minimum



cz_rz_rx

EXACT PHASE E=2 / L=20 / score ≈ 51.040 — upper bound



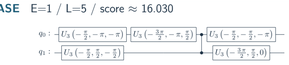
PROJECTIVE E=1 / L=6 / score ≈ 17.212 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 14.024 — proved minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.624 — proved minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.820 — upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 26.648 — proved minimum

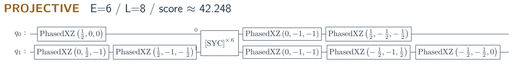


google_syc

EXACT PHASE E=6 / L=9 / score ≈ 44.254 — upper bound



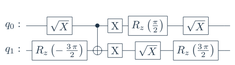
PROJECTIVE E=6 / L=8 / score ≈ 42.248 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=8 / score ≈ 20.606 — proved minimum



ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.404 — proved minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

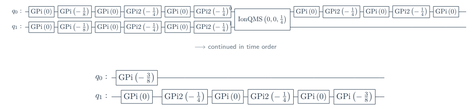
ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.406 — proved minimum



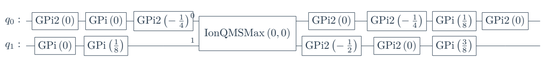
ionq_aria

EXACT PHASE E=1 / L=24 / score ≈ 58.054 — proved minimum



ionq_aria_full

EXACT PHASE E=1 / L=12 / score ≈ 30.828 — proved minimum



ionq_forte

EXACT PHASE E=1 / L=19 / score ≈ 47.240 — proved minimum



PROJECTIVE E=1 / L=18 / score ≈ 45.238 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=26 / score ≈ 65.304 — upper bound



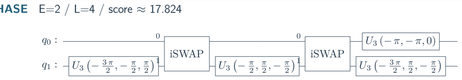
PROJECTIVE E=1 / L=6 / score ≈ 17.624 — proved minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.824 — proved minimum

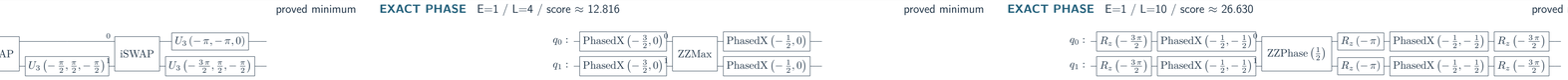


quantinuum_zzmax

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved minimum

quantinuum_zzphase

EXACT PHASE E=1 / L=10 / score ≈ 26.630 — proved minimum

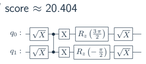


rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.404 — proved minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

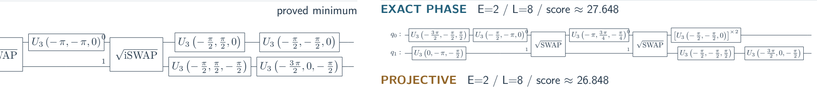
sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — proved minimum



sqrt_swap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.648 — proved minimum



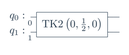
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

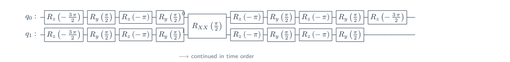
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved minimum



xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved minimum

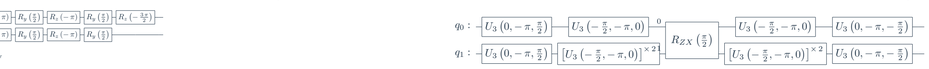


xx_rz_ry

EXACT PHASE E=1 / L=18 / score ≈ 42.638 — proved minimum

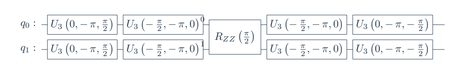
zx_u

EXACT PHASE E=1 / L=10 / score ≈ 26.262 — proved minimum



zz_u

EXACT PHASE E=1 / L=8 / score ≈ 21.850 — proved minimum

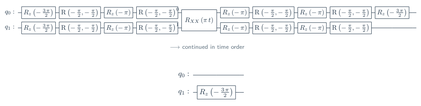


YYPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=18 / score ≈ 44.054 — conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

berkeley_b

EXACT PHASE E=4 / L=19 / score ≈ 58.914 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.902 — upper bound



$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=15 / score ≈ 38.818 — upper bound



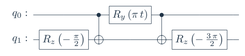
PROJECTIVE E=2 / L=7 / score ≈ 22.810 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 14.406 — upper bound



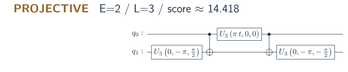
$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — upper bound



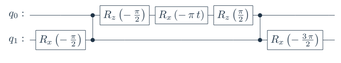
PROJECTIVE E=2 / L=3 / score ≈ 14.418 — upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

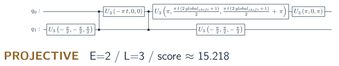
PROJECTIVE E=2 / L=5 / score ≈ 18.810 — upper bound



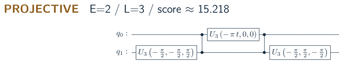
$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 15.218 — upper bound



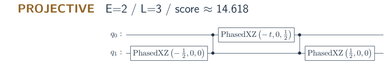
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.618 — upper bound



$U = e^{i\pi t(global_{shift}+1)} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=4 / score ≈ 16.632 — upper bound



PROJECTIVE E=2 / L=2 / score ≈ 12.420 — upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=19 / score ≈ 58.914 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.902 — upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 90.114 — upper bound



PROJECTIVE E=12 / L=17 / score ≈ 86.102 — upper bound

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

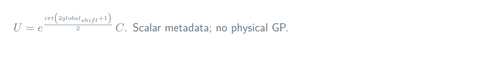
PROJECTIVE E=2 / L=6 / score ≈ 20.408 — upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

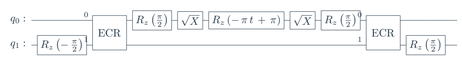
PROJECTIVE E=2 / L=7 / score ≈ 22.004 — upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

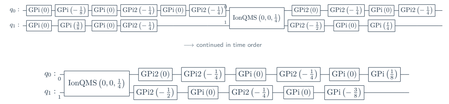
PROJECTIVE E=2 / L=7 / score ≈ 22.810 — upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

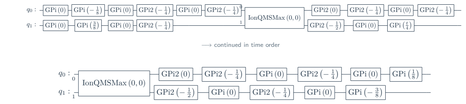
PROJECTIVE E=2 / L=28 / score ≈ 71.268 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=2 / L=28 / score ≈ 70.464 — upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

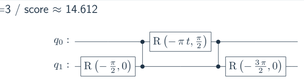
PROJECTIVE E=1 / L=16 / score ≈ 40.234 — conditional minimum



$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

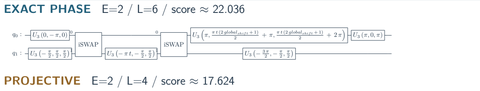
PROJECTIVE E=2 / L=3 / score ≈ 14.612 — upper bound



$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — conditional minimum



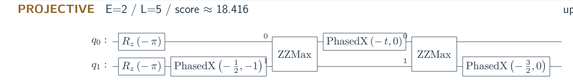
PROJECTIVE E=2 / L=4 / score ≈ 17.624 — conditional minimum



$U = \eta_{101} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=5 / score ≈ 18.416 — upper bound



$U = \eta_{54} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

quantinuum_zzphase

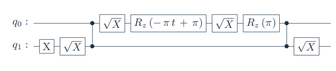
PROJECTIVE E=1 / L=10 / score ≈ 26.430 — conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

rigetti_cz

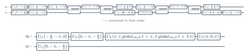
PROJECTIVE E=2 / L=7 / score ≈ 22.004 — upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=4 / L=19 / score ≈ 58.114 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound



$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=19 / score ≈ 58.514 — upper bound



PROJECTIVE E=4 / L=17 / score ≈ 54.102 — upper bound



$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

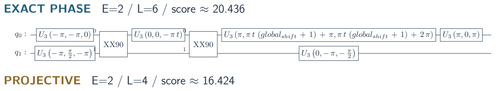
PROJECTIVE E=1 / L=0 / score ≈ 4.006 — conditional minimum



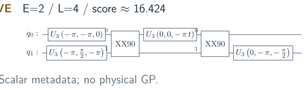
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 20.436 — upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound



$U = \eta_{102} C$. Scalar metadata; no physical GP.

xx_rz_ry

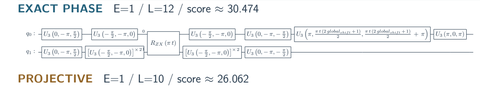
PROJECTIVE E=1 / L=18 / score ≈ 42.438 — conditional minimum



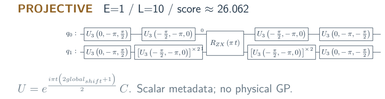
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

zx_u

EXACT PHASE E=1 / L=12 / score ≈ 30.474 — conditional minimum



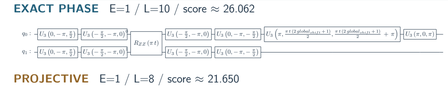
PROJECTIVE E=1 / L=10 / score ≈ 26.062 — conditional minimum



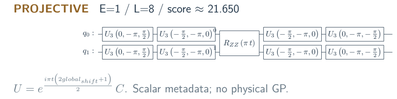
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=10 / score ≈ 26.062 — conditional minimum



PROJECTIVE E=1 / L=8 / score ≈ 21.650 — conditional minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

Z

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved minimum



cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

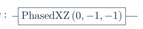
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



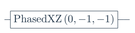
google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



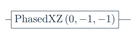
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum



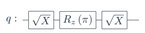
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum



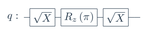
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

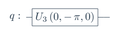
PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



quantinuum_zzmax

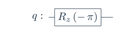
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

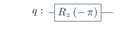
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum

tket_tk2

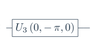
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

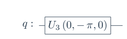
zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved minimum



Z_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



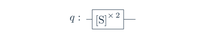
$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{i\frac{\pi}{4}} C$. Scalar metadata; no physical GP.

Z_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.406 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.208 proved minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

Z_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



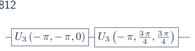
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

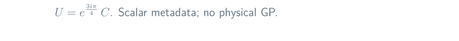
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

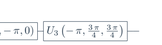
PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Z_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.006 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved minimum

Z_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum

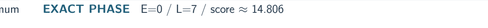


PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.806 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Z_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.406 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved minimum



iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.208 proved minimum



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

rigetti_cz

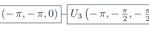
EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum



xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = -iC$. Scalar metadata; no physical GP.

Z_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



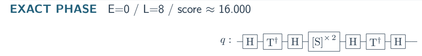
PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

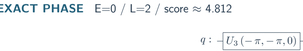
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



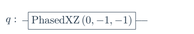
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



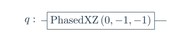
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ZPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved minimum

$q:$ — $\mathrm{P}(\pi\,t\ (global_{shift}+1))$ — H — $\mathrm{P}(\pi)$ — H — $\mathrm{P}(\pi\,global_{shift}\,t)$ — H — $\mathrm{P}(\pi)$ — H —

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $\mathrm{P}(\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ — $\mathrm{PhasedXZ}(1, 2\,global_{shift}\,t, -global_{shift}\,t)$ — $\mathrm{PhasedXZ}(1,0,0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ — $\mathrm{PhasedXZ}(1, 2\,global_{shift}\,t, -global_{shift}\,t)$ — $\mathrm{PhasedXZ}(1,0,0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ — $\mathrm{PhasedXZ}(1, 2\,global_{shift}\,t, -global_{shift}\,t)$ — $\mathrm{PhasedXZ}(1,0,0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ — $\mathrm{PhasedXZ}(1, 2\,global_{shift}\,t, -global_{shift}\,t)$ — $\mathrm{PhasedXZ}(1,0,0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $\mathrm{PhasedXZ}(0,t,0)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$q:$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$q:$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved minimum

$q:$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved minimum

$q:$ — $\mathrm{R}\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ — $\mathrm{R}(\pi\,t, 0)$ — $\mathrm{R}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved minimum

$q:$ — $R_z(\pi\,t)$ —

$U = e^{\frac{i\pi t\left(2global_{shift}+1\right)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ — $U_3(\pi, \pi\,global_{shift}\,t, \pi\,global_{shift}\,t+\pi)$ — $U_3(\pi,0,\pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved minimum

$q:$ — $U_3(0,0,\pi\,t)$ —

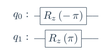
$U = e^{i\pi global_{shift}\,t}\,C$. Scalar metadata; no physical GP.

ZZ

2 qubits / 32 preserved native bases / alphabetical panel order

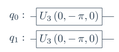
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 upper bound



clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 upper bound



clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 upper bound



cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



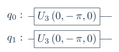
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



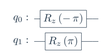
cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



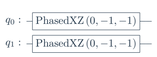
cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



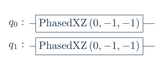
google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



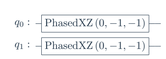
google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



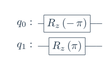
ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



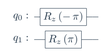
ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 8.808 upper bound



ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 8.808 upper bound



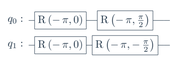
ionq_forte

EXACT PHASE E=0 / L=4 / score ≈ 8.808 upper bound



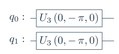
iqm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.416 upper bound



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



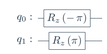
quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



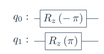
quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



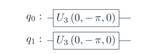
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



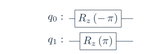
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



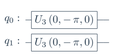
tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



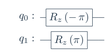
xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



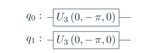
xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 upper bound



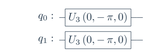
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 upper bound

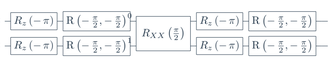


ZZ90

2 qubits / 32 preserved native bases / alphabetical panel order

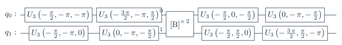
aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved minimum

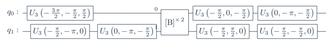


berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — proved minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.442 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



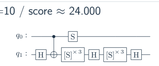
PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved minimum



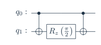
PROJECTIVE E=1 / L=4 / score ≈ 12.404 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.202 — upper bound



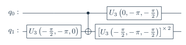
PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved minimum



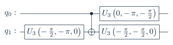
$U = \frac{\sqrt{2}i(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved minimum



PROJECTIVE E=1 / L=3 / score ≈ 10.818 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.222 — upper bound



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=3 / score ≈ 11.018 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved minimum



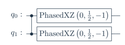
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=3 / score ≈ 10.618 — proved minimum



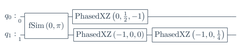
PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=3 / score ≈ 10.622 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.416 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

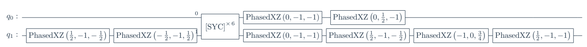
google_sqrt_iswap

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved minimum



google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — upper bound



PROJECTIVE E=6 / L=7 / score ≈ 40.042 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum

ibm_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=7 / score ≈ 18.404 — proved minimum



ionq_aria

EXACT PHASE E=1 / L=8 / score ≈ 22.022 — proved minimum



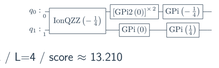
ionq_aria_full

EXACT PHASE E=1 / L=11 / score ≈ 28.626 — proved minimum



ionq_forte

EXACT PHASE E=1 / L=5 / score ≈ 15.212 — proved minimum



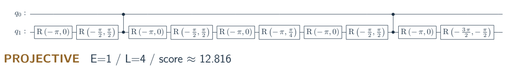
PROJECTIVE E=1 / L=4 / score ≈ 13.210 — proved minimum



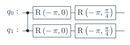
$U = -i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.040 — upper bound



PROJECTIVE E=1 / L=4 / score ≈ 12.816 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved minimum



PROJECTIVE E=2 / L=3 / score ≈ 14.618 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum



rigetti_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.842 — proved minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved minimum



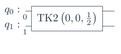
PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved minimum



$U = \eta_l\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved minimum



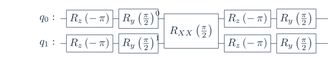
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved minimum



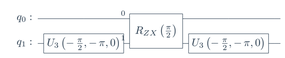
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 21.018 — proved minimum



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.614 — proved minimum



zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum

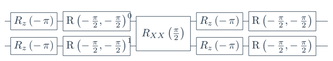


ZZMax

2 qubits / 32 preserved native bases / alphabetical panel order

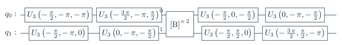
aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved minimum



berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — proved minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.442 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



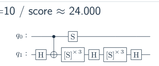
PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.000 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved minimum



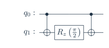
PROJECTIVE E=1 / L=4 / score ≈ 12.404 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.202 — upper bound



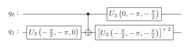
PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved minimum



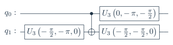
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved minimum



PROJECTIVE E=1 / L=3 / score ≈ 10.818 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.222 — upper bound



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=3 / score ≈ 11.018 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved minimum



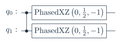
$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=3 / score ≈ 10.618 — proved minimum



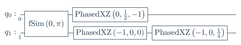
PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved minimum



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=3 / score ≈ 10.622 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.416 — proved minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

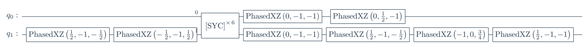
google_sqrt_iswap

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved minimum



google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — upper bound



PROJECTIVE E=6 / L=7 / score ≈ 40.042 — upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum

ibm_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=7 / score ≈ 18.404 — proved minimum



ionq_aria

EXACT PHASE E=1 / L=8 / score ≈ 22.022 — proved minimum



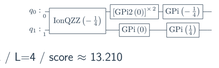
ionq_aria_full

EXACT PHASE E=1 / L=11 / score ≈ 28.626 — proved minimum



ionq_forte

EXACT PHASE E=1 / L=5 / score ≈ 15.212 — proved minimum



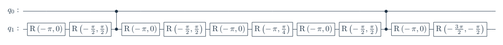
PROJECTIVE E=1 / L=4 / score ≈ 13.210 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.040 — upper bound



PROJECTIVE E=1 / L=4 / score ≈ 12.816 — proved minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved minimum



PROJECTIVE E=2 / L=3 / score ≈ 14.618 — proved minimum



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved minimum



quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum



rigetti_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved minimum



$U = \frac{\sqrt{2}i(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.842 — proved minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved minimum



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved minimum



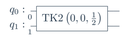
PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved minimum



$U = \eta_l\, C$. Scalar metadata; no physical GP.

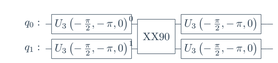
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved minimum



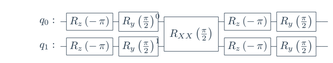
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved minimum



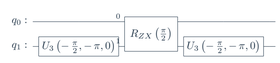
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 21.018 — proved minimum



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.614 — proved minimum



zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved minimum



ZZPhase (α)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — conditional minimum

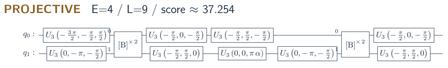


berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.666 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 37.254 — upper bound



$U = -e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — upper bound

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound



$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

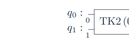
cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — upper bound

PROJECTIVE E=2 / L=1 / score ≈ 10.006 — upper bound



$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — upper bound



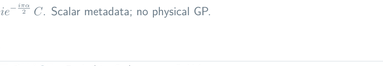
cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — upper bound



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 19.030 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 — upper bound

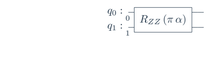


$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.228 — conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.016 — conditional minimum



$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 40.666 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 71.866 — upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — upper bound

$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — upper bound

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=12 / score ≈ 35.636 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.236 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 34.832 — upper bound

PROJECTIVE E=2 / L=12 / score ≈ 34.432 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — conditional minimum

iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — upper bound

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — conditional minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.274 — upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — upper bound

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — upper bound

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — conditional minimum

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — conditional minimum

zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — conditional minimum

zz_u

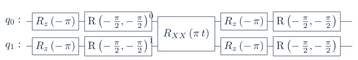
EXACT PHASE E=1 / L=0 / score ≈ 4.002 — conditional minimum

ZZPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

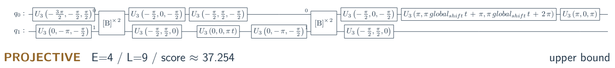
PROJECTIVE E=1 / L=8 / score ≈ 21.626 — conditional minimum



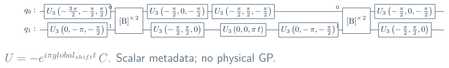
$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

berkeley_b

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 37.254 — upper bound



$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Open: time limit reached
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=9 / score ≈ 26.010 — upper bound



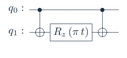
PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.018 — upper bound



PROJECTIVE E=2 / L=1 / score ≈ 10.006 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=2 / L=7 / score ≈ 23.214 — upper bound



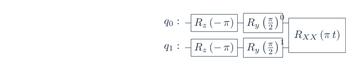
$U = -e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 — upper bound



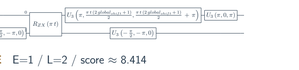
$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — upper bound



PROJECTIVE E=2 / L=3 / score ≈ 14.818 — upper bound



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.028 — conditional minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.016 — conditional minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=4 / L=11 / score ≈ 40.466 — upper bound



PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 71.666 — upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — upper bound

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — upper bound

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound



$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ibm_ecr

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — upper bound

$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria

PROJECTIVE E=2 / L=12 / score ≈ 35.236 — upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_aria_full

PROJECTIVE E=2 / L=12 / score ≈ 34.432 — upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

ionq_forte

PROJECTIVE E=1 / L=0 / score ≈ 4.202 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iqm_cz

PROJECTIVE E=2 / L=5 / score ≈ 18.820 — upper bound

$U = -e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — conditional minimum

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — upper bound

$U = -ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

quantinuum_zzphase

PROJECTIVE E=1 / L=0 / score ≈ 4.002 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

rigetti_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — upper bound

$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

sqrt_iswap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.466 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — upper bound

$U = -ie^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.006 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — upper bound

$U = ie^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 20.818 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Open: construction not found

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 — conditional minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.414 — conditional minimum

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — upper bound

PROJECTIVE E=1 / L=0 / score ≈ 4.002 — upper bound

$U = e^{\frac{i\pi t\left(2 global_{shift} t+1\right)}{2}} C$. Scalar metadata; no physical GP.

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$\eta_{1} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(-1+i)}{2}$

$\eta_{2} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$

$\eta_{3} = \frac{2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$

$\eta_{4} = -\frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$

$\eta_{5} = -\frac{2\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)}{\pi}$

$\eta_{6} = \frac{\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)}{\pi}$

$\eta_{7} = -\pi + \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$

$\eta_{8} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$

$\eta_{9} = -\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$

$\eta_{10} = \pi - \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$

$\eta_{11} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}\right)$

$\eta_{12} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}\right)$

$\eta_{13} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(-1+i)}{2}$

$\eta_{14} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + \pi$

$\eta_{15} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1+i)^{3}}{4}$

$\eta_{16} = \frac{\sqrt{2}(-1+i)e^{i(a+b+c)}}{2}$

$\eta_{17} = \sin(a+b+c) - i\cos(a+b+c)$

$\eta_{18} = \pi - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{19} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + 2\pi$$

$$\eta_{20} = \frac{(1+i)^6 e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}}{8}$$

$$\eta_{21} = -(-1)^{\frac{3}{8}} e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}$$

$$\eta_{22} = \frac{(1+i)^6 e^{\frac{i(\gamma_3+\pi)}{4}}}{8}$$

$$\eta_{23} = -\frac{(1+i)^2 e^{\frac{i(\gamma_3+\pi)}{4}}}{2}$$

$$\eta_{24} = \frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-\sqrt{-2+\sqrt{2}}\right)}{2}$$

$$\eta_{25} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{26} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right) + \pi$$

$$\eta_{27} = \frac{(-1)^{\frac{3}{8}}\left(-\sqrt{\sqrt{2}+2}-\sqrt{-2+\sqrt{2}}\right)}{2}$$

$$\eta_{28} = -\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right) + 2\pi$$

$$\eta_{29} = \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{30} = \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{31} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(1-i)}{2}$$

$$\eta_{32} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{33} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{34} = \frac{\sqrt[4]{-1}\left(-\sqrt{\sqrt{2}+2}+\sqrt{-2+\sqrt{2}}\right)}{2}$$

$$\eta_{35} = \frac{\sqrt[4]{-1}\left(-\sqrt{2-\sqrt{2}}+\sqrt{-2-\sqrt{2}}\right)}{2}$$

$$\eta_{36} = \frac{\sqrt[4]{-1}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{37} = -\frac{(1+i)^2 e^{\frac{i(2\gamma+ibd\alpha+\phi)}{4}}}{2}$$

$$\eta_{38} = (-1)^{\frac{3}{4}} e^{\frac{i(2\gamma+ibd\alpha+\phi)}{4}}$$

$$\eta_{39} = \frac{(1+i)^2 e^{\frac{i(2\gamma+ibd\alpha+\phi)}{4}}}{2}$$

$$\eta_{40} = -\frac{(1+i)^2 e^{\frac{i(ibd\alpha+\phi)}{4}}}{2}$$

$$\eta_{41} = (-1)^{\frac{3}{4}} e^{\frac{i(ibd\alpha+\phi)}{4}}$$

$$\eta_{42} = \frac{(1+i)^2 e^{\frac{i(ibd\alpha+\phi)}{4}}}{2}$$

$$\eta_{43} = -(-1)^{\frac{3}{4}} e^{\frac{i\pi t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}$$

$$\eta_{44} = -\frac{2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{45} = \frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{46} = \frac{-2\pi+2\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)}{\pi}$$

$$\eta_{47} = \frac{\pi-\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)}{\pi}$$

$$\eta_{48} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i(\pi-\phi)}{4}}}{4}$$

$$\eta_{49} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(\pi-\phi)}{4}}}{4}$$

$$\eta_{50} = \frac{\sqrt{2}i(1+i)e^{\frac{i(\phi-2\theta)}{4}}}{2}$$

$$\eta_{51} = \frac{-\frac{\pi t\left(2\mathrm{global}_{\mathrm{shift}}-1\right)}{2}-\pi}{\pi}$$

$$\eta_{52} = \frac{(1-i)^2 e^{i\pi t\left(\mathrm{global}_{\mathrm{shift}}-\frac{1}{2}\right)}}{2}$$

$$\eta_{53} = \frac{\sqrt{2}(1-i)e^{i\pi t\left(\mathrm{global}_{\mathrm{shift}}-\frac{3}{4}\right)}}{2}$$

$$\eta_{54} = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)e^{\frac{i\pi t\left(2\mathrm{global}_{\mathrm{shift}}+1\right)}{2}}}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{55} = \frac{\sqrt{2}i(-1+i)e^{i\pi\left(global_{shift}t-\frac{1}{4}\right)}}{2}$$

$$\eta_{56} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1-i)}{2}$$

$$\eta_{57} = -\frac{\pi t\left(2global_{shift}+1\right)}{\frac{\pi}{2}} - \pi$$

$$\eta_{58} = \frac{(1-i)^2 e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}}{2}$$

$$\eta_{59} = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)e^{i\pi global_{shift}t}}{2}$$

$$\eta_{60} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{61} = -(-1)^{\frac{3}{4}} e^{-\frac{i(4\gamma+\phi)}{4}}$$

$$\eta_{62} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{\pi}{2}$$

$$\eta_{63} = \frac{\frac{\chi}{2}-\gamma-\frac{\phi}{2}-\frac{\zeta}{2}}{\pi}$$

$$\eta_{64} = \frac{-\frac{\chi}{2}-\gamma-\frac{\phi}{2}+\frac{\zeta}{2}}{\pi}$$

$$\eta_{65} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{3\pi}{2}$$

$$\eta_{66} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{67} = \frac{\pi(-2phase_{exponent}+t(2global_{shift}-1))}{2}$$

$$\eta_{68} = \pi\left(axis_{phaseexponent} + z_{exponent} - \frac{1}{2}\right)$$

$$\eta_{69} = \pi\left(axis_{phaseexponent} + z_{exponent} + \frac{1}{2}\right)$$

$$\eta_{70} = ie^{\frac{i\pi(x_{exponent}+z_{exponent})}{2}}$$

$$\eta_{71} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(1+i)}{2}$$

$$\eta_{72} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(-1-i)}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{73} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(1-i)}{2}$$

$$\eta_{74} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(-1+i)}{2}$$

$$\eta_{75} = \frac{\sqrt{2}i(-1+i)e^{-\frac{i(2\phi+\pi)}{4}}}{2}$$

$$\eta_{76} = \frac{\sqrt{2}i(-1+i)e^{\frac{i(2\theta-\pi)}{4}}}{2}$$

$$\eta_{77} = -\pi + \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{78} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(1+i)}{2}$$

$$\eta_{79} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i\pi\left(t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{80} = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)e^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}}{2}$$

$$\eta_{81} = -e^{\frac{i\pi\left(t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)+1\right)}{4}}$$

$$\eta_{82} = \frac{\sqrt{2}(-1+i)e^{i\pi t\left(g\mathrm{lobal}_{\mathrm{shift}}-\frac{1}{2}\right)}}{2}$$

$$\eta_{83} = e^{\frac{i\pi\left(t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)-3t-2\right)}{4}}$$

$$\eta_{84} = -ie^{\frac{i\pi\left(t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)+1\right)}{4}}$$

$$\eta_{85} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i\pi\left(t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{86} = \frac{\sqrt{2}i(1+i)e^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}}{2}$$

$$\eta_{87} = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}}{2}$$

$$\eta_{88} = \frac{(-1)^{\frac{5}{24}}\sqrt{2}(1-i)}{2}$$

$$\eta_{89} = \frac{(-1)^{\frac{5}{24}}\sqrt{2}(1+i)}{2}$$

$$\eta_{90} = \operatorname{atan}\left(\frac{-\sqrt{2}\sqrt{2-\sqrt{2}}+\sqrt{6}\sqrt{2-\sqrt{2}}+\sqrt{2}\sqrt{\sqrt{2}+2}+\sqrt{6}\sqrt{\sqrt{2}+2}}{-\sqrt{2}\sqrt{\sqrt{2}+2}-\sqrt{6}\sqrt{2-\sqrt{2}}-\sqrt{2}\sqrt{2-\sqrt{2}}+\sqrt{6}\sqrt{\sqrt{2}+2}}\right) + \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{91} = \operatorname{atan}\left(\frac{-\sqrt{2}\sqrt{2-\sqrt{2}}+\sqrt{6}\sqrt{2-\sqrt{2}}+\sqrt{2}\sqrt{\sqrt{2}+2}+\sqrt{6}\sqrt{\sqrt{2}+2}}{-\sqrt{2}\sqrt{\sqrt{2}+2}-\sqrt{6}\sqrt{2-\sqrt{2}}-\sqrt{2}\sqrt{2-\sqrt{2}}+\sqrt{6}\sqrt{\sqrt{2}+2}}\right) + 2\pi$$

$$\eta_{92} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1+i)}{2}$$

$$\eta_{93} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{94} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{95} = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)e^{-\frac{i\pi c}{2}}}{2}$$

$$\eta_{96} = -\frac{\frac{\pi(2a+2b+2c+1)}{2}}{\pi} + 2\pi$$

$$\eta_{97} = \frac{\frac{\pi(2a+2b+2c+1)}{4}}{\pi} - \pi$$

$$\eta_{98} = \frac{\sqrt{2}(1-i)^3 e^{-\frac{i\pi(a+b+c+1)}{2}}}{4}$$

$$\eta_{99} = \frac{\sqrt{2}i(1+i)e^{-\frac{i\pi b}{2}}}{2}$$

$$\eta_{100} = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{-\frac{i\pi c}{2}}}{2}$$

$$\eta_{101} = \frac{\sqrt{2}(1-i)e^{\frac{i\pi\left(2t\left(2\mathrm{global_shift}+1\right)-3\right)}{4}}}{2}$$

$$\eta_{102} = \frac{\sqrt{2}i(-1+i)e^{\frac{i\pi\left(2t\left(2\mathrm{global_shift}+1\right)+2t-1\right)}{4}}}{2}$$